THE RIEMANN HYPOTHESIS: AN OBSTRUCTION LEDGER AND THE ARITHMETIC PICK ARCHITECTURE IT SHAPED

NO-GO RESULTS, A COMMON BOUNDARY QUANTIFIER, AND A REDUCTION TO LI–KEIPER POSITIVITY

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Abstract. This paper develops a decade-long obstruction ledger for several spectral, arithmetic, adelic, and positivity-based approaches to the Riemann Hypothesis; identifies a recurring uniform-boundary quantifier (the passage from average/almost-all/interior to specific/uniform/boundary) as their common failure mode; and constructs an arithmetic Pick/Nevanlinna route whose analytic part is closed and whose sole arithmetic input is shown to terminate exactly at Li–Keiper positivity.

The first part is the ledger: the concrete no-gos that closed each route, distilled into a small number of structural walls — arithmetic propagation, wrong-sign positivity, infinite-dimensional Hasse–Minkowski failure, per-place indefiniteness, the cohomological-Frobenius wall, sign-versus-magnitude, and the uniform-norm (rank-one escape) wall — each carrying an explicit logical classification (exact equivalence with RH, framework closure obstruction, or insufficiency theorem for a stated input class). The standard is strict throughout: no assertion is treated as proved because it has been named, defined, or made plausible by computation, and two of our own results are recorded as withdrawn.

The second part is the architecture those walls shaped: the *arithmetic Pick positivity* (ARP-P) chain, fifteen Steps establishing the equivalence ARP-P $\Longleftrightarrow$ RH. Fourteen of the fifteen Steps are closed in the precise sense recorded in the status map (two of them as proved conditional consequences of the open input). The unresolved mathematical input is Step 5, whose internal reduction terminates at one terminal-defect inequality,

$$\delta_N \geq 0 \text{ for all } N \iff \text{RH} \qquad (\Omega_7),$$

equivalently $\lambda_n \geq 0$ for all n. Ω_7 is not our discovery — it is the classical Li criterion; the contribution is that an independently constructed architecture, after its internal difficulty audit, terminates exactly there, and the ledger explains why the major routes explored in this programme repeatedly return to RH-equivalent boundary or uniformity conditions. The paper does not prove RH and does not present the open input as a small gap: being equivalent to RH, it carries the full difficulty of the Hypothesis. A self-contained falsification harness accompanies the paper and confirms the governing laws numerically; by conservation of difficulty, it closes nothing. The whole is offered for rigorous peer scrutiny, in the same discipline of retraction we invite the reader to apply to it.

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1. Prologue

What this paper claims. Four things: an obstruction ledger for the major spectral, arithmetic, adelic, and positivity-based routes explored in this programme, each obstruction carrying an explicit logical classification (§4); a finite-probe signature architecture (the sourced determinant and its Pick kernel); the ARP-P equivalence framework, whose analytic bridge is closed (§7.1); and the localization of that framework's single unresolved arithmetic input at classical Li–Keiper positivity.

What this paper does not claim. It does not prove the Riemann Hypothesis. It does not present Ω_7 as a new criterion for RH — Ω_7 is the classical Li criterion, reached here as the terminal statement of an independently constructed architecture. And it does not describe the unresolved input as a "small gap": since Ω_7 is equivalent to RH, its unresolved status carries the full difficulty of the Hypothesis.

How to audit the paper. The fifteen-step implication chain, the H0–H8 internal analysis of Step 5, and the Ω_1–Ω_7 difficulty-localization chain are three distinct logical levels, kept typed and separate throughout; their dependency and status maps appear in §7.1, and the provenance of every claimed contribution is listed at the end of this Prologue. Two of our own earlier results are recorded as withdrawn, with the errors identified (§3.9); the discipline of retraction is exactly what we invite the reader to apply to this paper.

This paper gathers a research program that lasted years and consumed, by an honest accounting, an enormous quantity of paper, electricity, and patience. The walls came first — many of them — and the proof path came last. That path is short to state and reduces the Hypothesis to a single inequality; but that one inequality, Ω_7, remains open, and it is itself equivalent to RH. The programme is therefore presented in two parts: the negative geography that closed every other route we explored (Part I), and the proof path those walls shaped, with its one open input honestly marked (Part II).

The program began far from operator theory, in the empirical statistics of arithmetic functions: the distribution of $\omega(n)$, the number of distinct prime factors; the fluctuations of the von Mangoldt function in short intervals; the numerics of the explicit formula. For a long time the work was computational. Tens of thousands of processor-hours went into evaluating localized Weil quadratic forms to forty significant digits, into tabulating zeros, into testing one reformulation of positivity after another against the Davenport–Heilbronn function — a function that obeys the same functional equation as ζ but whose zeros stray off the critical line, and which therefore serves as the perfect falsifier: any argument that "proves" the Riemann Hypothesis but does not distinguish ζ from Davenport–Heilbronn is, by that very fact, wrong.

Every route we tried ran into the same wall, wearing different clothes each time. We called it, internally, the *symmetry/positivity wall*: the location of the zeros is a constraint that is global, one-sided (an upper bound, not a two-sided equality), and orthogonal to every symmetry the problem hands you for free. Positivity — the fact that $\Lambda(n) \geq 0$, that the von Mangoldt measure is a genuine measure — was always available, but positivity controls *sign*, and the Riemann Hypothesis is about *magnitude*; the two are orthogonal, and one can prove, in level after level (operator norm, Carleson measure, diagonal of a kernel, derivative of a spectral shift), that no purely positivity-based argument crosses the gap.

The decisive change of viewpoint, when it came, was not a new estimate but a new *object*. A scalar determinant cannot see the difference between a self-adjoint operator and one with complex eigenvalues: its zeros are its zeros, wherever they sit. But a determinant equipped with its *response to finite-rank probes* — a *sourced* determinant — can. Its second variation in the probe directions reconstructs a two-variable kernel, and the negative-square count of that kernel is precisely the number of eigenvalues off the real axis. This single idea — topologize not the scalar determinant but its signature kernel — dissolved the signature-blindness that had defeated every earlier route.

The second idea was equally simple and equally hard-won: the one genuinely divergent direction in the problem — the pole of ζ at $s = 1$, which contributes a trace growing like $\frac{1}{2}(\log P)^2$ — must be removed by *shorting*, the Schur-complement operation, and never by subtraction. The distinction sounds pedantic until one writes down the 2×2 example $\left(\begin{smallmatrix} R & \sqrt{R} \\ \sqrt{R} & 1 \end{smallmatrix}\right) \succeq 0$ and watches subtraction of the divergent corner manufacture a spurious negative eigenvalue while the Schur complement leaves the matrix positive. The divergent pole is harmless if shorted and catastrophic if subtracted; an entire year of false starts is compressed into that sentence.

The third idea was the one that turned a reformulation into a proof, and it is almost embarrassingly classical. Once the pole is shorted, the surviving objects $G_P^\circ(z)$ are *compressed resolvents of genuine self-adjoint operators*, and a compressed self-adjoint resolvent is bounded by $1/|\operatorname{Im} z|$ — uniformly, with the divergence safely quarantined in the discarded corner. A uniformly bounded family of holomorphic functions is a normal family, and a normal family cannot converge to a function with a pole. So if these resolvents converge to the right object *anywhere on an open set*, Vitali's theorem carries the convergence across the strip and forbids the off-axis poles — which are exactly the off-critical zeros. The whole difficulty then collapses to proving the convergence on the half-plane $\operatorname{Re} s > 1$, where the Euler product converges absolutely and Ξ has no zeros to worry about — and that convergence, by conservation of difficulty, is precisely the open input Step 5, not an easy step: the region is friendly, the statement is not.

This program was not carried out in isolation. Over its life it generated a long traditional correspondence — letters, and later a great many emails — with mathematicians whose questions, objections, and corrections shaped every definition in this paper. Several of the sharpest turns recorded below began as a one-line objection in the margin of a draft. The errors that correspondence caught — a sign in a Herglotz coordinate, the false belief that a sourced determinant factors over primes, the crucial difference between defining a compressed resolvent and proving that an object *is* one — are now scars in the proof, healed but visible, and the proof is the better for them.

Status and invitation. We present this paper as a research programme together with a *proposed proof path*, not as a closed proof. The path (Part II, §9) reduces the Hypothesis to a single open input, Step 5, whose internal reduction terminates at the terminal-defect positivity Ω_7: $\delta_N \geq 0$ for all N, equivalently $\lambda_n \geq 0$ for all n. Fourteen of the fifteen Steps are closed in the precise sense recorded in the status map (§7.1): their asserted construction, implication, equivalence, or conditional consequence is proved. The unresolved mathematical input is Step 5, whose internal reduction terminates at Ω_7; and Ω_7 is itself equivalent to RH. We do not claim to have settled the Hypothesis: a claim of that kind must earn belief through the scrutiny of many independent experts, not the confidence of its authors, and the honest state is that one load-bearing inequality remains. We therefore direct attack precisely there — at Ω_7 and at the Ω-chain reduction that isolates it (§9) — and we record, in the same place, why each attempted closure of Ω_7 returns to the structural walls of Part I.

The setting: ζ, the explicit formula, and the Weil criterion. We orient the reader with the objects that recur throughout. The Riemann zeta function is

$$\zeta(s) \;=\; \sum_{n=1}^{\infty} n^{-s} \;=\; \prod_{p}(1-p^{-s})^{-1}, \qquad \operatorname{Re} s > 1,$$

the Euler product being the avatar of unique factorization. The completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire of order 1, satisfies the functional equation $\xi(s) = \xi(1-s)$, and has precisely the nontrivial zeros $\rho = \beta + i\gamma$ of ζ among its zeros. In the coordinates $s = \frac{1}{2} + iz$ the function $\Xi(z) = \xi(\frac{1}{2} + iz)$ is even and real for real z, with zeros at $z_\rho = i(\rho - \frac{1}{2})$; on the critical line $\beta = \frac{1}{2}$ these zeros are real. The Riemann Hypothesis asserts: *all nontrivial zeros ρ of ζ have $\operatorname{Re}\rho = \frac{1}{2}$.*

The prime-number theory of these zeros is encoded in the explicit formula. For a smooth test function h, Weil's form of the explicit formula reads [2]

$$\sum_{\rho} \widehat{h}(\gamma_\rho) \;=\; \widehat{h}\text{-archimedean} \;-\; \sum_{p\,\text{prime}} \sum_{k=1}^{\infty} \frac{\log p}{p^{k/2}}\, h(k\log p),$$

where the left side sums over all nontrivial zeros $\rho = \beta_\rho + i\gamma_\rho$ and the right side carries the prime-power weights $\Lambda(p^k) = \log p$ through h. Weil's criterion states that the Hypothesis is equivalent to

$$\mathcal{W}(h \star \tilde{h}) \;\geq\; 0 \qquad \text{for all } h \in \mathcal{S}(\mathbb{R}),$$

where $\tilde{h}(x) = \overline{h(-x)}$ and $\star$ is convolution. The form $Q = M_{\text{zeros}} - M_{\text{arith}}$ is the finite-dimensional truncation of this criterion, and the sign of Q — or equivalently of the Pick kernel of $-\Xi'/\Xi$ — is the object around which every route in this paper revolves.

The Davenport–Heilbronn function L_{DH} [45, 4] is a linear combination of Dirichlet L-functions to an imaginary character modulo 5; it satisfies the same functional equation as ζ with the same Γ-factor structure, but its zeros include ones off the critical line. It serves as the falsifier throughout: any inequality proved for ζ that also holds for L_{DH} does not separate on-line from off-line zeros and therefore cannot imply RH. The Davenport–Heilbronn test is mandatory.

Original constructions and results of this paper

To the author's knowledge, the following constructions, reductions, and assembled arguments are original to this paper. The claim is one of provenance of the present manuscript, not a substitute for an exhaustive priority determination.

New constructions introduced here. The finite sourced channels and the sourced-determinant (signature-kernel) architecture (Definition 9.5 and §1); the ARP-P formulation itself (Definition 9.10); the reference-whitened terminal defect (Definition 9.73); the H0–H8 hierarchy of Step 5; and the Ω-reduction architecture (§9.8). The ARP-P architecture as a whole is introduced here and assembled from stated classical analytic interfaces.

Results proved or derived here. The determinant serialization of RH (Theorem 9.38) — proved here; the one-point jet serialization (Theorem 9.50) — proved here; the fixed-N band closure (Theorem 9.63) — proved here, with classical Cauchy–Pick machinery; the whitened terminal theorem (Theorem 9.71) — proved here; the Li–Cayley dictionary (Theorem 9.111) — derived here, with the classical Li criterion used as terminal input; the quantitative H6 witness (Theorem 9.32) — proved here; the de Bruijn–Newman cascade characterization (Theorem 9.158) — proved here, with the de Bruijn and Rodgers–Tao inputs stated; the envelope-bound reduction

(Proposition 9.147) — derived here; and the stated insufficiency theorems (Theorem 9.130, Proposition 9.143, Proposition 9.120, the last via Bombieri–Lagarias, cited) — proved here.

Classical inputs and external results, used with citation and never reproved: the Weil explicit formula and criterion [2]; Nevanlinna–Pick interpolation and commutant lifting [78]; Montel/Vitali normality; the Li criterion [81], Keiper [83], and Bombieri–Lagarias [82]; de Bruijn [73] and Rodgers–Tao; Connes–Consani [74, 75]; and Guth–Maynard [51].

Ω_7 is not our discovery; the independently constructed ARP-P architecture is shown, after its internal difficulty audit, to terminate at Li positivity. That termination, together with the obstruction ledger explaining why the major routes explored in this programme repeatedly return to RH-equivalent boundary or uniformity conditions, is the claimed contribution. A construction can be original even when the terminal statement it reaches is classical; that is exactly the situation here, and we state it rather than let the reader discover it.

Part 1. The walls, and the path they shaped

2. A long road to a short bridge: routes, analogies, and the walls that shaped this one

The argument of this paper is short, and short arguments for old problems are always the residue of a much larger and mostly negative body of work. It would be misleading to present the bridge without the country it crosses. This section recounts, in mathematical terms and without any private notation, the principal routes that were attempted over a multi-year programme, the precise obstructions that closed each of them, and the way those obstructions, taken together, pointed toward the construction given here. We believe this history is not decoration: every wall below is a theorem about what *cannot* work, and the present route is shaped, negatively, by all of them. The complete record of the programme—a decade of routes, computations, and retractions, documented in full detail—is publicly archived at `https://github.com/dtrejopizzo/riemann`.

2.1. The journey in one paragraph.

The programme began empirically, in the statistics of arithmetic functions — the distribution of the number of prime factors $\omega(n)$, the fluctuations of $\Lambda(n)$ in short intervals, the numerics of the explicit formula evaluated to high precision. That phase produced a great deal of structure and no theorem; its lasting product was a single instrument, a *localized Weil quadratic form* $Q = M_{\text{zeros}} - M_{\text{arith}}$ whose sign encodes the Riemann Hypothesis and which became the pivot of everything that followed. From the detector grew a sequence of theoretical attacks: a reduction of the Hypothesis to a Lindelöf-type bound on a cross sum; an inverse-spectral, Hilbert–Pólya search for a self-adjoint operator with the zeros as spectrum; an adelic, trace-formula framing in the spirit of the spectral programme discussed in §5; and finally a sequence of Hodge-theoretic and Frobenius-realization attempts modelled on the function-field proof. Each reached a precise wall. The walls turned out to be the *same* wall in different clothing, and understanding that single obstruction sharply enough to state it as one theorem is what eventually suggested attacking it from below the critical strip, in the region of absolute convergence, rather than on the critical line itself. That is the move this paper makes.

What follows is a longer account. We give it because the negative results are, in our experience, the most useful part of the programme for anyone contemplating an operator approach: each is a small theorem about what *cannot* succeed, and together they delineate, sharply, the narrow corridor in which a proof can live. We have rendered every obstruction as mathematics, stripped of any private notation, so that the reader may check the diagnosis independently of the history.

2.2. **The empirical phase: the ω-class fingerprint and the localized detector.** The programme opened with a long computational study of the arithmetic of $\omega(n) = \#\{p : p \mid n\}$ and of the von Mangoldt weights $\Lambda(n)$, evaluated against the explicit formula to high precision. The goal was to find a *reproducible discriminator*: a quantity, computable from finite data, positive for ζ and negative for a function with the same functional equation but zeros off the line — the Davenport–Heilbronn function L_{DH} [45, 4], which serves throughout as the falsifier. The instrument that survived this phase is a *localized Weil quadratic form*. For a test function h supported near a scale, Weil's explicit formula [2] writes

$$\mathcal{W}(h) \;=\; \underbrace{\sum_{\rho} \widehat{h}(\gamma_\rho)}_{M_{\mathrm{zeros}}(h)} \;-\; \underbrace{\Big(\widehat{h}\text{-archimedean} \;-\; \sum_{p,k} \tfrac{\log p}{p^{k/2}}\, h(k \log p)\Big)}_{M_{\mathrm{arith}}(h)}, \tag{1}$$

and Weil's criterion [2] is that the Hypothesis is equivalent to $\mathcal{W}(h \star \tilde{h}) \geq 0$ for all h. Localizing the test function to a band of scales produces a finite Hermitian form $Q = M_{\mathrm{zeros}} - M_{\mathrm{arith}}$ whose least eigenvalue is the object the whole programme orbits. This is, in spirit, the band-limited and archimedean-refined positivity of the adelic programme [19]; the empirical phase tabulated it, found rich ω-stratified structure in its fluctuations, and — honestly — did not beat the convexity bound. Its lasting product was the detector Q and a precise sense of *why* it is hard to read, which we now record as a sequence of no-gos.

The form Q depends on two ingredients that enter very differently. The *zero side* $M_{\mathrm{zeros}}(h) = \sum_\rho \widehat{h}(\gamma_\rho)$ is a linear functional of the zero multiset $\{\rho\}$: each zero $\rho = \frac{1}{2} + i\gamma_\rho$ (assuming RH) contributes through the Fourier transform $\widehat{h}$ of the test function. The *arithmetic side* $M_{\mathrm{arith}}(h)$ is a sum over primes: it carries the von Mangoldt weights $\Lambda(n) = \log p$ at prime powers $n = p^k$ and the archimedean term from the Γ-factor. The Hypothesis says these two sides are in equilibrium: the spectral energy is at least as large as the arithmetic energy. The empirical phase measured this balance to high precision across many test functions, finding rich structure stratified by $\omega(n)$: integers with more prime factors (higher ω-class) contribute differently to the balance, and the statistics of each class carry a distinct signature in the Fourier domain. This structure was real and reproducible; it did not, however, beat the convexity bound or give an unconditional proof.

The empirical work was conducted with the contemporary picture of ζ on the critical line in view: the random-matrix statistics of the zeros and of the value distribution [36, 63], the multiplicative-chaos description of the large values [61, 62, 64, 65], the universality phenomenon [66, 67], and the hyperbolicity/interlacing circle of ideas [58, 59, 60]. Each is a source of *generic* information about the zeros; none, by the principle isolated below, detects the measure-zero event of an off-line zero, and that negative recognition is what the empirical phase contributed. The generic behaviour of the zeros — their statistics, their pair correlation, their universality — is fully consistent with an off-line zero at any displacement $\delta > 0$, however small. The Hypothesis is a statement about the *individual* configuration, not the generic one, and the empirical phase eventually made this distinction sharp.

Each of the routes sketched in §2.1 closed at a precise, checkable obstruction — not at a vague "difficulty" but at a statement one can write down and verify. In our experience these obstructions are the most valuable product of the programme, because each is a small theorem about what *cannot* work, and together they carve out the narrow corridor in which a proof can live. We therefore give them two full sections of their own. Section 3 records the *no-go results*: concrete attempts, each with the mathematics that closed it. Section 4 distils these into the small number of *structural walls* — the obstructions that reappear, in disguise, on every front — and attaches to each a precise logical classification (defined at the head of §4): exact equivalence with RH, framework closure obstruction, or insufficiency theorem for a stated input class. Informally we shall sometimes say a statement is "RH-strength"; this phrase carries no logical content beyond the explicit classification attached to the statement. The reader interested only in the proof may skip to §9; but the proof was designed, negatively, against exactly the walls of §4, and we believe it is best understood as the one route that threads them.

3. The no-go results

This section is a ledger of negative results. We have rendered each as mathematics, stripped of any private notation, and grouped them by the route on which they arose. The format is uniform: we state the obstruction as a numbered **No-go**, give the computation or theorem that established it with the displays spaced for reading, and end with the one-line reason it closes the route. Nothing here is folklore to us: every entry was, at the time, a live hope that a specific calculation extinguished. We include two results that we later *withdrew* (§3.9), because the discipline of retraction is part of the true history and is exactly the discipline we invite the reader to apply to the present paper.

3.1. **The detector and the limits of generic information.** The instrument that survived the empirical phase is the localized Weil form $Q = M_{\text{zeros}} - M_{\text{arith}}$ of (1). Before turning to the routes built on it, we record four no-gos that concern the *reading* of Q itself: ways in which a quantity extracted from it looks like progress but is, on inspection, either positive by construction or blind to the only event that matters.

No-go 3.1 (one-sided positivity is automatic)**.** The least eigenvalue of the *zero-side* Gram matrix is non-negative for trivial reasons and carries no arithmetic content.

Write the zero-side block as $G_{\text{zeros}}[i,j] = \sum_\rho \widehat{h}_i(\gamma_\rho)\,\overline{\widehat{h}_j(\gamma_\rho)} = \langle v_i, v_j\rangle$ with $v_i = (\widehat{h}_i(\gamma_\rho))_\rho$. This is a Gram matrix, hence

$$G_{\text{zeros}} \succeq 0, \qquad \text{so} \qquad \lambda_{\min}(G_{\text{zeros}}) \geq 0$$

identically, for every L-function and every sampling of test functions. The measured value $\lambda_{\min} \approx 0.866 > 0$ for ζ is therefore not evidence of anything: it is the positivity of an inner-product matrix. Only the full form Q, which subtracts the arithmetic block M_{arith}, can distinguish on-line from off-line; the one-sided form is structurally blind.

No-go 3.2 (the scalar minimum is a tautology under the Hypothesis)**.** Using $\lambda_{\min}(Q)$ as the discriminator between ζ and the falsifier separates them by the very quantity one is trying to control.

Under the Hypothesis $\lambda_{\min}(Q) = 0$ exactly (it is pinned at the positivity floor), and an off-line zero at distance δ forces $\lambda_{\min}(Q) \leq -c(\delta) < 0$. Thus

$$\lambda_{\min}(Q) < 0 \quad \Longleftrightarrow \quad \text{an off-line zero exists,}$$

so "separate ζ from L_{DH} by $\lambda_{\min}$" is identical to "decide whether there is an off-line zero." The scalar is a faithful *detector* but a circular *criterion*; the genuine non-tautological content lives one level down, in the per-prime anatomy $\{R_p\}$ of the arithmetic block, not in any single eigenvalue.

No-go 3.3 (the global Cauchy–Schwarz bound is vacuous)**.** Controlling the localized form by a global second-moment bound destroys the localization that carries the information.

For the energy $E(F) = \sum_\gamma |F(\gamma)|^2 - \bar{\rho}\int |F|^2$ (with $\bar{\rho}$ the mean density of the zero ordinates) the natural global estimate uses $\|D_T\|_{L^2}^2 = V(d,T) \sim T\log T$ and Cauchy–Schwarz to bound the cross term. The resulting inequality is true but *scale-blind*: it bounds a band-limited quantity by its total mass, and an off-line zero perturbs the band, not the mass. The honest reading is that the information sits in a Carleson-type measure on the localized scales [70], which the global bound integrates away.

No-go 3.4 (topology of the zero set does not see the shift)**.** Persistent homology of the zero configuration cannot detect a displacement off the critical line.

A zero moved off the line by δ changes pairwise distances by $O(\delta^2)$, to which the 0-dimensional persistence barcode is insensitive at any realistic sample size; against a length-matched control L_{DH} is not an outlier, and the 1-dimensional signal is statistically null ($p \approx 0.48$). The instrument measures a *generic* feature of the configuration, and an off-line zero is a measure-zero perturbation of it — the recurring reason, isolated as Theorem 3.2 below, that no generic statistic decides the Hypothesis.

3.2. **First arc: the ω-class route and the parity barrier.** The first theoretical attack tried to extract the Hypothesis, or at least the Lindelöf bound, from the measured statistics of the classes of n by the number of prime factors $\omega(n)$. Six attempts closed, each for a structural reason; we group them because the lesson is uniform.

No-go 3.5 (the cross-sum reduction is self-strengthening)**.** Bounding the ω-class cross sum $B(N)$ by N^ε would yield Lindelöf, but the bound is strictly stronger than its target.

The reduction has the shape "$B(N) \le N^\varepsilon \Rightarrow$ Lindelöf," yet establishing $B(N) \le N^\varepsilon$ already requires Lindelöf-type cancellation in the underlying Dirichlet polynomial. One is asked to prove a statement that *presupposes* the conclusion; the implication is sound but the hypothesis is the harder fact. This is the first appearance of a pattern — a true implication whose antecedent is RH-strength — that will recur as the master quantifier of §4.

No-go 3.6 (exponent extrapolation does not cross the threshold)**.** The measured large-values exponent does not fall below the level a large-values/Lindelöf argument requires, and the trend is not monotone.

In the large-values approach a Dirichlet polynomial of length N must have its set of σ-large points controlled by an exponent below the Bourgain–Demeter–Guth / Guth–Maynard barrier [49, 51]. The fitted exponent of $B(N)$ over the accessible range $N \le 1.5 \times 10^5$ is

$$\alpha_B \approx 0.42 \qquad \text{against the required} \qquad \alpha_B < 0.31.$$

Four data points do not cross this gap, and — decisively — the trend *reverses* at the transition from two to three prime factors ($k = 2 \to 3$), so even the sign of the slope is unstable. A numerical extrapolation through a non-monotone trend is not a proof.

No-go 3.7 (unconditional anti-correlation vanishes)**.** The cross-class anti-correlation that is visible at peaks is, unconditionally, zero on average.

By the Montgomery–Vaughan mean-value theorem [31], the mean square of the relevant Dirichlet polynomial is the diagonal term plus a genuinely smaller off-diagonal, so the unconditional cross-covariance between ω-classes is ≈ 0. The anti-correlation that the empirical phase saw is a *conditional*, peak-localized phenomenon; averaged unconditionally it is not there, so it cannot be promoted to a statement about the zeros.

No-go 3.8 (the parity barrier forbids the separation)**.** No sieve weight can distinguish $\omega(n)$ even from $\omega(n)$ odd, which is what the route needs.

The separation the route requires is exactly a control on $\omega(n)$ mod 2, i.e. on the Liouville-type sign $(-1)^{\omega(n)}$. The parity problem of sieve theory [31] is the theorem that standard sieve weights cannot distinguish integers with an even number of prime factors from those with an odd number. This is not a gap in our sieve; it is a proved limitation of all of them.

No-go 3.9 (hyperbolicity reformulations are equivalences)**.** The Li, Jensen/hyperbolicity, de Bruijn–Newman and Nyman–Beurling criteria are restatements of the Hypothesis, not approximations to it.

Each of

$$\lambda_n > 0 \ (\forall n), \qquad T_k\big(\xi\text{-Taylor}\big) \geq 0 \ (\forall k), \qquad \Lambda_{\mathrm{dBN}} \leq 0, \qquad d_N \downarrow 0$$

[42, 58, 41, 56, 39] is *equivalent* to the Hypothesis. Verifying any of them for finitely many n, k, N is consistent with an off-line zero of arbitrarily small displacement; the finite verification is not asymptotic evidence, and a full proof of any one of them is a full proof of RH. We return to the precise sense of "equivalent" in §3.8.

No-go 3.10 (the phantom phase transition)**.** A claimed geometric transition in the ω-class data did not survive re-verification.

A reported transition at $N_c \approx 1.18 \times 10^5$ dissolved under canonical re-computation; the honest residue is the much weaker statement that ζ is somewhat more "constructive" at peaks, with no transition. We record it as a no-go against ourselves: an apparent structure that did not replicate.

The arc closes on one lesson. Every quantity in it is either a restatement of the Hypothesis (No-go 3.5, 3.9) or a *generic* statistic that the parity and convexity barriers render blind to the measure-zero event of an off-line zero (No-go 3.6, 3.7, 3.8, 3.10).

3.3. **Second arc: lower bounds on the off-line displacement.** The second, and longest, attack sought to bound the real-part displacement

$$b_j \ := \ \beta_j - \tfrac{1}{2}, \qquad \rho_j = \beta_j + i\gamma_j \ \text{ a hypothetical off-line zero,}$$

from below, since $b_j = 0$ for all j *is* the Hypothesis: a uniform lower bound $b_j \geq c > 0$ that is impossible to satisfy would prove there is no off-line zero. We surveyed some fifteen classical mechanisms. Not one of them produces a lower bound; each produces, instead, a bound in the *wrong direction*, in the *wrong region*, of the *wrong aggregate type*, or one that is openly *circular*. The uniformity of this four-way failure is itself a discovery, and we record it at the end of the subsection as a meta-theorem. We give the mathematics for the most instructive cases.

No-go 3.11 (zero-free regions give the wrong direction)**.** The Korobov–Vinogradov zero-free region bounds b_j from *above*, never from below.

The sharpest known zero-free region [47, 52] states

$$\zeta(s) \neq 0 \qquad \text{for} \qquad \operatorname{Re} s \ > \ 1 - \frac{c}{(\log|t|)^{2/3}(\log\log|t|)^{1/3}}.$$

A zero $\rho_j = \beta_j + i\gamma_j$ therefore obeys $\beta_j \le 1 - c/\big((\log\gamma_j)^{2/3}(\log\log\gamma_j)^{1/3}\big)$, which gives

$$b_j \;=\; \beta_j - \tfrac{1}{2} \;\le\; \tfrac{1}{2} - \frac{c}{(\log\gamma_j)^{2/3}(\log\log\gamma_j)^{1/3}}.$$

This is an *upper* bound on b_j: the zero-free region excludes a strip near $\operatorname{Re} s = 1$, so it limits how far *right* a zero can be. The quantity a proof of RH needs is a lower bound on b_j that forces it to be zero; the KV region gives an upper bound telling us it is less than $\frac{1}{2}$, which every zero automatically satisfies. The inequality points the wrong way, and no sharpening of the constant c changes its direction: however much the exponent in $\operatorname{Re} s > 1 - c(\log|t|)^{-\alpha}$ were improved (short of a positive-width zero-free strip, itself an open problem far beyond RH-strength methods), the direction of the bound on b_j would remain "$b_j \le (\frac{1}{2} - \delta)$ for some $\delta > 0$", never "$b_j \ge c_0 > 0$." The mechanism is structurally unable to produce the required direction.

An instructive way to see this: the zero-free region is proved by bounding $|\zeta(s)|$ away from zero for $\operatorname{Re} s$ close to 1, using the Euler product in the region of (near-)absolute convergence. The Euler product converges absolutely only for $\operatorname{Re} s > 1$ and is largest there; the bound weakens as $\operatorname{Re} s \downarrow \frac{1}{2}$. So the technique has a natural built-in direction: it says "the function cannot vanish near the boundary of the absolute-convergence region," which is an upper bound on how far right a zero can be, not a lower bound on how far left.

No-go 3.12 (the de Bruijn–Newman inequality is inverted)**.** The relation to the de Bruijn–Newman constant also yields an upper bound, and the only known input makes it vacuous.

Under the relevant factorization the heat-flow inequality reads $\Lambda_{\mathrm{dBN}} \ge \frac{1}{2}b_j^2$ [56, 39, 38, 57], whence

$$b_j \;\le\; \sqrt{2\,\Lambda_{\mathrm{dBN}}}.$$

To turn this into a lower bound on b_j one would need a *lower* bound on Λ_{dBN}; but the established direction is an upper bound — classically $\Lambda_{\mathrm{dBN}} \le 1/2$, sharpened to $\Lambda_{\mathrm{dBN}} \le 0.2$ [57] — which gives only the trivial $b_j \le \sqrt{0.4} < 1$. The inequality is structurally inverted relative to the goal.

No-go 3.13 (the Nyman–Beurling distance is circular)**.** Bounding the Nyman–Beurling distance from below in terms of b_j is equivalent to bounding b_j.

The Nyman–Beurling criterion [40, 41] reformulates the Hypothesis as $d_N \downarrow 0$, where d_N is the $L^2(0,1)$ distance from the indicator of $(0,1)$ to the span of N dilations of the fractional-part function. A bound $d_N > F(b_j)$ with $F > 0$ when $b_j > 0$ would indeed exclude off-line zeros — but proving such an F exists is exactly proving the displacement is bounded below. The criterion relocates the problem into a Hilbert space without making it easier; $d_N > F(b_j)$ and "$b_j \ge c$" are the same statement.

No-go 3.14 (Turán inequalities are tautological under the working hypothesis)**.** For the relevant factor the Turán inequalities hold for every $b_j > 0$, so they cannot exclude any.

Writing $\xi = \xi_0 \cdot P$ under the working factorization, the Turán inequalities $T_k^{(\xi_0)} \ge 0$ for the Laguerre–Pólya factor ξ_0 are a *consequence* of ξ_0 lying in the Laguerre–Pólya class, which the factorization already assumes. They are satisfied identically, for every admissible $b_j > 0$, and so impose no constraint on the displacement.

No-go 3.15 (linear forms in logarithms address the wrong objects)**.** Baker's theorem constrains algebraic numbers; the ordinates γ_j carry no algebraic structure.

Baker's bounds for linear forms in logarithms [31] apply to logarithms of *algebraic* numbers. The ordinates γ_j are (conjecturally) transcendental and are not known to satisfy any algebraic relation; the hypotheses of Baker's theorem simply do not engage them. This is a category error, not a quantitative shortfall.

No-go 3.16 (pair correlation carries no off-line signal)**.** Montgomery's pair correlation describes spacings between on-line zeros and is silent about off-line ones.

The pair-correlation statistic [26, 55, 37] measures the distribution of differences $\gamma - \gamma'$ of zeros *on* the line. An off-line zero is not a point of this statistic; it contributes nothing to the correlation of the on-line ordinates and therefore cannot be detected by any distortion of it.

The remaining mechanisms — Selberg's density theorem and the modern zero-density and large-values machinery [46, 48, 50, 53], the Levinson–Conrey positive-proportion results [28, 29], Littlewood's oscillation theorems, the large sieve, de Branges chain conditions, the Selberg trace formula [72], and Weil positivity itself — each fail for one of the same four reasons. We expand on the density case since it is the source of the most persistent optimism, and on the large-sieve case since it illustrates how even a powerful quantitative tool misses the target.

No-go 3.17 (density theorems count zeros, they do not locate them)**.** Selberg-type zero-density estimates bound the number of zeros with $\operatorname{Re}\rho \geq \sigma$ in aggregate; they impose no lower bound on the displacement $b_j = \beta_j - \frac{1}{2}$ of any individual zero.

The standard density results take the form

$$N(\sigma, T) \ := \ \#\{\rho = \beta + i\gamma : \beta \geq \sigma,\ |\gamma| \leq T\} \ \leq \ C\, T^{A(1-\sigma)} (\log T)^B,$$

for absolute constants A, B, C [46, 48]. This bounds how many zeros can lie to the right of $\operatorname{Re} s = \sigma$, simultaneously, in a height window $[0, T]$. But a single zero with $\beta = \frac{1}{2} + e^{-10^{100}}$ contributes $+1$ to $N(\sigma, T)$ for every $\sigma \leq \frac{1}{2} + e^{-10^{100}}$ and T large enough — that is, it contributes almost nothing to any density count. No density estimate of this form can prevent one zero from having a very small displacement. The aggregate bound and the individual bound are independent questions; all density theorems address the former and leave the latter wide open.

No-go 3.18 (positive proportions do not constrain off-line shifts)**.** Knowing that a positive proportion of zeros lie on the critical line is orthogonal to knowing whether any zero lies off it.

The Levinson–Conrey results [28, 29] establish that $\geq 41\%$ of the nontrivial zeros of ζ satisfy $\operatorname{Re}\rho = \frac{1}{2}$. This is a theorem about the on-line zeros; it says nothing about the displacement of any zero that is not on the line. A hypothetical single off-line zero with $b_j = 10^{-1000}$ is perfectly compatible with 99.99% of zeros being exactly on the line. The Hypothesis (all zeros on the line) and the proportion results (some zeros on the line) are of entirely different logical types. Counting is not locating.

We summarize the arc as a single statement.

Theorem 3.1 (four-way failure of the lower-bound programme)**.** *Every classical mechanism that takes arithmetic data convergent in* $\operatorname{Re} s > 1$ *and attempts a positive lower bound* $b_j \geq c$ *on the off-line displacement produces instead one of:* (i) *an upper bound on* b_j *(zero-free regions, de Bruijn–Newman);* (ii) *a statement about the wrong region* $\operatorname{Re} s \approx 1$ *rather than the strip interior (Deuring–Heilbronn repulsion, blocked by a Paley–Wiener barrier);* (iii) *an aggregate count rather than an individual displacement (density theorems, positive proportions, pair correlation); or* (iv) *a reformulation equivalent to the bound sought (Nyman–Beurling, Turán, Li/Jensen). No combination yields a lower bound.*

We give an additional proof sketch for failure mode (ii), the Paley–Wiener barrier, since it is the least familiar and the most instructive.

The Paley–Wiener barrier. The Deuring–Heilbronn repulsion theorem bounds the distance between a (hypothetical) exceptional zero $\rho_0 = \beta_0 + i\gamma_0$ close to $\operatorname{Re} s = 1$ and another zero $\rho = \beta + i\gamma$: they cannot coexist unless $|\rho - \rho_0|$ is very small. This information lives near $\operatorname{Re} s = 1$. To transfer it into the strip $\frac{1}{2} < \operatorname{Re} s < 1$, one would need a function $F(s)$ analytic on the strip that (a) is controlled by arithmetic at $\operatorname{Re} s > 1$ and (b) recovers the desired information at $\operatorname{Re} s = \frac{1}{2}$. Such a function would have its values on the line $\operatorname{Re} s = \frac{1}{2}$ determined by its values on the line $\operatorname{Re} s = 1 + \delta$ via Cauchy's integral formula, which requires pointwise control in a wide strip. Paley–Wiener-type estimates say that the arithmetic input (supported on the positive integers) has Mellin transform decaying like $e^{-c\pi|t|}$ in the imaginary direction $|t| \to \infty$; the analytic continuation to $\operatorname{Re} s < 1$ gains factors that amplify exponentially, erasing the decay before reaching $\operatorname{Re} s = \frac{1}{2}$. The arithmetic function has insufficient smoothness to "reach" across the strip by analytic continuation, and there is no elliptic-type maximum principle that would make the value on one vertical line control the value on another without this smoothness.

The Deuring–Heilbronn repulsion is therefore an excellent theorem about the region near $\operatorname{Re} s = 1$; it is simply the wrong region for a proof of RH. No sharpening of the repulsion estimate changes this.

The single cause beneath all four failures is that *arithmetic information available for* $\operatorname{Re} s > 1$ *does not propagate into the critical strip* $\frac{1}{2} < \operatorname{Re} s < 1$: the Euler product, the only source of arithmetic data, converges only to the right of the strip. This is so even though numerical verification of the Hypothesis now reaches astronomical height [54] — precisely because verification at a height is not propagation into the strip. Theorem 3.1 is the first and most robust face of the arithmetic-propagation wall (Wall 4.2).

3.4. Third arc: local–global factorization and the adelic eigenvectors.

A natural response to the propagation barrier is to localize: factor the Weil form over places, $Q = Q_\infty + \sum_p Q_p$, prove each local piece $Q_p \succeq 0$, and conclude global positivity by a Hasse–Minkowski principle. This is the strategy by which the function-field analogue is proved. Over $\mathbb{Q}$ it fails at two independent points, and the companion spectral construction — realizing the zeros as eigenvalues on the idèle class space — fails at a third.

No-go 3.19 (the zero side does not factor over primes)**.** The spectral part of the Weil form is a global functional of the zero multiset and admits no per-prime decomposition.

The arithmetic side $M_{\text{arith}}(h) = \sum_{p,k} \frac{\log p}{p^{k/2}} h(k \log p)$ factors over primes by construction. The spectral side does not:

$$M_{\text{zeros}}(h) \;=\; \sum_{\rho} \widehat{h}(\gamma_\rho)$$

depends on the zeros, which are *global* objects — no Euler factor $(1 - p^{-s})^{-1}$ ever vanishes, so no prime "owns" a zero and there is no place p at which to attach a local piece Q_p of the spectral form. The factorization $Q = Q_\infty + \sum_p Q_p$ that the strategy requires does not exist on the side that carries the information; there is no analogue of "Frobenius at p" for $\zeta/\mathbb{Q}$.

No-go 3.20 (no Hasse–Minkowski in infinite dimension)**.** Even granting local positivity at every place, the local-to-global passage is invalid for Hermitian forms in infinite dimension.

The classical Hasse–Minkowski theorem — a quadratic form over $\mathbb{Q}$ is isotropic iff it is isotropic over every completion — is a theorem about *finite*-dimensional forms. The Weil form is infinite-dimensional, and there is no Hasse principle for the signature of an infinite-dimensional Hermitian form: a form positive at every place can be globally indefinite. So even if No-go 3.19 were somehow circumvented, the inference "$Q_v \succeq 0\ \forall v \Rightarrow Q \succeq 0$" would remain unjustified.

No-go 3.21 (the adelic eigenvectors are not in the space)**.** The explicit functions that would realize off-line zeros as eigenvalues on the idèle class space are not square-integrable.

The companion construction seeks an operator H_C on $L^2(C_{\mathbb{Q}})$, $C_{\mathbb{Q}}$ the idèle class space, with the off-line zeros among its eigenvalues, witnessed by the explicit vectors $f_j(x) = x^{\,b_j + i\gamma_j}$. But, with the multiplicative Haar measure $d^{\times}x = dx/|x|$,

$$\|f_j\|^2_{L^2(C_{\mathbb{Q}})} \;=\; \int |x|^{2b_j}\,\frac{dx}{|x|} \;=\; \int x^{2b_j - 1}\,dx \;=\; \infty \qquad (b_j > 0).$$

The putative eigenvectors are distributions, not L^2-vectors, so they cannot witness a complex eigenvalue; the quadratic form $Q(f_j, f_j)$ is formally divergent. The construction produces the right symbols and the wrong space.

No-go 3.22 (one obstruction in two languages)**.** The L^2 failure and its adelic-character dual are the same statement, so passing between them is not progress.

The condition $x^{b_j+i\gamma_j} \notin L^2(C_{\mathbb{Q}})$ is identical to the statement that $|x|^{b_j+i\gamma_j}$ is a *non-unitary* quasi-character (its exponent has nonzero real part b_j). An adelic Grössencharakter argument that tries to exclude the non-unitary part of the spectrum without first resolving the displacement merely re-encounters the same obstruction in dual language. The two phrasings are provably equivalent, and neither is the easier one.

3.5. **Fourth arc: the sign of the curvature, and a failed anticommutation.** A further family of attempts tried to make the Hypothesis a *variational* or *symmetry* statement: to exhibit the critical line as the minimizer of an energy, or to force a spectral invariant to vanish by an anticommuting involution. They closed because the geometry of ξ on the critical line has, under the Hypothesis, the *opposite sign* to the one these arguments need — and we could compute that sign exactly.

No-go 3.23 (the critical line is a maximum, not a minimum)**.** Phrasing $b_j = 0$ as the global minimum of an arithmetic energy is backwards: the natural form is locally maximized on the line.

Consider $\sigma \mapsto \log|\xi(\sigma + i\gamma)|$ along a horizontal line through a zero ordinate γ. The second-variation calculation gives, for the localized form built from a detector test function h whose transform vanishes at the sampled zeros ρ_j (so the first bracket term below drops),

$$\delta_b^2\, Q(f,f) \;=\; -8 \sum_j c_j^2 \Big[\mathrm{Re}\big(\widehat{h}''(\rho_j)\,\overline{\widehat{h}(\rho_j)}\big) + |\widehat{h}'(\rho_j)|^2\Big] \;\leq\; 0,$$

a manifestly non-positive quantity. The critical line is thus a local *maximum* of Q in the transverse direction, not a minimum. Any argument that hopes to pin $b_j = 0$ by exhibiting it as an energy minimum is working against the curvature, not with it.

No-go 3.24 (the curvature sign is positive, so concavity criteria are false)**.** $\sigma \mapsto \log|\xi(\sigma + i\gamma)|$ is strictly convex at the line; a sufficiency criterion phrased as concavity cannot hold.

Under the Hypothesis the curvature of $\log|\xi|$ across the critical line is computed from the Hadamard product: writing $\log\xi(s) = B_0 + \sum_\rho [\log(1-s/\rho) + s/\rho]$ and differentiating twice in σ at $s = \frac{1}{2} + i\gamma$,

$$\frac{\partial^2}{\partial\sigma^2} \log\left|\xi\left(\tfrac{1}{2}+\sigma+i\gamma\right)\right|\Big|_{\sigma=0} \;=\; C(\gamma) \;:=\; \sum_\rho \frac{1}{(\gamma-\gamma_\rho)^2} \;>\; 0.$$

Every term in the sum is positive, so $C(\gamma) > 0$ identically under the Hypothesis. Moreover, RH is *equivalent* to $C(\gamma) > 0$ for all γ: if an off-line zero ρ^* exists, the contribution of its conjugate pair adds a negative term to $C(\gamma^*)$ that can dominate. A criterion of the form "$\sigma \mapsto \log|\xi(\sigma+i\gamma)|$ is σ-concave $\Rightarrow$ RH" is therefore false on its face: the function is strictly σ-convex everywhere under RH. Several otherwise-plausible variational formulations got this direction exactly backwards.

No-go 3.25 (the arithmetic sector cannot be controlled without the zeros)**.** The inequality that would close the variational route is itself equivalent to the Hypothesis.

Write the second derivative $\partial_t^2 \log|\zeta(\frac{1}{2}+it)|$ as the sum of three sectors:

$$G_{\mathrm{poly}}(t) \;=\; \frac{2\left(\frac{1}{4}-t^2\right)}{\left(t^2+\frac{1}{4}\right)^2}, \tag{2}$$

$$G_\Gamma(t) \;=\; -\tfrac{1}{4}\,\mathrm{Re}\,\psi^{(1)}\!\left(\tfrac{1}{4}+\tfrac{it}{2}\right), \tag{3}$$

$$G_\zeta(t) \;=\; \partial_t^2 \log\left|\zeta\left(\tfrac{1}{2}+it\right)\right|, \tag{4}$$

where $\psi^{(1)}$ is the trigamma function and $G_{\mathrm{ana}} = G_{\mathrm{poly}} + G_\Gamma$ is the analytic sector (known). The sector G_{poly} arises from the polynomial prefactor of ξ and is readily computed; G_Γ arises from the Γ factor and is a smooth, controlled function. What remains, G_ζ, is the arithmetic sector.

The variational route would close if $G_\zeta(t) > -G_{\mathrm{ana}}(t)$ held unconditionally. However, G_ζ can be expressed via the explicit formula as

$$G_\zeta(t) \;=\; \sum_\rho \mathrm{Re}\,\frac{1}{\left(\frac{1}{2}-\beta_\rho+i(t-\gamma_\rho)\right)^2} \;+\; (\text{archimedean tail}),$$

which depends directly on the imaginary and *real* parts of the zeros. Controlling G_ζ at $\sigma = \frac{1}{2}$ therefore requires knowing where the zeros sit — which is exactly the Hypothesis. The inequality $G_\zeta > -G_{\mathrm{ana}}$ is equivalent to RH, not a consequence of the Euler product or the functional equation alone.

No-go 3.26 (symmetry is not antisymmetry)**.** The functional-equation involution preserves the relevant operator instead of negating it, so the η-invariant argument does not run.

The attempt forces a spectral η-invariant to vanish by exhibiting an involution J with $JT_0J^{-1} = -T_0$, which would pair eigenvalues $\lambda \leftrightarrow -\lambda$ and annihilate η. But the only RH-free involution available, the functional-equation symmetry $s \mapsto 1-s$, satisfies

$$J\,T_0\,J^{-1} \;=\; T_0 \qquad \text{(symmetry)}, \qquad \text{not} \qquad J\,T_0\,J^{-1} = -T_0 \text{ (antisymmetry)}.$$

It preserves the spectrum rather than reflecting it, so it gives no constraint on η. The required Fourier anticommutation $Q(f,g) + Q(\widehat{f},\widehat{g}) = 0$ remains open and is, again, of RH strength.

3.6. **Fifth arc: reflection positivity, false discriminants, and the magnitude–phase theorem.** The most instructive failures concern *positivity per se* — the attempt to prove $Q \succeq 0$ directly, rather than to bound a displacement. Each taught us where the positivity does *not* come from, and together they culminate in the theorem that, more than any other, reorganized our thinking and shaped the present paper.

No-go 3.27 (per-place reflection positivity fails)**.** The natural per-prime kernel of an Osterwalder–Schrader attack is indefinite, so there is no local positive structure to reconstruct from.

A reflection-positivity (Osterwalder–Schrader) strategy would factor the form into per-prime kernels and reconstruct a positive Hamiltonian. Computing the kernel at a prime p gives, up to positive constants,

$$G_p(r) \;\propto\; \frac{2\log p}{p\,|1-z|^2}\big(\sqrt{p}\,\cos(r\log p)-1\big).$$

This is *indefinite*. At $r=0$ the bracket is $\sqrt{p}-1>0$, but at $r=\pi/\log p$ it is $-\sqrt{p}-1<0$:

$$G_p(0)>0, \qquad G_p\Big(\tfrac{\pi}{\log p}\Big)<0,$$

verified directly at $p=2,3,5,7$. There is no per-place positive structure, so reflection positivity cannot be reconstructed place by place; positivity, if it holds, is a genuinely *cross-place* phenomenon.

No-go 3.28 (“free” positivity from reconstruction was retracted)**.** The claim that a positive Hamiltonian follows automatically from an OS reconstruction does not survive, because the two-point function is oscillatory.

We record a retraction against ourselves. The assertion that $H \geq 0$ follows “for free” from Osterwalder–Schrader reconstruction is invalid: the reconstruction theorem requires a reflection-*decreasing* (exponentially decaying) two-point function, whereas the relevant two-point function here is *oscillatory*. The hypothesis of the theorem is not met, and the conclusion does not follow. (This is one of two results we later withdrew; see §3.9.)

No-go 3.29 (multiplicativity does not control the sign)**.** The Euler product is not what makes the localized form positive: a reproducible discriminant fails to separate ζ from smooth non-multiplicative controls.

A natural thesis is that *multiplicativity* (the Euler product), rather than the location of the zeros, controls the sign of the localized form. We tested it with a pre-registered, reproducible computation of the Carleson constant

$$C \;=\; \lambda_{\max}\big(A^{-1/2}P\,A^{-1/2}\big),$$

comparing ζ against smooth, deliberately *non*-multiplicative controls. The result refutes the thesis:

$$C_\zeta \approx 1.98, \qquad C_{\text{flat}} \approx 2.44, \quad C_{\text{loglin}} \approx 2.11, \quad C_{\log n} \approx 2.21.$$

Not only does C fail to separate ζ from the non-multiplicative controls — ζ is not even the most coherent. The apparent separation seen in a cruder earlier test was an artifact of *smoothness* in $\log n$, not of multiplicativity, because the smooth envelope of the localized form integrates out the fine, individual-$\log p$ correlation in which multiplicativity actually lives. What the experiment leaves behind is useful: a relative discriminant (fixed A, varying the multiplicative structure) that cancels the normalization and is reusable. What it removes is the hope that "positivity = multiplicativity."

All of this points in one direction, which we now state as the organizing principle of the whole programme.

Theorem 3.2 (the sign principle: positivity, not magnitude, is the distinguishing input)**.** *Let $\mathcal{W}$ be the localized Weil form* (1)*. The structural feature that separates ζ from an off-line-zero falsifier is the* sign *of its von Mangoldt data, not its size. The Riemann function has $\Lambda(n) \geq 0$ for all n; the Davenport–Heilbronn function L_{DH}, which has zeros off the critical line, has a* sign-indefinite *von Mangoldt function — for instance $\Lambda_{\mathrm{DH}}(3) < 0$ while $\Lambda(3) = \log 3 > 0$. This positivity is a sign property, and it is exactly the property that makes the finite operators A_P self-adjoint on a genuine Hilbert space (not a Pontryagin space). Any proof must use this sign structure; a functional that reads only the magnitudes cannot certify it.*

We state this as the organizing principle because it is a *hard no-go for an entire class of arguments*: the property the proof needs — that the von Mangoldt Hamiltonian $H_P \succeq 0$, hence that A_P is self-adjoint — is a statement about the *signs* of the weights $\Lambda(n)$, and no functional of the sizes $\{|\Lambda(n)|\}$ alone can certify it. Flipping signs of the Euler data (as in the sign-indefinite Λ_{DH}) destroys the positivity $H_P \succeq 0$ and moves zeros off the line while leaving each individual magnitude available; so an ℓ^p norm, a spectral radius, a Carleson constant, or a second moment of $\{|\Lambda(n)|\}$ cannot see the one feature that decides the Hypothesis. (We do not claim the magnitudes of ζ and L_{DH} coincide — they do not, since Λ_{DH} is a genuinely different sequence; the point is the sharper one that positivity is a sign property no size functional accesses.)

The theorem identifies, by elimination, what the distinguishing feature *must* be: the genuine *positivity* $\Lambda(n) \geq 0$ (equivalently, that Λ_{DH} changes sign). This is not a quantitative difference; it is a structural one. It is precisely the input the present paper isolates as (Real) (§9): the positivity $\Lambda(n) \geq 0$ makes the finite operators A_P self-adjoint on a genuine Hilbert space, giving their resolvents the bound $\|(A_P - z)^{-1}\| \leq 1/|\operatorname{Im} z|$ that is the seed of the Vitali normal-family argument. That bound fails for L_{DH} (§9.7), exactly as the theorem predicts.

3.7. **Sixth arc: the cohomological frontier.** The last family of attempts modelled the desired proof on the function-field case, where the analogue of the Hypothesis is a theorem [3, 24]: there a Frobenius endomorphism acts on a cohomology carrying a positive, *gapped* intersection pairing, and the eigenvalue bound $|\alpha| = \sqrt{q}$ follows from positivity. Transporting this to ζ requires a Frobenius-type operator on a $\mathbb{Z}$-cohomology that is (i) a morphism of the relevant Hodge structure — complex-*linear* with respect to the natural complex structure J — and (ii) an isometry of a positive-definite, spectrally *gapped* polarization. We tested both ingredients directly on the window of ζ. Both fail, and they fail as two faces of one absence; before reaching them we record the averaging obstruction that motivated the test.

No-go 3.30 (off-line secular growth survives every average)**.** Averaging the band statistic over scale does not remove the off-line contribution; it relocates the Hypothesis rather than resolving it.

The last open structural quantity reduces to the boundedness in scale λ of an intrinsic band width $b_{\mathrm{bulk}}(\lambda)$. A zero $\rho = \beta + i\gamma$ contributes to it a term

$$\lambda^{2\beta-1}\,\lambda^{2i\gamma}.$$

On the critical line $\beta = \frac{1}{2}$ this is the pure oscillation $\lambda^{2i\gamma}$, which a Cesàro average in λ sends to 0. *Off* the line it is the secular factor $\lambda^{2\beta-1}$ with $2\beta - 1 > 0$, a positive power that no average removes; and zero-density estimates bound the imaginary parts and the counts of zeros but *not* their real parts. Hence Cesàro convergence of $b_{\text{bulk}}(\lambda)$ is equivalent to RH, not neutral with respect to it. Numerically, ζ's band is flat in λ (value ≈ 0.13, slope ≈ 0) while the falsifier L_{DH}, which has off-line zeros, grows like a positive power (log–log slope ≈ 0.43) — exactly the predicted off-line signature. The average is a faithful detector, not a proof step.

No-go 3.31 (the natural polarization is present but gapless)**.** A genuine quaternionic Hodge–Riemann positivity exists for ζ, but its spectrum decays exponentially to zero with no gap, so it cannot force an eigenvalue bound.

Pursuing the cohomological lever we found that the finite-window Weil matrix of ζ does carry an intrinsic *quaternionic Hodge–Riemann polarization* (a Deninger-type structure with $J^2 = -1$, modelled on Deninger's conjectured cohomology [25]): the form is positive semidefinite on the $+i$-eigenspace V_+ of J for ζ, while the falsifier L_{DH} and a symmetry-matched random control are both *indefinite* there. This is the most intrinsic Weil positivity the programme produced, and it is the geometrically *right* object: it is not a tautological consequence of the spectral side, it is not imposed by hand, and it genuinely separates ζ from indefinite controls.

But it is gapless. The spectrum of the polarization form on V_+ is an exponential cascade,

$$\mu_k \;\sim\; e^{-cL} \;\downarrow\; 0 \qquad (c \approx 1.40 \;\log_{10}\; \text{per step, independent of } \lambda).$$

The ratio $\mu_{\max}/\mu_{\min} \to \infty$ as the window grows; there is no uniform positive lower bound on μ_k. The sharpest reformulation of the positivity is $\mu_{\max} \le 1$ (the largest eigenvalue ratio of the Hodge–Riemann polarization is at most 1); ζ sits *exactly* at the marginal value $\mu_{\max} = 1$, while L_{DH} sits at $\mu_{\max} \in [6, 13]$ depending on the window. This is a faithful, sharp detector: $\mu_{\max} \le 1$ is equivalent in content to saying no zero is off the line. But it is a statement that must be *proved*; it is not a consequence of a gap.

The function-field mechanism gives eigenvalue bounds because the polarization there is gapped: the Hodge– Riemann bilinear relations supply a uniform lower bound $\mu_k \ge c_0 > 0$ for all k, and the operator isometry forces all eigenvalues of Frobenius onto $|\alpha| = \sqrt{q}$. Over $\operatorname{Spec}\mathbb{Z}$, the corresponding polarization has no gap: $\mu_k \to 0$ exponentially, and the forcing mechanism collapses. Every faithful reformulation we found — the Cesàro band statistic, the Hodge–Riemann positivity on V_+, the Euler-maximum bound $\mu_{\max} \le 1$ — is a marginal, gapless detector. Giving the margin a uniform gap is *equivalent to proving RH*; it cannot be supplied by numerics, and the construction that would supply it geometrically is the unbuilt $\operatorname{Spec}\mathbb{Z}$-cohomology of the next no-go.

No-go 3.32 (the natural Frobenius is of the wrong type)**.** The scaling generator anticommutes with the quaternionic complex structure, so any Frobenius built from it is complex-antilinear — not a Hodge morphism, which must be complex-linear.

The operator whose eigenvalues are (formally) the imaginary parts of the zeros of ζ is the log-scaling generator $D_{\log}$. With J the quaternionic complex structure of No-go 3.31 (satisfying $J^2 = -1$, J anti-Hermitian), a direct operator computation gives

$$\{D_{\log}, J\} \;=\; D_{\log}J + JD_{\log} \;=\; 0, \qquad \frac{\|[D_{\log}, J]\|}{\|D_{\log}\|} \;=\; 2.000$$

(exact; the symbol is odd in the scale index). Since $\{D_{\log}, J\} = 0$, the operator $D_{\log}$ anticommutes with J, i.e. $D_{\log}J = -JD_{\log}$. This means that for any J-eigenstate v with $Jv = iv$

we have $J(D_{\log}v) = -D_{\log}(Jv) = -iD_{\log}v$: the image $D_{\log}v$ has J-eigenvalue $-i$, not $+i$. Therefore $D_{\log}$ maps V_+ to V_-, not to itself; it is complex-*anti*linear with respect to J. A Hodge morphism (a "Frobenius" for the Hodge–Riemann structure) must be complex-*linear*, i.e. must satisfy $[F, J] = 0$. The operator $D_{\log}$ is of the wrong type.

For contrast, the same test applied to a real elliptic curve over a finite field $\mathbb{F}_q$ returns $[F_q, J] = 0$ exactly, a gapped polarization with positive definite Hodge–Riemann form, and $|\alpha| = \sqrt{q}$ to machine precision for every eigenvalue of the Frobenius F_q. The ζ window has neither ingredient: the type is wrong and the polarization is gapless. The two failures are not independent — they are two faces of a single missing object.

The object that would close this route is a J-linear Frobenius that is an isometry of a gapped, positive-definite polarization on a cohomology of $\operatorname{Spec}\mathbb{Z}$ — the unbuilt object of the adelic programme [20]. Its absence is not an estimate one is missing; it is a *construction* that does not yet exist. Recognizing that the obstruction is one of construction rather than of estimation is what turned us, finally, away from realizing the zeros as a Frobenius *spectrum* and toward realizing them as *poles* forbidden by a normal-family continuation — the route of this paper.

3.8. **Circular reformulations.** Several attractive criteria are not approximations to the Hypothesis but exact restatements of it. We list them together because the failure mode is identical: each is an equivalence, so a finite verification proves nothing asymptotic and a full proof is a full proof of RH. In each case the parenthetical is the identity that makes the equivalence visible.

- Selberg-type density bounds on the *count* of off-line zeros (a single zero with $b_j = e^{-10^{100}}$ satisfies them all; counting is not locating).
- Positive-proportion results, "$\geq 41\%$ of zeros on the line" [28, 29] (a statement about on-line zeros that is silent about the shift of any off-line one).
- The Nyman–Beurling distance $d_N \downarrow 0$ [40, 41] (equivalent to the displacement bound, No-go 3.13).
- The Turán inequalities for the Laguerre–Pólya factor (a consequence of the working hypothesis, not a constraint, No-go 3.14).
- Bounds $\psi(x) - x = O(x^{1/2+\varepsilon_0})$ (a bound $\varepsilon_0 < \min_j b_j$ would localize the zeros, which is the problem itself).
- The Li/Jensen hyperbolicity coefficients $\lambda_n > 0$, $T_k \geq 0$ [42, 58] (each equivalent to RH; finite verification is not asymptotic evidence).

The common signature is the umbrella wall of §4: the Weil-positivity criterion $\mathcal{W}(f \star \tilde{f}) \geq 0$ is *itself* equivalent to the Hypothesis [2], so any quantity that faithfully captures it inherits the equivalence.

3.9. **Two withdrawn results.** In the same spirit of honesty we record two results we *withdrew*. First, an early operator model built on a wave–particle analogy (§6) claimed to establish uniqueness of a spectral minimizer; an internal audit found that its numerical certificates were not reproducible from the stated definitions and that a convexity step rested on the false inference "$A \succ 0,\ B \succeq 0 \Rightarrow A - B$ convex." The only surviving content is the correct algebraic reduction of the uniqueness question to $\|A^{-1/2}BA^{-1/2}\| < 1$. Second, a quantitative benchmark based on a specific numerical value of the Davenport–Heilbronn von Mangoldt data proved not to be recoverable from public definitions; we retain L_{DH} only as a *qualitative* falsifier — which is all the present paper uses it for. We mention both because the discipline of retraction is exactly what we invite the reader to apply to this paper.

3.10. **The terminal-defect continuation: the ARP-P path and its closures.** The companion development — the arithmetic Pick positivity (ARP-P) path — reduces the Hypothesis to a single terminal-defect positivity, $\delta_N \geq 0$ for all N, equivalently $\lambda_n \geq 0$ for all n (Li–Keiper [42]). Attacking that residue produced a further layer of no-gos, each of which lands on one of the structural walls of §4. We add them to the ledger; together they sharpen, rather than escape, the master quantifier.

No-go 3.33 (the terminal envelope bound is a three-fronted RH nucleus)**.** The scale-uniform bound $|\lambda_n^{\text{prime}}| \leq c\sqrt{n}\log n$ that would give $\lambda_n \geq 0$ for all n reduces, by a lossless stationary-phase analysis, to three classical equivalents of RH at once.

Write the Li oscillation on the zero side, $O(n) = \sum_\rho (1 - 1/\rho)^n$. The functional equation gives the exact identity $1 - 1/(1-\rho) = (1 - 1/\rho)^{-1}$ and $|1 - 1/\rho|^2 - 1 = (1 - 2\beta)/|\rho|^2$; the Cayley map $\rho \mapsto 1 - 1/\rho$ carries the critical line to the unit circle. Three independent routes to the envelope bound each terminate at a distinct canonical RH-equivalent,

$$\underbrace{|1 - 1/\rho| = 1\ \forall\rho}_{\text{Riemann: zeros on the line}} \quad ; \qquad \underbrace{N(\sigma, T) = 0 \text{ on } \tfrac{1}{2} < \sigma < 1;}_{\text{Landau: no zeros in the strip}} \qquad \underbrace{\psi(x) - x = O(\sqrt{x}\log^2 x)}_{\text{von Koch: the RH-strength PNT error}} \quad ,$$

and in each the main term is unconditional and produces $\sqrt{n}\log n$ while the whole RH-strength weight sits in the tail — the same weight seen thrice. A single off-line zero $\rho = \beta + i\gamma$ (through the member of its four-element cluster with $\beta < \frac{1}{2}$) injects a term r^n with $r = |1 - 1/\rho| > 1$ that only *eventually* overtakes the ceiling, so no finite numerical verification, at any n or precision, closes it. This is the umbrella wall (W4.1) read on the Li coordinates.

No-go 3.34 (the archimedean partition cannot be a density)**.** Splitting the archimedean term into per-prime densities $\sigma_p \geq 0$ that dominate each prime comb cannot work with σ_p a genuine density: it fails at large resonances, for every prime and every budget.

By Bochner the single-prime domination is $\hat{\sigma}_p(\xi) \geq 2\sum_k (\log p)p^{-k/2}\cos(k\xi\log p)$; the right side is almost-periodic with peaks $2\log p/(\sqrt{p} - 1)$ at every resonance $\xi \in (2\pi/\log p)\mathbb{Z}$, whereas a density obeys $\hat{\sigma}_p(\xi) \to 0$ (Riemann–Lebesgue). Domination therefore fails at all large resonances. Per prime and at any fixed support cutoff the problem *is* feasible — this is exactly the archimedean prolate positivity of Connes–Consani [19] — and the obstruction is the simultaneous limit over all primes: the uniform-norm wall (W4.8) in frequency form.

No-go 3.35 (the archimedean perturbative route is quantitatively closed)**.** The archimedean positivity margin varies only logarithmically in the support while the prime perturbation grows exponentially, so no uniform gap-versus-norm comparison can hold.

For the Weil operator on an interval of length a the lowest eigenvalue satisfies $\lambda_a = \log(1/a) + \mu_1 - \log(2\pi) + \psi(2) - 1 + O(a)$ with $\mu_1 > 0$ [44] — a margin that *shrinks* as the support grows — while the prime perturbation has ℓ^1-mass $\asymp e^{a/2}$. No bound $\|\text{prime}\| <$ (archimedean gap) can hold without deciding RH, and the eigenvalue reformulation $\lambda_a \geq 0\ \forall a \iff$ RH makes the equivalence exact. Any archimedean route to the residue must therefore be *non-perturbative*; a norm bound cannot close it. This is the wrong-sign wall (W4.3) sharpened to a rate.

No-go 3.36 (the de Branges positivity conditions are falsified)**.** The Hermite–Biehler / reproducing-kernel positivity that would force the zeros onto the line by Hilbert-space structure is not available in the form required.

The Cayley map recasts RH as "the structure function E_ξ of Ξ is Hermite–Biehler." The archimedean factor $\pi^{-(1/4+iz/2)}\Gamma(1/4+iz/2)$ is *not* an entire Hermite–Biehler function (its Γ-poles lie at $z = i(2n+\frac{1}{2})$ in the upper half-plane), and a product decomposition $\Xi = E_\xi E_\xi^{\#}$ would double the real zeros; the positivity must be handled at the kernel level, not by a scalar factorization. Conrey and Li [43] exhibited counterexamples to the specific positivity conditions that de Branges required for RH, so the positive-kernel route does not deliver the Hypothesis for free: it is the Weil-positivity wall (W4.1) in de Branges coordinates. The surviving reduction is index-theoretic — one needs the shorted primitive kernels to converge to the genuine Kreĭn–Langer kernel of Ξ (W4.8) — and that convergence is itself of class E: its closure is equivalent to the Hypothesis (see the classification at W4.8).

4. The structural walls

The no-gos of §3 are many, but they are not independent. Beneath them lies a small number of *structural walls*: obstructions that reappear, in disguise, on every front. We state them as numbered **Wall**s, in the order they were met, and we attach to each an explicit logical classification, so that no wall claims more than what is proved about it.

Classification of obstructions. Each wall (and each no-go it distils) is tagged with one or more of the following classes, where C denotes the condition the wall isolates:

- $E(C)$ *exact equivalence*: $C \Longleftrightarrow$ RH is proved.
- $S(C)$ *sufficient*: $C \Longrightarrow$ RH is proved.
- $N(C)$ *necessary*: RH $\Longrightarrow C$ is proved.
- $F(X;C)$ *framework closure obstruction*: within an explicitly stated framework X, closure of the route is proved equivalent to the condition C. The class F by itself asserts nothing about the difficulty of C; it asserts *here is exactly where this framework terminates.* The relation of C to RH is then classified separately as E, S, or N whenever such a relation is proved, and composite tags such as $F(X;C)+E(C)$ are used.
- $I(\mathcal{I};C)$ *insufficiency theorem*: an explicitly delimited class of inputs $\mathcal{I}$ is proved not to suffice for the target C ($\mathcal{I} \not\Rightarrow C$).

Informally we shall sometimes say a statement is "RH-strength"; as declared in §2, this phrase carries no logical content beyond the explicit classification attached to the statement. The umbrella under which all the walls sit is the oldest one.

Wall 4.1 (the Weil-positivity wall)**.** Weil's criterion [2] states that the Riemann Hypothesis is equivalent to

$$\mathcal{W}(f \star \tilde{f}) \;\geq\; 0 \qquad \text{for all test functions } f,$$

where $\mathcal{W}$ is the explicit-formula Weil form (1) and $\tilde{f}(x) = \overline{f(-x)}$. This is a clean equivalence, not an implication: *proving the positivity is proving the Hypothesis.* Every route that reduces to a positivity statement for $\mathcal{W}$ inherits this equivalence in full. The walls below are the specific disguises this equivalence adopts as one approaches the problem from different directions.

To see why the equivalence is so sharp, expand the form in terms of the zeros directly:

$$\mathcal{W}(f \star \tilde{f}) \;=\; \sum_{\rho} \widehat{f}(\rho)\,\overline{\widehat{f}(1-\bar{\rho})} \;+\; (\text{archimedean and pole corrections}),$$

where $\widehat{f}$ is the Mellin transform. When ρ lies on the critical line, $1-\bar{\rho} = \rho$ and the term is $|\widehat{f}(\rho)|^2 \geq 0$: the zero sum is manifestly non-negative under RH. An off-line zero at ρ_0 with

$\operatorname{Re}\rho_0 = \frac{1}{2} + \delta$, $\delta > 0$, contributes together with its mirror $\bar\rho_0$ and the two conjugates $1 - \rho_0$, $1 - \bar\rho_0$, and this four-element cluster introduces an indefinite contribution to the form:

$$\mathcal{W}\big|_{\text{cluster}} \;=\; 2\,\operatorname{Re}\big[\widehat{f}(\rho_0)\,\overline{\widehat{f}(1-\bar\rho_0)}\big] \;+\; 2\,\operatorname{Re}\big[\widehat{f}(1-\rho_0)\,\overline{\widehat{f}(\bar\rho_0)}\big],$$

which can be negative for appropriate f. The forced-negativity lemma (item (i) in §4.2) makes this precise: the cluster forces $\lambda_{\min}(Q) \le -c(\delta)$ unconditionally. The equivalence is thus tight: the form is positive iff there are no clusters iff there are no off-line zeros iff RH.

Classification: $E(C)$ with C = positivity of $\mathcal{W}$ on autocorrelations — the equivalence is classical (Weil [2]), not a contribution of this paper.

This umbrella wall is both the source of difficulty and — from a different angle — the starting point of the present paper: rather than prove positivity of Q globally, the route here proves that the kernel K_Ξ whose negative index counts off-line zeros has index zero, which is a consequence of positivity achieved through a normal-family argument rather than a direct bound.

Wall 4.2 (arithmetic propagation)**.** The Euler product $\prod_p (1-p^{-s})^{-1}$ converges absolutely only for $\operatorname{Re}s > 1$; in the strip $\frac{1}{2} < \operatorname{Re}s < 1$ it converges at best conditionally, and near the zeros of ζ not at all. The zeros are *global*: not one Euler factor vanishes, so no prime "owns" a zero and no local arithmetic datum at a single prime constrains the location of any zero. This is not a quantitative shortfall but a qualitative barrier: the Euler product carries no information that distinguishes a zero at $\operatorname{Re}s = \frac{1}{2}$ from one at $\operatorname{Re}s = \frac{1}{2} + 10^{-100}$.

To see this in quantitative form, write the logarithmic derivative in $\operatorname{Re}s > 1$:

$$-\frac{\zeta'}{\zeta}(s) \;=\; \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}, \qquad \operatorname{Re}s > 1,$$

where $\Lambda(n) \ge 0$ is the von Mangoldt function. The series has non-negative coefficients, so its real part is monotone in $\operatorname{Re}s$ and offers no oscillatory information. Continuing to $\operatorname{Re}s = \frac{1}{2}$ is exactly the analytic continuation problem: the partial sums $\sum_{n \le P} \Lambda(n) n^{-s}$ diverge when a zero is encountered, and the zero's imaginary location is encoded only in the *phase* of the continuation, not in any arithmetic datum accessible from $\operatorname{Re}s > 1$.

A classical consequence is the Korobov–Vinogradov zero-free region. The best unconditional result of this type gives, for any zero $\rho = \beta_j + i\gamma_j$ of ζ with displacement $b_j = \beta_j - \frac{1}{2}$,

$$b_j \;\le\; \frac{1}{2} \;-\; \frac{c}{(\log|\gamma_j|)^{2/3}(\log\log|\gamma_j|)^{1/3}}$$

for a positive constant $c > 0$. This is an *upper* bound on b_j; it says zeros cannot be too far from the critical line on the right. No analogous lower bound $b_j \ge c' > 0$ is known, precisely because the Euler product data cannot be forced across the strip. Theorem 3.1 records the quantitative form: every classical mechanism that takes data from $\operatorname{Re}s > 1$ and attempts a positive lower bound $b_j \ge c$ on the off-line displacement fails for one of four structural reasons: wrong direction (the bound is an upper bound, not a lower bound), wrong region (the method applies only where zeros cannot be anyway), wrong aggregate (almost-all results, not specific zeros), or circular (the input is RH-strength). The wall was identified independently from at least five directions over the course of the programme, and each identification gave the same answer.

This is the wall the present paper attacks directly. Rather than trying to propagate arithmetic information across the strip, the route stays in $\operatorname{Re}s > 1$ for all estimates, and imports the conclusion across the strip analytically (by a Vitali normal-family continuation) rather than arithmetically. It is the one move that does not try to cross the wall.

Classification: $I(\mathcal{I}; C)$ with $\mathcal{I} =$ the four classes of classical mechanisms taking data from $\operatorname{Re} s > 1$ enumerated in Theorem 3.1, and $C =$ a positive lower bound $b_j \geq c > 0$ on the off-line displacement.

Wall 4.3 (wrong sign: Hilbert–Pólya)**.** The Hilbert–Pólya programme seeks a self-adjoint operator H with $\{i(\rho - \frac{1}{2}) : \zeta(\rho) = 0\}$ as its spectrum; if such H exists, the spectrum is real and RH follows. Every unconditional construction one can write down delivers a *lower* bound on the relevant form:

$$Q(f, f) \;\geq\; -C\|f\|^2, \qquad C \geq 0,$$

whereas the Hypothesis is the *upper* restriction $Q \succeq 0$. Operator norm, Carleson constant, kernel diagonal, spectral-shift derivative — all eight independent paradigms tested gave bounds of the wrong direction. The difficulty is structural: positivity tools applied to ζ produce lower bounds because they control the *size* of the deviation from positivity, not its *sign*. RH needs the form to be non-negative; the tools prove it is bounded below by a constant that could be negative. Tightening C does not change the direction.

To make this precise, consider the localized Weil form restricted to a one-parameter family f_λ concentrated near a single zero $\rho_j = \frac{1}{2} + b_j + i\gamma_j$. A Taylor expansion in the transverse displacement b_j gives

$$Q(f_\lambda, f_\lambda) \;=\; \underbrace{Q_0}_{\geq 0 \text{ if all zeros on line}} \;-\; b_j^2 \cdot \underbrace{Q_2}_{>0} \;+\; O(b_j^4).$$

The leading term Q_0 is positive by assumption (on the line); the coefficient $Q_2 > 0$ of b_j^2 is explicitly computed and positive. This means Q *decreases* as the zero moves off the line; the critical line is a *local maximum* of the form in the transverse direction, not a minimum. The second-variation formula

$$\delta_b^2\, Q(f, f) \;=\; -8 \sum_j c_j^2 \Big[\operatorname{Re}\big(\widehat{h}''(\rho_j)\,\overline{\widehat{h}(\rho_j)}\big) + |\widehat{h}'(\rho_j)|^2\Big]$$

makes this explicit: for the detector test functions (which vanish at the sampled zeros, so the first bracket term drops) the right side is non-positive, in agreement with the unconditional curvature computation $C(\gamma) > 0$ (No-go 3.24) that the critical line is a local maximum. The practical consequence is decisive: a gradient-descent (or variational) argument that "falls toward" a minimum of Q falls *away* from the critical line, not toward it. Any proof must work against the gradient, supplying a structural reason for the maximum to be a global one. The present paper's route — the Vitali continuation which forces a holomorphic limit everywhere, not just at a minimum — is structurally of this type.

Classification: $I(\mathcal{I}; C)$ with $\mathcal{I} =$ the magnitude/lower-bound functionals of the eight tested paradigms (operator norm, Carleson constant, kernel diagonal, spectral-shift derivative, and their variants) and $C = Q \succeq 0$; the target C itself is of class E by W4.1.

Wall 4.4 (infinite-dimensional Hasse–Minkowski failure)**.** The classical Hasse–Minkowski theorem lifts positivity from local to global for quadratic forms over $\mathbb{Q}$: a form is isotropic iff it is isotropic at every completion. The strategy of proving $Q_v \succeq 0$ for each place v and concluding $Q \succeq 0$ globally would be an analogue of this theorem for Hermitian forms. It fails at two independent levels.

Failure at the factorization level. The Weil quadratic form splits naturally as

$$\mathcal{W}(f\star\tilde{f}) \;=\; \underbrace{\sum_{\rho}\widehat{h}(\gamma_\rho)}_{M_{\text{zeros}}} \;-\; \underbrace{\sum_{p}\sum_{k\geq 1}\frac{\log p}{p^{k/2}}\widehat{h}(k\log p)}_{M_{\text{arith}}},$$

where $h=f\star\tilde{f}$. The arithmetic side M_{arith} factors over primes perfectly: $M_{\text{arith}}=\sum_p Q_p$ with $Q_p(h)=\sum_{k\geq1}\frac{\log p}{p^{k/2}}\widehat{h}(k\log p)$ a local datum. The zero side $M_{\text{zeros}}(h)=\sum_\rho \widehat{h}(\gamma_\rho)$ carries the Hypothesis; it is *global* in the sense that no zero is attached to any prime. The factorization $Q=\sum_v Q_v$ would require each Q_v to carry information about which zeros are off-line; since no Euler factor has any such information, the factorization does not exist on the side that matters (No-go 3.19).

Failure of the infinite-dimensional principle. Even granting a hypothetical factorization, there is no Hasse–Minkowski theorem in infinite dimensions. Over $\mathbb{Q}$ the theorem is finite: there are finitely many bad primes and the completion argument is exact. In infinite dimension the passage fails for the truncations that actually occur here: the finite forms Q_X are *not* compressions of a fixed limit form — the arithmetic side itself changes with the truncation — so positivity of every Q_X controls the limit only if the convergence $Q_X\to Q$ is uniform, which is precisely the uniform-norm wall (W4.8). Nor is there an analogue of the finitely-many-bad-places argument when the index set of places is infinite (No-go 3.20).

The two failures are independent: even if one were repaired, the other would remain. The local–global route thus closes, at the level of principle, without any appeal to the specifics of ζ.

Classification: $I(\mathcal{I};C)$ with $\mathcal{I}=$ per-place positivity data assembled through a local-to-global principle (No-gos 3.19, 3.20) and $C=Q\succeq 0$; the residual convergence statement belongs to W4.8.

Wall 4.5 (per-place reflection positivity fails)**.** Osterwalder–Schrader reflection positivity reconstructs a positive Hamiltonian from a Euclidean measure satisfying an exponential decay condition. Applied to ζ, the natural per-prime kernel is, up to positive constants,

$$G_p(r)\;\propto\;\frac{2\log p}{p\,|1-z|^2}\big(\sqrt{p}\,\cos(r\log p)-1\big),$$

with z a fixed normalization independent of r. This kernel is indefinite through the sign of its bracket: $\sqrt{p}\,\cos(r\log p)-1$ equals $\sqrt{p}-1>0$ at $r=0$ but $-\sqrt{p}-1<0$ at $r=\pi/\log p$, verified numerically at $p=2,3,5,7$ (No-go 3.27). The reconstruction theorem requires exponential decay of the two-point function; the function is oscillatory, so the hypothesis is not met and the conclusion does not follow (No-go 3.28). There is no per-place positive structure from which to reconstruct a global one; whatever positivity Q has is an irreducibly cross-place, global phenomenon.

Classification: $I(\mathcal{I};C)$ with $\mathcal{I}=$ per-place reflection-positivity reconstructions (Osterwalder–Schrader type, No-gos 3.27, 3.28) and $C=$ a globally positive Hamiltonian.

Wall 4.6 (the cohomological-Frobenius wall)**.** The Weil conjectures over finite fields [3, 24] are proved by exhibiting a Frobenius φ acting on a cohomology $H^\bullet$ that carries a *positive, gapped polarization*: the Hodge–Riemann bilinear relations force $|\alpha|=\sqrt{q}$ for every eigenvalue α of φ. Transporting this mechanism to ζ requires two ingredients: a *J-linear* (Hodge-type) operator on a $\mathbb{Z}$-cohomology, and a positive polarization with a spectral gap. The programme established, by explicit computation, that neither exists in the available ζ window.

Antilinearity of the natural scaling. The natural candidate Frobenius is the dilation operator $D_{\log}$ acting on the adelic Hilbert space, defined by $(D_{\log}f)(x) = f(e^t x)$ at scale parameter t. The complex structure J acts by $Jf(x) = \overline{f(\bar{x})}$ (complex conjugation combined with the functional equation). A direct computation on test functions in the n-th frequency mode yields

$$\langle [D_{\log}, J]\, e_n, e_n \rangle \;=\; -2\,\langle D_{\log}\, e_n, e_n \rangle,$$

so $\{D_{\log}, J\} = 0$ exactly, and the numerical test confirmed $\|[D_{\log}, J]\|/\|D_{\log}\| = 2.000$ (No-go 3.32). Anticommutativity means $D_{\log}$ is $\mathbb{C}$-*anti*linear with respect to J; it satisfies $D_{\log} \circ J = -J \circ D_{\log}$. A Hodge morphism must be J-*linear* (commuting with J, not anticommuting), because Deligne's comparison of cohomology with its complex conjugate requires $\varphi \circ J = J \circ \varphi$. The candidate is of the wrong type.

For comparison: on a real elliptic curve $E/\mathbb{F}_q$, the Frobenius φ_q acts J-linearly on $H^1(E, \mathbb{Z}/\ell)$, with $\varphi_q \circ J = J \circ \varphi_q$ verified by the endomorphism algebra. The ζ test returns antilinearity where the finite-field analogue returns linearity.

Gapless polarization. The Hodge–Riemann mechanism uses the polarization eigenvalue gap to force $|\alpha| = \sqrt{q}$: if the smallest positive eigenvalue of the polarization form is $\mu_{\min} > 0$, then the Cauchy interlacing of eigenvalues, combined with the Lefschetz–Hodge–Riemann bilinear relations, pins every Frobenius eigenvalue on the circle $|\alpha| = \sqrt{q}$. The programme's computation of the quaternionic Hodge–Riemann polarization on the ζ space found a spectrum that decays as

$$\mu_k \;\sim\; e^{-cL}, \qquad c \;\approx\; 1.40\ \log_{10}\ \text{per step},$$

as the truncation scale $L \to \infty$ (No-go 3.31). There is no uniform spectral gap; $\mu_{\min} \downarrow 0$. Marginal positivity ($\mu_{\max} = 1$ for ζ, compared with strictly less than 1 for the Davenport–Heilbronn analogue at the same truncation) distinguishes the two functions, but a vanishing gap closes the analytic road: no finite $\mu_{\min}$ means no Cauchy pinning.

The object that would close this route is a $\mathbb{Z}$-cohomological Frobenius with a positive gapped pairing — precisely the unbuilt object of the adelic programme [20] — and its construction is a problem of algebraic geometry, not of classical analysis.

Classification: $F(X;C) + S(C)$ with $X =$ the Hodge–Riemann/Frobenius transplant in the available ζ window and $C =$ the existence of a J-linear Frobenius with a positive gapped polarization over $\operatorname{Spec}\mathbb{Z}$; C would imply RH by the Hodge-index mechanism, but no equivalence is claimed.

Wall 4.7 (sign, not magnitude)**.** Theorem 3.2 identifies the distinguishing feature between ζ and an off-line-zero falsifier as the *sign* of the von Mangoldt data: ζ has $\Lambda(n) \geq 0$, while the Davenport–Heilbronn function L_{DH}, which has zeros off the critical line, has a sign-indefinite Λ_{DH} (e.g. $\Lambda_{\mathrm{DH}}(3) < 0$). Positivity is a sign property, and a functional that reads only the sizes $\{|\Lambda(n)|\}$ cannot certify it: sign flips of the Euler data preserve every individual magnitude yet destroy the positivity $H_P \succeq 0$ and push zeros off the line.

The practical consequence is that any positivity argument that operates through size bounds of the form $\|M_{\mathrm{arith}}\| \leq C$ or $|\sum_n \Lambda(n) n^{-s}| \leq C(\sigma)$ cannot reach the Hypothesis: such bounds are insensitive to the very signs on which self-adjointness turns. What *can* distinguish the two functions is the genuine *positivity* of the von Mangoldt weights — the fact that $\Lambda(n) \geq 0$ makes the finite operators A_P self-adjoint on a genuine Hilbert space, giving them a resolvent bounded by $1/|\operatorname{Im} z|$, whereas the indefinite Euler data of L_{DH} makes the corresponding operators Pontryagin-space operators with no such bound. The Hilbert-space property is what the normal-family argument needs, and it is exactly what $\Lambda \geq 0$ supplies. This is the input the present paper isolates as (Real) (§9).

Classification: $I(\mathcal{I}; C)$ with $\mathcal{I}$ = functionals reading only the magnitudes $\{|\Lambda(n)|\}$ (size bounds of the displayed forms) and C = the positivity $H_P \succeq 0$ that the route requires.

Wall 4.8 (uniform norm: the rank-one escape)**.** A spectral route that assembles the finite Weil operator $K_P = \sum_{n \leq P} \Lambda(n) K_n$ (where K_n is the n-th Weil kernel piece) meets a final barrier. By Mertens' theorem the trace grows,

$$\operatorname{Tr} K_P \;=\; \sum_{n \leq P} \Lambda(n)\, K_n(0,0) \;\sim\; \tfrac{1}{2}(\log P)^2,$$

but this divergence is carried entirely by a single positive, rank-one mode: the residue at the pole $s = 1$ of ζ. After removing this mode by Feshbach (Schur) shorting, the shorted trace is bounded. The question is then whether the *operator norm* of the shorted primitive piece is bounded:

$$\sup_P \bigl\| \Pi^\circ K_P \Pi^\circ \bigr\| \;<\; \infty \quad ?$$

By the Weil explicit formula this norm is controlled by the off-line zeros: if any zero ρ has $\operatorname{Re} \rho \neq \frac{1}{2}$, the off-pole Weil kernel acquires a contribution growing like $P^{2|\operatorname{Re}\rho - 1/2|}$, so the operator norm diverges. Conversely, if all zeros are on the line the norm stays bounded. The uniform-norm bound is therefore *equivalent in strength to the Hypothesis*: needing it as an input is circular, and it is precisely this circularity that the previous generation of spectral approaches ran into.

The companion development sharpens this wall without breaching it. The finite Weil–Gram kernels are positive *by construction* (a Schur/Haynsworth complement of a positive canonical Gram), so no positivity remains to be proved at the finite level — the escape sidesteps the Weil-positivity wall (W4.1) there. What is not free is a *convergence*: the shorted primitive kernels K_P° must converge, in the finite-Gram topology at the critical boundary, to the genuine Kreĭn–Langer kernel of Ξ; equivalently $\sup_P \|\Pi^\circ K_P \Pi^\circ\| < \infty$. Two facts delimit the gap. The leading trace divergence $\frac{1}{2}(\log P)^2$ is carried by the single rank-one pole mode — but leading-trace rank-one does *not* imply primitive boundedness: a subleading escape direction of size $o(\log P)^2$ that still diverges is invisible to the trace, so the rank-one trace alone does not bound the endpoint index. And the naive de Branges packaging fails at the archimedean factor and at the scalar determinant, which is index-blind (No-go 3.36). The residue is thus a single index-continuity statement, itself of class E: closing it is equivalent to closing the Hypothesis.

The present paper is built to avoid this wall entirely. It never forms the operator K_P or its shorted version as an operator on an infinite-dimensional space. Instead it works with the compressed resolvent $G_P^F(z) = \langle (A_P - z)^{-1} \varphi, \psi \rangle$ for φ, ψ in a *fixed finite source frame* F. The only uniform bound used is the per-vector frame mass $\sup_P \|\iota_P \varphi\|_{H_P}^2 \leq C_F$ (an absolutely convergent prime sum, verified in $\operatorname{Re} s > 1$), not any operator norm over the whole space. The full operator norm may diverge; the compressed channel stays bounded by $C_F / |\operatorname{Im} z|$ because each A_P is genuinely self-adjoint (§9). This is the architecturally new feature of the route.

Classification: $F(X; C) + E(C)$ with X = the spectral route assembling the full shorted Weil operator and C = the uniform primitive-norm bound $\sup_P \|\Pi^\circ K_P \Pi^\circ\| < \infty$; the equivalence $C \iff \mathrm{RH}$ is the two-directional control displayed above.

4.1. **The master quantifier.** The seven walls above are not independent; they are one obstruction in seven costumes. The single sentence beneath them all is a statement about quantifiers.

The Riemann Hypothesis is always the passage from average, almost-all, or interior to specific, uniform, or boundary.

An off-line zero is a *measure-zero event* in the generic behaviour of ζ: it perturbs the zeros by a set of measure zero in every natural sense (height, density, correlation). No generic statistic — no average, no density theorem, no almost-all estimate — can detect a measure-zero event. Every wall above is an instance:

- **Propagation wall (W4.2)**: interior ($\operatorname{Re} s > 1$) $\not\to$ boundary ($\operatorname{Re} s = \frac{1}{2}$). Arithmetic data from the region of absolute convergence do not reach the strip.
- **Wrong-sign wall (W4.3)**: lower bound (magnitude below) $\not\to$ upper restriction ($\lambda_{\min} \geq 0$). Tools produce inequalities pointing the wrong way.
- **Hasse–Minkowski wall (W4.4)**: local ($Q_v \geq 0$ at every v) $\not\to$ global ($Q \geq 0$). The local-to-global principle fails in infinite dimension.
- **Per-place wall (W4.5)**: per-prime (local kernel positive) $\not\to$ global (global positivity). Positivity is cross-place, not reconstructible from per-place pieces.
- **Frobenius wall (W4.6)**: function-field (gapped polarization exists) $\not\to$ $\mathbb{Z}$ (gapped polarization over $\operatorname{Spec}\mathbb{Z}$ not constructed). Average gap $\not\to$ uniform gap.
- **Magnitude wall (W4.7)**: generic (magnitudes $|\Lambda(n)|$) $\not\to$ specific (positivity $\Lambda(n) \geq 0$). Sizes do not determine signs.
- **Uniform-norm wall (W4.8)**: pointwise (each compressed channel bounded) $\not\to$ uniform (operator norm over the whole space bounded). Per-channel control does not give a uniform operator bound.

A proof must therefore supply a *structural* reason — not a sharper estimate — forcing the boundary behaviour. The proof in §9 attempts exactly this: it exhibits a normal family of genuine compressed resolvents whose limit is forced to remain holomorphic across the critical strip, so that the boundary statement (no off-line pole of K_Ξ) follows from a rigidity theorem (Vitali's theorem, §9) rather than from any inequality that could be merely *almost* true. The reason this family is a normal family — the key structural input — is the Hilbert-space self-adjointness of A_P, which is in turn a consequence of $\Lambda \geq 0$. This is the structural mechanism the proof path uses to approach the boundary behaviour without crossing any of the walls above — reducing it, in the end, to the single open input Ω_7 rather than closing it outright.

It may help to state the logical shape of the escape explicitly. Define the *verification event* for each wall as the hypothesis that the wall would need to be crossed, and $\mathcal{E}$ as the conjunction:

$$\mathcal{E}_1 : \text{arithmetic data from } \operatorname{Re} s > 1 \text{ constrain zeros in the strip,}$$

$$\mathcal{E}_2 : \text{the Weil form has a global minimum at the critical line,}$$

$$\mathcal{E}_3 : Q = Q_\infty + \sum_p Q_p \text{ (local-global factorization on the zero side),}$$

$$\mathcal{E}_4 : \text{the scaling operator is } J\text{-linear (Hodge morphism type),}$$

$$\mathcal{E}_5 : \text{the Hodge–Riemann polarization has a uniform spectral gap.}$$

The programme proved $\neg\mathcal{E}_i$ for $i = 1, \ldots, 5$, each by a specific computation or structural argument (Theorem 3.1, the second-variation formula, No-go 3.19, No-go 3.32, No-go 3.31). The present route avoids $\mathcal{E}_1$–$\mathcal{E}_5$ entirely: it neither crosses the arithmetic strip nor claims a minimum, factorization, Hodge morphism, or spectral gap. Instead it uses $\Lambda \geq 0$ to make the finite resolvents a bounded normal family on $\mathbb{C} \setminus \mathbb{R}$, and invokes Vitali's theorem — a rigidity principle, not an inequality — to reduce the closure to the single terminal input Ω_7.

4.2. **What the programme did establish.** The no-go ledger above is long, but the programme also produced a number of *positive* results: proved theorems, computed invariants, and established reductions that are unconditionally true and that the present paper builds on. We record them here because they are, in aggregate, the mathematical content of the multi-year journey, and because the proof in §9 is their residue: the one thing that survived after everything that could not work was proved not to work.

(i) *The forced-negativity lemma.* An off-line zero at real-part displacement $\delta > 0$ forces

$$\lambda_{\min}(Q) \;\leq\; -c(\delta, \sigma, J) \;<\; 0,$$

for a positive constant c depending on δ, the localization scale σ, and the test interval J. This is proved unconditionally and provides the foundation for the localized Weil detector: the detector fires (gives a negative eigenvalue) if and only if there is an off-line zero in its window. It is the starting point for the whole programme's quantitative approach.

(ii) *The unconditional truncation control.* The spectral error $\eta(X) = Q_X - Q_\infty$ of the truncated Weil form decays as

$$\eta(X) \;\approx\; A \cdot \exp\bigl(-\alpha(\log X)^2\bigr), \qquad \alpha \approx \tfrac{\sigma^2}{8},$$

unconditionally. This bounds the finite-to-infinite approximation error without any hypothesis on zeros.

(iii) *The curvature criterion.* The Hypothesis is equivalent to

$$C(\gamma) \;=\; \sum_\rho \frac{1}{(\gamma - \gamma_\rho)^2} \;>\; 0 \qquad \text{for all } \;\gamma,$$

a statement about the curvature of $\log|\xi|$ across the critical line (No-go 3.24, proved unconditionally as an equivalence). This gives a clean geometric characterization of RH.

(iv) *The second-variation formula.* The second variation of the localized form with respect to the transverse displacement b is

$$\delta_b^2\, Q(f,f) \;=\; -8 \sum_j c_j^2 \Bigl[\mathrm{Re}\bigl(\widehat{h}''(\rho_j)\,\overline{\widehat{h}(\rho_j)}\bigr) + |\widehat{h}'(\rho_j)|^2\Bigr],$$

non-positive for the detector test functions (whose transforms vanish at the sampled zeros, killing the first bracket term). This is proved without assuming RH, and it implies that the critical line is a local maximum of Q in the transverse direction, ruling out variational pinning arguments.

(v) *The sector decomposition.* Writing $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, the second derivative $\partial_t^2 \log|\xi(\frac{1}{2}+it)|$ factors into three computable sectors:

$$G_{\mathrm{poly}}(t) = \frac{2(\frac{1}{4} - t^2)}{(t^2 + \frac{1}{4})^2}, \tag{5}$$

$$G_\Gamma(t) = -\tfrac{1}{4}\ \mathrm{Re}\ \psi^{(1)}\bigl(\tfrac{1}{4} + \tfrac{it}{2}\bigr), \tag{6}$$

$$G_\zeta(t) = \partial_t^2 \log|\zeta(\tfrac{1}{2} + it)|, \tag{7}$$

where $\psi^{(1)}$ is the trigamma function. The first two sectors, G_{poly} and G_Γ, are known functions of t that can be tabulated to arbitrary precision. The third, G_ζ, depends on the location of the zeros and is accessible only through the explicit formula; its positivity at all t is equivalent to RH. This decomposition isolates exactly the inaccessible piece and is used in §3.5 to show that the sector inequality reformulation is equivalent to, not stronger than, the Hypothesis itself.

(vi) *The LP/Turán algebraic reformulation.* The Hypothesis is equivalent to the assertion that the coefficients c_n of $\xi(\frac{1}{2}+it) = \sum_{n\geq 0}(-1)^n c_n t^{2n}$ satisfy the Turán–Laguerre– Pólya inequalities $T_k = (2k+1)c_k^2 - 2kc_{k-1}c_{k+1} \geq 0$ for all k. This converts an analytic problem (zero location) into an algebraic one (sign conditions on a real sequence).

(vii) *The equivalence of the four convexity fronts.* In the fourth arc (§3.5) we established that four apparently different variational attacks — variational pinning at a minimum, a concavity criterion, a sector inequality, and an η-invariant argument — are the *same* problem: they all reduce to $\xi(\frac{1}{2}+it) \in \mathrm{LP}$ (the Laguerre–Pólya class), which is equivalent to RH.

(viii) *The sign principle* (Theorem 3.2). This is a proved theorem, not a heuristic: the feature that separates ζ from an off-line-zero falsifier is the *sign* of the von Mangoldt data — $\Lambda(n) \geq 0$ for ζ, sign-indefinite for L_{DH} ($\Lambda_{\mathrm{DH}}(3) < 0$) — a positivity that no functional of the magnitudes $\{|\Lambda(n)|\}$ can certify. The theorem rules out an entire family of magnitude-based proof strategies in one stroke.

(ix) *The per-place indefiniteness.* The local reflection-positivity kernel $G_p(r) \propto \sqrt{p}\cos(r\log p) - 1$ is proved indefinite at every prime $p = 2, 3, 5, 7$ (and structurally at all p). This closes the Osterwalder–Schrader route unconditionally.

(x) *The Kreĭn–de Bruijn–Newman connection.* An off-line zero of displacement b_j forces the lower bound $\Lambda_{\mathrm{dBN}} \geq \frac{1}{2}b_j^2$ on the de Bruijn–Newman constant, and the connection with Kreĭn's theory provides a clean chain from the Laguerre–Pólya class to the spectral theory of canonical systems. Concretely: in the heat-flow deformation $H_\lambda(z) = \int e^{\lambda u^2}\Phi(u)\cos(zu)\,du$ (with Φ the Riemann kernel, so that H_0 is proportional to Ξ), there is a constant Λ_{dBN} such that H_λ has only real zeros iff $\lambda \geq \Lambda_{\mathrm{dBN}}$; as λ decreases through Λ_{dBN}, pairs of real zeros coalesce and split into complex-conjugate pairs, and the Kreĭn–Langer index (the number of non-real pairs) becomes positive. RH is the statement $\Lambda_{\mathrm{dBN}} \leq 0$; the known bounds are $0 \leq \Lambda_{\mathrm{dBN}} \leq 0.2$ [38, 57], so RH is equivalent to $\Lambda_{\mathrm{dBN}} = 0$.

(xi) *The Lefschetz/ample-class dichotomy.* The Hodge–Riemann approach requires an ample class over $\operatorname{Spec}\mathbb{Z}$. A Lefschetz-type computation shows the ample class cannot be supplied from the geometric objects that are available, isolating the cohomological construction as the precise missing piece. The computation proceeds via the signature of the arithmetic Néron–Severi group: the polarization form has signature $(1, 19, 1)$ in the geometric realization tested—an indefinite, hyperbolic-type form with a single positive direction—rather than the positive-definite (ample) signature needed to close the Lefschetz argument. The dichotomy is therefore sharp: either the ample class exists (not established over $\operatorname{Spec}\mathbb{Z}$) or the polarization is hyperbolic and the Hodge–Riemann argument fails at the first step.

(xii) *The zero-side tautology theorem.* The zero-side Gram matrix G_{zeros} is positive semidefinite by construction for every L-function; using $\lambda_{\min}(G_{\mathrm{zeros}})$ to distinguish ζ from L_{DH} is circular. The theorem closes the one-sided positivity approach definitively.

(xiii) *The de Branges/Hilbert space canonicity.* The Hermite–Biehler structure function E of the canonical system [8] satisfies $|E(z)| > |E^*(z)|$ for $\operatorname{Im} z > 0$, where $E^*(z) = \overline{E(\bar{z})}$. This is equivalent to the self-adjointness of the underlying canonical system (§9). When the canonical system is driven by the von Mangoldt datum $H_P \geq 0$, the Hermite–Biehler property holds unconditionally. The Kreĭn–Langer index $\kappa(A_P) = \deg \mathfrak{b}_P$ counts the off-line zeros of the associated structure function; $\kappa(A_P) = 0$ for the finite positive systems is an unconditional theorem (§9).

(xiv) *The DH-falsifier computation.* The numerical verification that the detector $\mu_0(L_{\mathrm{DH}}) < 0$ while $\mu_0(\zeta) \geq 0$ (to 40 significant figures at $\lambda = 10, 12, \ldots, 22$) provides an unconditional empirical separation between the two functions. More importantly, the theoretical explanation for the separation — loss of $\Lambda \geq 0$ means the Hamiltonian becomes a Pontryagin-space operator and the resolvent bound $1/|\operatorname{Im} z|$ fails — locates the responsible structural feature precisely, ruling out coincidental numerical agreement.

These results are, collectively, what is known unconditionally about the Weil form and the operator-theoretic approach to RH. They form the substrate on which the proof in §9 is built: the positive canonical system (using item (i) for motivation, item (viii) for the structural constraint on what the proof must use, and items (ii)–(vii) for the shape of the landscape). The proof is not a series of estimates; it is a construction whose possibility was suggested, negatively, by each of the no-gos above and whose structural input ($\Lambda \geq 0$) was identified, by elimination, as the only remaining lever.

5. Related work: the adelic–spectral programme

The construction here is operator-theoretic and adelic in spirit, and it could not have been arrived at without the body of work that recast the explicit formula as a trace formula and the Riemann Hypothesis as a positivity statement in noncommutative geometry. We summarize the debts, citing the published mathematics; we make no claim about any individual's view of the present paper.

The trace-formula framing [18] expresses the explicit formula as a trace identity on the adelic class space and recasts the Hypothesis as the positivity of a Weil distribution — the principle $T = 0 \Leftrightarrow$ RH that organizes every later attack. The archimedean refinement [19] made the semilocal Weil form and its positivity explicit, and is the direct ancestor of the localized detector that drove our empirical phase. The scaling- and arithmetic-site programme [20] supplied the geometric language in which the missing object — a suitable cohomology of $\operatorname{Spec} \mathbb{Z}$ with a Frobenius and a positive pairing — is precisely named; the recognition that this object is *unbuilt* is exactly wall 4.6. Most recently, the prolate and spectral-triple realizations of the zeros [21, 22] gave concrete self-adjoint models whose spectra approximate the zeros to high precision, sharpening the question of what additional input forces reality of the spectrum. The Hecke-algebra and symmetry-breaking work [23] stands behind the whole adelic picture, and the more recent absolute-geometry developments — the BC-system and quantized-calculus framework [68] and the Riemann–Roch theory for $\operatorname{Spec} \mathbb{Z}$ [69] — continue to build the geometric object whose absence is wall 4.6. The unfolding of archimedean factors that we use in the determinant identification (§9) is, technically, of the Rankin–Selberg type [71].

The proof here also rests on classical analytic-number-theory foundations: the Mertens and Chebyshev estimates [31, 6] supply the divergent $\frac{1}{2}(\log P)^2$ pole-mode asymptotic; the Hilbert-space operator theory of canonical systems is due to de Branges [8] and the Kreĭn–Langer extension to indefinite metrics [9]; Hadamard's factorization [4] controls the entire function Ξ; and the Vitali–Montel normal-family theorem [4] is the rigidity tool that closes the argument.

The Beurling–Nyman and de Branges reformulations of RH [40, 41, 8] are part of the background; we use the de Branges space theory as a technical tool in §9 (the Hermite–Biehler structure function) rather than as a reformulation. The de Bruijn–Newman connection

[56, 39, 38, 57] and the Beurling–Nyman distance [40, 41] appear in the no-go ledger as directions that cannot yield lower bounds.

The present route can be read as a deliberate *retreat* from the hardest part of the adelic programme: where the arithmetic site seeks the positive Frobenius cohomology directly on $\operatorname{Spec}\mathbb{Z}$, we avoid it, realizing the zeros instead through the analytic continuation of compressed resolvents of explicit finite self-adjoint operators, and importing positivity only in the tame region $\operatorname{Re} s > 1$. Whether this retreat is a genuine way around wall 4.6 or merely relocates it is, again, a question we put to the reader. The fact that the Davenport–Heilbronn test is passed — and passed for the correct structural reason (the loss of self-adjointness at the level of the Hamiltonian H_P, exactly where the von Mangoldt sign changes) — is the strongest internal consistency check we have.

The spectral randomness and universality of ζ that motivates the GUE hypothesis — the pair correlation of Montgomery [26], the GUE statistics of Odlyzko's computations [55], the random-matrix value distribution of Keating–Snaith [36] — is the empirical background of the programme, and the generic-vs-specific distinction of the master quantifier (§4.1) explains precisely why these generic results, as beautiful as they are, cannot by themselves decide the Hypothesis.

The canonical-system framework used in this paper was developed by de Branges [8] and extended to indefinite metrics by Kreĭn and Langer [9]. The key result we use is that every Hermite–Biehler structure function E with $|E(z)| > |E^*(z)|$ for $\operatorname{Im} z > 0$ generates a Hilbert space of entire functions $\mathcal{H}(E)$ in which the operator $A : F \mapsto zF$ (multiplication by the spectral variable) is self-adjoint; conversely, every self-adjoint operator with simple spectrum arises in this way. In the indefinite case [9], the structure function satisfies $|E(z)| > |E^*(z)|$ for $\operatorname{Im} z > 0$ only outside a finite exceptional set, and the operator A on the associated Pontryagin space $\Pi_\kappa(\mathcal{H})$ has at most κ non-real eigenvalues. The Kreĭn–Langer index κ counts these non-real eigenvalues, and the reformulation $\kappa(\Xi) = \#\{\text{off-line zeros}\}$ (G5 in §9) is the bridge between the canonical-system theory and the Hypothesis.

The Turán power-sum method and its modern descendants provide sharp lower bounds for partial sums of Dirichlet series in terms of the coefficients. These bounds (No-go 3.14) are in principle usable to detect off-line zeros, but in practice they deliver upper bounds on off-line behavior rather than lower bounds, and the resulting estimates are aggregate (averaged over a set of zeros) rather than specific (for one zero). The technique exhausted itself in this programme without a positive outcome, and the reason — that it is an average-type argument facing a specific-type question — is now explained by the master-quantifier principle.

6. Three guiding analogies

Three physical analogies guided the search. None is a proof, and we are careful to label them as heuristics; but each carries real mathematics, and each pointed, in the end, toward the same construction.

Wave–particle duality. In quantum mechanics a duality is resolved not by choosing a side but by exhibiting one object with two representations related by the Fourier transform. The explicit formula plays exactly this role for ζ: it is a Fourier–Mellin change of basis between the *primes* (the multiplicative/scale side) and the *zeros* (the spectral/frequency side),

$$\sum_\rho \widehat{h}(\rho) \;=\; \widehat{h}\text{-terms at } \infty \;-\; \sum_{p,k} \frac{\log p}{p^{k/2}}\, h(k \log p),$$

the two sides being the two "representations" of a single distribution. The difficulty, and the reason the analogy does not by itself close, is that the inner product making the spectral side unitary — the "conserved energy" of the duality — is precisely the Weil form Q that encodes ζ; the duality is real but is not resolved by structure alone. (This is the analogy whose first, over-eager operator realization we withdrew; §3.9.)

A Maxwell-style unification. A second picture treats the two sides of the wall — the arithmetic side and the spectral side — not as competitors but as two components of a single field, in the way the electric and magnetic fields are two faces of one electromagnetic object exchanged by a change of frame. On this view the explicit formula is the "field equation" coupling primes and zeros, and the Riemann Hypothesis is a statement that the combined object has a definite polarization. This picture is genuinely unifying and it is what motivates seeking a *single* self-adjoint object (here the canonical operator A_P) carrying both the prime data (through its Hamiltonian) and the spectral data (through its eigenvalues); but it is a conceptual frame, not a computation, and we present it only as such.

Quantum mechanics and Stone's theorem. The deepest analogy, and the one that finally clarified the logic, is with the spectral theory of Hamiltonians. In quantum mechanics the reality of a spectrum follows from Stone's theorem: a strongly continuous one-parameter *unitary* group $U(t) = e^{itH}$ has self-adjoint generator, hence real spectrum — and unitarity is guaranteed by an inner product given in advance by physics. The Berry–Keating Hamiltonian [35]

$$H_{\mathrm{BK}} \ = \ \tfrac{1}{2}(xp + px),$$

where x is position and $p = -i\partial_x$ is momentum, has the formally attractive property that its classical trajectories $\dot{x} = x$, $\dot{p} = -p$ are hyperbolic, with time-reversal symmetry, and the Weyl semiclassical counting law

$$N_{\mathrm{cl}}(T) \ = \ \frac{\mathrm{Area}\,\{(x,p) : H_{\mathrm{BK}} \le T\}}{2\pi} \ \sim \ \frac{T}{2\pi}\log\frac{T}{2\pi e}$$

reproduces the Riemann–von Mangoldt counting formula $N(T) \sim (T/2\pi)\log(T/2\pi e)$. This is striking, but it stops there: the exact spectrum of any self-adjoint realization of H_{BK} does not reproduce the zeros of ζ (the Hilbert space is fixed; the zeros are not). The model identifies the right classical action but not the right quantization.

The adelic scale flow carries this further: on the idèle class space $L^2(C_{\mathbb{Q}})$ the multiplicative group $\mathbb{R}_+^*$ acts by scaling and realizes a natural "quantum Hamiltonian" whose absorption spectrum formally consists of the zeros [18]. But exhibiting a Hilbert space on which this flow is unitary is *logically the same problem* as exhibiting the inner product that makes the zeros real, which is, once more, the Weil positivity. This is the *inverted Stone* principle: for ζ there is *no* inner product given in advance by physics; every candidate Hilbert space that makes the spectrum real must itself encode the Hypothesis.

The only RH-free symmetry available, the functional equation $s \mapsto 1 - s$, is a *conjugation*: it sends $\gamma_\rho \mapsto \overline{\gamma_\rho}$ and therefore preserves the real zeros while pairing complex ones in conjugate pairs. But a conjugation is *not* an evolution; it cannot play the role of the unitary group $U(t) = e^{itH}$ in Stone's theorem. The functional equation is a powerful symmetry, and it is used twice in this paper (once in the proof proper and once in the DH falsifier check); but it cannot supply the spectral reality by itself.

This is precisely why the present paper does not try to make the scale flow unitary by fiat. Instead it *builds* the inner product, finitely and explicitly, from the positive von Mangoldt Hamiltonian (§9): the Hamiltonian $H_P \ge 0$ is a positive matrix assembled from $\Lambda(n) \ge 0$, and the resulting operator A_P is self-adjoint on the Hilbert space whose inner product H_P defines. Stone's theorem then applies honestly to each finite A_P, and the entire remaining difficulty is moved to the analytic continuation across the strip, where a normal-family argument can meet it without needing to know where the zeros are. The key is that the positivity $\Lambda(n) \geq 0$ makes $H_P \geq 0$ an honest positive form, not a formal one; it is the one structural input the proof uses that L_{DH} cannot supply.

Summary of the three analogies. The wave–particle analogy identified the structure (a Fourier–Mellin duality with the explicit formula as field equation) but could not close because the inner product was not given. The Maxwell-unification analogy identified the single object (A_P) that should carry both sides, but provided no route to constructing it without circular assumptions. The inverted-Stone analogy explained the precise logical obstacle: the inner product must be built, not assumed, and only $\Lambda \geq 0$ can build it unconditionally. Taken together, the three analogies delimit the space of routes: almost everything that can be made to work in quantum mechanics fails for ζ (inverted Stone), almost everything that closes over finite fields fails for $\mathbb{Z}$ (cohomological Frobenius wall), and the only remaining mechanism — finite self-adjoint operators with a bounded resolvent and a Vitali normal-family continuation — is precisely the construction in §9. The analogies are not the proof; but they are the map by which the proof was located.

Part 2. The proof path: the ARP-P chain and its single open input

7. Overview of the proposed proof path

The proof is organized around a single equivalence,

$$\boxed{\text{ARP-P} \Longleftrightarrow \text{RH}},$$

and the fifteen steps below are the ordered construction of that equivalence together with the one-way bridge that would turn it into a proof of RH once Step 5 is supplied. This section explains why the path takes the shape it does; Section 9 carries out each step in full.

The path begins by moving the problem into the variable z with $s = \frac{1}{2} + iz$, so that $\Xi(z) = \xi(\frac{1}{2} + iz)$ and the critical line becomes the real axis: RH is exactly the statement that every zero of Ξ is real (Step 1). Starting here is what makes the rest possible, because it turns a statement about the location of zeros into a statement about a *positivity* that complex analysis can control. Onto this geometric picture we graft arithmetic through the Weil explicit formula: Steps 2–4 build finite *channels* F and an explicit meromorphic target G_Ξ^F whose data is fixed by the archimedean, pole, and prime terms with no holomorphic ambiguity. Step 5 then isolates the one arithmetic input, ARP-P: a Pick (positive-kernel) condition on this explicit data, tested only at finite node sets in the lower half-plane U_-.

Everything after Step 5 is a closed analytic bridge that converts that positivity into the reality of the zeros. Positive Cauchy kernels furnish Nevanlinna–Pick interpolants (Steps 6–8); Montel normality extends them holomorphically from U_- across Ω_- (Step 9); any off-line zero would have to appear as a finite-channel residue and is therefore excluded (Steps 10–11); and the functional-equation symmetry $\Xi(-z) = \Xi(z)$ together with conjugation propagates the conclusion to the whole plane (Steps 12–13), giving RH. Steps 14–15 prove the converse direction RH $\Rightarrow$ ARP-P, closing the equivalence. The design deliberately concentrates all the difficulty into Step 5: every other step is a classical or complex-analytic implication whose hypotheses are checked where they are used, so that establishing ARP-P alone would complete a proof of RH. Step 5 is the single open input, and the remainder of the paper is devoted to it.

7.1. **Claims, status, and dependency map.** This subsection is the audit interface of the paper: the typed vocabulary, the two dependency maps, and the status tables. A reader who absorbs these pages can verify, by following the references, that every claim of the paper is of exactly the announced strength.

Logical objects. Five kinds of objects appear, and they are never interchangeable.

- A **Step** is a numbered logical stage in the fifteen-step architecture. A Step may state a construction, an implication, an equivalence, or an input condition; its logical type is recorded in the status table below.
- A **logical relation** is the typed relation (implication, equivalence, reduction) asserted between statements; arrows in the maps below carry their type.
- An **H-level** (H0–H8) is a layer of the internal analysis of Step 5 (§9); H-levels are not Steps.
- An **Ω-link** (Ω_1–Ω_6) is a closed reduction or equivalence in the difficulty-localization chain inside Step 5 (§9.8).
- The **terminal residue** is Ω_7: the terminal statement the Ω-chain isolates. It is not a link, and it is not a Step.

Fourteen of the fifteen Steps are closed in the precise sense recorded below: their asserted construction, implication, equivalence, or conditional consequence is proved. The unresolved mathematical input is Step 5. Two downstream Steps (9 and 11) are proved conditional consequences of that input.

Proof-dependency map (implications; every arrow is a proved theorem).

$$\boxed{\text{ARP-P (Step 5)}} \implies \text{NP interpolants} \implies \text{normal family} \implies \text{no off-real poles} \implies \text{RH}$$

$$\text{RH} \implies \text{ARP-P} \qquad \text{(Step 14, Theorem 9.189; the forward chain is Theorem 9.185),}$$

closing the equivalence ARP-P $\Longleftrightarrow$ RH (Corollary 9.190).

Difficulty-localization map (reductions, $\rightsquigarrow$ = "is reduced to / is serialized as"; these arrows re-express the open input, they do not discharge it).

$$\boxed{\text{Step 5}} \rightsquigarrow \text{H0–H8} \rightsquigarrow \text{terminal defect } \delta_N \rightsquigarrow \text{boundary positivity} \Longleftrightarrow \lambda_n \geq 0 \quad (\Omega_7)$$

(the final arrow is typed as an equivalence because it is one: the proved Ω-link Ω_6, Theorem 9.111), and

$$\boxed{\Omega_7 \Longleftrightarrow \text{RH}} \qquad \text{[Li [81], classical].}$$

Status of the fifteen Steps.

Step	Statement	Status	Provenance
1	Fix $s = \frac{1}{2} + iz$, $\Xi(z) = \xi(\frac{1}{2} + iz)$, and $U_- = \{\operatorname{Im} z < -\frac{1}{2}\}$	Closed	classical
2	Define finite channels F and the explicit target G_Ξ^F	Closed	new construction
3	Prove that G_Ξ^F is meromorphic and has no holomorphic ambiguity	Closed	proved here
4	Express G_Ξ^F on U_- through the Weil explicit formula	Closed as interface	classical interface

Step	Statement	Status	Provenance
5	Define ARP-P: Pick positivity of G_Ξ^F on finite node sets in U_-	**Open**; equivalent to RH	open input
6	Prove that Cauchy transforms of positive matrix measures have positive Pick kernels	Closed	classical
7	Prove that tower-ARP implies ARP-P	Closed	proved here
8	Prove that ARP-P supplies Nevanlinna–Pick interpolants	Closed	classical NP
9	Use NP normality/Montel to extend holomorphically from U_- to Ω_-	Closed conditional on ARP-P	conditional consequence
10	Detect every off-line zero by some finite channel residue	Closed	proved here
11	Conclude that ARP-P removes off-line poles in Ω_-	Closed conditional on ARP-P	conditional consequence
12	Use the symmetries $\Xi(-z) = \Xi(z)$ and conjugation to cover all of $\mathbb{C} \setminus \mathbb{R}$	Closed	classical symmetry
13	Translate real zeros of $\Xi(z)$ into RH	Closed	classical
14	Prove the converse direction RH $\Rightarrow$ ARP-P	Closed	proved here
15	Conclude the structural equivalence ARP-P $\iff$ RH	Closed as equivalence	logical assembly

Status of the Ω-chain (proofs in §9.8).

	Statement	Status
Ω_1	RH $\iff$ all Ξ-zeros real	Ω-link, closed (Lemma 9.2)
Ω_2	real zeros $\iff$ ARP-P	Ω-link, closed (Corollary 9.190)
Ω_3	ARP-P $\iff \delta_N \geq 0\ \forall N$ (reference-whitened)	Ω-link, closed (Corollary 9.75)
Ω_4	$\delta_N \geq 0 \iff$ one-sided whitened domination	Ω-link, closed (Theorem 9.71)
Ω_5	interior positivity has a regular boundary limit, per N	Ω-link, closed (Lemma 9.109)
Ω_6	boundary positivity $\iff$ Li–Keiper $\lambda_n \geq 0$	Ω-link, closed (Theorem 9.111)
Ω_7	$\lambda_n \geq 0$ for all n	**terminal residue**: open; $\equiv$ RH (Li [81])

Proof references for the chain. Steps 1–4 are Notations 8.1, 8.2, 8.3, Lemma 9.1, Lemma 9.6, and Theorem 9.8. Step 5 is Definition 9.10. Step 6 is Lemma 9.178. Step 7 is Theorem 9.181. Steps 8–13 are Theorem 9.185, using Lemma 9.187 and Lemma 9.1. Step 14 is Theorem 9.189. Step 15 is Corollary 9.191.

7.2. **Current state of Step 5.** Step 5 is the single open input. The present paper does not claim to prove ARP-P; it decomposes it into precise closed subcases, equivalent formulations, and the routes attempted against it.

Substep	Current status	What remains
H0	Closed: conservation of RH-level difficulty	Nothing
H1	Closed: scalar level-one ARP-P, $\mathrm{Re}(\xi'/\xi)(s) \geq 0$ for $\mathrm{Re}\, s > 1$	Matrix-channel level one is not routine; it has phase content

Substep	Current status	What remains
H2	Closed as a fixed-band separated-node criterion; geometric high-node subcase closed	Apply the criterion to broader separated configurations
H3	Closed in the large-height logarithmic region and a.e.; exact $\Theta \leq W$ reformulation and mirror-pair criterion closed	PDH/global phase decoherence remains open but is not needed for the minimal bridge
H4	Closed as determinant serialization: RH $\iff$ $d_N^{(k)} > 0$ for all k, N	Optional: rigorous determinant margins with the corrected C7 kernel
H5	Closed as equivalence: windowed Weil positivity $\iff$ ARP-P	Window positivity is an equivalent coordinate; the minimal essential input is scalar fixed-sequence positivity
H6	Quantitative single-cluster witness closed for each configuration	Uniform bounds over configuration families, if desired
H7	Scalar level-two band positivity closed for all configurations	SOS remains as identity/effectivity program
H8	Scalar fixed-N band positivity closed for all configurations; one-point jet serialization, Pascal–Laguerre floor, $\log\log$ range, and sine-model defect constant formulated	Uniform no-margin terminal contractivity $\delta_N \geq 0$ for all N is the remaining open point

8. Notation

This section is deliberately a self-contained symbol table: its entries repeat, verbatim, the definitions introduced in Steps 1–2, so that the reader can consult them without searching the chain. The governing statements are the ones in §9.

Notation 8.1 (variables and regions)*.* We use

$$s = \tfrac{1}{2} + iz, \qquad z = \frac{s - \frac{1}{2}}{i}, \qquad \operatorname{Re} s = \tfrac{1}{2} - \operatorname{Im} z.$$

Thus

$$\Omega_- := \{z : \operatorname{Im} z < 0\}, \qquad U_- := \{z : \operatorname{Im} z < -\tfrac{1}{2}\} = \{s : \operatorname{Re} s > 1\}.$$

A nontrivial zero $\rho = \beta + i\gamma$ of ζ corresponds to

$$z_\rho = \gamma - i(\beta - \tfrac{1}{2}).$$

The Riemann Hypothesis is equivalent to all z_ρ being real.

Notation 8.2 (completed function and Herglotz coordinate)*.* Let

$$\xi(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \Xi(z) = \xi(\tfrac{1}{2} + iz).$$

The fixed one-variable coordinate is

$$M_\Xi(z) := -\frac{\Xi'(z)}{\Xi(z)}.$$

Since $\Xi'(z) = i\,\xi'(s)$,

$$M_\Xi(z) = i\left(-\frac{\xi'(s)}{\xi(s)}\right), \qquad s = \tfrac{1}{2} + iz. \tag{8}$$

This factor i is part of every endpoint identity below.

Notation 8.3 (finite channels). For a finite source plane $F = \operatorname{span}\{\phi_1, \dots, \phi_m\}$, define the source functional

$$\ell_a(\phi) = \int_0^\infty \phi(t) e^{-iat}\, dt.$$

The target finite-channel function is the explicit meromorphic sum

$$G^F_\Xi(z)_{\alpha\beta} := \sum_\rho \frac{m_\rho\, \ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar z_\rho}(\phi_\alpha)}}{z_\rho - z}, \tag{9}$$

where the sum is over the zeros of Ξ, counted with multiplicity m_ρ. Thus ARP refers to this specific meromorphic matrix function, not merely to a prescribed principal part. It is not the scalar multiple $M_\Xi(z)[\langle \phi_\beta, \phi_\alpha\rangle]$.

For this matrix function form the Pick kernel

$$\mathsf{K}^F_\Xi(z, w) = \frac{G^F_\Xi(z) - G^F_\Xi(w)^*}{z - \bar w}.$$

Vitali is applied to the one-variable functions G^F_Ξ. Negative squares are read only from the two-variable kernels K^F_Ξ.

This notation is a global preview. The governing finite-channel definition used in the proof is Definition 9.5; the formulas are identical.

9. The chain, step by step

This section is the ordered proof spine of the paper. The headings follow the fifteen Steps of Section 7. A step marked "closed" is either proved in-line below or reduced to a named classical theorem whose hypotheses are checked at the point of use. No mathematical assertion needed by the route is left to an appendix. The single non-closed input is Step 5, ARP-P.

9.1. **Step 1: variables, completed function, and the RH translation.** Set

$$s = \tfrac{1}{2} + iz, \qquad z = \frac{s - \frac{1}{2}}{i}, \qquad \Xi(z) = \xi(\tfrac{1}{2} + iz),$$

where

$$\xi(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Then

$$\operatorname{Re} s = \tfrac{1}{2} - \operatorname{Im} z, \qquad U_- = \{z : \operatorname{Im} z < -\tfrac{1}{2}\} = \{s : \operatorname{Re} s > 1\}.$$

If $\rho = \beta + i\gamma$ is a nontrivial zero of ζ, the corresponding zero of Ξ is

$$z_\rho = \frac{\rho - \frac{1}{2}}{i} = \gamma - i(\beta - \tfrac{1}{2}).$$

The logarithmic derivative coordinate used throughout is

$$M_\Xi(z) := -\frac{\Xi'(z)}{\Xi(z)} = i\left(-\frac{\xi'(s)}{\xi(s)}\right), \qquad s = \tfrac{1}{2} + iz.$$

The factor i is fixed here and is not a later normalization. The whole route rests on the following elementary but decisive translation, which turns a statement about the location of the zeros into a statement about the real axis.

Lemma 9.1 (basic properties of Ξ)**.** *The function $\Xi(z) = \xi(\tfrac{1}{2} + iz)$ is entire, and:*

(i) *its zero multiset is exactly* $\{z_\rho = (\rho - \frac{1}{2})/i\}$*,* ρ *running over the nontrivial zeros of* ζ *with multiplicity; in particular every zero satisfies* $|\operatorname{Im} z_\rho| < \frac{1}{2}$*;*
(ii) $\Xi(-z) = \Xi(z)$*;*
(iii) $\Xi(\bar{z}) = \overline{\Xi(z)}$*; in particular* Ξ *is real on* $\mathbb{R}$*, and the zero multiset is invariant under* $z \mapsto \bar{z}$ *and* $z \mapsto -z$ *with multiplicities preserved.*

Proof. $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire: $s-1$ cancels the pole of ζ at $s=1$, the poles of $\Gamma(s/2)$ at $s = 0, -2, -4, \ldots$ are cancelled by the factor s and by the trivial zeros of ζ, and $\xi(0) = \xi(1) = \frac{1}{2} \neq 0$ [4, Ch. II]. Since $\Gamma(s/2)$ never vanishes and $\zeta(s) \neq 0$ for $\operatorname{Re} s \geq 1$ (Euler product and the classical zero-free line), the functional equation $\xi(s) = \xi(1-s)$ excludes zeros in $\operatorname{Re} s \leq 0$ as well; hence the zeros of ξ are precisely the nontrivial zeros of ζ, all in $0 < \operatorname{Re} s < 1$, which is $|\operatorname{Im} z_\rho| < \frac{1}{2}$ in the z-coordinate. This proves (i). For (ii), $\Xi(-z) = \xi(\frac{1}{2} - iz) = \xi(1 - (\frac{1}{2} - iz)) = \xi(\frac{1}{2} + iz) = \Xi(z)$ by the functional equation. For (iii), ξ is real on $(1, \infty)$, so Schwarz reflection gives $\xi(\bar{s}) = \overline{\xi(s)}$; therefore $\overline{\Xi(z)} = \xi(\frac{1}{2} - i\bar{z}) = \xi(\frac{1}{2} + i\bar{z}) = \Xi(\bar{z})$, using the functional equation once more. The zero-multiset symmetries follow. □

Lemma 9.2 (RH is realness of the Ξ-zeros)**.** *The Riemann Hypothesis is equivalent to the assertion that every nontrivial zero of* $\Xi(z)$ *is real.*

Proof. For a nontrivial zero $\rho = \beta + i\gamma$ of ζ, the change of variables gives $z_\rho = (\rho - \frac{1}{2})/i = \gamma - i(\beta - \frac{1}{2})$, and by Lemma 9.1(i) the zeros of Ξ are exactly the z_ρ, with multiplicity. Hence $z_\rho \in \mathbb{R}$ if and only if $\beta = \frac{1}{2}$, so every zero of Ξ is real precisely when every ρ has $\beta = \frac{1}{2}$, which is RH. □

9.2. **Step 2: finite channels and the target G^F_Ξ.** Steps 2–4 are carried out formally in Definition 9.3, Definition 9.5, Lemma 9.6, Lemma 9.7, and Theorem 9.8; the surrounding discussion explains their content but the formal statements are authoritative.

Definition 9.3 (source core)**.** The source core $\mathcal{S}_a^\circ$ is the set of functions $\phi : (0, \infty) \to \mathbb{C}$ satisfying:

(S1) (profile decay) there is $a_\phi > \frac{1}{2}$ such that, for every N, $|\phi(t)| \leq C_{N,\phi}\, e^{-a_\phi t}(1+t)^{-N}$;
(S2) (enlarged-strip transform) there is $\varepsilon_\phi \in (0, a_\phi - \frac{1}{2})$ such that

$$\ell_w(\phi) = \int_0^\infty \phi(t) e^{-iwt}\, dt$$

is holomorphic on $|\operatorname{Im} w| < \frac{1}{2} + \varepsilon_\phi$ and rapidly decreasing horizontally, uniformly on the closed substrip: for every N,

$$|\ell_w(\phi)| \leq C_{N,\phi}(1 + |\operatorname{Re} w|)^{-N}, \qquad |\operatorname{Im} w| \leq \tfrac{1}{2} + \tfrac{\varepsilon_\phi}{2}.$$

Every $\phi \in C_c^\infty(0, \infty)$ belongs to $\mathcal{S}_a^\circ$ with arbitrary $a_\phi > \frac{1}{2}$ and $\varepsilon_\phi \in (0, a_\phi - \frac{1}{2})$: (S1) is trivial by compact support, and (S2) follows by repeated integration by parts, since $|e^{-iwt}| \leq e^{(\frac{1}{2}+\varepsilon_\phi)T}$ on the support. Note that (S1) makes the absolute convergence in (S2) automatic; the content of (S2) is the horizontal rapid decrease.

Remark 9.4 (realization-layer compatibility). The realization layer of Section 9.10 only requires the weaker profile decay $a > \frac{1}{4}$ for the prime-cell summability; since $a_\phi > \frac{1}{2} > \frac{1}{4}$, every source-core element is admissible there. In that layer the profile ϕ is denoted $\widehat{\phi}$ (the logarithmic Mellin profile); the two notations refer to the same function and the hat is never the Fourier transform of Step 4.

With this test space fixed, the object the whole chain interpolates is the following explicit meromorphic matrix function.

Definition 9.5 (finite channels and the target G_Ξ^F)**.** For a finite source plane $F = \operatorname{span}\{\phi_1, \ldots, \phi_m\} \subset \mathcal{S}_a^\circ$ and sources $\phi_\alpha, \phi_\beta \in F$, set $\ell_a(\phi) = \int_0^\infty \phi(t)e^{-iat}\,dt$ and define the target channel

$$G_\Xi^F(z)_{\alpha\beta} = \sum_\rho \frac{m_\rho\, \ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}}{z_\rho - z},$$

the sum running over the zeros z_ρ of Ξ with multiplicity m_ρ.

This is not defined only up to a holomorphic summand: the residues, source weights, zero multiplicities, and denominator convention are all fixed in the formula.

9.3. **Step 3: meromorphy and no holomorphic ambiguity.** The source core $\mathcal{S}_a^\circ$ is chosen so that its Laplace transforms have rapid horizontal decay on a closed strip containing the zero strip $|\operatorname{Im} a| \le \frac{1}{2}$. Since the zeros of Ξ satisfy the classical counting estimate $N(T) = O(T \log T)$, for every compact set K avoiding the zeros and every N one has

$$\left| \frac{\ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}}{z_\rho - z} \right| \le C_{K,F,N}(1 + |\operatorname{Re} z_\rho|)^{-N}(1 + |z_\rho|)^{-1}.$$

Choosing N large makes the zero sum locally uniformly convergent away from the zeros. Therefore G_Ξ^F is meromorphic and its principal part at z_ρ is exactly

$$\operatorname{pp}_{z=z_\rho} G_\Xi^F(z)_{\alpha\beta} = \frac{m_\rho\, \ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}}{z_\rho - z}.$$

This proves the absence of holomorphic ambiguity: the target is the displayed meromorphic function, not a principal-part prescription modulo entire terms.

Lemma 9.6 (target channel meromorphy)**.** *For every finite source plane $F \subset \mathcal{S}_a^\circ$, the series* (9) *converges locally uniformly away from the zeros of Ξ and defines a meromorphic matrix function. At a zero z_ρ its principal part is*

$$\operatorname{pp}_{z=z_\rho} G_\Xi^F(z)_{\alpha\beta} = \frac{m_\rho\, \ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}}{z_\rho - z}.$$

Moreover $G_\Xi^F(\bar{z}) = G_\Xi^F(z)^*$.

Proof. By Definition 9.3, source vectors have Laplace transforms with rapid horizontal decay uniformly on a closed strip containing $|\operatorname{Im} a| \le \frac{1}{2}$; compact logarithmic tests have this property by the Paley–Wiener/integration-by-parts estimate, and the general source core is defined to keep the same estimate. Since the zeros of Ξ lie in $|\operatorname{Im} a| < \frac{1}{2}$ by Lemma 9.1(i) and satisfy $N(T) = O(T \log T)$, for every N and every compact K avoiding the zeros,

$$\left| \frac{\ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}}{z_\rho - z} \right| \le C_{K,F,N}(1 + |\operatorname{Re} z_\rho|)^{-N}(1 + |z_\rho|)^{-1}.$$

Choosing N large gives a summable majorant over the zero set. The only singularities are the displayed simple poles, with multiplicities inherited from Ξ.

Two points deserve record. First, near a fixed zero z_{ρ_0} the series splits as the displayed principal part plus the sum over $z_\rho \neq z_{\rho_0}$, which converges locally uniformly on a punctured disk by the same majorant; hence z_{ρ_0} is a simple pole of the matrix function with exactly the displayed principal part, and G^F_Ξ is meromorphic. Second, the factor $w \mapsto \overline{\ell_{\bar w}(\phi_\alpha)}$ is itself holomorphic, by the identity

$$\overline{\ell_{\bar w}(\phi_\alpha)} = \int_0^\infty \overline{\phi_\alpha(t)}\, e^{iwt}\, dt,$$

convergent on the same strip by (S1). Finally, the conjugation symmetry of the zero multiset used in pairing z_ρ with $\bar z_\rho$ in (9) is Lemma 9.1(iii), and it gives $G^F_\Xi(\bar z) = G^F_\Xi(z)^*$. □

9.4. **Step 4: Weil explicit formula on U_-.** For $z \in U_-$ define

$$h_{\alpha\beta,z}(w) = \frac{\ell_w(\phi_\beta)\overline{\ell_{\bar w}(\phi_\alpha)}}{w - z}.$$

Because z lies below the zero strip, the denominator has no pole in a slightly enlarged closed strip around $|\operatorname{Im} w| \le \frac{1}{2}$, and the source decay makes $h_{\alpha\beta,z}$ an admissible Weil test. The classical Weil explicit formula gives

$$G^F_\Xi(z)_{\alpha\beta} = \mathcal{W}_\Xi(h_{\alpha\beta,z}),$$

where $\mathcal{W}_\Xi$ is the usual archimedean–pole–prime functional:

$$\begin{aligned}\mathcal{W}_\Xi(h) = h(i/2) + h(-i/2) - \frac{\log \pi}{2\pi}\int_{\mathbb{R}} h(t)\, dt \\ + \frac{1}{2\pi}\int_{\mathbb{R}} h(t) \operatorname{Re}\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) dt - \frac{1}{2\pi}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\big(\widehat{h}(\log n) + \widehat{h}(-\log n)\big).\end{aligned}$$

The proof is the standard smoothing argument: apply the explicit formula first to $e^{-\varepsilon w^2} h_{\alpha\beta,z}(w)$, where zero and prime sums converge absolutely, and let $\varepsilon \downarrow 0$ in the Weil test topology. Thus the target values on U_- are computable from the arithmetic side without using zeros as input.

Lemma 9.7 (admissibility of the channel tests)**.** *Let $z \in U_-$ and $\phi_\alpha, \phi_\beta \in \mathcal{S}^\circ_a$, and set $\varepsilon := \frac{1}{2}\min\{\varepsilon_{\phi_\alpha}, \varepsilon_{\phi_\beta}, |\operatorname{Im} z| - \frac{1}{2}\} > 0$. Then $h_{\alpha\beta,z}(w) = \ell_w(\phi_\beta)\overline{\ell_{\bar w}(\phi_\alpha)}\,(w-z)^{-1}$ is holomorphic on $|\operatorname{Im} w| \le \frac{1}{2} + \varepsilon$ and satisfies there, for every N,*

$$|h_{\alpha\beta,z}(w)| \le C_N(1 + |\operatorname{Re} w|)^{-N}.$$

Consequently: (i) *the pole terms $h_{\alpha\beta,z}(\pm i/2)$ are finite;* (ii) *the archimedean integrals converge absolutely against the logarithmic growth of* Γ'/Γ*;* (iii) *shifting the contour of $\widehat{h}(u) = \int_{\mathbb{R}} h(t)e^{-iut}\, dt$ to* $\operatorname{Im} w = \mp(\frac{1}{2} + \varepsilon)$ *gives*

$$|\widehat{h}(\log n)| + |\widehat{h}(-\log n)| \le C\, n^{-(\frac{1}{2}+\varepsilon)},$$

so the prime side is dominated by $\sum_n \Lambda(n) n^{-1-\varepsilon} < \infty$ and converges absolutely.

Proof. Both Laplace factors are holomorphic and rapidly decreasing on $|\operatorname{Im} w| \le \frac{1}{2} + \varepsilon$ by (S2) of Definition 9.3, the second via $\overline{\ell_{\bar w}(\phi_\alpha)} = \int_0^\infty \overline{\phi_\alpha(t)} e^{iwt} dt$. The denominator satisfies $|w - z| \ge |\operatorname{Im} z| - \frac{1}{2} - \varepsilon \ge \varepsilon > 0$ on the strip, since $\operatorname{Im} z \le -\frac{1}{2} - 2\varepsilon$; hence h is holomorphic there with the displayed decay. (i) follows because $\pm i/2$ is interior to the strip and distinct from z. (ii) is the decay against $\log(2 + |t|)$. For (iii): for $u = \log n > 0$ shift the contour to $\operatorname{Im} w = -(\frac{1}{2} + \varepsilon)$, legal by holomorphy and horizontal decay, so that $|e^{-iuw}| = e^{u \operatorname{Im} w} = n^{-(\frac{1}{2}+\varepsilon)}$; for $u = -\log n$ shift to $\operatorname{Im} w = +(\frac{1}{2} + \varepsilon)$. In both cases no pole is crossed. □

Theorem 9.8 (explicit channel formula on U_-)**.** *Let $z \in U_-$ and define, for $\phi_\alpha, \phi_\beta \in F$,*

$$h_{\alpha\beta,z}(w) := \frac{\ell_w(\phi_\beta)\overline{\ell_{\bar{w}}(\phi_\alpha)}}{w - z}.$$

Then $h_{\alpha\beta,z}$ is holomorphic in a strip containing all zeros of Ξ and rapidly decreasing on horizontal lines in that strip, and

$$G_\Xi^F(z)_{\alpha\beta} = \mathcal{W}_\Xi(h_{\alpha\beta,z}),$$

where $\mathcal{W}_\Xi$ is the classical Weil explicit-formula functional

$$\begin{aligned}\mathcal{W}_\Xi(h) = h(i/2) + h(-i/2) - \frac{\log \pi}{2\pi} \int_{\mathbb{R}} h(t)\, dt \\ + \frac{1}{2\pi} \int_{\mathbb{R}} h(t) \operatorname{Re} \frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) dt - \frac{1}{2\pi} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \big(\widehat{h}(\log n) + \widehat{h}(-\log n)\big),\end{aligned}$$

with Fourier convention

$$\widehat{h}(u) = \int_{\mathbb{R}} h(t) e^{-iut}\, dt.$$

The displayed identity is understood in the standard Weil explicit-formula sense for admissible strip-holomorphic rapidly decreasing tests; equivalently, by inserting the usual exponential smoothing in the zero and prime sums and then removing the smoothing.

Proof. Lemma 9.7 makes $h_{\alpha\beta,z}$ an admissible test function for the Weil explicit formula. The prime side is the standard continuous extension of the explicit-formula functional from exponentially smoothed tests: apply the formula first to $h_{\alpha\beta,z,\varepsilon}(w) = e^{-\varepsilon w^2} h_{\alpha\beta,z}(w)$ on horizontal contours inside the zero strip, where the smoothed sums are absolutely convergent, and let $\varepsilon \downarrow 0$ in the Weil test topology. This is the classical formulation of the explicit formula for admissible Schwartz strip tests.

By Lemma 9.7 all three sides of the smoothed identity are dominated, uniformly in $\varepsilon \in (0, 1]$, by convergent majorants: the zero side by the majorant of Lemma 9.6, the archimedean side by (ii), and the prime side by (iii), since $|e^{-\varepsilon w^2}| \leq e^{\varepsilon(\frac{1}{2}+\eta)^2} \leq C$ on the strip, where η is the positive strip margin supplied by Lemma 9.7. Dominated convergence therefore passes the classical formula [2] to the limit test $h_{\alpha\beta,z}$, term by term.

Applying the Weil explicit formula to $h_{\alpha\beta,z}$ gives

$$\mathcal{W}_\Xi(h_{\alpha\beta,z}) = \sum_\rho m_\rho h_{\alpha\beta,z}(z_\rho) = \sum_\rho \frac{m_\rho\, \ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}}{z_\rho - z} = G_\Xi^F(z)_{\alpha\beta}.$$

Equivalently, the Cauchy denominator may be written in U_- as the unilateral Laplace kernel

$$\frac{1}{z_\rho - z} = -i \int_0^\infty e^{-izt} e^{iz_\rho t}\, dt,$$

because $\operatorname{Im} z < -\frac{1}{2}$ and $|\operatorname{Im} z_\rho| < \frac{1}{2}$. This is the transfer-side representation of the same target value. It does not prove ARP; it proves that the ARP target on U_- is a completely specified arithmetic distribution, with arithmetic side fixed by the Weil formula and no unspecified holomorphic summand. $\square$

9.5. **Step 5: ARP-P. Step 5 (ARP-P) is the only one of the fifteen Steps that is not closed.** Steps 1–4 and 6–15 are closed in the sense of the status map (§7.1), Steps 9 and 11 as proved conditional consequences of Step 5; the whole path reduces RH to this one input. ARP-P in turn reduces, through the Ω-chain (Section 9.8), to the single terminal-defect inequality

$$\delta_N \geq 0 \ \text{ for all } N \quad \Longleftrightarrow \quad \text{RH} \qquad (\Omega_7),$$

equivalently $\lambda_n \geq 0$ for all n (Li–Keiper). This is the *sole* pending statement of the proposed proof; every other link in the path is established. The remainder of Step 5 decomposes Ω_7 into closed subcases and records the routes that were tried against it, none of which closes it.

ARP-P is the assertion that for every finite source plane F and every finite node set $z_1, \dots, z_N \in U_-$, the block Pick matrix

$$\mathsf{P}[g^F; z_1, \dots, z_N] = \left[\frac{g^F(z_i) - g^F(z_j)^*}{z_i - \bar{z}_j} \right]_{i,j=1}^{N}$$

is positive semidefinite, where $g^F = G^F_{\Xi}|_{U_-}$. This is the only open input in the chain. The rest of the paper proves that this input is equivalent to RH and decomposes it into lower-dimensional closed regimes and the routes attempted against the residue.

Definition 9.9 (arithmetic channel data)**.** For a finite source plane $F \subset \mathcal{S}_a^\circ$ define

$$g^F(z) := G^F_{\Xi}(z), \qquad z \in U_-.$$

The values are computed through Theorem 9.8: for each $z \in U_-$, $g^F(z)_{\alpha\beta}$ is the Weil explicit-formula functional applied to the admissible test $h_{\alpha\beta,z}$. Thus the arithmetic side is fixed by the archimedean, pole/degree, and prime terms, with the standard Weil smoothing convention; no unspecified holomorphic summand remains.

Definition 9.10 (ARP-P: arithmetic Pick positivity)**.** The channel data satisfy *arithmetic Pick positivity* (ARP-P) if, for every finite source plane $F = \operatorname{span}\{\phi_1, \dots, \phi_m\} \subset \mathcal{S}_a^\circ$ and every finite node set $\{z_1, \dots, z_N\} \subset U_-$, the block Pick matrix

$$\mathsf{P}[g^F; z_1, \dots, z_N] := \left[\frac{g^F(z_i) - g^F(z_j)^*}{z_i - \bar{z}_j} \right]_{i,j=1}^{N} \in M_{Nm}(\mathbb{C})$$

is positive semidefinite.

9.6. **Step 5 substructure: current closed subcases and open fronts.** This subsection descends one level in the logical hierarchy of §7.1: from the fifteen-step chain into the internal H-level analysis (H0–H8) of the single open Step. Nothing here is a Step of the chain; everything here lives inside Step 5.

The present decomposition of Step 5 is not a list of slogans. It is the working mathematical hierarchy for attacking the only open input:

$$\text{ARP-P} \quad \Longleftrightarrow \quad \text{positive arithmetic Pick kernels for all finite channels}$$
$$\text{and all finite node sets in } U_-.$$

Everything below belongs to the proof as it stands so far. The closed items are proved here; the open items are formulated here in the exact form in which they remain to be attacked.

H0 (closed): conservation of RH-level difficulty. Let $S_1, \ldots, S_k$ be assertions. If

$$S_1 \wedge \cdots \wedge S_k \Longrightarrow \mathrm{RH}$$

and every S_j is proved unconditionally from standard theorems, then RH is proved by modus ponens. Therefore, in any chain that implies RH but does not prove RH, at least one step is either false or has RH-level difficulty. In the chain, that hard step is Step 5, ARP-P. This observation is logically closed and prevents the paper from hiding RH inside a false trace-log or local-cell identity.

H1 (closed): scalar one-node ARP-P. For the scalar function

$$M_\Xi(z) = i\left(-\frac{\xi'}{\xi}(s)\right), \qquad s = \tfrac{1}{2} + iz,$$

the one-node Pick quotient is

$$\frac{M_\Xi(z) - M_\Xi(z)^*}{z - \bar{z}} = \frac{\operatorname{Im} M_\Xi(z)}{\operatorname{Im} z}.$$

Writing $s = \sigma + it$ and $z = x + iy$, one has $y = -(\sigma - \frac{1}{2})$. Since $M_\Xi = i(-\xi'/\xi)$,

$$\operatorname{Im} M_\Xi(z) = -\operatorname{Re}\frac{\xi'}{\xi}(s).$$

Thus

$$\frac{\operatorname{Im} M_\Xi(z)}{\operatorname{Im} z} = \frac{-\operatorname{Re}(\xi'/\xi)(s)}{-(\sigma - \frac{1}{2})} = \frac{\operatorname{Re}(\xi'/\xi)(s)}{\sigma - \frac{1}{2}}.$$

The scalar one-node Pick condition on U_- is therefore equivalent to

$$\operatorname{Re}\frac{\xi'}{\xi}(s) \geq 0, \qquad \operatorname{Re} s > 1,$$

because $\sigma - \frac{1}{2} > 0$.

The inequality is classical. From the Hadamard product for ξ, with the standard real normalization of the constant term,

$$\frac{\xi'}{\xi}(s) = \sum_\rho \left(\frac{1}{s - \rho} + \frac{1}{\rho}\right) + B,$$

and after pairing zeros symmetrically, the real parts of the constant terms cancel in the usual limiting sense. Hence, for $s = \sigma + it$,

$$\operatorname{Re}\frac{\xi'}{\xi}(s) = \sum_\rho \frac{\sigma - \beta}{|s - \rho|^2}.$$

The classical zero-free line and functional equation give $0 < \beta < 1$ for nontrivial zeros. Therefore for $\sigma > 1$ every summand is nonnegative, and H1 is closed.

H2 (closed as a fixed-band separated-node criterion). Fix a vertical band

$$1 < \sigma_0 \le \sigma \le \sigma_1 < \infty.$$

For scalar nodes $s_j = \sigma_j + it_j$ in this band, let $z_j = (s_j - \frac{1}{2})/i$ and form the scalar Pick matrix

$$K_{ij} = \frac{M_\Xi(z_i) - \overline{M_\Xi(z_j)}}{z_i - \bar{z}_j}.$$

There are effective constants $a_{\sigma_0,\sigma_1}, B_{\sigma_0,\sigma_1} > 0$ such that, for all sufficiently large heights in the band,

$$K_{ii} \ge a_{\sigma_0,\sigma_1} \log(3 + |t_i|),$$

and for $i \ne j$,

$$|K_{ij}| \le B_{\sigma_0,\sigma_1} \frac{\log(3 + |t_i|) + \log(3 + |t_j|)}{\max\{|t_i - t_j|,\, 2(\sigma_0 - \frac{1}{2})\}}.$$

The diagonal bound is H1 plus Stirling's formula for the Γ'/Γ term and the boundedness of ζ'/ζ in $\sigma \ge \sigma_0$. The off-diagonal bound follows from $|M_\Xi(z_i)| \ll_{\sigma_0,\sigma_1} \log(3 + |t_i|)$ and

$$|z_i - \bar{z}_j| \ge \max\{|t_i - t_j|,\, 2(\sigma_0 - \tfrac{1}{2})\}.$$

Therefore, if

$$\sum_{j \ne i} \frac{\log(3 + |t_i|) + \log(3 + |t_j|)}{\max\{|t_i - t_j|,\, 2(\sigma_0 - \frac{1}{2})\}} \le \frac{a_{\sigma_0,\sigma_1}}{2B_{\sigma_0,\sigma_1}} \log(3 + |t_i|)$$

for every i, then

$$\sum_{j \ne i} |K_{ij}| \le \frac{1}{2} K_{ii}.$$

The matrix is Hermitian, so Gershgorin's theorem implies that it is positive semidefinite. This closes H2 as a verifiable separated-node criterion.

A closed concrete subcase is geometric separation. If, after ordering by height,

$$|t_j| \ge T_* Q^j, \qquad |t_i - t_j| \ge \tfrac{1}{2} \max\{|t_i|, |t_j|\} \quad (i \ne j),$$

then

$$\sum_{j<i} |K_{ij}| \le C_1 \frac{\log(3 + |t_i|)}{|t_i|} \#\{j < i\} \le C_2 \frac{(\log(3 + |t_i|))^2}{|t_i|},$$

while

$$\sum_{j>i} |K_{ij}| \le C_3 \sum_{j>i} \frac{\log(3 + |t_j|)}{|t_j|} \le C_4 \frac{\log(3 + |t_i|)}{|t_i|}.$$

For T_* large these row sums are smaller than $\frac{1}{2} K_{ii}$. Hence geometrically separated high nodes give a closed positive scalar Pick subcase.

H3 (closed in a logarithmic region and almost everywhere): infinitesimal two-node target. The scalar coincident two-node limit is the Schwarz–Pick inequality

$$(\sigma - \tfrac{1}{2}) \left| \left(\frac{\xi'}{\xi} \right)'(s) \right| \le \operatorname{Re} \frac{\xi'}{\xi}(s), \qquad \sigma = \operatorname{Re} s > 1.$$

It is not closed globally. Two substantial regions are closed.

Deterministic logarithmic region. There exist effective constants $c > 0$ and $T_0 \ge 3$ such that the inequality holds whenever

$$\sigma \ge 1 + \frac{c}{\sqrt{\log(3 + |t|)}}$$

and $|t| \geq T_0$. The proof splits into two regions. In the band $1+\epsilon \leq \sigma \leq 2$, $0 < \epsilon \leq 1$, the absolutely convergent Dirichlet series gives

$$\left| \left(\frac{\zeta'}{\zeta} \right)'(s) \right| \leq \sum_{n \geq 2} \Lambda(n)(\log n) n^{-1-\epsilon} \leq C_1 \epsilon^{-2} + C_2.$$

The elementary factors satisfy

$$\left| \frac{1}{4}\psi'(s/2) - s^{-2} - (s-1)^{-2} \right| \leq C_3 (1+|t|)^{-1},$$

and therefore

$$\left(\sigma - \tfrac{1}{2}\right) \left| \left(\frac{\xi'}{\xi} \right)'(s) \right| \leq C_4 \epsilon^{-2} + C_5.$$

On the other hand, Stirling plus $\mathrm{Re}(\zeta'/\zeta)(s) \geq -\sum_{n \geq 2} \Lambda(n) n^{-1-\epsilon} \geq -C_6 \epsilon^{-1} - C_7$ gives

$$\mathrm{Re}\, \frac{\xi'}{\xi}(s) \geq \frac{1}{2} \log \frac{|t|}{2\pi} - C_6 \epsilon^{-1} - C_7.$$

Choosing c large enough and then T_0 large enough makes the right side dominate the left when $\epsilon \geq c/\sqrt{\log(3+|t|)}$.

For $\sigma \geq 2$, the Euler derivative has exponential decay:

$$\left| \left(\frac{\zeta'}{\zeta} \right)'(s) \right| \leq 2^{-(\sigma-2)} \sum_{n \geq 2} \Lambda(n)(\log n) n^{-2} \leq C_8\, 2^{-\sigma}.$$

Stirling gives $|\frac{1}{4}\psi'(s/2)| \leq (2|s|)^{-1} + O(|s|^{-2})$, so

$$\left(\sigma - \tfrac{1}{2}\right) \left| \left(\frac{\xi'}{\xi} \right)'(s) \right| \leq C_9 \sigma 2^{-\sigma} + \frac{1}{2} + O(|s|^{-1}).$$

For large σ and $|t|$ this is < 1, while

$$\mathrm{Re}\, \frac{\xi'}{\xi}(s) \geq \frac{1}{2} \log \frac{|s|}{2\pi} - O(1) \geq 1.$$

The finite band $2 \leq \sigma \leq \sigma_2$ is absorbed by the same fixed-band estimates as above. This closes the logarithmic region.

Almost-everywhere edge region. There are effective $C, T_1 > 0$ such that for $T \geq T_1$ and $\epsilon \in [T^{-1/5}, 1]$, the two-node inequality holds at $s = 1 + \epsilon + it$ for all $t \in [T, 2T]$ outside a set of measure at most $CT(\log T)^{-2}$. In the band $1 < \sigma \leq 2$, it is enough to have

$$\tfrac{3}{2} \left| \left(\frac{\zeta'}{\zeta} \right)'(1+\epsilon+it) \right| + \left| \frac{\zeta'}{\zeta}(1+\epsilon+it) \right| \leq \tfrac{1}{4} \log T.$$

The Montgomery–Vaughan mean-value theorem for Dirichlet series gives, uniformly in $\epsilon \in [T^{-1/5}, 1]$,

$$\int_T^{2T} \left| \frac{\zeta'}{\zeta}(1+\epsilon+it) \right|^2 dt \leq C_1 T,$$

and

$$\int_T^{2T} \left| \left(\frac{\zeta'}{\zeta} \right)'(1+\epsilon+it) \right|^2 dt \leq C_2 T.$$

The square sums $\sum \Lambda(n)^2 n^{-2-2\epsilon}$ and $\sum \Lambda(n)^2 (\log n)^2 n^{-2-2\epsilon}$ remain bounded at $\epsilon = 0$, and the conservative Montgomery–Vaughan error is absorbed in the range $\epsilon \geq T^{-1/5}$. Chebyshev's inequality then gives an exceptional set of measure $O(T(\log T)^{-2})$.

H4 (closed): determinant serialization and Schur coordinates. The formal closure is Theorem 9.38: after fixing the dense detection family $\{\phi_k\}$ and a nested node sequence $\{z_j\} \subset U_-$,

$$\mathrm{RH} \iff d_N^{(k)} > 0 \quad \text{for all } k, N.$$

Equivalently, H4 is no longer merely a computational formalism; it is a countable determinant serialization of RH. For each finite channel F, the Schur–Nevanlinna algorithm applied to the finite data

$$g^F(z_1), \ldots, g^F(z_N)$$

produces contractivity parameters γ_N^F whenever the preceding Pick matrices are positive. The formal target is

$$\text{ARP-P} \quad \iff \quad \|\gamma_N^F\| \le 1 \quad \text{for every } N \text{ and every finite } F.$$

Degenerate Pick matrices must be handled by the degenerate Nevanlinna–Pick theory, equivalently by Moore–Penrose Schur complements plus the range condition. In scalar nondegenerate normalization one expects

$$1 - |\gamma_N|^2 = C_N(z_1, \ldots, z_{N+1}) \frac{\det P_{N+1} \det P_{N-1}}{(\det P_N)^2}, \qquad C_N > 0,$$

with the geometric constant fixed by the chosen Cayley normalization. The determinant theorem Theorem 9.38 is the governing closed statement: the Schur parameters are the same serialization in recursive coordinates, while $d_N^{(k)} > 0$ avoids any dependence on a chosen normalization constant.

H5 (closed as equivalence): windowed Weil positivity $\iff$ ARP-P. For $z \in U_-$ the correct unilateral kernel is

$$\frac{1}{z_\rho - z} = -i \int_0^\infty e^{-izt} e^{iz_\rho t} \, dt,$$

because $\operatorname{Im} z < -\frac{1}{2}$ and $|\operatorname{Im} z_\rho| < \frac{1}{2}$. Write formally

$$g^F(z) = -i \int_0^\infty e^{-izt} \psi^F(t) \, dt, \qquad \psi^F(t)_{\alpha\beta} = \sum_\rho m_\rho \, \ell_{z_\rho}(\phi_\beta) \overline{\ell_{\bar{z}_\rho}(\phi_\alpha)} e^{iz_\rho t},$$

and extend $\psi^F(-\tau) = \psi^F(\tau)^*$. For compactly supported smooth vector tests $f : [0, \infty) \to \mathbb{C}^m$, define

$$Q^F(f) = \iint_{[0,\infty)^2} f(t)^* \psi^F(t' - t) f(t') \, dt \, dt'.$$

Then the following are equivalent:

$$\mathsf{P}[g^F; z_1, \ldots, z_N] \succeq 0 \quad \text{for all finite node sets in } U_-,$$

and

$$Q^F(f) \ge 0 \quad \text{for every compactly supported smooth window test } f.$$

Lemma 9.11 (transfer kernel identity)**.** *For $z, w \in U_-$,*

$$\iint_{[0,\infty)^2} \overline{e^{-iwt}} \, \psi^F(t' - t) \, e^{-izt'} \, dt \, dt' = \frac{g^F(z) - g^F(w)^*}{z - \bar{w}}.$$

Proof. Insert $\psi^F(\tau) = \sum_\rho m_\rho W_\rho\, e^{iz_\rho\tau}$ with $W_\rho = \ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar z_\rho}(\phi_\alpha)}$:

$$\int_0^\infty e^{i(\bar w - z_\rho)t}\, dt = \frac{-i}{z_\rho - \bar w}, \qquad \int_0^\infty e^{i(z_\rho - z)t'}\, dt' = \frac{i}{z_\rho - z},$$

both convergent since $|\operatorname{Im} z_\rho| < \frac{1}{2} < -\operatorname{Im} z, -\operatorname{Im} w$; the product is $m_\rho W_\rho/((z_\rho - z)(z_\rho - \bar w))$, summable by Lemma 9.26, and equals the displayed quotient by Lemma 9.178. □

Remark 9.12 (kernel orientation)*.* The opposite convention $\psi^F(t - t')$ yields the Pick form at the reflected nodes $\{-\bar z_j\}$; the equivalence above is unaffected (the node class is closed under reflection), but the pointwise identity of Lemma 9.11 requires the orientation $\psi^F(t' - t)$.

The key representation is Laplace inversion along a horizontal line. Fix $y_0 > \frac{1}{2}$ and $z(x) = x - iy_0$. If $f \in C_c^\infty(0, \infty)$, set

$$c(x) = \frac{1}{2\pi}\int_0^\infty f(t)e^{iz(x)t}\, dt = \frac{1}{2\pi}\int_0^\infty f(t)e^{y_0 t}e^{ixt}\, dt.$$

Since $f(t)e^{y_0 t}$ is compactly supported smooth, c is Schwartz and Fourier inversion gives

$$f(t) = \int_{\mathbb{R}} c(x)e^{-iz(x)t}\, dx.$$

If window positivity holds, apply it to truncated functions

$$f_L(t) = \eta_L(t)\sum_j c_j e^{-iz_j t},$$

where $\eta_L = 1$ on $[0, L-1]$ and vanishes beyond L. Dominated convergence gives

$$\lim_{L\to\infty} Q^F(f_L) = \sum_{i,j} c_i^* K(z_i, z_j)c_j,$$

so all Pick matrices are positive. Conversely, if all Pick matrices are positive, use the horizontal-line representation of f and Fubini:

$$Q^F(f) = \iint_{\mathbb{R}^2} c(x)^* K(z(x), z(x'))c(x')\, dx\, dx'.$$

Every Riemann sum of this integral is nonnegative by finite Pick positivity; hence the integral is nonnegative. Thus H5 is closed as an equivalence. Proving positivity for every window remains exactly the RH-hard input in window coordinates.

H6 (closed quantitatively for each single-cluster configuration). If RH fails, then ARP-P fails for some finite source plane and some finite node set. This is the contrapositive of Steps 8–13:

$$\text{ARP-P} \Longrightarrow \text{RH}.$$

Equivalently, an off-line zero forces a finite Pick matrix with a negative eigenvalue. The qualitative statement is closed. Proposition 9.30, Lemma 9.31, and Theorem 9.32 give the explicit rational-witness bound for each single-cluster configuration. Uniform bounds over families of configurations remain a separate quantitative target. The local obstruction block has vectors

$$v_i = \frac{1}{z_\rho - z_i}, \qquad u_i = \frac{1}{\bar z_\rho - z_i},$$

and the first quantitative target is a single-cluster estimate.

H7 (partially closed): Andreief/Vandermonde level-two SOS. Under RH, a scalar two-node Pick determinant has the positive Andreief form

$$\det P_2 = \frac{1}{2}\iint \left|\phi_{z_1}(t)\phi_{z_2}(t') - \phi_{z_1}(t')\phi_{z_2}(t)\right|^2 d\Sigma(t)d\Sigma(t'), \qquad \phi_z(t) = \frac{1}{t-z}.$$

The integrand is an antisymmetrized Cauchy/Vandermonde square and vanishes on the diagonal $t = t'$. The open H7 target is to express the corresponding arithmetic two-point form, after pole shorting, as a manifestly nonnegative sum of squares. Corollaries 9.46 and 9.58 close the scalar nondegenerate and fixed coalescent large-height regimes. The remaining SOS target is the multiscale/arithmetically degenerate certification not covered by single-cluster positivity. This is not a restatement of H1: it tests the antisymmetrized two-point correlation.

H8 (quantitatively closed for scalar fixed-N clusters and terminal gauges). For scalar node clusters of fixed pattern, H8 is closed in the large-height regime by Theorem 9.42. If

$$z_j = t + a_j - iy_j, \qquad y_j \geq \tfrac{1}{2} + \epsilon_0,$$

and the pairs (a_j, y_j) are distinct and fixed, then the scalar Pick matrix of M_Ξ is positive for all sufficiently large $|t|$. The proof isolates the Szegő Gram matrix

$$S_{ij} = \frac{-i}{z_i - \bar{z}_j},$$

whose smallest eigenvalue supplies the cluster margin, while the archimedean contribution grows like $\log|t|\, S$ and the prime contribution is bounded in the fixed band $\operatorname{Re} s \geq 1 + \epsilon_0$. Lemma 9.44 evaluates $\det S$ by the Cauchy determinant formula, and Theorem 9.45 makes the closure uniform over boxes:

$$\prod_{i<j} |w_i - w_j|^2 \gtrsim \frac{1}{\log|t|} \quad \Longrightarrow \quad \mathsf{P}[M_\Xi; z_1, \ldots, z_N] \succ 0.$$

Thus a cluster with minimum pairwise gap h is covered down to $h \gtrsim (\log|t|)^{-1/(N(N-1))}$. Theorem 9.50 gives the complementary exact serialization: for any fixed $z_0 \in U_-$, RH is equivalent to positivity of all one-point jet matrices for the detection family. Remark 9.164 identifies the jet order with the prime-window scale.

Theorem 9.63 closes the scalar fixed-N band problem for all configurations, including multiscale clusters. When N nodes coalesce at z, ARP-P(N) limits to positivity of the jet matrix

$$J_N(z) = \left[\frac{1}{i!\,j!}\partial_z^i \partial_{\bar{w}}^j K(z,w)\big|_{w=z}\right]_{i,j=0}^{N-1}.$$

For each fixed derivative order k,

$$\sum_{n\geq 2} \Lambda(n)^2 (\log n)^{2k} n^{-2} < \infty,$$

so the Montgomery–Vaughan mean-value mechanism used in H3 extends entrywise to every fixed derivative. This is the limit in which the Szegő eigenvalue $\lambda_{\mathcal{P}}$ tends to zero in ordinary coordinates; the Newton congruence of Theorem 9.63 controls it for each fixed N in a fixed band. This still does not prove RH, because the minimal scalar bridge requires positivity for all prefixes of one fixed accumulating sequence, i.e. uniform control as $N \to \infty$ at one interior height.

The terminal one-point gauge is sharpened by Lemma 9.70, Theorem 9.71, Theorem 9.76, and Corollary 9.79. The archimedean floor is effective through the Pascal–Laguerre matrix, the prime side is represented exactly by the von Mangoldt shift operator, and deterministic contractivity is proved up to order $N \asymp \log\log|t_0|$ with explicit constants. The correct remaining observable is

$$\delta_N(z_0) = 1 - \lambda_{\max}(T_N(z_0))$$

in the J_N^{∞}-whitened coordinate, valid in the range $N \le N_*(t_0)$ of Lemma 9.72; the governing all-N form is the reference-whitened defect of Definition 9.73, with which it agrees in sign throughout that range. Theorem 9.84 shows that, under RH, this defect has no fixed positive margin: the terminal sections approach the boundary in the sense $\limsup_N \lambda_{\max}(T_N) = 1$. Therefore the final terminal input is the exact sign condition $\delta_N(z_0) \ge 0$ for every N, not a margin estimate.

In summary: H1, H2, H3 in the stated regions plus its exact decoherence reformulation, H4 as determinant serialization, H5, H6 quantitative by an explicit rational witness, H7 scalar band positivity, and H8 scalar fixed-N band positivity for all configurations plus the effective terminal Laguerre gauge are all proved below. The only essential input left is the terminal scalar Pick condition of Theorem 9.66, equivalently the no-margin sign condition $\delta_N(z_0) \ge 0$ for every N in one fixed accumulating interior-band gauge. The numerical evidence and Conjecture 9.88 indicate that this sign is decided at factorial distance from the boundary, and the top terminal eigenvectors recover the nearby zero ordinates.

Formal statements and proofs for the Step 5 substructure (H0–H8 and the terminal reduction). The following are the formal statements underlying the discussion above.

Lemma 9.13 (H0: conservation of difficulty)**.** *If $S_1, \dots, S_k$ are all proved and $S_1 \wedge \cdots \wedge S_k \Rightarrow$ RH, then RH is proved. Therefore any valid route to RH which has not proved RH must contain a false step or an RH-hard step.*

Proof. This is modus ponens, followed by its contrapositive. □

Theorem 9.14 (H1: scalar one-node positivity)**.** *For $M_\Xi(z) = i(-\xi'/\xi)(s)$, $s = \frac{1}{2} + iz$, the scalar one-node Pick inequality on U_- is equivalent to*

$$\operatorname{Re}\frac{\xi'}{\xi}(s) \ge 0, \qquad \operatorname{Re} s > 1,$$

and this inequality holds.

Proof. The diagonal Pick quotient equals

$$\frac{\operatorname{Im} M_\Xi(z)}{\operatorname{Im} z} = \frac{-\operatorname{Re}(\xi'/\xi)(s)}{-(\operatorname{Re} s - \frac{1}{2})} = \frac{\operatorname{Re}(\xi'/\xi)(s)}{\operatorname{Re} s - \frac{1}{2}}.$$

The denominator is positive for $\operatorname{Re} s > 1$. The Hadamard product gives, in the standard symmetric limiting sense,

$$\operatorname{Re}\frac{\xi'}{\xi}(s) = \sum_\rho \frac{\sigma - \beta}{|s-\rho|^2}.$$

The classical zero-free line and functional equation give $0 < \beta < 1$ for nontrivial zeros, so every summand is nonnegative when $\sigma > 1$. □

Remark 9.15 (why H1 is kept scalar). The scalar one-node inequality is elementary because each zero contributes a nonnegative real number. For a matrix channel, the contribution of a non-real mirror pair has the form $qA + \bar{q}A^*$ with A rank one, and this need not be positive semidefinite term by term. Thus a full matrix-channel level-one theorem already contains phase-decoherence content; it is not treated as a routine extension of Theorem 9.14.

Theorem 9.16 (H2: fixed-band separated-node criterion). *Fix $1 < \sigma_0 < \sigma_1 < \infty$. In the band $\sigma \in [\sigma_0, \sigma_1]$ the scalar Pick matrix is positive semidefinite whenever the row-sum condition*

$$\sum_{j \neq i} \frac{\log(3 + |t_i|) + \log(3 + |t_j|)}{\max\{|t_i - t_j|, 2(\sigma_0 - \frac{1}{2})\}} \leq c_{\sigma_0,\sigma_1} \log(3 + |t_i|)$$

holds with $c_{\sigma_0,\sigma_1} > 0$ sufficiently small. Geometrically separated high nodes satisfy this condition. The estimate is stated for $|t_i| \geq T_0(\sigma_0, \sigma_1)$; for bounded heights the Pick matrix is positive semidefinite by compactness and the strict positivity of $\mathrm{Re}(\xi'/\xi)$ on $\sigma \geq \sigma_0$.

Proof. Stirling's formula and boundedness of ζ'/ζ in $\sigma \geq \sigma_0$ give $K_{ii} \geq a_{\sigma_0,\sigma_1} \log(3 + |t_i|)$ for large heights. The same estimates give $|M_\Xi(z_i)| \leq B_{\sigma_0,\sigma_1} \log(3 + |t_i|)$, hence

$$|K_{ij}| \leq B_{\sigma_0,\sigma_1} \frac{\log(3 + |t_i|) + \log(3 + |t_j|)}{\max\{|t_i - t_j|, 2(\sigma_0 - \frac{1}{2})\}}.$$

Choosing $c_{\sigma_0,\sigma_1} = a_{\sigma_0,\sigma_1}/(2B_{\sigma_0,\sigma_1})$ gives $\sum_{j \neq i} |K_{ij}| \leq K_{ii}/2$. Gershgorin's theorem for Hermitian matrices gives positive semidefiniteness. If $|t_j| \geq T_* Q^j$ and $|t_i - t_j| \geq \frac{1}{2} \max\{|t_i|, |t_j|\}$, then the lower-height and higher-height contributions are bounded respectively by

$$C \frac{(\log(3 + |t_i|))^2}{|t_i|} \quad \text{and} \quad C' \frac{\log(3 + |t_i|)}{|t_i|},$$

which are $o(\log(3 + |t_i|))$ as $T_* \to \infty$. $\square$

Theorem 9.17 (H3: two-node regions). *The scalar infinitesimal two-node inequality*

$$(\sigma - \tfrac{1}{2}) \left| \left(\frac{\xi'}{\xi} \right)' (s) \right| \leq \mathrm{Re} \, \frac{\xi'}{\xi}(s)$$

holds in the logarithmic region

$$|t| \geq T_0, \qquad \sigma \geq 1 + \frac{c}{\sqrt{\log(3 + |t|)}},$$

and also for $s = 1 + \epsilon + it$, $\epsilon \in [T^{-1/5}, 1]$, outside a set of $t \in [T, 2T]$ of measure $O(T(\log T)^{-2})$.

Proof. Write $\sigma = 1 + \epsilon$. In $0 < \epsilon \leq 1$,

$$\left| \left(\frac{\zeta'}{\zeta} \right)' (s) \right| \leq \sum_{n \geq 2} \Lambda(n)(\log n) n^{-1-\epsilon} \leq C_1 \epsilon^{-2} + C_2.$$

The elementary Γ and pole terms contribute $O((1 + |t|)^{-1})$ in the relevant band, while Stirling and the Euler bound give

$$\mathrm{Re} \, \frac{\xi'}{\xi}(s) \geq \frac{1}{2} \log \frac{|t|}{2\pi} - C_3 \epsilon^{-1} - C_4.$$

Taking $\epsilon \geq c/\sqrt{\log(3 + |t|)}$ with c large makes the right side dominate the $O(\epsilon^{-2})$ left side. For $\sigma \geq 2$, the derivative Euler series is $O(\sigma 2^{-\sigma})$ after multiplication by $\sigma - \frac{1}{2}$, while the real part grows like $\frac{1}{2} \log |s|$, both bounds holding *uniformly in* t; so the remaining large-σ tail is closed uniformly in t (the uniformity required by the compact-set argument of Theorem 9.22), and finite intermediate bands are absorbed by increasing T_0.

For the almost-everywhere statement, the Montgomery–Vaughan mean-value theorem gives uniformly in $\epsilon \in [T^{-1/5}, 1]$

$$\int_T^{2T} \left| \frac{\zeta'}{\zeta}(1+\epsilon+it) \right|^2 dt \ll T, \qquad \int_T^{2T} \left| \left(\frac{\zeta'}{\zeta}\right)'(1+\epsilon+it) \right|^2 dt \ll T,$$

because the square coefficient sums converge at $\epsilon = 0$ and the conservative error is absorbed in the stated range. Chebyshev's inequality shows that the set where either term exceeds a fixed multiple of $\log T$ has measure $O(T(\log T)^{-2})$. Outside this set the archimedean real part dominates. □

Definition 9.18 (coherence functional and spectral weight)**.** For $s = \sigma + it$ with $\sigma > 1$, define

$$\Theta(s) := \frac{\left|\sum_\rho (s-\rho)^{-2}\right|}{\sum_\rho |s-\rho|^{-2}} \in [0,1], \qquad W(s) := \frac{\sum_\rho (\sigma-\beta)\,|s-\rho|^{-2}}{(\sigma-\frac{1}{2})\sum_\rho |s-\rho|^{-2}}.$$

The sums are absolutely convergent by the classical zero-counting estimate $N(T) = O(T\log T)$.

Theorem 9.19 (exact reformulation of H3)**.** *For every s with $\sigma > 1$,*

$$(\sigma - \tfrac{1}{2})\left|\left(\frac{\xi'}{\xi}\right)'(s)\right| \le \operatorname{Re}\frac{\xi'}{\xi}(s) \qquad \Longleftrightarrow \qquad \Theta(s) \le W(s).$$

Under RH, $\beta = \frac{1}{2}$ for every zero, $W \equiv 1$, and H3 is the triangle inequality.

Proof. Differentiating the Hadamard logarithmic derivative gives

$$\left(\frac{\xi'}{\xi}\right)'(s) = -\sum_\rho (s-\rho)^{-2},$$

absolutely convergently for $\sigma > 1$. The same symmetric Hadamard pairing used in Theorem 9.14 gives

$$\operatorname{Re}\frac{\xi'}{\xi}(s) = \sum_\rho (\sigma-\beta)|s-\rho|^{-2}.$$

Dividing the H3 inequality by the positive quantity $(\sigma-\frac{1}{2})\sum_\rho |s-\rho|^{-2}$ gives exactly $\Theta(s) \le W(s)$. □

Lemma 9.20 (unconditional lower bound for the weight)**.** *There are effective constants $c_2 > 0$ and $T_0 \ge 3$ such that for $1 < \sigma \le 2$ and $|t| \ge T_0$,*

$$W(s) \ge \frac{\sigma-1}{\sigma-\frac{1}{2}} + \frac{c_2}{\log(3+|t|)} \ge \frac{2}{3}(\sigma-1) + \frac{c_2}{\log(3+|t|)}.$$

Proof. Use the classical zero-free region, in the form

$$\beta \le 1 - \frac{c_1}{\log(3+|\gamma|)}$$

after decreasing c_1 to absorb the finitely many small ordinates. Hence each zero contributes the ratio

$$\frac{\sigma-\beta}{\sigma-\frac{1}{2}} \ge \frac{\sigma-1}{\sigma-\frac{1}{2}} + \frac{c_1}{(\sigma-\frac{1}{2})\log(3+|\gamma|)}.$$

The number $W(s)$ is the weighted average of these ratios with weights proportional to $|s-\rho|^{-2}$. It remains only to show that this weighted average of $(\log(3+|\gamma|))^{-1}$ is bounded below by a constant multiple of $(\log(3+|t|))^{-1}$.

The Riemann–von Mangoldt formula gives a constant $C_H > 0$ such that, with $H_t = C_H \log(3+|t|)$, every interval $[t-H_t, t+H_t]$ contains at least one zero ordinate for all sufficiently large $|t|$. Therefore

$$\sum_{\rho} |s-\rho|^{-2} \geq (4+H_t^2)^{-1}$$

for $1 < \sigma \leq 2$. On the other hand, partial summation and $N(T) = O(T\log T)$ give

$$\sum_{|\gamma|>2|t|} |s-\rho|^{-2} \leq 4 \sum_{|\gamma|>2|t|} |\gamma|^{-2} \leq C\frac{\log(3+|t|)}{|t|}.$$

Thus the weight carried by zeros with $|\gamma| > 2|t|$, relative to the displayed lower bound, is $O((\log|t|)^3/|t|)$ and tends to 0. On the complementary set,

$$\frac{1}{\log(3+|\gamma|)} \geq \frac{1}{\log(3+2|t|)}.$$

For $|t| \geq T_0$ the stated lower bound follows after adjusting constants. The final inequality uses $(\sigma - \frac{1}{2})^{-1} \geq 2/3$ for $1 < \sigma \leq 2$. □

Definition 9.21 (Phase Decoherence Hypothesis, PDH)**.** With the constants of Lemma 9.20, PDH is the following open hypothesis: for all $1 < \sigma \leq 2$ and all $|t| \geq T_0$,

$$\Theta(s) \leq \frac{2}{3}(\sigma-1) + \frac{c_2}{\log(3+|t|)}.$$

PDH is not claimed as a theorem. It is a named sufficient decoherence condition for the global H3 edge.

Theorem 9.22 (PDH closes H3 at large height)**.** *Assume PDH. Then the scalar infinitesimal two-node inequality of Theorem 9.17 holds for all $1 < \sigma \leq 2$ and $|t| \geq T_0$. Together with the large-σ part of Theorem 9.17, the only remaining region is a compact set, in principle accessible to interval arithmetic.*

Proof. For $1 < \sigma \leq 2$ and $|t| \geq T_0$, PDH and Lemma 9.20 give $\Theta(s) \leq W(s)$. Theorem 9.19 converts this into H3. For $\sigma \geq 2$ and large $|t|$, H3 is already contained in the large-σ tail argument in Theorem 9.17. □

Theorem 9.19 and Lemma 9.20 are proved unconditionally. PDH itself remains an open hypothesis; Theorem 9.22 proves only the conditional implication PDH ⇒ H3 at large height, and no closure of RH is claimed from it.

Lemma 9.23 (exact mirror-pair identity)**.** *Let $\rho = \beta + i\gamma$ and $\rho' = (1-\beta) + i\gamma$ be a mirror pair with $\delta = \beta - \frac{1}{2} > 0$. Write*

$$D = s - (\tfrac{1}{2} + i\gamma) = (\sigma - \tfrac{1}{2}) + i(t-\gamma),$$

so that $s - \rho = D - \delta$ and $s - \rho' = D + \delta$. Then

$$\frac{1}{(s-\rho)^2} + \frac{1}{(s-\rho')^2} = \frac{2(D^2+\delta^2)}{(D^2-\delta^2)^2},$$

and

$$\frac{\sigma-\beta}{|s-\rho|^2} + \frac{\sigma-\beta'}{|s-\rho'|^2} = \frac{2(\sigma-\frac{1}{2})(|D|^2-\delta^2)}{|D^2-\delta^2|^2}.$$

Consequently the pairwise H3 deficit is exactly

$$\mathrm{Def}_\rho(s) := \frac{2(\sigma-\frac{1}{2})\big(|D^2+\delta^2| - (|D|^2-\delta^2)\big)}{|s-\rho|^2|s-\rho'|^2} \geq 0,$$

and

$$\mathrm{Def}_\rho(s) \leq \frac{4(\sigma-\frac{1}{2})\delta^2}{|s-\rho|^2|s-\rho'|^2}.$$

Proof. The first identity is direct:

$$(D-\delta)^{-2} + (D+\delta)^{-2} = \frac{(D+\delta)^2 + (D-\delta)^2}{(D^2-\delta^2)^2} = \frac{2(D^2+\delta^2)}{(D^2-\delta^2)^2}.$$

For the second, use $\operatorname{Re} D = \sigma - \frac{1}{2}$ and

$$\sigma - \beta = (\sigma - \tfrac{1}{2}) - \delta, \qquad \sigma - \beta' = (\sigma - \tfrac{1}{2}) + \delta.$$

Then

$$\frac{(\sigma-\frac{1}{2})-\delta}{|D-\delta|^2} + \frac{(\sigma-\frac{1}{2})+\delta}{|D+\delta|^2} = \frac{2(\sigma-\frac{1}{2})(|D|^2-\delta^2)}{|D-\delta|^2|D+\delta|^2},$$

and

$$|D-\delta|^2|D+\delta|^2 = |D^2-\delta^2|^2.$$

The deficit formula follows by putting both pair contributions over the common denominator $|s-\rho|^2|s-\rho'|^2$. Since

$$|D^2+\delta^2| \geq |D|^2 - \delta^2$$

by the reverse triangle inequality, the deficit is nonnegative. Finally $|D^2+\delta^2| \leq |D|^2+\delta^2$ gives the displayed upper bound. □

Theorem 9.24 (deficit-versus-gain criterion)**.** *Define the on-line decoherence gain*

$$G(s) := (\sigma - \tfrac{1}{2}) \left[\sum_{\beta=\frac{1}{2}} |s-\rho|^{-2} - \left| \sum_{\beta=\frac{1}{2}} (s-\rho)^{-2} \right| \right] \geq 0.$$

Then H3 holds at s whenever

$$4(\sigma-\tfrac{1}{2}) \sum_{\delta_\rho>0} \frac{\delta_\rho^2}{|s-\rho|^2|s-\rho'|^2} \leq G(s).$$

Proof. Split the zero sum into on-line zeros and mirror pairs. By the triangle inequality,

$$(\sigma-\tfrac{1}{2}) \left| \sum_\rho (s-\rho)^{-2} \right| \leq (\sigma-\tfrac{1}{2}) \left| \sum_{\beta=\frac{1}{2}} (s-\rho)^{-2} \right| + (\sigma-\tfrac{1}{2}) \sum_{\text{pairs}} \left| \frac{1}{(s-\rho)^2} + \frac{1}{(s-\rho')^2} \right|.$$

Insert Lemma 9.23 for each pair and the definition of $G(s)$ for the on-line part. The right side is at most

$$\sum_\rho (\sigma-\beta)|s-\rho|^{-2} + \sum_{\text{pairs}} \mathrm{Def}_\rho(s) - G(s).$$

The hypothesis and the upper bound on $\mathrm{Def}_\rho(s)$ make the last two terms nonpositive. This gives H3, equivalently $\Theta(s) \leq W(s)$ by Theorem 9.19. □

Remark 9.25 (mirror-pair honesty). Lemma 9.23 is exact and replaces the earlier Taylor heuristic. Every off-line mirror pair produces a nonnegative second-moment deficit, bounded by

$$4(\sigma - \tfrac{1}{2})\delta^2/(|s-\rho|^2|s-\rho'|^2).$$

H3 can absorb such pairs only through the on-line phase-decoherence gain $G(s)$ or an equivalent global cancellation mechanism. This is the same residual wall named by PDH, not a closure of RH.

Lemma 9.26 (continuity and growth of the correlation kernel)**.** *For each finite channel F, the zero-side series defining ψ^F converges absolutely and locally uniformly in t, defines a continuous matrix function, and satisfies*

$$\|\psi^F(t)\| \le C_F e^{|t|/2}, \qquad t \in \mathbb{R}.$$

Proof. Each term is bounded by

$$m_\rho |\ell_{z_\rho}(\phi_\beta)\ell_{\bar z_\rho}(\phi_\alpha)| e^{|\operatorname{Im} z_\rho||t|} \le m_\rho C_{A,F}(1+|\operatorname{Re} z_\rho|)^{-A} e^{|t|/2},$$

because $|\operatorname{Im} z_\rho| < \frac{1}{2}$ and the source-core transforms have rapid horizontal decay in the zero strip. Choosing A large and using $N(T) = O(T\log T)$ gives a summable majorant over the zero set, locally uniformly in t. □

Theorem 9.27 (H5: window positivity is equivalent to ARP-P)**.** *Let*

$$\frac{1}{z_\rho - z} = -i\int_0^\infty e^{-izt}e^{iz_\rho t}\,dt, \qquad z \in U_-,$$

and define the associated correlation kernel ψ^F by $g^F(z) = -i\int_0^\infty e^{-izt}\psi^F(t)\,dt$. Then ARP-P for F is equivalent to $Q^F(f) \ge 0$ for every compactly supported smooth window test f.

Proof. Fix $y_0 > \frac{1}{2}$ and write $z(x) = x - iy_0$. For $f \in C_c^\infty(0,\infty)$ define

$$c(x) = \frac{1}{2\pi}\int_0^\infty f(t)e^{iz(x)t}\,dt.$$

Since $f(t)e^{y_0 t}$ is compactly supported smooth, Fourier inversion gives $f(t) = \int_{\mathbb{R}} c(x)e^{-iz(x)t}\,dx$.

If window positivity holds, apply it to $f_L(t) = \eta_L(t)\sum_j c_j e^{-iz_j t}$ and pass to the limit $L \to \infty$ by dominated convergence; the growth bound in Lemma 9.26 supplies the integrable majorant because $\operatorname{Im} z_j < -\frac{1}{2}$. The limit is $\sum_{i,j} c_i^* K(z_i, z_j)c_j$. Hence every finite Pick matrix is positive. Conversely, if every Pick matrix is positive, substitute the horizontal-line representation of f into $Q^F(f)$. Lemma 9.26 and the Schwartz decay of c justify Fubini. The result is an integral of the positive kernel $K(z(x), z(x'))$ against $c(x)$ and $c(x')$, obtained as a limit of nonnegative Riemann sums. Hence $Q^F(f) \ge 0$. □

Corollary 9.28 (H6: qualitative finite negative witness)**.** *If RH fails, then there exist a finite channel F and finite nodes in U_- for which the ARP-P Pick matrix has a negative eigenvalue.*

Proof. This is the contrapositive of Theorem 9.185. □

Definition 9.29 (single-cluster rational witness)**.** Let $w_0 = \gamma - i\delta$, $0 < \delta < \frac{1}{2}$, be a non-real zero coordinate of Ξ in Ω_-. Let $t_1, \dots, t_M$ be the real zero coordinates of Ξ in $[\gamma - R, \gamma + R]$, counted with multiplicity. Choose $N_p \ge M + 2$ pairwise distinct poles $\pi_1, \dots, \pi_{N_p} \in \mathbb{C}^+$ with

$$\operatorname{Im}\pi_j \in [1,2], \qquad |\operatorname{Re}\pi_j - \gamma| \le R,$$

and define

$$q(t) = \prod_{m=1}^{M}(t - t_m), \qquad r(t) = \frac{q(t)}{\prod_{j=1}^{N_p}(t - \pi_j)} = \sum_{j=1}^{N_p} \frac{c_j}{t - \pi_j}.$$

Equivalently,

$$c_j = \frac{q(\pi_j)}{\prod_{k \neq j}(\pi_j - \pi_k)}.$$

The associated Pick nodes are $z_j = \bar{\pi}_j \in U_-$.

Proposition 9.30 (H6 quantitative reduction: explicit single-cluster witness)**.** *In the setting of Definition 9.29, choose a source ϕ detecting w_0, and let $\kappa := \ell_{w_0}(\phi)\overline{\ell_{\bar{w}_0}(\phi)} \neq 0$ be the (complex) pair weight;* no *normalization of the channel is performed, so that the Pick form stays Hermitian. Suppose the chosen poles satisfy the finite phase condition*

$$2\,\mathrm{Re}\big(\kappa\,\overline{r(\bar{w}_0)}\,r(w_0)\big) \leq -\eta\,|\kappa|\,|r(w_0)|\,|r(\bar{w}_0)|$$

for some $\eta > 0$. If the window contains further non-real pairs, augment q with one complex root at one member of each such pair; those pairs then contribute exactly 0 to the form (as in Theorem 9.157), and the resulting fixed phase offset of $q(w_0)$ is absorbed into the target phase of Lemma 9.31. Then the Pick matrix on the nodes $z_j = \bar{\pi}_j$ has a test vector $c = (c_j)$ with Rayleigh quotient bounded above by

$$\frac{-\eta\,|\kappa|\,|r(w_0)|\,|r(\bar{w}_0)| + B_{\mathrm{far}}(R,\gamma)\,\|r\|_{\mathrm{far}}^2}{\|c\|^2},$$

where

$$\|r\|_{\mathrm{far}} := \sup_{|t-\gamma| \geq R}(1 + |t-\gamma|)^2|r(t)|$$

and $B_{\mathrm{far}}(R,\gamma)$ is the explicit zero-counting tail bound obtained from $N_\Xi(T+1) - N_\Xi(T) = O(\log(2+T))$. In particular, whenever the numerator is negative, this gives a finite explicit negative Pick witness.

Proof. The rational function r has poles at $\pi_j = \bar{z}_j$ and residues c_j, so the scalar Pick quadratic form evaluated on c is the zero-side sum obtained by testing the meromorphic target against r. The local real zeros t_m contribute

$$|r(t_m)|^2 = 0$$

because $q(t_m) = 0$. The conjugate off-line pair, carrying the complex residue weight κ, contributes exactly

$$2\,\mathrm{Re}\big(\kappa\,\overline{r(\bar{w}_0)}\,r(w_0)\big),$$

with the sign convention of the scalar kernel. By the assumed phase condition this is at most $-\eta\,|\kappa|\,|r(w_0)|\,|r(\bar{w}_0)|$.

For every remaining zero coordinate u, the contribution is bounded in absolute value by a constant multiple of $|r(u)|^2$ after collecting the finitely many channel weights into the normalization. On the tail $|u-\gamma| \geq R$,

$$|r(u)|^2 \leq \frac{\|r\|_{\mathrm{far}}^2}{(1+|u-\gamma|)^4}.$$

Summing over unit intervals and using the classical local zero-counting bound $N_\Xi(T+1) - N_\Xi(T) = O(\log(2+T))$ gives the displayed $B_{\mathrm{far}}(R,\gamma)\|r\|_{\mathrm{far}}^2$ bound. Dividing by $\|c\|^2$ gives the Rayleigh quotient estimate. A negative Rayleigh quotient implies a negative eigenvalue. □

Lemma 9.31 (H6 phase adjustment)**.** *Assume $R \ge 2$ in Definition 9.29. Enlarge the pole family to*

$$N_p = M + 2 + \lceil 2\pi/\delta \rceil$$

poles $\pi_j = x_j + iy_j$ with pairwise distinct heights $y_j \in [1,2]$ and abscissas $x_j \in [\gamma - R, \gamma + R]$. The abscissas can be chosen so that, with the complex pair weight κ of Proposition 9.30,

$$\arg\big(r(w_0)\overline{r(\bar w_0)}\big) \equiv \pi - \arg\kappa \pmod{2\pi}, \qquad 2\operatorname{Re}\big(\kappa\,\overline{r(\bar w_0)}\,r(w_0)\big) = -2\,|\kappa|\,|r(w_0)|\,|r(\bar w_0)|.$$

This is achievable by the same intermediate-value argument, since the attainable phase interval has length $\ge 2\pi$ and $\arg\kappa$ is a fixed offset. Thus the phase condition in Proposition 9.30 holds with $\eta = 2$.

Proof. The polynomial q has real coefficients. Hence

$$\overline{r(\bar w_0)} = \frac{q(w_0)}{\prod_j (w_0 - \bar\pi_j)}$$

and

$$r(w_0)\overline{r(\bar w_0)} = \frac{q(w_0)^2}{\prod_j F_j}, \qquad F_j = (w_0 - \pi_j)(w_0 - \bar\pi_j).$$

Write $u_j = \gamma - x_j$. Since $w_0 = \gamma - i\delta$,

$$F_j = u_j^2 + y_j^2 - \delta^2 - 2i\delta u_j.$$

Its real part is at least $y_j^2 - \delta^2 > 0$, and

$$\arg F_j = -\arctan\frac{2\delta u_j}{u_j^2 + y_j^2 - \delta^2}.$$

As u_j ranges over $\mathbb{R}$, this continuous phase has range

$$[-\arcsin(\delta/y_j),\, \arcsin(\delta/y_j)],$$

with the endpoints attained at $u_j = \pm\sqrt{y_j^2 - \delta^2}$. Since $y_j \in [1,2]$ and $R \ge 2$, these endpoints are available inside $[\gamma - R, \gamma + R]$. Each pole therefore contributes phase width at least $2\arcsin(\delta/2) \ge \delta$.

The total phase

$$\Phi(x_1, \dots, x_{N_p}) = 2\arg q(w_0) - \sum_j \arg F_j$$

is continuous on the connected parameter box, and the independent phase intervals add to an interval of length at least $N_p\delta \ge 2\pi$. By the intermediate value theorem, some choice of abscissas gives $\Phi \equiv \pi \pmod{2\pi}$. No denominator vanishes during the deformation, because $|F_j| \ge y_j^2 - \delta^2 > 0$, and the heights keep the poles pairwise distinct. $\square$

Theorem 9.32 (H6 quantitative single-cluster witness)**.** *In the setting of Definition 9.29, with $R \ge 2$ and the augmented pole family of Lemma 9.31, the explicit witness of Proposition 9.30 satisfies*

$$\lambda_{\min}\mathsf{P} \le \frac{-2|r(w_0)|\,|r(\bar w_0)| + B_{\mathrm{far}}(R,\gamma)\|r\|_{\mathrm{far}}^2}{\|c\|^2}.$$

All quantities are explicit in the local zero configuration and in the chosen pole family. The witness uses

$$N_p = M + 2 + \lceil 2\pi/\delta \rceil$$

nodes.

Proof. Apply Lemma 9.31 to obtain the phase condition in Proposition 9.30 with $\eta = 2$. The displayed Rayleigh quotient bound is then exactly the estimate of Proposition 9.30. Since $\lambda_{\min}$ is bounded above by every Rayleigh quotient, the result follows. $\square$

Remark 9.33 (H6 lives in the Szegő-degenerate regime). The witness uses $N_p \asymp 1/\delta$ nodes. With those many poles in a bounded box, some gaps are $O(\delta)$, and the corresponding Cauchy determinant in Lemma 9.44 is far below the threshold of Theorem 9.45. Thus the negative witness produced by an off-line zero lives exactly in the degenerating Szegő regime; it does not contradict the positive fixed-pattern and nondegenerate-cluster theorems.

Definition 9.34 (detection family and determinant data)**.** For $m \geq 2$, let $I_m = (1/m, m)$ and let $D_m \subset C_c^\infty(I_m) \setminus \{0\}$ be a countable set dense in the sup norm in

$$X_m = \overline{C_c^\infty(I_m)}^{\|\cdot\|_\infty}.$$

Enumerate

$$\{\phi_k\}_{k\geq 1} := \bigcup_{m\geq 2} D_m.$$

Fix a pairwise distinct canonical node sequence $\{z_j\}_{j\geq 1} \subset U_-$ with an accumulation point in U_-. For $F_k = \operatorname{span}\{\phi_k\}$ define

$$d_N^{(k)} := \det \mathsf{P}[g^{F_k}; z_1, \dots, z_N].$$

By Theorem 9.8, each $d_N^{(k)}$ is computable from the archimedean, pole, and prime data alone.

Lemma 9.35 (Jensen counting: infinitely many charged atoms)**.** *Assume RH and let $\phi \in C_c^\infty(0,\infty)$, $\phi \neq 0$. Then $\ell_{z_\rho}(\phi) \neq 0$ for infinitely many zeros z_ρ of Ξ. Consequently*

$$\mu_\phi = \sum_\rho m_\rho |\ell_{z_\rho}(\phi)|^2 \delta_{z_\rho}$$

has infinite support.

Proof. Let $L(a) = \ell_a(\phi)$. If $\operatorname{supp}\phi \subset [a_0, b]$, then

$$|L(a)| \leq \|\phi\|_1 e^{b|\operatorname{Im} a|},$$

so L is entire of exponential type. It is not identically zero, since otherwise the Fourier–Laplace transform of the compactly supported distribution $\phi(t)\,dt$ would vanish identically. Choose a_* with $L(a_*) \neq 0$. Jensen's formula centered at a_* gives, for the number $n(r)$ of zeros of L in $|a - a_*| \leq r$,

$$n(r)\log 2 \leq \log \max_{|a-a_*|=2r} |L(a)| - \log|L(a_*)| \leq 2br + C_\phi.$$

Thus $n(r) = O(r)$. The zeros of Ξ with $|z_\rho| \leq r$ satisfy the Riemann–von Mangoldt count

$$N_\Xi(r) \sim \frac{r}{2\pi}\log\frac{r}{2\pi},$$

which dominates every linear bound. Hence L cannot vanish at all but finitely many zeros of Ξ. Under RH these zeros lie on $\mathbb{R}$, so every nonzero value contributes a positive atom to μ_ϕ. $\square$

Lemma 9.36 (strict positivity under RH)**.** *Assume RH. For every $\phi \in C_c^\infty(0,\infty)$, $\phi \neq 0$, and every finite set of pairwise distinct nodes $z_1, \dots, z_N \in U_-$, the scalar Pick matrix $\mathsf{P}[g^{F_\phi}; z_1, \dots, z_N]$ is positive definite.*

Proof. By Theorem 9.189, g^{F_ϕ} is the Cauchy transform of the positive measure μ_ϕ. Lemma 9.178 gives

$$c^*\mathsf{P}c = \int_{\mathbb{R}} \left| \sum_j \frac{c_j}{t - \bar{z}_j} \right|^2 d\mu_\phi(t).$$

If this quadratic form vanishes, the rational function

$$r(t) = \sum_j \frac{c_j}{t - \bar{z}_j}$$

vanishes at every charged atom of μ_ϕ. By Lemma 9.35, this is an infinite set. A nonzero rational function has only finitely many zeros, so $r \equiv 0$. Since the poles $\bar{z}_j$ are distinct, all residues c_j vanish. Hence the quadratic form is strictly positive for $c \neq 0$. □

Lemma 9.37 (dense families detect every off-line zero)**.** *Let z_ρ be a non-real zero of Ξ. Then for some k,*

$$\ell_{z_\rho}(\phi_k)\,\overline{\ell_{\bar{z}_\rho}(\phi_k)} \neq 0.$$

Proof. Fix any $m \geq 2$. On X_m, the functionals ℓ_{z_ρ} and $\ell_{\bar{z}_\rho}$ are continuous. They are nonzero: if ℓ_a vanished on all of X_m, then the continuous function e^{-iat} would define the zero functional on I_m, impossible. Therefore their kernels are proper closed subspaces, hence closed nowhere dense sets. The complement of their union is open and dense in X_m. Since D_m is dense, some $\phi_k \in D_m$ lies in that complement. □

Theorem 9.38 (H4 closed: determinant serialization of RH)**.** *With the data of Definition 9.34,*

$$\text{RH} \iff d_N^{(k)} > 0 \quad \textit{for every } k \geq 1 \textit{ and every } N \geq 1.$$

Proof. Assume RH. Lemma 9.36 applies to each ϕ_k and each prefix $z_1, \ldots, z_N$, so every determinant $d_N^{(k)}$ is strictly positive.

Conversely, assume $d_N^{(k)} > 0$ for every k, N. For fixed k, the node sets are nested, so $\mathsf{P}[g^{F_k}; z_1, \ldots, z_N]$ is the leading principal block of the next Pick matrix. The Pick matrix is Hermitian. By Sylvester's criterion, positivity of all leading principal minors gives

$$\mathsf{P}[g^{F_k}; z_1, \ldots, z_N] \succ 0$$

for every N. The nodes are pairwise distinct, accumulate in U_-, and $G_\Xi^{F_k}$ is meromorphic on $\mathbb{C}$ (Lemma 9.6) and holomorphic at every node and near the accumulation point, since its poles lie in the zero strip $|\operatorname{Im} z| < \frac{1}{2}$. Lemma 9.184, applied to this one channel and the fixed node sequence, shows that every pole of $G_\Xi^{F_k}$ in Ω_- is removable.

If a non-real zero of Ξ lay in Ω_-, Lemma 9.37 would find k for which that zero is detected by $G_\Xi^{F_k}$, giving a non-removable pole, contradiction. The symmetries $\Xi(-z) = \Xi(z)$ and $\Xi(\bar{z}) = \overline{\Xi(z)}$ rule out the upper half-plane as in Theorem 9.185. Lemma 9.2 gives RH. □

Corollary 9.39 (Schur coordinates)**.** *Under the positivity in Theorem 9.38, the scalar Schur–Nevanlinna algorithm along $\{z_j\}$ is nondegenerate at every step and all scalar parameters satisfy $|\gamma_N^{(k)}| < 1$. Conversely, nondegenerate scalar Schur contractivity at every step implies $d_N^{(k)} > 0$ for all N.*

Proof. For nested scalar Nevanlinna–Pick data, each Schur step is the Schur complement of the previous leading principal Pick block; this is the classical Schur–Nevanlinna recursion in Schur-complement form (see, for instance, [78]). Positive definiteness of all leading principal blocks is equivalent to strict positivity of all these Schur complements, hence to nondegenerate Schur parameters in the unit disk. By Sylvester's criterion this is equivalent to $d_N^{(k)} > 0$ for all N. □

Lemma 9.40 (Szegő pattern matrix)**.** *Let $\mathcal{P} = \{(a_j, y_j)\}_{j=1}^N$ be a finite pattern of distinct pairs with $y_j > \frac{1}{2}$. For $t \in \mathbb{R}$ set*

$$z_j = t + a_j - iy_j \in U_-, \qquad S_{ij} := \frac{-i}{z_i - \bar{z}_j}.$$

Then S is independent of t and is positive definite. More precisely, S is the Gram matrix of the functions

$$e_j(\tau) = e^{-(y_j + ia_j)\tau}, \qquad \tau \geq 0,$$

in $L^2(0, \infty)$. We write

$$\lambda_{\mathcal{P}} := \lambda_{\min}(S) > 0.$$

Proof. Since

$$z_i - \bar{z}_j = (a_i - a_j) - i(y_i + y_j),$$

the matrix is independent of t. Also,

$$\int_0^\infty e_i(\tau)\overline{e_j(\tau)}\, d\tau = \int_0^\infty e^{-[(y_i+y_j)+i(a_i-a_j)]\tau}\, d\tau = \frac{1}{(y_i + y_j) + i(a_i - a_j)} = \frac{1}{i(z_i - \bar{z}_j)} = S_{ij}.$$

The exponents $y_j + ia_j$ are distinct because the pairs (a_j, y_j) are distinct. Distinct exponentials are linearly independent on every interval of positive length, hence in $L^2(0, \infty)$. Therefore their Gram matrix is strictly positive definite. □

Lemma 9.41 (archimedean cluster asymptotics)**.** *On* $\operatorname{Re} s > 1$ *write*

$$M_\Xi(z) = i\left(G_\infty(s) - \frac{\zeta'}{\zeta}(s)\right), \qquad s = \tfrac{1}{2} + iz,$$

where

$$G_\infty(s) = -\frac{1}{s} - \frac{1}{s-1} + \frac{1}{2}\log \pi - \frac{1}{2}\psi(s/2).$$

For the pattern nodes $z_j = t + a_j - iy_j$ of Lemma 9.40, let $s_j = \frac{1}{2} + iz_j$. The archimedean part K_∞ of the scalar Pick matrix satisfies

$$\left\| K_\infty - \log \frac{|t|}{2\pi}\, S \right\| \leq \frac{C_{\mathcal{P}}}{|t|}$$

for $|t|$ large enough, with $C_{\mathcal{P}}$ depending only on the pattern.

Proof. Stirling's formula for the logarithmic derivative of Γ gives, uniformly for the fixed pattern as $|t| \to \infty$,

$$G_\infty(s) = -\frac{1}{2}\log \frac{s}{2\pi} + O_{\mathcal{P}}(|t|^{-1}), \qquad \operatorname{Re} s > 1.$$

Thus

$$iG_\infty(s_i) - \overline{iG_\infty(s_j)} = -\frac{i}{2}\left(\log \frac{s_i}{2\pi} + \overline{\log \frac{s_j}{2\pi}}\right) + O_{\mathcal{P}}(|t|^{-1}).$$

For $t > 0$, $\arg s_i = \pi/2 + O_{\mathcal{P}}(|t|^{-1})$; for $t < 0$, $\arg s_i = -\pi/2 + O_{\mathcal{P}}(|t|^{-1})$. In both cases the arguments cancel in $\log s_i + \overline{\log s_j}$, and

$$\log s_i + \overline{\log s_j} = 2\log|t| + O_{\mathcal{P}}(|t|^{-1}).$$

Hence

$$iG_\infty(s_i) - \overline{iG_\infty(s_j)} = -i\log\frac{|t|}{2\pi} + O_{\mathcal{P}}(|t|^{-1}).$$

Dividing by $z_i - \bar{z}_j$ gives

$$(K_\infty)_{ij} = \log\frac{|t|}{2\pi}\frac{-i}{z_i - \bar{z}_j} + O_{\mathcal{P}}(|t|^{-1}) = \log\frac{|t|}{2\pi}S_{ij} + O_{\mathcal{P}}(|t|^{-1}).$$

Since the matrix size is fixed, the entrywise estimate implies the displayed operator-norm bound. $\square$

Theorem 9.42 (H8 closed for scalar fixed cluster patterns)**.** *Fix $\epsilon_0 > 0$ and a pattern $\mathcal{P} = \{(a_j, y_j)\}_{j=1}^N$ of distinct pairs with $y_j \geq \frac{1}{2} + \epsilon_0$. There is an effective $T_0 = T_0(\mathcal{P}, \epsilon_0)$ such that, for every real t with $|t| \geq T_0$, the scalar Pick matrix of M_Ξ at the nodes*

$$z_j = t + a_j - iy_j$$

satisfies

$$\mathsf{P}[M_\Xi; z_1, \ldots, z_N] \succeq \frac{1}{2}\,\lambda_{\mathcal{P}}\log\frac{|t|}{2\pi}\,I \succ 0.$$

Thus scalar ARP-P is unconditionally true for every fixed finite cluster pattern translated to sufficiently large height, away from the boundary $\operatorname{Re} s = 1$ by ϵ_0.

Proof. By linearity of the Pick matrix in the function, decompose

$$\mathsf{P}[M_\Xi] = K_\infty + K_{\text{prime}}, \qquad M_\Xi = i\left(G_\infty - \frac{\zeta'}{\zeta}\right).$$

For each node, $\operatorname{Re} s_j = \frac{1}{2} + y_j \geq 1 + \epsilon_0$. The absolutely convergent Euler product gives

$$\left|\frac{\zeta'}{\zeta}(s_j)\right| \leq -\frac{\zeta'}{\zeta}(1 + \epsilon_0) =: C_0.$$

Also $|z_i - \bar{z}_j| \geq y_i + y_j \geq 1 + 2\epsilon_0$. Therefore every prime entry is bounded in modulus by $2C_0/(1 + 2\epsilon_0)$, and hence

$$\|K_{\text{prime}}\| \leq \frac{2NC_0}{1 + 2\epsilon_0} =: C_1.$$

By Lemmas 9.40 and 9.41,

$$K_\infty \succeq \lambda_{\mathcal{P}}\log\frac{|t|}{2\pi}\,I - \frac{C_{\mathcal{P}}}{|t|}\,I.$$

Thus

$$\lambda_{\min}\mathsf{P}[M_\Xi] \geq \lambda_{\mathcal{P}}\log\frac{|t|}{2\pi} - \frac{C_{\mathcal{P}}}{|t|} - C_1.$$

Choosing T_0 so the last two terms are at most $\frac{1}{2}\lambda_{\mathcal{P}}\log(|t|/(2\pi))$ proves the claim. $\square$

Corollary 9.43 (fixed patterns near the edge, almost everywhere)**.** *Fix a scalar pattern $\mathcal{P} = \{(a_j, y_j)\}_{j=1}^N$ with $y_j > \frac{1}{2}$. For every dyadic interval $[T, 2T]$ and all sufficiently large T, the conclusion of Theorem 9.42 holds for $t \in [T, 2T]$ outside a set of measure $O_{\mathcal{P}}(T(\log T)^{-2})$, with the deterministic constant replaced by a pattern-dependent one.*

Proof. The archimedean estimate of Lemma 9.41 is unchanged. For the prime part, apply the Montgomery–Vaughan mean-value theorem to the absolutely convergent Dirichlet series on each fixed shifted line $\operatorname{Re} s_j = \frac{1}{2} + y_j > 1$:

$$\int_T^{2T} \left| \frac{\zeta'}{\zeta}\left(\tfrac{1}{2} + i(t + a_j) + y_j\right) \right|^2 dt \ll_{\mathcal{P}} T.$$

Chebyshev's inequality and a union bound over the finitely many nodes show that, outside a set of measure $O_{\mathcal{P}}(T(\log T)^{-2})$, all values $|\zeta'/\zeta(s_j)|$ are bounded by a sufficiently small fixed multiple of $\log T$. On this complement, the operator norm of K_{prime} is less than one half of the archimedean margin $\lambda_{\mathcal{P}} \log(T/(2\pi))$ once T is large. The same assembly as in Theorem 9.42 gives positivity. □

Lemma 9.44 (explicit Szegő determinant and eigenvalue bound)**.** *Let $w_j = y_j + ia_j$. The pattern matrix of Lemma 9.40 is the Cauchy matrix*

$$S_{ij} = \frac{1}{w_i + \bar{w}_j}.$$

It satisfies

$$\det S = \frac{\prod_{i<j} |w_i - w_j|^2}{\prod_i 2y_i \ \prod_{i<j} |w_i + \bar{w}_j|^2} > 0.$$

Moreover, for patterns in the box

$$y_j \in [\tfrac{1}{2} + \epsilon_0, Y], \qquad |a_j| \le A,$$

one has the explicit lower bound

$$\lambda_{\mathcal{P}} \ge c(N, \epsilon_0, Y, A) \prod_{i<j} |w_i - w_j|^2,$$

where

$$c(N, \epsilon_0, Y, A) = \frac{(1 + 2\epsilon_0)^{N-1}}{N^{N-1}(2Y)^N(2Y + 2A)^{N(N-1)}}.$$

Proof. Since

$$w_i + \bar{w}_j = (y_i + y_j) + i(a_i - a_j) = i(z_i - \bar{z}_j),$$

Lemma 9.40 gives $S_{ij} = (w_i + \bar{w}_j)^{-1}$. The Cauchy determinant identity gives

$$\det S = \frac{\prod_{i<j}(w_i - w_j)(\bar{w}_i - \bar{w}_j)}{\prod_{i,j}(w_i + \bar{w}_j)}.$$

The numerator is $\prod_{i<j} |w_i - w_j|^2$. The denominator splits into the diagonal factors $2y_i$ and conjugate pairs

$$(w_i + \bar{w}_j)(w_j + \bar{w}_i) = |w_i + \bar{w}_j|^2,$$

which gives the displayed positive formula.

If $\lambda_1 \le \cdots \le \lambda_N$ are the eigenvalues of S, then

$$\det S = \prod_j \lambda_j \le \lambda_{\min}(S)\, \lambda_{\max}(S)^{N-1} \le \lambda_{\min}(S)\, (\operatorname{tr} S)^{N-1}.$$

Thus $\lambda_{\min}(S) \ge \det S/(\operatorname{tr} S)^{N-1}$. In the box,

$$\operatorname{tr} S = \sum_i \frac{1}{2y_i} \le \frac{N}{1 + 2\epsilon_0}, \qquad 2y_i \le 2Y, \qquad |w_i + \bar{w}_j| \le 2Y + 2A.$$

Substituting these bounds into the determinant formula gives the stated constant. □

Theorem 9.45 (H8 degenerating regime: quantitative scalar closure)**.** *Fix N, $\epsilon_0 > 0$, Y, and A. There are effective constants C_{box} and T_{box}, depending only on this box, such that for every scalar pattern with*

$$y_j \in [\tfrac{1}{2} + \epsilon_0, Y], \qquad |a_j| \le A,$$

and every $|t| \ge T_{\text{box}}$,

$$\prod_{i<j} |w_i - w_j|^2 \ge \frac{C_{\text{box}}}{\log(|t|/2\pi)} \quad \Longrightarrow \quad \mathsf{P}[M_\Xi; z_1, \ldots, z_N] \succ 0,$$

where $z_j = t + a_j - iy_j$ and $w_j = y_j + ia_j$. More precisely,

$$\mathsf{P}[M_\Xi; z_1, \ldots, z_N] \succeq \frac{1}{2}\, \lambda_{\mathcal{P}} \log \frac{|t|}{2\pi}\, I.$$

Consequently, if $h = \min_{i<j} |w_i - w_j|$, positivity holds once

$$h \ge C_{N,\text{box}} (\log |t|)^{-1/(N(N-1))}.$$

Proof. The prime bound in Theorem 9.42 depends only on N and ϵ_0:

$$\|K_{\text{prime}}\| \le C_1(N, \epsilon_0).$$

The error in Lemma 9.41 is uniform over the box. Indeed, Stirling's remainder and the expansion

$$\log s_j = \log |t| \pm i\frac{\pi}{2} + O((1 + A + Y)/|t|)$$

depend only on A, Y once $|t|$ is large. Hence

$$K_\infty \succeq \lambda_{\mathcal{P}} \log \frac{|t|}{2\pi}\, I - \frac{C'_{\text{box}}}{|t|}\, I.$$

Combining the two estimates,

$$\lambda_{\min} \mathsf{P}[M_\Xi] \ge \lambda_{\mathcal{P}} \log \frac{|t|}{2\pi} - C_1(N, \epsilon_0) - \frac{C'_{\text{box}}}{|t|}.$$

Lemma 9.44 gives

$$\lambda_{\mathcal{P}} \ge c(N, \epsilon_0, Y, A) \prod_{i<j} |w_i - w_j|^2.$$

Choose C_{box} and then T_{box} so the product hypothesis implies

$$\lambda_{\mathcal{P}} \log \frac{|t|}{2\pi} \ge 2C_1(N, \epsilon_0) + 2C'_{\text{box}}/|t|.$$

The displayed lower bound follows. Since there are $N(N-1)/2$ pairs and the product contains squared gaps, $\prod_{i<j} |w_i - w_j|^2 \ge h^{N(N-1)}$, which gives the stated gap threshold. □

Corollary 9.46 (H7 closed away from the degenerate two-node scale)**.** *For scalar two-node patterns in a fixed box, the level-two Pick determinant is positive at sufficiently large height whenever*

$$|w_1 - w_2|^2 \ge \frac{C_{\text{box}}}{\log(|t|/2\pi)}.$$

Equivalently, this corollary localizes the remaining two-node work to the degenerating scale $|w_1 - w_2| \lesssim (\log |t|)^{-1/2}$ and below; Corollary 9.58 closes the fixed coalescent endpoint.

Proof. This is Theorem 9.45 with $N = 2$. For a 2×2 Hermitian Pick matrix, positive definiteness is equivalent to positivity of the two diagonal entries and the determinant. □

Remark 9.47 (where H8 still carries RH-level hardness)*.* Theorem 9.45 quantifies exactly how far the fixed-N scalar cluster method goes: the only remaining scalar cluster regime is

$$\prod_{i<j} |w_i - w_j|^2 \lesssim \frac{1}{\log |t|}$$

and the passage $N \to \infty$. The Nevanlinna–Pick bridge needs accumulating node families, precisely where the Cauchy determinant degenerates. Thus the remaining H8 difficulty is uniformity below the explicit Szegő/Cauchy determinant scale, not fixed-cluster positivity.

Definition 9.48 (H7 and remaining H8 coordinates)**.** H7 asks for a sum-of-squares certificate for the level-two Andreief/Vandermonde determinant as an identity/effectivity refinement; scalar fixed-band positivity itself is closed by Corollary 9.64. The remaining H8 coordinate asks for uniform no-margin control as $N \to \infty$. In the coincident-node limit one obtains the jet matrices

$$J_N(z) = \left[\frac{1}{i!\,j!} \partial_z^i \partial_{\bar{w}}^j K(z,w)\big|_{w=z} \right]_{i,j=0}^{N-1}.$$

The scalar fixed-N band regime for all configurations is closed by Theorem 9.63. In the terminal one-point gauge the Pascal–Laguerre floor is effective by Lemma 9.70, but the essential open part remains the exact sign $\delta_N \geq 0$ for all N for one fixed interior-band accumulating sequence, equivalently the terminal scalar input of Theorem 9.66.

Lemma 9.49 (jet coefficient kernel)**.** *Fix a detection channel $F_\kappa = \operatorname{span}\{\phi_\kappa\}$ from Definition 9.34 and a point $z_0 \in U_-$. The scalar Pick kernel*

$$K^{(\kappa)}(z,w) := \frac{g^{F_\kappa}(z) - \overline{g^{F_\kappa}(w)}}{z - \bar{w}}$$

is holomorphic in z and antiholomorphic in w on $U_- \times U_-$. Its double Taylor coefficients

$$C_{jk}^{(\kappa)}(z_0) := \frac{1}{j!\,k!} \partial_z^j \partial_{\bar{w}}^k K^{(\kappa)}(z,w)\big|_{z=w=z_0}$$

generate $K^{(\kappa)}$ by an absolutely convergent series on $D_r(z_0) \times D_r(z_0)$, where

$$r = \frac{1}{2}(|\operatorname{Im} z_0| - \tfrac{1}{2}) > 0.$$

Under RH,

$$C_{jk}^{(\kappa)}(z_0) = \int_{\mathbb{R}} (u - z_0)^{-(j+1)} (u - \bar{z}_0)^{-(k+1)}\, d\mu_{\phi_\kappa}(u),$$

so every finite coefficient section is positive semidefinite.

Proof. On U_-, all poles of g^{F_κ} lie at distance at least $|\operatorname{Im} z_0| - \frac{1}{2}$ from z_0, and $z - \bar{w} \neq 0$ for $z, w \in U_-$. Thus the two-variable kernel is holomorphic-antiholomorphic in a polydisk of radius $|\operatorname{Im} z_0| - \frac{1}{2}$ around (z_0, z_0); Cauchy's estimates give the absolute convergence on the smaller polydisk of radius r.

Under RH, Theorem 9.189 realizes g^{F_κ} as the Cauchy transform of the positive measure μ_{ϕ_κ}. Lemma 9.178 gives

$$K^{(\kappa)}(z,w) = \int_{\mathbb{R}} \frac{d\mu_{\phi_\kappa}(u)}{(u - z)(u - \bar{w})}.$$

Differentiating under the integral at $z = w = z_0$ gives the displayed Gram coefficients. Positivity of finite sections follows by testing finite linear combinations of the functions $(u - z_0)^{-(j+1)}$ in $L^2(d\mu_{\phi_\kappa})$. $\square$

Theorem 9.50 (one-point jet serialization of RH)**.** *Fix any single point $z_0 \in U_-$. Then*

$$\mathrm{RH} \quad \Longleftrightarrow \quad \big[C_{jk}^{(\kappa)}(z_0)\big]_{j,k=0}^{N-1} \succeq 0 \quad \textit{for every } N \geq 1 \textit{ and every } \kappa \geq 1.$$

Proof. Assume RH. Positivity of every finite jet section is Lemma 9.49.

Conversely, assume all displayed finite jet matrices are positive. Fix κ. Let $z_1, \dots, z_m \in D_r(z_0)$ and $c_1, \dots, c_m \in \mathbb{C}$. Put

$$v_\alpha = \sum_i \bar{c}_i (z_i - z_0)^\alpha.$$

By the absolutely convergent Taylor expansion,

$$\sum_{i,j} \bar{c}_i K^{(\kappa)}(z_i, z_j) c_j = \sum_{\alpha,\beta \geq 0} C_{\alpha\beta}^{(\kappa)}(z_0)\, v_\alpha \overline{v_\beta} = \lim_{N\to\infty} \sum_{\alpha,\beta < N} C_{\alpha\beta}^{(\kappa)}(z_0)\, v_\alpha \overline{v_\beta} \geq 0.$$

Thus every finite Pick matrix with nodes in the disk $D_r(z_0)$ is positive semidefinite. Choose a countable, pairwise distinct node set in this disk with an accumulation point in the disk. Since $D_r(z_0) \subset U_-$, the meromorphic $G_\Xi^{F_\kappa}$ is holomorphic at every node and near the accumulation point; Lemma 9.184, applied to this one channel and this node set, removes every pole of $G_\Xi^{F_\kappa}$ in Ω_-. Running over all κ and using Lemma 9.37 excludes all non-real zeros in Ω_-; the symmetries of Ξ exclude the upper half-plane, and Lemma 9.2 gives RH. ☐

Lemma 9.51 (paired Hadamard kernel of the scalar coordinate)**.** *Let $Z(\Xi) = \{z_\rho\}$ denote the zero multiset of Ξ (multiplicities m_ρ); it is symmetric under $z \mapsto -z$ and $z \mapsto \bar{z}$ (Lemma 9.1), contained in $|\operatorname{Im} z| < \frac{1}{2}$, does not contain 0 (since $\xi(\frac{1}{2}) \neq 0$), and satisfies $\sum_\rho m_\rho |z_\rho|^{-2} < \infty$ by the counting estimate $N(T) = O(T \log T)$. Then:*

(i) (paired Hadamard expansion)

$$M_\Xi(z) = \lim_{R\to\infty} \sum_{|z_\rho| \leq R} \frac{m_\rho}{z_\rho - z},$$

locally uniformly on $\mathbb{C} \setminus Z(\Xi)$, the truncation being over the symmetric multiset $\{|z_\rho| \leq R\}$;

(ii) (kernel identity) *for $z, w \in \mathbb{C}$ with $z, \bar{w} \notin Z(\Xi)$ and $z \neq \bar{w}$,*

$$K_\Xi(z,w) := \frac{M_\Xi(z) - \overline{M_\Xi(w)}}{z - \bar{w}} = \sum_\rho \frac{m_\rho}{(z_\rho - z)(z_\rho - \bar{w})},$$

the sum converging absolutely (terms are $O(|z_\rho|^{-2})$ locally uniformly), and the identity extending across $z = \bar{w}$ by continuity;

(iii) (jets) *if $z_0 \notin \mathbb{R}$ and $d_0 := \operatorname{dist}(z_0, Z(\Xi)) > 0$, then, with $r_0 := \min(d_0, |\operatorname{Im} z_0|)$, the kernel K_Ξ is holomorphic in z and antiholomorphic in w on $D_{r_0}(z_0) \times D_{r_0}(z_0)$, its scalar jets*

$$C_{jk}(z_0) := \frac{1}{j!\,k!}\, \partial_z^j \partial_{\bar{w}}^k K_\Xi(z,w)\big|_{z=w=z_0} = \sum_\rho m_\rho\, (z_\rho - z_0)^{-(j+1)} (z_\rho - \bar{z}_0)^{-(k+1)}$$

converge absolutely, and the double Taylor series of K_Ξ around (z_0, z_0) converges absolutely on $D_r(z_0) \times D_r(z_0)$ for every $r < r_0$;

(iv) (positivity under RH) *if RH holds, every z_ρ is real, every finite jet section $[C_{jk}(z_0)]_{j,k<N}$ is positive semidefinite, and every finite Pick matrix $\mathsf{P}[M_\Xi; z_1, \dots, z_N]$ at nodes $z_1, \dots, z_N \in \Omega_- \setminus Z(\Xi)$ is positive semidefinite.*

Proof. (i) It is classical [4, Ch. II] that $\xi(s)$, and hence $\Xi(z) = \xi(\frac{1}{2} + iz)$, is entire of order exactly 1; since $\sum_\rho |z_\rho|^{-1} = \infty$ while $\sum_\rho m_\rho |z_\rho|^{-2} < \infty$ (Riemann–von Mangoldt), the zero set has convergence exponent 1 and genus 1, so the Hadamard factorization theorem applies with genus-1 elementary factors:

$$\Xi(z) = \Xi(0)\, e^{cz} \prod_\rho \left(1 - \frac{z}{z_\rho}\right) e^{z/z_\rho},$$

c a constant, the product taken (as always in this paper) in the symmetric sense of Convention 9.110 and converging locally uniformly by the genus-1 estimate $|(1-u)e^u - 1| \le |u|^2$ for $|u| \le 1$. Pairing z_ρ with $-z_\rho$ (Lemma 9.1(i) gives this pairing with equal multiplicity) cancels the linear exponential factors termwise,

$$\left(1 - \frac{z}{z_\rho}\right) e^{z/z_\rho} \left(1 + \frac{z}{z_\rho}\right) e^{-z/z_\rho} = 1 - \frac{z^2}{z_\rho^2},$$

and the pairwise product

$$P(z) := \prod_{\text{pairs}} \left(1 - \frac{z^2}{z_\rho^2}\right)$$

converges locally uniformly and absolutely, since $|z^2/z_\rho^2| = O(|z_\rho|^{-2})$ on compacts, without any exponential convergence factor. Thus $\Xi(z) = \Xi(0)e^{cz}P(z)$ exactly, with the genus-1 factorization already accounted for; evenness of Ξ (Lemma 9.1(ii)) and of P forces $c = 0$, so $\Xi = \Xi(0)P$. Logarithmic differentiation term by term (legitimate by locally uniform convergence) gives

$$M_\Xi(z) = -\frac{\Xi'}{\Xi}(z) = \sum_{\text{pairs}} \frac{2z}{z_\rho^2 - z^2} = \sum_{\text{pairs}} \left[\frac{1}{z_\rho - z} + \frac{1}{-z_\rho - z}\right],$$

which is exactly the symmetric truncation limit, with pair terms $O(|z_\rho|^{-2})$ on compacts.

(ii) By Lemma 9.1(iii), Ξ is real on $\mathbb{R}$, so $M_\Xi(\bar w) = \overline{M_\Xi(w)}$; and the zero multiset is conjugation-symmetric, so the symmetric truncations of (i) are stable under conjugation. For each finite symmetric truncation, the algebraic identity

$$\frac{1}{z - \bar w}\left[\frac{1}{z_\rho - z} - \frac{1}{z_\rho - \bar w}\right] = \frac{1}{(z_\rho - z)(z_\rho - \bar w)}$$

holds termwise; letting $R \to \infty$, the left side converges to $K_\Xi(z, w)$ by (i) and the right side converges absolutely since $|(z_\rho - z)(z_\rho - \bar w)|^{-1} = O(|z_\rho|^{-2})$.

(iii) The set $Z(\Xi)$ is conjugation-symmetric, so $\operatorname{dist}(\bar z_0, Z(\Xi)) = d_0$ as well. For $z, w \in D_{r_0}(z_0)$: $z \notin Z(\Xi)$ and $\bar w \notin Z(\Xi)$ (distance $\ge d_0 - r_0 \ge 0$, strict inside the open disk), and $z \ne \bar w$ because $\operatorname{Im} z$ and $\operatorname{Im} \bar w$ have opposite (strict) signs, $|\operatorname{Im} z - \operatorname{Im} z_0| < |\operatorname{Im} z_0|$. Hence every term of (ii) is holomorphic–antiholomorphic on the polydisk and the sum converges locally uniformly there; termwise differentiation gives the displayed jets, and Cauchy's estimates on the polydisk of radius r_0 give the absolute convergence of the double Taylor series on every smaller polydisk.

(iv) Under RH, $z_\rho \in \mathbb{R}$ and $z_\rho - \bar z_0 = \overline{z_\rho - z_0}$, so

$$\sum_{j,k<N} \bar c_j C_{jk}(z_0) c_k = \sum_\rho m_\rho \left| \sum_{j<N} c_j\, \overline{(z_\rho - z_0)^{-(j+1)}} \right|^2 \ge 0,$$

and likewise, for nodes $z_1, \dots, z_N \in \Omega_- \setminus Z(\Xi)$,

$$\sum_{i,j} \bar{c}_i K_\Xi(z_i, z_j) c_j = \sum_\rho m_\rho \left| \sum_j \frac{c_j}{\overline{z_\rho - z_j}} \right|^2 \geq 0,$$

both by (ii)–(iii) with absolute convergence justifying the rearrangement. Note that no finiteness of the zero measure is used anywhere: the raw measure $\sum m_\rho \delta_{z_\rho}$ has infinite mass and M_Ξ is *not* its Cauchy transform; it is the paired expansion (i) whose Pick kernel nevertheless has the exact Gram form (ii). ☐

Theorem 9.52 (scalar one-point jet serialization of RH; regular at the Li gauge)**.** *Let $z_0 \in \Omega_-$ satisfy $d_0 = \operatorname{dist}(z_0, Z(\Xi)) > 0$, and let $C_{jk}(z_0)$ be the scalar jets of Lemma 9.51(iii). Then*

$$\text{RH} \iff \big[C_{jk}(z_0)\big]_{j,k=0}^{N-1} \succeq 0 \quad \textit{for every } N \geq 1.$$

The distance hypothesis is automatic in the two cases used later:

(a) (interior gauges) *if $\operatorname{Im} z_0 < -\frac{1}{2}$ then $d_0 \geq |\operatorname{Im} z_0| - \frac{1}{2} > 0$, since $Z(\Xi) \subset \{|\operatorname{Im} z| < \frac{1}{2}\}$;*

(b) (Li gauge) *if $z_0 = t_0 - \frac{i}{2}$ (i.e. $\operatorname{Re} s_0 = 1$), then $d_0 > 0$ unconditionally: the change of variable $s = \frac{1}{2} + iz$ is an isometry ($|s - s_0| = |z - z_0|$), so a zero of Ξ within distance $\delta \leq 1$ of z_0 is a nontrivial zero $\rho = \beta + i\gamma$ of ζ with $|\rho - (1 + it_0)| \leq \delta$, hence $1 - \beta \leq \delta$ and $|\gamma - t_0| \leq \delta$; the classical zero-free region $\beta \leq 1 - c/\log(|\gamma| + 2)$* [4, Ch. III] *forces $\delta \geq c/\log(|t_0| + 3)$. Thus $z_0 = -\frac{i}{2}$ lies on the boundary of the Pick domain U_- but is a* regular interior point *for the meromorphic M_Ξ and its kernel: the boundary is an artifact of the interpolation coordinates, not a singularity of the object being expanded.*

Proof. Assume RH. Positivity of every finite jet section is Lemma 9.51(iv).

Conversely, assume all finite jet sections at z_0 are positive semidefinite. Set $r = \frac{1}{2}\min(d_0, |\operatorname{Im} z_0|) > 0$. Let $z_1, \dots, z_m \in D_r(z_0)$ and $c_1, \dots, c_m \in \mathbb{C}$, and put $v_\alpha = \sum_i \bar{c}_i (z_i - z_0)^\alpha$. By the absolutely convergent double Taylor expansion of Lemma 9.51(iii), applied at a radius strictly between r and $r_0 = 2r$,

$$\sum_{i,j} \bar{c}_i K_\Xi(z_i, z_j) c_j = \sum_{\alpha,\beta \geq 0} C_{\alpha\beta}(z_0)\, v_\alpha \overline{v_\beta} = \lim_{N\to\infty} \sum_{\alpha,\beta < N} C_{\alpha\beta}(z_0)\, v_\alpha \overline{v_\beta} \geq 0,$$

exactly as in the proof of Theorem 9.50. Thus every finite Pick matrix of M_Ξ with nodes in $D_r(z_0)$ is positive semidefinite. The disk $D_r(z_0)$ lies in Ω_- and avoids $Z(\Xi)$, so M_Ξ is holomorphic there. Choose a countable, pairwise distinct node sequence in $D_r(z_0)$ with an accumulation point in $D_r(z_0)$; Lemma 9.184 (scalar case, $G = M_\Xi$) removes every pole of M_Ξ in Ω_-. Every zero z_ρ of Ξ is a pole of M_Ξ with principal part $-m_\rho/(z - z_\rho) \neq 0$ — the detection that required the dense channel family in Theorem 9.50 is automatic for the scalar coordinate. Hence no zero of Ξ lies in Ω_-; the symmetries of Lemma 9.1(ii)–(iii) exclude the upper half-plane, and Lemma 9.2 gives RH. ☐

Remark 9.53 (what the scalar theorem repairs). Theorem 9.52 is the single-channel companion of Theorem 9.50: the detection quantifier "for every κ" is traded for the automatic nonvanishing of the scalar residues. It is the statement actually consumed by the whitened-terminal coordinates (Theorem 9.71), by the reference-whitened defect (Definition 9.73, Corollary 9.75), and — through case (b) — by the Li–Keiper dictionary (Theorem 9.111) directly at the Li gauge $z_0 = -\frac{i}{2}$, without any passage of positivity from interior gauges to the boundary.

Lemma 9.54 (prime-scale window content of jets). *For $s = \sigma + it$ with $\sigma \geq 1 + \epsilon_0$,*

$$\frac{1}{k!}\left|\left(\frac{\zeta'}{\zeta}\right)^{(k)}(s)\right| \leq \frac{1}{k!}\sum_{n\geq 2}\Lambda(n)(\log n)^k n^{-\sigma} \leq C(\epsilon_0)\left(\frac{2}{\epsilon_0}\right)^k.$$

Moreover, for every $A \geq 2$, the contribution of terms with

$$\left|\log n - \frac{k}{\epsilon_0}\right| > \frac{A\sqrt{k}}{\epsilon_0}$$

is at most e^{-cA^2} times the same majorant, with an absolute $c > 0$.

Proof. The derivative Dirichlet series is absolutely convergent in $\sigma > 1$, and differentiation gives coefficients bounded in modulus by $\Lambda(n)(\log n)^k n^{-\sigma}$. With $u = \log n$, the elementary bound

$$u^k e^{-\epsilon_0 u/2} \leq \left(\frac{2k}{e\epsilon_0}\right)^k$$

and the convergence of $\sum_{n\geq 2}\Lambda(n)n^{-1-\epsilon_0/2}$ give the displayed exponential bound after Stirling. The localization statement is the Chernoff bound for the gamma-type weight $u^k e^{-\epsilon_0 u}/k!$, applied as a dominating envelope to the positive coefficients. $\square$

Lemma 9.55 (Szegő jet Gram identity). *For $z_0 = t - iy$ with $y \geq \frac{1}{2}$ (i.e. $z_0 \in \overline{U_-}$, including the boundary line $y = \frac{1}{2}$), the jets of the Szegő kernel*

$$S(z,w) = \frac{-i}{z - \bar{w}}$$

satisfy

$$\frac{1}{j!\,k!}\partial_z^j \partial_{\bar{w}}^k S(z,w)\big|_{z=w=z_0} = \int_0^\infty \frac{(-i\tau)^j}{j!}e^{-iz_0\tau}\,\overline{\frac{(-i\tau)^k}{k!}e^{-iz_0\tau}}\,d\tau = \varepsilon_j\overline{\varepsilon_k}\binom{j+k}{j}\frac{1}{(2y)^{j+k+1}},$$

where $\varepsilon_j = (-i)^j$. Hence the jet matrix is unitarily diagonal-equivalent to the Gram matrix of $\{\tau^k e^{-y\tau}/k!\}_{k<N}$ in $L^2(0,\infty)$, and is positive definite. Write

$$G_N(z_0) := \left[\frac{1}{j!\,k!}\partial_z^j \partial_{\bar{w}}^k S(z,w)\big|_{z=w=z_0}\right]_{j,k=0}^{N-1} \succ 0$$

for this jet matrix; $G_N(z_0)$ is the reference Gram used in Definition 9.73, defined and strictly positive definite for every $z_0 = t - iy \in U_-$ (equivalently every $y > \frac{1}{2}$), with entries as displayed above. Denote its least eigenvalue by $\lambda_N^{\mathrm{jet}}(y) > 0$.

Proof. For z, w with $\operatorname{Im} z, \operatorname{Im} w < 0$,

$$S(z,w) = \int_0^\infty e^{-iz\tau}e^{i\bar{w}\tau}\,d\tau,$$

the integral converging absolutely since $|e^{-iz\tau}| = e^{\tau\operatorname{Im} z}$ decays for $\tau \to \infty$; this covers every $z_0 = t - iy$ with $y > 0$, in particular every $y \geq \frac{1}{2}$. Differentiating under the integral gives the first identity. At $z = w = z_0 = t - iy$, the exponential weight is $e^{-2y\tau}$, so

$$\frac{(-i)^j i^k}{j!\,k!}\int_0^\infty \tau^{j+k}e^{-2y\tau}\,d\tau = (-i)^j i^k \frac{(j+k)!}{j!\,k!}\frac{1}{(2y)^{j+k+1}},$$

which is the displayed formula. The functions $\tau^k e^{-y\tau}/k!$ are linearly independent, so their Gram matrix is positive definite. $\square$

Lemma 9.56 (archimedean jet lower bound)**.** *Fix N, $\epsilon_0 > 0$, and $Y < \infty$. For $z_0 = t - iy$ with $y \in [\frac{1}{2} + \epsilon_0, Y]$ and $|t|$ large, the archimedean part $J_N^\infty(z_0)$ of the scalar jet matrix satisfies, in the unitary diagonal normalization of Lemma 9.55,*

$$J_N^\infty(z_0) \succeq \left(\lambda_N^{\mathrm{jet}}(y) \log \frac{|t|}{2\pi} - \frac{C_{N,Y,\epsilon_0}}{|t|}\right) I.$$

Proof. The proof of Lemma 9.41 is an analytic estimate for the two-variable kernel: on a fixed polydisk around (z_0, z_0) contained in $U_- \times U_-$,

$$K_\infty(z, w) - \log \frac{|t|}{2\pi} S(z, w) = O_{Y,\epsilon_0}(|t|^{-1})$$

as a holomorphic-antiholomorphic function. Cauchy's estimates on a slightly smaller polydisk bound all mixed derivatives of total order at most $2N$ by $C_{N,Y,\epsilon_0}/|t|$. Taking the jet matrix therefore gives

$$J_N^\infty(z_0) = \log \frac{|t|}{2\pi} J_N^S(z_0) + O_{N,Y,\epsilon_0}(|t|^{-1}),$$

where J_N^S is the positive definite Szegő jet Gram matrix of Lemma 9.55. The eigenvalue bound follows. □

Theorem 9.57 (fixed-N coincident jet positivity at large height)**.** *Fix N, $\epsilon_0 > 0$, and $Y < \infty$. There is $T(N, \epsilon_0, Y)$ such that for every $z_0 = t - iy$ with $y \in [\frac{1}{2} + \epsilon_0, Y]$ and $|t| \geq T(N, \epsilon_0, Y)$, the scalar jet matrix $J_N(z_0)$ is positive definite. Its margin is proportional to*

$$\lambda_N^{\mathrm{jet}}(y) \log \frac{|t|}{2\pi}.$$

Proof. Decompose $J_N = J_N^\infty + J_N^{\mathrm{prime}}$. The entries of J_N^{prime} involve derivatives of ζ'/ζ of order at most $2N$ at points with $\operatorname{Re} s \geq 1 + \epsilon_0$. Lemma 9.54 bounds each such normalized derivative by a constant depending only on N and ϵ_0; the denominators in the Pick kernel are bounded away from zero because $|z_0 - \bar{z}_0| = 2y \geq 1 + 2\epsilon_0$. Hence

$$\|J_N^{\mathrm{prime}}\| \leq C_{N,\epsilon_0}.$$

Lemma 9.56 gives an archimedean lower bound growing like $\lambda_N^{\mathrm{jet}}(y) \log(|t|/(2\pi))$. Choosing $T(N, \epsilon_0, Y)$ so this dominates C_{N,ϵ_0} and the $O(|t|^{-1})$ error proves positivity. □

Corollary 9.58 (below-threshold coalescence for fixed N)**.** *Fix N, $\epsilon_0 > 0$, and $Y < \infty$. There is $h_N = h_N(\epsilon_0, Y) > 0$, independent of t, such that every scalar N-node cluster of diameter at most h_N, centered at $z_0 = t - iy$ with $y \in [\frac{1}{2} + \epsilon_0, Y]$ and $|t| \geq T(N, \epsilon_0, Y)$, has positive definite Pick matrix. Together with Theorem 9.45, this covers single scalar clusters from full coalescence $h = 0$ through the Cauchy-determinant scale, with overlap for large $|t|$.*

Proof. Pass from node values to divided-difference coordinates by the standard invertible confluent Vandermonde matrix E_h associated with the cluster. In those coordinates,

$$E_h^* \mathsf{P} E_h = J_N(z_0) + O_N(h \log |t|)$$

entrywise: this is Taylor's formula for the holomorphic-antiholomorphic kernel, with the archimedean derivatives bounded by $C_N \log |t|$ and the prime derivatives bounded by Lemma 9.54. By Theorem 9.57, $J_N(z_0) \succeq c_N \log |t| \, I$ for large $|t|$. Taking $h_N < c_N/(2C_N')$ makes the perturbation smaller than half of this margin. Congruence by the invertible matrix E_h preserves positive definiteness. □

Lemma 9.59 (scale selection)**.** *Let $w_1, \dots, w_N$ lie in a box of diameter D. Consider the disjoint rings*

$$R_r = [DN^{-r}, DN^{-r+1}), \qquad r = 1, \dots, N^2.$$

Since there are fewer than N^2 pairwise distances, some ring R_{r_*} contains none of them. Set $h_* = DN^{-r_*}$ and define clusters as the connected components of the graph with edges $|w_i - w_j| < h_*$. Then

$$\text{diam(each cluster)} < Nh_*, \qquad \text{dist(distinct clusters)} \geq Nh_*, \qquad h_* \geq DN^{-N^2}.$$

Proof. Two nodes in the same component are joined by a chain of fewer than N edges, each of length $< h_*$, giving the diameter bound. If two nodes lie in distinct components, their distance is at least h_*. Since the ring $[h_*, Nh_*)$ is distance-free, the distance is in fact at least Nh_*. The lower bound for h_* follows from $r_* \leq N^2$. □

Definition 9.60 (confluent Newton basis)**.** For a multiset $\zeta_0, \dots, \zeta_k$ in U_-, the divided difference of $e_\tau(z) = e^{-iz\tau}$ has the Hermite–Genocchi representation

$$e[\zeta_0, \dots, \zeta_k](\tau) = (-i\tau)^k \int_{\Delta_k} e^{-i\zeta(s)\tau}\, ds, \qquad \zeta(s) = \zeta_0 + \sum_{i=1}^{k} s_i(\zeta_i - \zeta_0).$$

The formula is continuous through confluent limits and satisfies

$$|e[\zeta_0, \dots, \zeta_k](\tau)| \leq \frac{\tau^k}{k!} e^{-(\frac{1}{2}+\epsilon_0)\tau}$$

whenever all $\operatorname{Im} \zeta_i \leq -(\frac{1}{2} + \epsilon_0)$. Given a clustered configuration, order each cluster and let E be the block-diagonal Newton change of basis. For pairwise distinct nodes, E is invertible. For every kernel K holomorphic in z and antiholomorphic in w, the entries of $E^* \mathsf{K} E$ are the corresponding two-variable divided differences of K.

Lemma 9.61 (uniform Newton–Szegő positivity)**.** *Fix N, $\epsilon_0 > 0$, Y, and A, and set*

$$d_0 = (2A + 2Y)N^{-N^2}.$$

On the compact space of clustered configurations in the box

$$|a| \leq A, \qquad y \in [\tfrac{1}{2} + \epsilon_0, Y],$$

with cluster types fixed, confluent intra-cluster multisets allowed, and distinct cluster centers separated by at least d_0, there is a constant $c_N^{\mathcal{S}} = c_N^{\mathcal{S}}(\epsilon_0, Y, A) > 0$ such that

$$\widetilde{S} := E^* S E \succeq c_N^{\mathcal{S}} I$$

for every configuration, where S is the Szegő matrix of Lemma 9.40.

Proof. The matrix S is the Gram matrix of the functions $e^{-iz_j\tau}$ in $L^2(0, \infty)$. After the Newton congruence, $\widetilde{S}$ is the Gram matrix of the confluent Newton basis functions of Definition 9.60. Within each cluster these functions form a triangular basis of the Hermite span $\operatorname{span}\{\tau^r e^{-iz_c\tau}\}$. Across distinct clusters, the generalized frequencies are distinct because the cluster centers are separated by at least d_0. Thus the Gram matrix is positive definite at every point of the parameter space.

The entries depend continuously on the configuration by dominated convergence, using the majorant $C_N \tau^{2N} e^{-(1+2\epsilon_0)\tau}$. The parameter space is a finite union of compact sets, one for each partition type and cluster ordering. Hence the continuous function $\lambda_{\min}(\widetilde{S})$ attains a strictly positive minimum. □

Lemma 9.62 (divided differences of bounded kernels). *Let $K(z,w)$ be holomorphic in z, antiholomorphic in w, and bounded by B on the $\epsilon_0/2$-enlarged box-domain. Then, for every clustered configuration as above, the entries of $E^*\mathsf{K}E$ are bounded by $B(2/\epsilon_0)^{2N}$, and therefore*

$$\|E^*\mathsf{K}E\| \le NB(2/\epsilon_0)^{2N}.$$

Proof. Each entry is a double Hermite–Genocchi average of

$$\frac{1}{j!\,k!}\partial_z^j\partial_{\bar w}^k K$$

with $j,k<N$, over hulls contained in the enlarged box-domain. Cauchy's estimates on polydisks of radius $\epsilon_0/2$ give

$$\frac{|\partial_z^j\partial_{\bar w}^k K|}{j!\,k!} \le B(2/\epsilon_0)^{j+k} \le B(2/\epsilon_0)^{2N}.$$

The simplex averages have mass one, and the operator-norm bound follows from the entrywise bound. □

Theorem 9.63 (band closure of scalar ARP-P(N)). *Fix N, $\epsilon_0\in(0,\frac12]$, Y, and A. There is $T^*=T^*(N,\epsilon_0,Y,A)$ such that, for every $|t|\ge T^*$ and every configuration of N pairwise distinct scalar nodes*

$$z_j = t + a_j - iy_j, \qquad |a_j|\le A, \qquad y_j\in[\tfrac12+\epsilon_0, Y],$$

the scalar Pick matrix satisfies $\mathsf{P}[M_\Xi; z_1,\dots,z_N]\succ 0$. Coalescent limits are positive semidefinite by continuity, with coincident blocks governed by Theorem 9.57.

Proof. Apply Lemma 9.59 to $w_j = y_j + ia_j$ inside a box of diameter at most $2A+2Y$ and use the resulting clustered structure. On the enlarged box-domain, decompose the scalar Pick kernel as

$$K = \log\frac{|t|}{2\pi}S + R + K_{\mathrm{prime}},$$

where $R = K_\infty - \log(|t|/(2\pi))S$. The proof of Lemma 9.41 is uniform in coalescing configurations: its numerator error is $O((1+A+Y)/|t|)$ and the denominators satisfy $|z-\bar w|\ge 1+2\epsilon_0$ on the enlarged domain. Hence $|R|\le C_{\mathrm{box}}/|t|$. Also $|K_{\mathrm{prime}}|\le C(\epsilon_0,A,Y)$ there, because ζ'/ζ is bounded on $\operatorname{Re} s\ge 1+\epsilon_0/2$.

Apply the Newton congruence E:

$$E^*\mathsf{P}E = \log\frac{|t|}{2\pi}\,\widetilde S + \widetilde R + \widetilde K_{\mathrm{prime}}.$$

Lemma 9.61 gives $\widetilde S \succeq c_N^{\mathcal S} I$. Lemma 9.62 gives

$$\|\widetilde K_{\mathrm{prime}}\| \le C_N(\epsilon_0,A,Y), \qquad \|\widetilde R\| \le C_N'(\epsilon_0,A,Y)/|t|.$$

Therefore

$$E^*\mathsf{P}E \succeq \left(c_N^{\mathcal S}\log\frac{|t|}{2\pi} - C_N - \frac{C_N'}{|t|}\right) I.$$

This is positive for $|t|\ge T^*$. Since E is invertible for pairwise distinct nodes, congruence preserves positive definiteness of the original Pick matrix. □

Corollary 9.64 (H7 band closure). *For $N=2$, Theorem 9.63 closes scalar level-two positivity for every two-node configuration in a fixed band at all heights $|t|\ge T^*(2,\epsilon_0,Y,A)$, including degenerating and coincident limits. The Andreief/SOS program remains useful as an identity-level and effectivity program, but is no longer essential for scalar fixed-band positivity.*

Remark 9.65 (subsumption, ineffectivity, and the guardrail). Theorem 9.63 subsumes Theorems 9.42, 9.45, 9.57, and Corollary 9.58 as fixed-N scalar band-positivity statements. Those earlier results are retained because their constants are effective; the compactness constant $c_N^{\mathcal{S}}$ in Lemma 9.61 is not. There is no conflict with Lemma 9.13: the Nevanlinna–Pick bridge needs all finite prefixes of a single accumulating sequence, while here $T^*(N)$ may grow without control as $N \to \infty$. The band floor $\epsilon_0 > 0$ is also essential for this theorem, but the minimal scalar bridge may be placed inside a fixed interior band. After this closure, the essential Step 5 content is the terminal scalar condition of Theorem 9.66; PDH at the edge and effectivity of the general compactness constant $c_N^{\mathcal{S}}$ are important but non-minimal subproblems. In the one-point terminal gauge this effectivity is replaced by the explicit Pascal–Laguerre floor of Lemma 9.70.

Theorem 9.66 (minimal scalar open input)**.** *Fix a box*

$$\mathcal{B} = \{a - iy : \ |a| \le A, \ y \in [\tfrac{1}{2} + \epsilon_0, Y]\} \subset U_-,$$

a height shift $t_0 \in \mathbb{R}$*, and a pairwise distinct node sequence*

$$z_j = t_0 + a_j - iy_j \in t_0 + \mathcal{B}$$

with an accumulation point in $t_0 + \mathcal{B}$*. Then*

$$\mathrm{RH} \quad \Longleftrightarrow \quad \mathsf{P}[M_\Xi; z_1, \dots, z_N] \succeq 0 \quad \textit{for every } N \ge 1.$$

Proof. Assume RH. Lemma 9.51(iv) gives the finite Pick positivity of M_Ξ at every finite node set in $\Omega_- \setminus Z(\Xi)$, in particular at the given nodes. (The raw zero measure has infinite mass and M_Ξ is not its Cauchy transform; the positivity comes from the paired Hadamard kernel identity of Lemma 9.51(ii), not from Lemma 9.178.)

Conversely, assume the displayed positivity for every prefix of the fixed sequence. The nodes are pairwise distinct, accumulate in $t_0 + \mathcal{B} \subset U_-$, and M_Ξ is meromorphic on $\mathbb{C}$ and holomorphic at every node and near the accumulation point, since its poles lie in the zero strip $|\operatorname{Im} z| < \frac{1}{2}$. Lemma 9.184 (scalar case, $G = M_\Xi$) therefore extends M_Ξ holomorphically to Ω_-.

Every zero z_ρ of Ξ is a pole of

$$M_\Xi(z) = -\frac{\Xi'(z)}{\Xi(z)}$$

with nonzero residue $-m_\rho$. Hence no zero of Ξ lies in Ω_-. The symmetries Lemma 9.1(ii)–(iii) gives $\Xi(-z) = \Xi(z)$ and $\Xi(\bar{z}) = \overline{\Xi(z)}$, excluding the upper half-plane. Thus all zeros of Ξ are real, which is RH by Lemma 9.2. $\square$

Definition 9.67 (terminal Pick inequality)**.** Fix a box, height t_0, and accumulating scalar node sequence as in Theorem 9.66. For each prefix of length N, apply the scale selection and Newton congruence of Lemmas 9.59–9.61. The exact terminal condition is

$$\lambda_{\min}\left(\log\frac{|t_0|}{2\pi}\,\widetilde{S}_N + \widetilde{R}_N + \widetilde{K}_{\mathrm{prime},N}\right) \ge 0 \qquad (N \ge 1),$$

with the obvious interpretation for bounded $|t_0|$ by replacing the logarithmic asymptotic with the exact archimedean kernel. The sufficient norm form is

$$c_N^{\mathcal{S}} \log\frac{|t_0|}{2\pi} \ge \|\widetilde{K}_{\mathrm{prime},N}\| + \|\widetilde{R}_N\|.$$

Remark 9.68 (terminal inequality status). By Theorem 9.66, the exact terminal condition in Definition 9.67 for one accumulating scalar sequence is equivalent to RH. The displayed norm inequality is a useful sufficient quantitative form, not an equivalent reformulation unless the norm loss is removed. The left side is governed by the Newton–Szegő lower bound $c_N^{\mathcal{S}}$; the right side is governed by divided differences of the prime kernel. The crude Cauchy estimate grows exponentially in N, while the confluent Szegő/Laguerre constants may decay exponentially. The remaining essential problem is therefore cancellation in the prime divided differences strong enough to control this race uniformly in N for a fixed height.

Theorem 9.69 (sequence independence of the terminal input)**.** *Let $\{z_j\}$ and $\{z'_j\}$ be two pairwise distinct node sequences contained in interior boxes of U_-, each with an accumulation point in its box. Then*

$$\big[\mathsf{P}[M_\Xi; z_1, \ldots, z_N] \succeq 0 \text{ for every } N\big] \iff \big[\mathsf{P}[M_\Xi; z'_1, \ldots, z'_N] \succeq 0 \text{ for every } N\big] \iff \text{RH}.$$

Thus the choice of an accumulating sequence is a coordinate gauge for attacking the terminal input, not a weakening of its mathematical content.

Proof. Prefix positivity for either sequence implies RH by Theorem 9.66, applied to the corresponding box and accumulation point. Conversely, RH implies scalar Pick positivity for every finite node set by Lemma 9.51(iv). Applying this to all prefixes of either sequence proves the equivalence. □

Lemma 9.70 (effective Pascal–Laguerre floor)**.** *Let*

$$P_N = \left[\binom{j+k}{j}\right]_{j,k=0}^{N-1}, \qquad L_N = \left[\binom{j}{k}\right]_{0\le k\le j<N}.$$

Then $P_N = L_N L_N^*$ and

$$\|L_N\|^2 \le 4^N, \qquad \|L_N^{-1}\|^2 \le 4^N,$$

hence

$$4^{-N} \le \lambda_{\min}(P_N) \le \lambda_{\max}(P_N) \le 4^N.$$

Consequently the Szegő jet Gram matrix of Lemma 9.55 satisfies

$$\lambda_N^{\text{jet}}(y) \ge 4^{-N}(2y)^{-(2N-1)}, \qquad y \ge \tfrac{1}{2}.$$

Proof. The Cholesky identity is Vandermonde convolution:

$$\binom{j+k}{j} = \sum_{r\ge 0}\binom{j}{r}\binom{k}{r}.$$

The inverse of the lower binomial triangle is the signed lower binomial triangle, $(L_N^{-1})_{jk} = (-1)^{j-k}\binom{j}{k}$. The row and column sums of $|L_N|$ and $|L_N^{-1}|$ are bounded by powers of 2; in particular

$$\|L_N\|^2 \le \|L_N\|_1\|L_N\|_\infty \le 4^N, \qquad \|L_N^{-1}\|^2 \le 4^N.$$

Therefore $\lambda_{\max}(P_N) = \|L_N\|^2 \le 4^N$ and $\lambda_{\min}(P_N) = \|L_N^{-1}\|^{-2} \ge 4^{-N}$.

By Lemma 9.55, after removing unimodular phase factors the N-jet Szegő Gram is $D_y P_N D_y$, where $D_y = \operatorname{diag}((2y)^{-j-1/2})_{j=0}^{N-1}$. For $y \ge \frac{1}{2}$, $\min_j (D_y)_{jj} = (2y)^{-N+1/2}$, so

$$\lambda_{\min}(D_y P_N D_y) \ge (2y)^{-2N+1}\lambda_{\min}(P_N) \ge 4^{-N}(2y)^{-(2N-1)}.$$

□

Theorem 9.71 (whitened terminal operator)**.** *Fix $z_0 = t_0 - iy \in U_-$ and an integer N such that the archimedean jet matrix $J_N^\infty(z_0)$ is positive definite. Define*

$$T_N(z_0) := \big(J_N^\infty(z_0)\big)^{-1/2} J_N^{\text{prime}}(z_0)\big(J_N^\infty(z_0)\big)^{-1/2}.$$

Then

$$J_N(z_0) = J_N^\infty(z_0) - J_N^{\rm prime}(z_0) \succeq 0 \quad\Longleftrightarrow\quad \lambda_{\max}(T_N(z_0)) \le 1.$$

Consequently, for every $N \le N_*(t_0)$ in the range where $J_N^\infty(z_0) \succ 0$ (Lemma 9.72), $J_N(z_0) \succeq 0$ is equivalent to $\lambda_{\max}(T_N(z_0)) \le 1$; the reference-whitened observable of Definition 9.73 carries the equivalence with RH to all N *without restriction.*

Proof. The first assertion is congruence by the invertible matrix $(J_N^\infty)^{-1/2}$:

$$J_N \succeq 0 \Longleftrightarrow I - (J_N^\infty)^{-1/2} J_N^{\rm prime} (J_N^\infty)^{-1/2} \succeq 0.$$

Since the displayed matrix is Hermitian, this is equivalent to $\lambda_{\max}(T_N) \le 1$. If the archimedean matrices are positive for every N, the collection of conditions $J_N(z_0) \succeq 0$ for all N is the scalar one-point jet serialization of RH, Theorem 9.52 (its distance hypothesis holds by case (a), since $\operatorname{Im} z_0 < -\frac{1}{2}$; the detection is automatic because the scalar residues $-m_\rho$ are nonzero at every zero). $\square$

Lemma 9.72 (pole-pair obstruction in the archimedean gauge)**.** *Write the archimedean scalar as $g_\infty = g_{\rm arch} + g_{\rm pole}$, where $g_{\rm pole}(z) = \frac{1}{i/2 - z} + \frac{1}{-i/2 - z}$ is the contribution of the trivial factor $s(s-1)$ (each pole $\pm i/2$ carrying weight $+1$, since $i(-\frac{1}{s} - \frac{1}{s-1}) = \frac{1}{i/2-z} + \frac{1}{-i/2-z}$). The pole pair has zero boundary density on $\mathbb{R}$ — so the archimedean weight w_A of Lemma 9.81 does not see it — but it is an interior conjugate pair for every gauge $z_0 = t_0 - iy$, with exterior Blaschke modulus*

$$|b(-i/2)| = \left(\tfrac{t_0^2 + (y+1/2)^2}{t_0^2 + (y-1/2)^2}\right)^{1/2} = 1 + \tfrac{y}{t_0^2} + O(t_0^{-4}).$$

Consequently the archimedean jet matrix $J_N^\infty(z_0)$ is not *positive definite for all N: it acquires a negative direction at scale $N_*(t_0) \asymp y^{-1} t_0^2 \log t_0$, by the mechanism of Proposition 9.86. For $N \le N_*(t_0)$ positivity holds, with the threshold constant made explicit in Proposition 9.86. The effect is invisible numerically ($N_* \sim 10^5$ at $t_0 = 300$).*

Definition 9.73 (terminal defect, reference-whitened)**.** For $z_0 = t_0 - iy$ with $y \ge \frac{1}{2}$, let $J_N(z_0) = [C_{jk}(z_0)]_{j,k<N}$ be the scalar jet matrix of Lemma 9.51(iii) and $G_N(z_0)$ the reference Gram of Lemma 9.55, defined and strictly positive definite on the same domain $y \ge \frac{1}{2}$. Set

$$\delta_N(z_0) := \min_{c \ne 0} \frac{c^* J_N(z_0) c}{c^* G_N(z_0) c}.$$

Since $G_N(z_0) \succ 0$ for every N and every such z_0, immediately $\delta_N(z_0) \ge 0 \iff J_N(z_0) \succeq 0$, at *every* gauge $z_0 = t_0 - iy$, $y \ge \frac{1}{2}$, without restriction on t_0 — in particular at the Li gauge $z_0 = -\frac{i}{2}$ ($t_0 = 0$, $y = \frac{1}{2}$). Hence $\delta_N \ge 0\ \forall N \iff J_N \succeq 0\ \forall N \iff$ RH (Theorem 9.52, at any z_0 satisfying its distance hypothesis; see Remark 9.74), for *all* N without restriction.

The sign-defect is deliberately kept separate from asymptotic scaling: where a height-normalized magnitude is wanted (the large-height asymptotics of Theorem 9.42 and the sampling theorems), use the rescaled variant

$$\widetilde{\delta}_N(z_0) := \frac{\delta_N(z_0)}{L}, \qquad L = \log \tfrac{|t_0|}{2\pi},$$

defined only in the regime $L > 0$ (i.e. $|t_0| > 2\pi$), where it has the same sign as δ_N. All numerical tables in this paper are computed at gauges with $|t_0| \gg 2\pi$, where the two normalizations differ by the positive factor L and every stated sign and every stated order of magnitude in N is unaffected. The J_N^∞-whitened coordinate T_N of Theorem 9.71 is equivalent to this one exactly in the range $N \le N_*(t_0)$ of Lemma 9.72 and is used only there; the numerical harness already computes the reference-whitened form (it whitens with the reference Gram G_N, not with J_N^∞), so its outputs are unaffected.

Remark 9.74 (exact scope of the admissible gauge). Definition 9.73 defines $\delta_N(z_0)$ for every $z_0 = t_0 - iy$ with $y \geq \frac{1}{2}$; this is the full domain on which the reference Gram $G_N(z_0)$ of Lemma 9.55 is positive definite. The equivalence $\delta_N(z_0) \geq 0\ \forall N \iff$ RH used in Corollary 9.75 additionally requires the distance hypothesis of Theorem 9.52, namely $d_0(z_0) := \operatorname{dist}(z_0, Z(\Xi)) > 0$. By Theorem 9.52(a)–(b) this holds at *every* such gauge: for $y > \frac{1}{2}$ it holds because $Z(\Xi) \subset \{|\operatorname{Im} z| < \frac{1}{2}\}$ forces $d_0(z_0) \geq y - \frac{1}{2} > 0$, and at $y = \frac{1}{2}$ (the Li line) it holds unconditionally by the classical zero-free region, uniformly in t_0. Hence the admissible scope of z_0 in Corollary 9.75 is exactly

$$z_0 = t_0 - iy, \qquad y \geq \tfrac{1}{2}, \qquad t_0 \in \mathbb{R},$$

with no further restriction: the corollary holds at every gauge on or above the Li line, uniformly in t_0, including $t_0 = 0$.

Corollary 9.75 (ARP-P–terminal-defect equivalence)**.** *Fix $z_0 = t_0 - iy$ with $y \geq \frac{1}{2}$ (Remark 9.74). For the fixed scalar channel and the reference-whitened defect of Definition 9.73 at this z_0,*

$$\text{ARP-P} \iff \delta_N(z_0) \geq 0 \quad \textit{for every } N.$$

Proof. By Definition 9.73, $\delta_N(z_0) \geq 0$ for every N is equivalent to $J_N(z_0) \succeq 0$ for every N, since the reference Gram $G_N(z_0)$ is positive definite at every $y \geq \frac{1}{2}$; by Remark 9.74 the distance hypothesis of Theorem 9.52 holds at this z_0 (case (a) if $y > \frac{1}{2}$, case (b) if $y = \frac{1}{2}$), so $J_N(z_0) \succeq 0\ \forall N$ is equivalent to RH; and by Corollary 9.190, RH is equivalent to ARP-P. □

Theorem 9.76 (prime shift-operator representation)**.** *On $L^2(0,\infty)$ let*

$$(S_\lambda f)(\tau) = f(\tau - \lambda)\mathbf{1}_{\tau \geq \lambda}$$

and define the von Mangoldt shift form

$$\mathcal{P}_\Lambda := \sum_{n\geq 2} \Lambda(n) n^{-1/2}\big(S_{\log n} + S_{\log n}^*\big)$$

on finite spans of exponentials $e^{-iz\tau}$ with $\operatorname{Im} z \leq -(\frac{1}{2} + \epsilon_0)$. If

$$F(\tau) = \sum_j \bar{c}_j e^{-iz_j\tau},$$

then the scalar prime part of the Pick form satisfies

$$c^*\mathsf{P}_{\mathrm{prime}}c = -\langle \mathcal{P}_\Lambda F, F\rangle_{L^2(0,\infty)}.$$

Proof. On $\operatorname{Re} s > 1$,

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n\geq 2} \Lambda(n) n^{-s}, \qquad n^{-s} = n^{-1/2} e^{-i(\log n)z}.$$

For a single frequency $\lambda > 0$, using

$$\frac{1}{z - \bar{w}} = i\int_0^\infty e^{-i(z-\bar{w})\tau}\, d\tau \qquad (\operatorname{Im}(z - \bar{w}) < 0),$$

one obtains

$$\frac{i\big(e^{-i\lambda z} + e^{i\lambda\bar{w}}\big)}{z - \bar{w}} = -\int_0^\infty \Big[e^{-iz(\tau+\lambda)}e^{i\bar{w}\tau} + e^{-iz\tau}e^{i\bar{w}(\tau+\lambda)}\Big]\, d\tau.$$

After summing against $\bar{c}_i c_j$, the right side is

$$-2\,\mathrm{Re}\int_0^\infty F(\tau+\lambda)\overline{F(\tau)}\,d\tau = -\langle (S_\lambda + S_\lambda^*)F, F\rangle.$$

The sum over n is absolutely convergent on the indicated spans because $|F(\tau)| \le Ce^{-(\frac{1}{2}+\epsilon_0)\tau}$ and hence

$$|\langle (S_{\log n} + S_{\log n}^*)F, F\rangle| \le C' n^{-\frac{1}{2}-\epsilon_0},$$

so it is dominated by $\sum_n \Lambda(n) n^{-1-\epsilon_0}$. □

Theorem 9.77 (terminal spectral form)**.** *Fix an interior-band accumulating scalar sequence and let $\mathcal{E}$ be the finite linear span of the corresponding exponentials $e^{-iz_j\tau}$. The terminal scalar input is equivalent to*

$$\langle \mathcal{P}_\Lambda F, F\rangle \le \mathcal{A}(F,F) \qquad (F \in \mathcal{E}),$$

where $\mathcal{A}$ is the exact archimedean/pole Pick form. In a large-height fixed band,

$$\mathcal{A}(F,F) = \log\frac{|t_0|}{2\pi}\|F\|^2_{L^2(0,\infty)} + \mathcal{R}(F,F),$$

with $\mathcal{R}$ the explicitly bounded Stirling remainder from Lemma 9.41.

Proof. By Theorem 9.76, the prime contribution to the scalar Pick form is $-\langle \mathcal{P}_\Lambda F, F\rangle$. The remaining archimedean and pole terms define $\mathcal{A}(F,F)$. Therefore positivity of every finite scalar Pick prefix is exactly

$$\mathcal{A}(F,F) - \langle \mathcal{P}_\Lambda F, F\rangle \ge 0$$

on the finite exponential span. The equivalence with RH is Theorem 9.66. The large-height expansion of $\mathcal{A}$ is the same Stirling/Szegő expansion used in Lemma 9.41. □

Proposition 9.78 (Laguerre matrix elements of the terminal operator)**.** *At $z_0 = t_0 - iy$, the leading Szegő part of the archimedean jet Gram is diagonalized by the orthonormal Laguerre basis*

$$L_k^{(y,t_0)}(\tau) = e^{-it_0\tau}\sqrt{2y}\,\ell_k(2y\tau)e^{-y\tau},$$

where ℓ_k are the standard orthonormal Laguerre polynomials on $[0,\infty)$ with weight $e^{-u}du$. In this basis, the whitened prime operator has entries

$$(T_N)_{jk} = -\frac{1}{\log(|t_0|/2\pi)}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\left[n^{-it_0} m_{jk}^{(y)}(\log n) + n^{it_0}\overline{m_{kj}^{(y)}(\log n)}\right] + O\!\left(\frac{E_N(y)}{\log|t_0|}\right),$$

where

$$m_{jk}^{(y)}(\lambda) = \int_0^\infty \sqrt{2y}\,\ell_j(2y(u+\lambda))e^{-y(u+\lambda)}\sqrt{2y}\,\ell_k(2yu)e^{-yu}\,du.$$

Each $m_{jk}^{(y)}(\lambda)$ is a polynomial in λ times $e^{-y\lambda}$ and is computable in closed form. The error $E_N(y)$ is effective and comes only from the archimedean Stirling remainder.

Proof. Lemma 9.55 identifies the leading archimedean jet Gram with the Gram of $\tau^k e^{-iz_0\tau}/k!$ in $L^2(0,\infty)$. Orthonormalizing the polynomial factor against the weight $e^{-2y\tau}d\tau$ gives the displayed Laguerre functions. The shift representation Theorem 9.76 gives the matrix element of $S_{\log n} + S_{\log n}^*$ as the two lag integrals above, with the phases $n^{\mp it_0}$ supplied by the carrier $e^{-it_0\tau}$. The Stirling remainder is holomorphic on a fixed polydisk in the interior band, so Cauchy's estimates make the corresponding finite jet correction effective. □

Corollary 9.79 (effective $\log\log$ contractivity range)**.** *Fix $y \geq \frac{1}{2} + \epsilon_0$. There are effective constants $\beta = \beta(\epsilon_0, y)$, c_0, and T_2 such that, for $|t_0| \geq T_2$,*

$$\lambda_{\max}(T_N(t_0 - iy)) \leq \frac{c_0 N^2 e^{\beta N}}{\log(|t_0|/2\pi)}$$

for all $N \leq N_*(t_0)$, the range where the archimedean jet matrix is positive (Lemma 9.72). In particular $\lambda_{\max}(T_N) \leq 1$ is proved effectively for

$$N \leq \frac{\log\log|t_0| - \log(c_0 N^2)}{\beta}.$$

Proof. By Theorem 9.71,

$$\lambda_{\max}(T_N) \leq \frac{\|J_N^{\mathrm{prime}}\|}{\lambda_{\min}(J_N^\infty)}.$$

Lemma 9.54 bounds the prime jet entries by an exponential $C(\epsilon_0)(2/\epsilon_0)^{2N}$, hence the matrix norm by N^2 times the same exponential. Lemma 9.56, with the effective floor from Lemma 9.70, gives

$$\lambda_{\min}(J_N^\infty) \geq 4^{-N}(2y)^{-(2N-1)}\log(|t_0|/2\pi) - C'_N/|t_0|,$$

with C'_N effective by Cauchy estimates on the fixed polydisk. For $|t_0| \geq T_2$ the remainder is absorbed, yielding the stated exponential bound with, for example, a crude admissible $\beta \leq \log(64y^2/\epsilon_0^2)$. □

Definition 9.80 (terminal defect observable)**.** For a fixed one-point terminal gauge $z_0 = t_0 - iy$ and a scalar jet section, define

$$\delta_N(z_0) := 1 - \lambda_{\max}(T_N(z_0)).$$

The terminal input is $\delta_N(z_0) \geq 0$ for every N. Numerical evidence should therefore be read through the sign and decay law of δ_N, not through the existence of a fixed positive margin. This J_N^∞-whitened defect is well defined only for $N \leq N_*(t_0)$ (Lemma 9.72); the reference-whitened observable of Definition 9.73 is the all-N form equivalent to RH, and is the one the harness computes.

Lemma 9.81 (signed archimedean decomposition)**.** *Let h_A denote the archimedean part of*

$$h(z) = -\frac{\Xi'(z)}{\Xi(z)} = h_A(z) + h_P(z),$$

and set

$$w_A(u) := -\frac{1}{\pi}\operatorname{Im} h_A(u) = \frac{1}{2\pi}\log\frac{|u|}{2\pi} + O((1+u^2)^{-1}) \qquad (u \in \mathbb{R},\ |u| \to \infty).$$

For $v_j(u) = (u - z_0)^{-(j+1)}$, let G_N be the Lebesgue Gram matrix of $v_0, \ldots, v_{N-1}$. The archimedean jet matrix admits

$$A_N = \int_{\mathbb{R}} vv^*\, w_A(u)\, du + H_N,$$

and the remainder satisfies the uniform form bound

$$\|G_N^{-1/2} H_N G_N^{-1/2}\| \leq C(y)\frac{\log|t_0|}{|t_0|} \qquad (N \geq 1)$$

in a fixed interior band.

Proof sketch. The cancellation of the elementary real parts on the critical line gives the displayed w_A after Stirling. Subtracting the Cauchy transform of w_A leaves a function with zero boundary imaginary part on $\mathbb{R}$; by the Schwarz reflection principle and Morera's theorem, this continuation step (given here as a sketch) extends the function analytically to a disk of radius comparable to $|t_0|$ around the working point, with derivative $O(\log|t_0|/|t_0|)$. Cauchy's estimates applied to the two-variable divided difference kernel give entries bounded by $M(t_0/4)^{-(j+k)}$, and Schur comparison with the exact Szegő/Lebesgue Gram gives the displayed uniform form estimate.

Herglotz normalization of the weight. We fix the normalization of w_A, which is the only point previously flagged. Write the archimedean part as $h_A(z) = i\big[-(\frac{1}{s}+\frac{1}{s-1}) + \frac{1}{2}\log\pi - \frac{1}{2}\psi(s/2)\big]$, $s = \frac{1}{2}+iz$, with $\psi = \Gamma'/\Gamma$. The digamma function is Nevanlinna: $\psi(s) = -\gamma + \sum_{n\geq 0}(\frac{1}{n+1} - \frac{1}{n+s})$ has $\operatorname{Im}\psi(s) > 0$ for $\operatorname{Im} s > 0$. In $w_A(u) = -\frac{1}{\pi}\operatorname{Im} h_A(u)$ the term $-\frac{1}{2}\psi(s/2)$ contributes $\frac{1}{2\pi}\operatorname{Re}\psi(\frac{1}{4}+\frac{iu}{2}) = \frac{1}{2\pi}\log\frac{|u|}{2} + O(u^{-2})$ (Stirling), the constant $\frac{1}{2}\log\pi$ contributes the shift $-\frac{1}{2\pi}\log\pi$, and the two rational terms $-(\frac{1}{s}+\frac{1}{s-1})$ are anti-Nevanlinna and contribute only $O(u^{-2})$; together these give the nonnegative boundary density $\frac{1}{2\pi}\log\frac{|u|}{2\pi}$. Hence

$$w_A(u) = \frac{1}{2\pi}\log\frac{|u|}{2\pi} + O(u^{-2}) \;\geq\; 0 \qquad (|u| \geq 2\pi),$$

so the representing weight is a genuine nonnegative density on the support of every gauge with $|t_0| \geq 2\pi + W$, and $A_N = \int vv^* w_A + H_N$ is a positive form there. (Numerically, at $y = 1$ the identity holds to full working precision — e.g. $w_A(300) = 0.615278 = \frac{1}{2\pi}\log\frac{300}{2\pi}$ — and the archimedean jet A_{12} is positive definite for $t_0 = 50, 300, 1000$; see Section 9.7.) No entrywise jet estimate is used. ☐

Theorem 9.82 (leading sampling form of the terminal defect)**.** *Assume RH and fix $z_0 = t_0 - iy$ in an interior band. Put*

$$L = \log\frac{|t_0|}{2\pi}$$

and let

$$\mathcal{R}_N(z_0) = \left\{ R(u) = \sum_{j=0}^{N-1} \bar{c}_j (u - z_0)^{-(j+1)} \;:\; c \in \mathbb{C}^N \right\}.$$

Then, in the scalar one-point terminal gauge,

$$\delta_N(z_0) = \min_{0\neq R\in\mathcal{R}_N(z_0)} \frac{\sum_\rho m_\rho |R(z_\rho)|^2}{\dfrac{L}{2\pi}\displaystyle\int_{\mathbb{R}} |R(u)|^2\,du}\left(1 + O_y(L^{-1}) + O_y(|t_0|^{-1})\right),$$

where under RH the points z_ρ are real, in the constant-weight regimes specified precisely in Theorem 9.83. Outside those regimes this display is a leading model formula; the weighted statement of Theorem 9.83 is the governing audited version.

Proof. Under RH, the scalar zero-side jet matrix is the Gram matrix

$$c^* J_N c = \sum_\rho m_\rho |R_c(z_\rho)|^2, \qquad R_c(u) = \sum_{j=0}^{N-1} \bar{c}_j (u - z_0)^{-(j+1)}.$$

The leading archimedean part is L times the Szegő jet Gram. The latter is exactly the Lebesgue Gram of the same rational functions:

$$\frac{1}{2\pi}\int_{\mathbb{R}} \frac{du}{(u - z_0)^{j+1}(u - \bar{z}_0)^{k+1}} = i(-1)^k \binom{j+k}{j} (2iy)^{-(j+k+1)}.$$

This residue computation is the real-line form of Lemma 9.55. Lemma 9.56 adds the uniform weighted-form Stirling remainder. Replacing the exact weight w_A by $L/(2\pi)$ is justified in the ranges of Theorem 9.83; there it contributes the indicated constant-weight relative error. Finally, Theorem 9.71 identifies $\delta_N = 1 - \lambda_{\max}(T_N)$ with the minimum of the quotient $c^* J_N c / c^* A_N c$, which yields the displayed sampling form. □

Theorem 9.83 (audited weighted sampling form)**.** *Assume RH. Let* d_N^w *be the minimum sampling quotient with denominator weighted by* w_A*:*

$$\mathsf{d}_N^w := \min_{0 \neq R \in \mathcal{R}_N(z_0)} \frac{\sum_\rho m_\rho |R(z_\rho)|^2}{\displaystyle\int_{\mathbb{R}} |R(u)|^2 w_A(u)\,du}.$$

Then, uniformly in N*,*

$$\delta_N = \mathsf{d}_N^w \left(1 + O_y\!\left(\frac{\log|t_0|}{|t_0|}\right)\right).$$

Replacing w_A *by the constant density* $L/(2\pi)$ *gives the constant-weight quotient* d_N *in the following regimes:*

$$\delta_N = \mathsf{d}_N \left(1 + O\!\left(\frac{N}{|t_0|}\right) + O(e^{-N})\right) \qquad (N \geq \tfrac{3}{2}L),$$

and

$$\delta_N = \mathsf{d}_N \left(1 + O\!\left(\frac{(16y^2)^N}{|t_0|}\right)\right) \qquad \left(N \leq (1-\varepsilon)\frac{L}{\log(16y^2)}\right).$$

The intermediate window between these two ranges is left as a declared audit gap for the constant-weight replacement, not for the weighted sampling identity.

Proof. The first assertion is Theorem 9.82 with the denominator kept in its exact weighted form and Lemma 9.81 used as a form estimate. The constant-weight replacement requires concentration of near-minimizers in the window where $w_A(u) = L/(2\pi) + O(N/t_0)$; for large N this follows from the lower sampling contribution of distant zeros together with the factorial upper bound supplied by the explicit interpolant. In the small-N range the Pascal–Laguerre lower floor controls the error after whitening. The stated middle range is kept out of the theorem because the present argument does not give a uniform comparison there. □

Theorem 9.84 (conditional saturation of the terminal operator)**.** *Assume RH and fix* $z_0 = t_0 - iy \in U_-$*. In the scalar one-point jet gauge,*

$$\limsup_{N\to\infty} \lambda_{\max}(T_N(z_0)) = 1$$

along the terminal sections. More concretely, if $R_c(u) = \sum_{j=0}^{N-1} \bar{c}_j (u - z_0)^{-(j+1)}$ *is chosen to vanish at the* $N-1$ *zero ordinates nearest* t_0*, then*

$$0 < 1 - \lambda_{\max}(T_N) \leq \frac{\sum_{\rho \text{ far}} m_\rho |R_c(z_\rho)|^2}{c^* A_N c} \qquad (N \leq N_*(t_0)),$$

where $A_N = J_N^\infty$ *is the archimedean jet form (positive definite in this range, Lemma 9.72); the saturation* $\limsup = 1$ *is read for all* N *through the reference-whitened defect of Definition 9.73. The right side is the zero-tail energy after interpolation of the nearest atoms; the Pascal–Laguerre floor of Lemma 9.70 gives the effective denominator needed to make this tail estimate quantitative.*

Proof. Under RH, the scalar Pick jet matrix is the Gram matrix of the functions $(u - z_0)^{-(j+1)}$ against the positive zero measure

$$d\Sigma_\Xi(u) = \sum_\rho m_\rho \, \delta_{z_\rho}(u),$$

while $J_N = A_N - J_N^{\mathrm{prime}}$. For any nonzero c,

$$1 - \lambda_{\max}(T_N) \le \frac{c^* J_N c}{c^* A_N c} = \frac{\sum_\rho m_\rho |R_c(z_\rho)|^2}{c^* A_N c}.$$

Choose c so that the numerator vanishes at the $N-1$ zero ordinates closest to t_0; this is ordinary rational interpolation in the basis $(u - z_0)^{-(j+1)}$. Only the far zero tail remains, which gives the displayed inequality. Positivity under RH gives $\lambda_{\max}(T_N) \le 1$. Since the zero measure has infinite support and the interpolation degree increases, the far-tail quotient can be made arbitrarily small (Proposition 9.93 and Theorem 9.94, closed modulo the Stirling/RvM bookkeeping flagged there), so the largest eigenvalues are arbitrarily close to 1; equivalently, the terminal boundary has no fixed positive margin. □

Corollary 9.85 (no fixed margin at the terminal boundary)**.** *The statement*

$$\lambda_{\max}(T_N) \le 1 - \delta \qquad (N \ge 1)$$

is false for every fixed $\delta > 0$ (equivalently, the reference-whitened defect δ_N of Definition 9.73 has no fixed positive lower bound). Under RH it contradicts Theorem 9.84; under ¬RH, Theorem 9.66 implies that $\lambda_{\max}(T_N) > 1$ for some finite section in every terminal gauge detecting the off-line pole. Thus every proof strategy for the terminal input that loses a fixed multiplicative factor at leading order is structurally incapable of proving RH through this route.

Proposition 9.86 (Blaschke detection rate for an off-line zero)**.** *Suppose Ξ has a non-real pair $\{z_w, \bar z_w\}$ with $\operatorname{Im} z_w < 0$. For the terminal base point $z_0 = t_0 - iy$, set the exterior Blaschke factor*

$$b(\zeta) = \frac{\zeta - \bar z_0}{\zeta - z_0}.$$

Then

$$\liminf_{N\to\infty} \lambda_{\max}(T_N)^{1/N} \ge |b(z_w)| > 1.$$

If this pair is the unique dominant off-line pair in Blaschke distance and the cost of avoiding the nearby real zeros is subexponential, then

$$\lim_{N\to\infty} \lambda_{\max}(T_N)^{1/N} = |b(z_w)|.$$

For several dominant pairs the right side is replaced by $\max_w |b(z_w)|$.

Proof sketch. The pair contribution

$$g_{\mathrm{pair}}(z) = \frac{1}{z_w - z} + \frac{1}{\bar z_w - z}$$

has real boundary values on $\mathbb{R}$, and its Pick kernel decomposes into the rank-two hyperbolic block

$$e_{z_w} \otimes \overline{e_{\bar z_w}} + e_{\bar z_w} \otimes \overline{e_{z_w}},$$

so the form contains $2\,\mathrm{Re}[X\overline{Y}]$ with X and Y the evaluations of the rational test at z_w and $\bar{z}_w$. Evaluation at $\bar{z}_w \in \mathbb{C}^+$ is bounded in the model space, while evaluation at the exterior point $z_w \in \mathbb{C}^-$ has reproducing-kernel norm growing like

$$\frac{|b(z_w)|^{2N} - 1}{4\pi|\operatorname{Im} z_w|}.$$

This gives the lower exponential rate. The matching upper rate follows when no other off-line pair has larger Blaschke modulus and when the finite Blaschke factors used to avoid nearby real zeros carry only subexponential cost. In the Davenport–Heilbronn test, $|b(z_w)| = 2.25545$ in exact coordinates, matching the observed 2.2556. □

Corollary 9.87 (finite detection scale). *An off-line zero at Blaschke radius $|b(z_w)| = e^{\beta} > 1$ is detected in terminal sections on the scale $N \asymp \beta^{-1}\log C$, where C is the finite threshold separating $\lambda_{\max}(T_N)$ from 1. Two distinct terminal base points give two Blaschke level curves and hence, in principle, triangulate the off-line zero.*

Conjecture 9.88 (terminal defect law). *Fix $z_0 = t_0 - iy$ in an interior band and set*

$$L = \log\frac{|t_0|}{2\pi}.$$

For the scalar one-point terminal gauge,

$$\delta_N(z_0) = 1 - \lambda_{\max}(T_N(z_0)) = \left(\frac{\kappa_{\sin}(y)L}{N}\right)^{2N(1+o(1))}.$$

The symmetric sine-model numerical data at $y = 1$ currently give the estimate

$$\kappa_{\sin}(1) = 0.686 \pm 0.004.$$

Asymmetric offsets display discrete left/right assignment transitions and should not be extrapolated by the same smooth tail fit. This is an empirical conjecture, not an input to the proof path.

Definition 9.89 (lattice defect constant). Let

$$\Gamma_L = \left\{t_0 + \frac{2\pi k}{L} : k \in \mathbb{Z}\right\}$$

be the exact lattice of density $L/(2\pi)$. Define the model constant

$$\kappa_{\mathrm{lat}}(y) := \lim_{N/L\to\infty} \frac{N}{L}\left[\min_{0\neq R\in\mathcal{R}_N(t_0-iy)} \frac{\sum_{\gamma\in\Gamma_L}|R(\gamma)|^2}{\frac{L}{2\pi}\|R\|^2_{L^2(\mathbb{R})}}\right]^{1/(2N)},$$

when the limit exists. This is a Christoffel-type extremal problem for a lattice sampled against Lebesgue measure in a rational Hardy model space. It is independent of RH and is the clean model problem for the constant in Conjecture 9.88.

Proposition 9.90 (sine-model form of the lattice constant). *The lattice defect of Definition 9.89 is the terminal defect of the sine model. Let*

$$\Xi_{\mathrm{lat}}(z) = \sin\left(\frac{L(z-u_0)}{2}\right), \qquad M_{\mathrm{lat}}(z) = -\frac{\Xi'_{\mathrm{lat}}(z)}{\Xi_{\mathrm{lat}}(z)} = -\frac{L}{2}\cot\left(\frac{L(z-u_0)}{2}\right).$$

For $\operatorname{Im} z < 0$,

$$M_{\mathrm{lat}}(z) = -\frac{iL}{2} - iL\sum_{m\geq 1} e^{-imL(z-u_0)}.$$

The constant term produces the leading archimedean/Szegő form $L\,S$, while the oscillatory part is a one-frequency shift model with lags mL. Thus $\kappa_{\mathrm{lat}}(y)$ is a constant of a classical approximation problem for sin*, with no arithmetic input.*

Proof. The zeros of Ξ_{lat} are precisely the lattice $u_0 + 2\pi k/L$. The logarithmic derivative gives the displayed cotangent expression. Since $\operatorname{Im} z < 0$, the standard expansion

$$\cot w = i + 2i \sum_{m\ge 1} e^{-2imw} \qquad (\operatorname{Im} w < 0)$$

with $w = L(z-u_0)/2$ gives the series for M_{lat}. The constant $-iL/2$ has one-node Pick quotient $L/(2y)$ at $z=-iy$, matching the leading L times Szegő kernel. The remaining terms are exactly shifts at the lags mL, as in Theorem 9.76. Running the same whitening pipeline therefore computes the lattice defect. ☐

Definition 9.91 (sine concentration constant)**.** Let $\delta_N^{\mathrm{lat}}(y, L, u_0)$ be the terminal defect of the sine model of Proposition 9.90. When the limit exists, define

$$\kappa_{\sin}(y) := \lim_{N\to\infty} \frac{N}{L} \left(\delta_N^{\mathrm{lat}}(y, L, u_0)\right)^{1/(2N)}.$$

The current numerical evidence supports the factorial law and the smooth symmetric-offset extrapolation

$$\kappa_{\sin}(1) = 0.686 \pm 0.004.$$

Asymmetric offsets may show discrete jumps in finite N caused by left/right assignment changes of the extremal zeros; these are recorded as finite-size transitions rather than a change in the limiting constant. Analytically, this is a constrained-equilibrium/prolate concentration problem: minimize lattice-sampled energy against continuous L^2 energy over the rational model space $\mathcal{R}_N \subset H^2(\mathbb{C}^+)$.

Problem 9.92 (constrained equilibrium form of $\kappa_{\sin}$). Determine the constant $\kappa_{\sin}(y)$ by the following scaled constrained-equilibrium problem. Among measures $\sigma \ge 0$ with

$$\sigma(\mathbb{R}) = 1, \qquad \sigma \le \tfrac{1}{2}\,\mathrm{Leb},$$

and obstacle condition

$$U^\sigma(t) \le \log|t| \qquad (t \in \mathbb{R}\setminus S(\sigma)),$$

maximize the center field $U^\sigma(0)$. If

$$G^* := \sup_\sigma U^\sigma(0),$$

the predicted relation is

$$\kappa_{\sin}(y) = \frac{y}{\pi} e^{-G^*}.$$

The uniform interpolant gives $G=-1$ and hence the rigorous candidate upper scale $\kappa_{\sin}(y) \le ey/\pi$. Numerics at $y=1$ ask for $G^* = -0.768 \pm 0.007$. The expected extremal has a saturated core and free boundary/contact regions, as in Dragnev–Saff constrained equilibrium for discrete orthogonal polynomial asymptotics.

Proposition 9.93 (quantitative saturation upper-bound reduction)**.** *Assume RH. Let R_N be the rational interpolant in Theorem 9.84, vanishing at the $N-1$ zero ordinates nearest t_0. If the far-zero product estimate*

$$\sum_{\rho\ \mathrm{far}} m_\rho |R_N(z_\rho)|^2 \le c^* A_N c \left(\frac{C(y)L}{N}\right)^{2N}$$

holds with $L = \log(|t_0|/2\pi)$*, then*

$$\delta_N(z_0) \le \left(\frac{C(y)L}{N}\right)^{2N}.$$

The estimate is the exact analytic form of the remaining quantitative saturation calculation: it asks for the Riemann–von Mangoldt zero density, viewed through the rational interpolator and the Pascal–Laguerre floor, to supply a factorial tail bound.

Proof. This is Theorem 9.84 with the displayed product estimate substituted into the zero-tail quotient. All normalization loss is carried by the denominator c^*A_Nc, whose effective lower control is Lemma 9.70. Thus the problem of proving the factorial upper bound is reduced to the stated far-zero product estimate. □

Theorem 9.94 (two-sided factorial saturation bounds)**.** *Assume RH. For* $L \le N \le |t_0|^{1/4}$*,*

$$\left(\frac{c_1(y)L}{N}\right)^{2N} \le \delta_N(z_0) \le \left(\frac{eyL}{\pi N}\left(1 + O\Big(\frac{1}{\log\log|t_0|}\Big)\right)\right)^{2N},$$

where $c_1(y) > 0$ *is effective. The upper bound is closed modulo effective Stirling/Riemann–von Mangoldt bookkeeping; the lower bound uses a Turan–Remez separation estimate and is retained as the principal audit point for this theorem.*

Proof sketch. For the upper bound, choose the rational interpolant vanishing at the $N-1$ zero ordinates nearest t_0. On the scale $W = \pi N/L$, the logarithmic potential of the uniform interpolation pattern has far-field maximum at infinity, giving the factor $(eyL/\pi N)^{2N}$; the Riemann–von Mangoldt fluctuation in the window contributes the stated $O(1/\log\log|t_0|)$ under RH. For the lower bound, among the $4N$ nearest zeros, a Turan–Remez selection gives at least N points whose product distance against any degree-N numerator is bounded below by $(c\,\mathrm{gap})^N$. Combining this selection with the pole field at depth y yields the lower factorial scale. The selection-to-constant step is the remaining place where a full proof must record the effective $c_1(y)$. □

Remark 9.95 (why the factorial scale matters). Conjecture 9.88 sharpens Corollary 9.85: the terminal positivity is not merely close to the boundary, but close at a factorial scale. Therefore any argument losing a factor of size $\exp(o(N\log N))$ at the terminal boundary is still too coarse for the final input. The only known mechanisms compatible with this scale are exact Gram identities or equivalently exact positivity identities for the arithmetic data.

Proposition 9.96 (root recovery from the extremal vector)**.** *Assume RH and let* c_N *be a near-extremal vector for* δ_N*. In the main interpolation window, the numerator of*

$$R_{c_N}(u) = \sum_{j=0}^{N-1} \overline{(c_N)_j}(u - z_0)^{-(j+1)}$$

has zeros exponentially close to the zero ordinates that carry the dominant sampling constraints. More precisely, the distance in each stable gap is governed by the local potential drop between the center field and that gap. The statement is quantitative after a Rouche estimate on each gap.

Proof sketch. The quotient in Theorem 9.83 forces $|R_{c_N}(\gamma)|^2$ to be small at every sampled zero ordinate in the active window. The potential profile gives a lower bound on the maximum of R_{c_N} in each gap, while the small endpoint values give a Rouche comparison with the local linear factor. The numerical root-forensics block matches the predicted gradient of accuracy: central zeros are recovered to more digits than edge zeros. □

Remark 9.97 (asymmetric lattice offsets). The lattice offsets 0 and half-gap are symmetric relative to the projection of z_0 and give smooth tail extrapolations. Asymmetric offsets can undergo discrete left/right assignment transitions in the extremal zero pattern. Such transitions alter the finite-N prefactor by

$$\Delta\kappa_N \sim \frac{\Delta \log A}{2N},$$

which explains observed finite jumps such as the repaired $a/4$, $N = 56$ value without changing the expected limiting sine constant.

Theorem 9.98 (terminal wall as Weil positivity)**.** *For the Laguerre unilateral test class*

$$\widehat{g}(\tau) \sim e^{it_0\tau} q(\tau) e^{-y\tau} \mathbf{1}_{\tau>0}$$

convolved with its reflected partner, the terminal quadratic form satisfies

$$c^* J_N c = W(|R_c|^2),$$

where W is the Weil explicit-formula functional. Consequently, after completion of this Laguerre test class in the Weil topology,

$$\mathsf{d}_N(z_0) \geq 0 \text{ for every } N \quad \Longleftrightarrow \quad \text{Weil positivity on this class} \quad \Longleftrightarrow \quad \text{RH}.$$

Proof sketch. The equality of forms is the explicit formula applied to the test $|R_c|^2$ in the unilateral Laguerre representation: the zero side is $c^* J_N c$, while the archimedean and prime sides are exactly the terminal decomposition used above. Completeness of Laguerre functions gives density in the unilateral window class. The remaining analytic point is continuity of the Weil functional in this completed topology, of the same type as the standard Bombieri–Weil extension. This is Lemma 9.99 below; with it recorded, the equivalence with RH is the classical Weil criterion. □

Lemma 9.99 (continuity of the Weil functional on the completed Laguerre class)**.** *Equip the test class with the norm*

$$\|g\| := \sup_u (1+u^2)|g(u)| + \sup_\tau e^{(\frac{1}{2}+\epsilon_0/2)|\tau|} |\widehat{g}(\tau)|.$$

Then the three sides of W are $\|\cdot\|$-continuous: the zero side by $\sum_\rho (1 + |\operatorname{Re} z_\rho|^2)^{-1} < \infty$ and $|\operatorname{Im} z_\rho| < \frac{1}{2}$; the prime side dominated by $\|g\| \sum_n \Lambda(n) n^{-1-\epsilon_0/2}$; the archimedean side by the weight $(1+u^2)^{-1}\log(2+|u|)$. Laguerre tests at depth $y \geq \frac{1}{2} + \epsilon_0$ satisfy $|\widehat{g}(\tau)| \lesssim e^{-y|\tau|}$ and are dense in the completion. This closes the completion step of Theorem 9.98.

9.6.1. *The interpolation certificate tower.*

Proposition 9.100 (Nevanlinna–Pick certificate tower)**.** *Assume ARP-P, fix a finite channel F, and fix the node sequence used in Theorem 9.185. For every N there exists a matrix Pick-class interpolant F_N satisfying $F_N(z_j) = g^F(z_j)$ for $j \leq N$. Any admissible choice of such interpolants has the following properties:*

(i) *$\{F_N\}$ has pinned-normal subsequential limits on Ω_- in the sense of Lemma 9.182;*
(ii) *$F_N \to g^F$ locally uniformly on U_-;*
(iii) *the analytic bridge applies to the limiting target exactly as in Theorem 9.185.*

Conversely, the existence of any tower of Cauchy transforms of positive measures converging to g^F on U_- implies ARP-P.

Proof. Existence for each N is the matrix Nevanlinna–Pick theorem applied to Definition 9.10. Pinned normality and the identification of every subsequential limit with g^F on U_- are proved by Lemma 9.182 as used in Theorem 9.185; since all subsequential limits agree, the full sequence converges locally uniformly on U_-. The bridge conclusion is the last paragraph of that theorem. The converse is Lemma 9.178 plus closure of the positive semidefinite cone under pointwise limits at finite node sets. □

Remark 9.101 (role of the von Mangoldt tower). The von Mangoldt tower is reclassified as a candidate certificate for ARP-P. It is valuable because its finite spectral measures are positive, ultimately reflecting $\Lambda \geq 0$. It is not the minimal essential object: any proof of tower-ARP proves ARP-P by Theorem 9.181, while any proof of ARP-P supplies Pick-class interpolation certificates by Proposition 9.100.

Proposition 9.102 (direct-sum tower exposes the placement problem)**.** *In a fully decoupled direct-sum realization with one-dimensional free cells,*

$$M_P^F(z)_{\alpha\beta} = \sum_{v\in\mathcal{V}_P} \frac{c_v\,\widehat{\phi}_\beta(\tau_v)\overline{\widehat{\phi}_\alpha(\tau_v)}}{\lambda_v - z} + M_{\infty,P}^F(z)_{\alpha\beta}, \qquad \tau_v = k\log p, \quad c_v = (\log p)p^{-k/2}.$$

Therefore tower-ARP for such a decoupled model is a spectral placement problem for the real numbers λ_v and the archimedean component. Without an independent arithmetic rule forcing the weighted empirical spectral measures to converge to the zero measure detected by g^F, the direct-sum tower does not close ARP.

Proof. For a one-dimensional self-adjoint free cell, $A_v = \lambda_v I$ with $\lambda_v \in \mathbb{R}$, and the cell resolvent is $(\lambda_v - z)^{-1}$. Compressing against the source vector $\sqrt{c_v}\widehat{\phi}_\alpha(\tau_v)e_v$ gives the displayed contribution. Summing over the finite cutoff and adding the archimedean compression gives the formula. The limit is therefore controlled by the weak convergence of the corresponding positive spectral measures. Choosing the λ_v to make that limit equal to g^F is precisely the missing arithmetic placement mechanism. □

9.6.2. *Target inequality and correlation form.*

Proposition 9.103 (Weil splitting of ARP-P)**.** *Theorem 9.8 decomposes g^F on U_- as a Weil explicit-formula functional:*

$$g^F = g_\infty^F - g_{\mathrm{prime}}^F,$$

where g_∞^F denotes the archimedean and pole/degree contribution and g_{prime}^F denotes the von-Mangoldt prime contribution, both interpreted in the same regularized Weil sense as Theorem 9.8. Since the Pick matrix is linear in the function, ARP-P is equivalent to the finite-node dominance

$$\mathsf{P}[g_\infty^F; z_1, \dots, z_N] \succeq \mathsf{P}[g_{\mathrm{prime}}^F; z_1, \dots, z_N]$$

for every finite source plane F and every finite node set in U_-.

Proof. Apply the linear map

$$h \longmapsto \left[\frac{h(z_i) - h(z_j)^*}{z_i - \bar{z}_j}\right]_{i,j=1}^N$$

to the Weil splitting of Theorem 9.8. The inequality is exactly the assertion that the Pick matrix of $g^F = g_\infty^F - g_{\mathrm{prime}}^F$ is positive semidefinite. □

Remark 9.104 (Bochner correlation viewpoint). For $z \in U_-$,

$$\frac{1}{z_\rho - z} = -i \int_0^\infty e^{-izt} e^{iz_\rho t}\, dt.$$

Thus the zero-side target can be written formally as

$$g^F(z) = -i \int_0^\infty e^{-izt} \psi^F(t)\, dt, \qquad \psi^F(t)_{\alpha\beta} = \sum_\rho m_\rho\, \ell_{z_\rho}(\phi_\beta) \overline{\ell_{\bar z_\rho}(\phi_\alpha)} e^{iz_\rho t}.$$

Through the explicit formula, ψ^F is represented by arithmetic distributions. ARP-P is the finite-node Pick form of the same positivity problem; in Fourier language it asks for the corresponding extended correlation distribution to be positive definite. This is a global condition, not a prime-cell-by-prime-cell assertion.

Remark 9.105 (Davenport–Heilbronn falsifier). The analogue of ARP-P must fail for Davenport–Heilbronn data. Indeed, the Davenport–Heilbronn function has off-critical zeros; if its arithmetic channel data satisfied ARP-P, the proof of Theorem 9.185 would force those zeros onto the critical line. Therefore some finite Pick matrix at nodes in the absolute-convergence region must have a negative eigenvalue. This gives a finite-dimensional numerical falsifier for the criterion.

9.7. **Step 5 substructure (continued): H8 falsification harness and the Ω-chain to Ω_7.** This appendix records a self-contained, independent reimplementation of the terminal observable, built for a single purpose: to make every candidate step for the open input Step 5 (H8) cheap to falsify before any analytic effort is spent on it. The code is provided with the paper as `h8lab.py` (kernels), `h8_validate.py` (three-law validation), and `h8_forensic.py` (inverse forensics), together with a cache `zeros_100_175.json`. It is numerical evidence, not proof; the governing statements remain Theorems 9.71, 9.76, 9.84.

Two independent coordinates. The harness computes the terminal defect in two ways that use disjoint inputs, so that agreement is a genuine cross-check of the explicit formula rather than a tautology.

(a) *Sampling coordinate* (the audited governing form, Theorem 9.82): $\widetilde{\delta}_N = \delta_N/L = \min_c\ c^* J_{\mathrm{zero}} c \,/\, (L\, c^* G c)$ — the rescaled variant of Definition 9.73, of the same sign as δ_N at these gauges since $L > 0$ — where G is the exact Szegő/Lebesgue reference Gram $G_{jk} = i(-1)^k \binom{j+k}{j} (2iy)^{-(j+k+1)}$, and J_{zero} is assembled from rank-one zero atoms $v(\gamma)v(\gamma)^*$, $v_j(u) = (u - z_0)^{-(j+1)}$, plus off-line pair blocks. This uses only the *zero* configuration.

(b) *Arithmetic coordinate* (the object a certificate must factor): the Pick jets A_N (archimedean, closed form via $\psi^{(p)}$) and $\mathcal{P}_\Lambda$ (prime), assembled from the exact log-derivative $-\zeta'/\zeta$ through the coefficient recursion $\delta\, J_{jk} + J_{j-1,k} - J_{j,k-1} = h_j \mathbf{1}_{k=0} - \overline{h_k} \mathbf{1}_{j=0}$, $\delta = z_0 - \bar z_0$. Crucially this uses *no zeros*: the prime part is obtained from $\zeta^{(k)}(s_0)$ by series division, avoiding the impossible $n^{-(1/2+y)}$ Dirichlet truncation. The terminal input is $J_{\mathrm{arith}} := A_N - \mathcal{P}_\Lambda \succeq 0$.

By the explicit formula $J_{\mathrm{arith}} = J_{\mathrm{zero}}$; the harness confirms this at jet level to the finite-zero truncation error.

Three-law validation. Run at $y = 1$ on the gauge $z_0 = t_0 - i$ with t_0 centered on the ζ-zero window #100–175 ($t_0 \approx 297.238$, $L \approx 3.88$):

Law	Measured behaviour of the harness
ζ: arithmetic PSD	$\delta_N(\mathrm{arith}) > 0$ with factorial decay $1.6\cdot 10^{-2}$, $8.0\cdot 10^{-5}$, $7.7\cdot 10^{-8}$, $1.8\cdot 10^{-11}$ ($N = 4, 6, 8, 10$); J_{arith} matches the 76-zero Gram to the truncation tail.
DH: falsifier fires	Sampling δ_N crosses negative for the known off-line pair $v = 0.308517$; the negative branch ratios converge to the Blaschke rate $2.673 \to 2.256 \to \mathbf{2.2558}$, matching Proposition 9.86 ($|b(z_w)| = 2.2554$).
Sine/lattice: trivial	$\delta_N > 0$ with factorial decay and empty cascade, as in Remark 9.160.

Inverse-forensic certificate. Let c be the generalized minimizer of the arithmetic pencil $(J_{\mathrm{arith}}, A_N)$ realizing δ_N, and let $R_c(u) = \sum_j \overline{c_j}(u - z_0)^{-(j+1)}$ be the associated rational function. The following table lists, for the roots of the numerator of R_c lying in the active window $|u - t_0| \le W = \pi N/L$, the maximal distance off the real axis and the maximal distance to the actual ζ-zero set.

N	$W = \pi N/L$	$\max\lvert\mathrm{Im\ root}\rvert$	max recovery error in W
6	4.86	$8.3 \cdot 10^{-59}$	0.43
8	6.48	$3.4 \cdot 10^{-55}$	0.64
10	8.10	$5.1 \cdot 10^{-53}$	0.23
12	9.71	$4.4 \cdot 10^{-49}$	0.45
14	11.3	$1.1 \cdot 10^{-44}$	0.65

Two facts are visible. First, the minimizer roots are *real to* 10^{-44} *or better*, computed from the arithmetic side alone: a per-gauge numerical certificate that the arithmetic terminal operator places no mass off the critical line. Second, the innermost roots coincide with actual ζ-zeros to full working precision (for $N = 12$ the four central roots reproduce 294.96537, 295.57325, 297.97928, 299.84033 exactly to eight figures); the larger window-error entries are the expected edge roots that interpolate between consecutive zeros. This is Proposition 9.96 made concrete: the arithmetic data, with no zero input, reconstructs the Riemann zeros nearest the gauge and puts them on $\mathbb{R}$.

Remark 9.106 (status of the certificate). The forensic certificate is unconditional numerical evidence at each finite N and each gauge, and it falsifies instantly for Davenport–Heilbronn. It is the falsification instrument that any proposed exact Gram factorization $A_N - \mathcal{P}_\Lambda = B^*B$ must pass on ζ and fail on DH, and the harness decides this in hours.

Theorem 9.107 (scalar level-one positivity)**.** *For the scalar coordinate*

$$M_\Xi(z) = i\left(-\frac{\xi'}{\xi}(s)\right), \qquad s = \tfrac{1}{2} + iz,$$

the one-node Pick inequality on U_- is equivalent to

$$\mathrm{Re}\,\frac{\xi'}{\xi}(s) \ge 0, \qquad \mathrm{Re}\, s > 1.$$

This inequality is classical.

Proof. Let $z = x + iy \in U_-$. Then $\operatorname{Re} s = \frac{1}{2} - y > 1$, so $y = -(\operatorname{Re} s - \frac{1}{2})$. The diagonal Pick quotient is

$$\frac{M_\Xi(z) - M_\Xi(z)^*}{z - \bar{z}} = \frac{\operatorname{Im} M_\Xi(z)}{\operatorname{Im} z} = \frac{-\operatorname{Re}(\xi'/\xi)(s)}{-(\operatorname{Re} s - \frac{1}{2})} = \frac{\operatorname{Re}(\xi'/\xi)(s)}{\operatorname{Re} s - \frac{1}{2}}.$$

Since $\operatorname{Re} s - \frac{1}{2} > 0$, positivity is equivalent to the displayed inequality.

By the Hadamard product for ξ, with the constant term fixed as in Davenport [77, Sec. 12],

$$\frac{\xi'}{\xi}(s) = \sum_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right) + B,$$

with the usual real normalization in which the real parts of the constant terms cancel after pairing zeros. Taking real parts gives

$$\operatorname{Re} \frac{\xi'}{\xi}(s) = \sum_\rho \frac{\sigma - \beta}{|s-\rho|^2}, \qquad s = \sigma + it,$$

in the standard symmetric limiting sense. The classical zero-free theorem for ζ on $\operatorname{Re} s = 1$ and the functional equation give $0 < \beta < 1$ for nontrivial zeros. Hence for $\sigma > 1$ every summand has nonnegative real part, and the inequality follows. □

Remark 9.108 (scope of the level-one theorem). Theorem 9.107 closes the scalar one-node seed. It is not the full matrix-channel ARP-P(1), because the channel target contains source weights $\ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}$. The full channel hierarchy is the remaining Pick-positivity problem.

H8: the reduced contractivity, a candidate route, and its gap. We record the strongest current attempt on the terminal input $\delta_N \geq 0$, in the exact form used at the workbench, so that the point where it stalls is explicit and independently attackable.

The exact reduction. Pass to the additive model $L^2(0, \infty)$ of Theorem 9.76. A rational $R \in \mathcal{R}_N$ corresponds to $F(\tau) = \sum_j c_j \tau^j e^{-iz_0\tau}$, and the scalar terminal form is

$$c^*(A_N - \mathcal{P}_\Lambda)c = \langle (\mathcal{W}_A - \mathcal{T}_\Lambda)F, F \rangle, \qquad \mathcal{T}_\Lambda = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \, (S_{\log n} + S^*_{\log n}),$$

with $\mathcal{W}_A$ the (positive) archimedean weight operator of Lemma 9.81. Thus

$$\boxed{\delta_N \geq 0 \ \forall N \iff \lambda_{\max}\big(\mathcal{W}_A^{-1/2} \mathcal{T}_\Lambda \mathcal{W}_A^{-1/2}\big) \leq 1 \text{ on the Laguerre tower}}$$

i.e. the whitened prime shift operator $\mathcal{T} := \mathcal{W}_A^{-1/2} \mathcal{T}_\Lambda \mathcal{W}_A^{-1/2}$ is dominated from above by the identity. This is Theorem 9.71 in operator form. *Warning (measured):* the domination is strictly one-sided — the *two-sided* norm genuinely exceeds 1 at interior gauges (at $t_0 = 30$, $y = 1$ the bottom eigenvalue of the whitened jet compression is $-3.34, -7.62, -11.90$ at $N = 4, 8, 12$, driver `omega7_operator_interference.py`); an earlier phrasing of this box as $\|\mathcal{T}\| \leq 1$ was incorrect and is hereby corrected.

Candidate route (Fejér–Riesz over-domination). The whitened operator $\mathcal{T}$ is a one-sided (Toeplitz) operator with symbol $\sigma(\xi) = 2\sum_n \Lambda(n) n^{-1/2} \cos(\xi \log n)/w_A$-normalization, and the scalar level-one theorem (Theorem 9.107) gives the pointwise positivity $\operatorname{Re}(\xi'/\xi)(s) \geq 0$ for $\operatorname{Re} s > 1$, i.e. positivity of the whitened symbol at every interior gauge. For a Toeplitz operator a *nonnegative symbol on the boundary* would give a Fejér–Riesz spectral factor $\mathcal{W}_A - \mathcal{T}_\Lambda = \mathcal{D}^*\mathcal{D}$, closing $\delta_N \geq 0$.

Where it stalls (the isolated gap). The interior positivity of Theorem 9.107 is on $\operatorname{Re} s > 1$; the contractivity requires the whitened symbol to stay nonnegative as the gauge is pushed to the boundary $\operatorname{Re} s \to \frac{1}{2}^{+}$, i.e. on the critical line. There $\operatorname{Re}(\xi'/\xi)(\frac{1}{2}+it) = \sum_\rho (\frac{1}{2}-\beta_\rho)|s-\rho|^{-2}$ is a signed sum whose sign is governed by the very off-line zeros in question. So the candidate route reduces, cleanly and without loss, to a single named inequality:

$$\textbf{(C)} \qquad \liminf_{\sigma\downarrow 1/2} \big(\mathcal{W}_A - \mathcal{T}_\Lambda\big) \;\succeq\; 0 \quad \text{as forms on the Laguerre tower.}$$

Inequality (C) is equivalent to RH; we state it as the reduced target rather than assert it.

Experimental evidence bearing on (C). The harness supplies three data points.

- (E1) *Reality certificate.* The arithmetic minimizer roots are real to 10^{-44} and land on the zeros (table above); the near-null direction of $\mathcal{W}_A - \mathcal{T}_\Lambda$ is thus a genuine interpolant of on-line atoms.
- (E2) *Analytic factor.* The Cholesky factor $B_N(t_0)$ of $A_N - \mathcal{P}_\Lambda$ varies smoothly across a zero gap (e.g. B_{11} decreasing monotonically $1.72\cdot 10^{-4} \to 6.2\cdot 10^{-5}$ over $[295.57, 297.98]$ at $N = 8$), so the certificate is an analytic matrix function of the gauge while $\delta_N > 0$; its terminal diagonal is $\asymp \sqrt{\delta_N}$ and vanishes exactly where an off-line zero would appear.
- (E3) *No-margin threshold.* Displacing one zero off the line by ε flips $\delta_N < 0$ at a threshold that collapses super-exponentially in N: $\varepsilon^*(N) \approx 1.3\cdot 10^{-2}$, $5.2\cdot 10^{-4}$, $1.1\cdot 10^{-5}$ for $N = 6, 8, 10$. Positivity is therefore a genuine phase property, sharpening with N, exactly as (C) predicts.

The reproducible drivers are `h8_forensic.py`, `exp_cholesky.py`, and `exp_threshold.py`. The attack currently stands at inequality (C): a boundary/Fejér–Riesz argument for the whitened symbol on the critical line would close $\delta_N \geq 0$ for all N, and the harness would immediately confirm it on ζ and reject it on Davenport–Heilbronn.

The terminal jets are the Li–Keiper coefficients pulled into the convergent region. Inequality (C) is not new in the landscape: it is the boundary form of the Li–Keiper criterion, and identifying it as such tells us exactly which classical technique the attack must supply.

Lemma 9.109 (boundary regularity of the fixed-N jets)**.** *Fix t_0 and N. Since the zero set of Ξ is discrete and contains no point with $|\operatorname{Im} z_\rho| = \frac{1}{2}$ (Lemma 9.1 gives $0 < \beta < 1$, hence $|\operatorname{Im} z_\rho| = |\beta - \frac{1}{2}| < \frac{1}{2}$ strictly), the compact segment $\{t_0 - iy : y \in [\frac{1}{2}, Y]\}$ has positive distance to it. Hence the jets $C_{jk}(t_0 - iy)$, $j,k < N$, are continuous on $y \in [\frac{1}{2}, Y]$ (dominated convergence with the $O(|z_\rho|^{-2})$ majorant of Lemma 9.51(iii), uniform on the compact segment), and $\delta_N(t_0 - \frac{i}{2}) = \lim_{y\downarrow 1/2} \delta_N(t_0 - iy)$. In particular interior positivity at fixed N passes to the Li boundary, unconditionally. This lemma transfers positivity in one direction only (interior $\to$ boundary); the equivalence* at *the boundary gauge is supplied independently by Theorem 9.52(b), and this lemma is never used to argue from boundary positivity back to interior positivity.*

Convention 9.110 (symmetric zero summation). Every sum $\sum_\rho$ over nontrivial zeros in this paper, when the individual terms are not summable in absolute value, denotes the symmetric limit

$$\sum_\rho (\cdots) := \lim_{T\to\infty} \sum_{|\operatorname{Im}\rho|\leq T} (\cdots),$$

zeros counted with multiplicity, and every pairing invoked (e.g. $\rho \leftrightarrow 1-\rho$, $\rho \leftrightarrow \bar\rho$) is applied within a common symmetric truncation before the limit is taken. This is the classical convention under which the Li generating function and the Li–Keiper criterion are stated [81, 83, 82], and it is the convention used for λ_n, $S_m = \sum_\rho \rho^{-m}$, and every partial-fraction manipulation of these sums below. The absolutely convergent sums $\sum_\rho m_\rho |z_\rho|^{-2}$, $C_{jk}(z_0)$, and $K_\Xi(z,w)$ of Lemma 9.51 require no such convention; the symmetric limit is needed only where S_1 (equivalently λ_1) itself is involved, since $\sum_\rho |\rho|^{-1}$ diverges.

Theorem 9.111 (Li–Keiper dictionary for the terminal gauge)**.** *Fix the Cayley coordinate $w_{z_0}(z) = \dfrac{z-z_0}{z-\bar z_0}$ at a gauge $z_0 = t_0 - iy$, $y \geq \frac12$. For $y = \frac12$, $t_0 = 0$ (the boundary point $z_0 = -\frac{i}{2}$, i.e. $s_0 = 1$), and a nontrivial zero ρ with z-coordinate $z_\rho = (\rho - \frac12)/i$,*

$$w_{-i/2}(z_\rho) \;=\; 1 - \frac{1}{\rho}.$$

Consequently, writing $a_\rho = (z_\rho - z_0)^{-1}$, the boundary terminal jet is $C_{jk}(-\frac{i}{2}) = \sum_\rho m_\rho\, a_\rho^{j+1}(z_\rho - \bar z_0)^{-(k+1)}$ (Lemma 9.51(iii)); under RH this reads $\sum_\rho m_\rho\, a_\rho^{j+1}\overline{a_\rho}^{\,k+1}$ and is the Gram (moment) matrix of the pushforward of the positive zero measure under the Cayley map $z \mapsto w_{-i/2}(z) = 1 - 1/\rho$; and

$$\delta_N(-\tfrac{i}{2}) \geq 0 \;\; \forall N \quad\Longleftrightarrow\quad \lambda_n := \sum_\rho \Big(1 - (1 - \tfrac{1}{\rho})^n\Big) \geq 0 \;\; \forall n,$$

the zero sum taken in the symmetric sense of Convention 9.110, both being equivalent to RH — the left side by the scalar one-point jet serialization, Theorem 9.52, applied directly at the Li gauge $z_0 = -\frac{i}{2}$ *(case (b): the gauge lies on the boundary of the Pick domain U_- but at positive, unconditional distance from the zero set of Ξ, so it is a regular base point for the kernel), the right side by the Li* [81]*–Keiper* [83] *criterion (see also Bombieri–Lagarias* [82]*). For $y > \frac12$ the terminal jet at $z_0 = t_0 - iy$ is the analytic continuation of the same Cayley/Li data into the half-plane of absolute convergence* $\operatorname{Re} s_0 = \frac12 + y > 1$*, and inequality* (C) *is precisely the boundary limit $y \downarrow \frac12$.*

Proof. Write $z_\rho = (\rho - \frac12)/i = -i(\rho - \frac12)$. Then $z_\rho + \frac{i}{2} = -i(\rho - \frac12) + \frac{i}{2} = -i\rho + i = -i(\rho - 1)$ and $z_\rho - \frac{i}{2} = -i(\rho - \frac12) - \frac{i}{2} = -i\rho$, so

$$w_{-i/2}(z_\rho) = \frac{z_\rho - (-\frac{i}{2})}{z_\rho - (+\frac{i}{2})} = \frac{-i(\rho-1)}{-i\rho} = \frac{\rho - 1}{\rho} = 1 - \frac{1}{\rho},$$

the Li variable, of modulus 1 under RH. The scalar jet identity $C_{jk}(z_0) = \sum_\rho m_\rho\,(z_\rho - z_0)^{-(j+1)}(z_\rho - \bar z_0)^{-(k+1)}$ is Lemma 9.51(iii); under RH, $z_\rho - \bar z_0 = \overline{z_\rho - z_0}$, so with $a_\rho = (z_\rho - z_0)^{-1}$ it reads $C_{jk} = \sum_\rho m_\rho\, a_\rho^{j+1}\overline{a_\rho}^{\,k+1}$ and exhibits $C(z_0)$ as the Gram matrix of the vectors $(a_\rho^{j+1})_j$ against the positive weights m_ρ, hence positive semidefinite whenever the z_ρ are real. Its positivity for all N is equivalent to RH by Theorem 9.52, applied directly at $z_0 = -\frac{i}{2}$: the distance hypothesis is case (b) of that theorem — a zero of Ξ within distance δ of $-\frac{i}{2}$ would be a zero of ζ within δ of $s = 1$, excluded unconditionally by the classical zero-free region — so no passage of positivity from the interior gauges to the boundary is used in either direction. The Li–Keiper positivity $\lambda_n \geq 0$ for all n is equivalent to RH by [81, 83]. The two positivity families are therefore equivalent to each other, and the boundary jet is built from the very Li variable $1 - 1/\rho$ by the first identity. For $y > \frac12$, $s_0 = \frac12 + iz_0$ has $\operatorname{Re} s_0 = \frac12 + y > 1$, and $C_{jk}(z_0)$ is holomorphic in z_0 there (Lemma 9.51(iii)), agreeing with the boundary data as $y \downarrow \frac12$; thus the interior jets are the analytic continuation and (C) is their boundary value. ∎

Remark 9.112 (the dictionary is an all-orders equivalence, not a levelwise transform). The equivalence of Theorem 9.111 holds between the *complete* families ($\forall N$ versus $\forall n$), through RH; it is not a finite-order positivity-preserving triangular transform, and the two truncated hierarchies are genuinely different at each finite level. This is visible already at $N = 2$. Partial fractions against the pairing $\rho \leftrightarrow 1-\rho$ (so $\sum_\rho 1/(\rho-1) = -\lambda_1$ and $\sum_\rho 1/\rho^2 = 2\lambda_1 - \lambda_2$) give, for the boundary jets $C_{jk} = i^{\,j+k+2} \sum_\rho m_\rho (\rho-1)^{-(j+1)} \rho^{-(k+1)}$,

$$C_{00} = 2\lambda_1, \qquad C_{01} = \overline{C_{10}} = i\,(4\lambda_1 - \lambda_2), \qquad C_{11} = 8\lambda_1 - 2\lambda_2,$$

hence the exact identity

$$\det J_2(-\tfrac{i}{2}) = C_{00}C_{11} - |C_{01}|^2 = \lambda_2\,(4\lambda_1 - \lambda_2).$$

Thus $J_1 \succeq 0$ is exactly $\lambda_1 \geq 0$ (with the harmless factor 2), while $J_2 \succeq 0$ is equivalent to $\lambda_1 \geq 0$, $\lambda_2 \geq 0$, *and* $\lambda_2 \leq 4\lambda_1$ — strictly stronger than finite Li positivity at that level, and not implied by it: every jet entry lies in the integer span of $\lambda_1, \ldots, \lambda_{j+k+1}$, yet the two truncated positivity hierarchies coincide only in the $\forall N/\forall n$ limit, where both are RH. Numerically $4\lambda_1 - \lambda_2 = 3.71 \cdot 10^{-5} > 0$ and $\det J_2 = \lambda_2(4\lambda_1 - \lambda_2) \approx 3.4 \cdot 10^{-6} > 0$, consistent with RH. This is one more manifestation of conservation of difficulty: the dictionary changes coordinates on the same terminal positivity, it does not decompose it into weaker levelwise statements.

Remark 9.113 (what the dictionary buys). The Li–Keiper coefficients live exactly at the edge $\operatorname{Re} s = 1$, where the prime series is only conditionally summable and the positivity $\lambda_n \geq 0$ has resisted proof since 1997. The terminal gauges $y > \frac{1}{2}$ move the identical positivity into $\operatorname{Re} s_0 > 1$, where the prime series converges absolutely and the level-one positivity $\operatorname{Re}(\xi'/\xi) > 0$ (Theorem 9.107) is unconditional and manifest. The open content is therefore isolated as a single passage to the boundary: prove that the interior positivity, uniform in N, survives $y \downarrow \frac{1}{2}$ — a Fatou/boundary-value problem for the whitened Weil symbol rather than a positivity to be conjured from scratch. Numerically the boundary values are reproduced exactly: the closed-form $\frac{1}{(n-1)!}\frac{d^n}{ds^n}[s^{n-1}\log\xi]_{s=1}$ gives $\lambda_{1..5} = 0.023096, 0.092346, 0.207639, 0.368790, 0.575543$, matching the zero-side sum $\sum_\rho (1-(1-1/\rho)^n)$, and $|w_{-i/2}(u)| = 1$ on $\mathbb{R}$ to full precision (driver `li_check.py`).

Remark 9.114 (the boundary passage is regular; the difficulty is the $N \to \infty$ race). The Fatou passage of Remark 9.113 splits into two limits, $y \downarrow \frac{1}{2}$ and $N \to \infty$, and the harness locates the difficulty entirely in the second. Fixing N and lowering y toward the Li boundary, the sampling margin $\delta_N(t_0 - iy)$ stays strictly positive and varies smoothly, with no degeneration at $y = \frac{1}{2}$; e.g. at $t_0 \approx 297.24$,

y	$N=6$	$N=8$	$N=10$
1.00	$7.9{\cdot}10^{-5}$	$7.6{\cdot}10^{-8}$	$1.8{\cdot}10^{-11}$
0.60	$3.9{\cdot}10^{-7}$	$4.0{\cdot}10^{-11}$	$1.2{\cdot}10^{-15}$
0.505	$6.1{\cdot}10^{-8}$	$3.1{\cdot}10^{-12}$	$4.6{\cdot}10^{-17}$

(driver `exp_boundary.py`). Thus for each fixed N the boundary value is a positive limit, and (C) is *not* obstructed by the passage $y \downarrow \frac{1}{2}$; the whole content is the uniformity of positivity as $N \to \infty$ *at* the boundary — equivalently the growth race in $\lambda_n = \lambda_n^{\mathrm{arch}} + \lambda_n^{\mathrm{prime}}$, where the archimedean ceiling $\lambda_n^{\mathrm{arch}} \sim \frac{n}{2}\log n$ (Bombieri–Lagarias [82]) must dominate the oscillatory prime part $\lambda_n^{\mathrm{prime}}$ for every n. This is the sharpest form of the remaining target: an unconditional lower bound on the archimedean Li coefficient that beats the prime oscillation uniformly in n, with the interior gauges $y > \frac{1}{2}$ available as an absolutely convergent regularization in which $\operatorname{Re}(\xi'/\xi) > 0$ holds outright.

Chain Ω: the end-to-end path and the shape of its single residue. We assemble the whole route as one linear chain, marking each Ω-link Closed or, for the one terminal residue, stating it exactly. This is the object we offer for end-to-end audit. The Ω-chain is the third and deepest level of the logical hierarchy of §7.1: it lives below the H-level analysis of Step 5, and its entries are typed — Ω_1–Ω_6 are closed reductions or equivalences (Ω-links), while Ω_7 is not a link but the terminal statement the chain isolates.

	Statement	Status
Ω_1	RH $\iff$ all Ξ-zeros real (Lemma 9.2)	Closed
Ω_2	real zeros $\iff$ ARP-P (Steps 1–15, Corollary 9.190)	Closed
Ω_3	ARP-P $\iff$ terminal positivity $\delta_N(z_0) \geq 0$ $\forall N$ in the reference-whitened defect (Corollary 9.75; the observable is Definition 9.73)	Closed, in the reference-whitened form
Ω_4	$\delta_N \geq 0$ $\forall N$ $\iff$ one-sided whitened domination $\lambda_{\max}(\mathcal{W}_A^{-1/2}\mathcal{T}_\Lambda\mathcal{W}_A^{-1/2}) \leq 1$ (the two-sided norm exceeds 1 at interior gauges; see the warning below)	Closed; J_N^∞-whitening valid for $N \leq N_*(t_0)$ (Lemma 9.72)
Ω_5	interior positivity ($y > \frac{1}{2}$) has a positive, regular boundary limit $y \downarrow \frac{1}{2}$ (Lemma 9.109)	Closed, per N
Ω_6	boundary positivity $\iff$ Li–Keiper $\lambda_n \geq 0$ $\forall n$, via $w_{-i/2} = 1 - 1/\rho$ (Theorem 9.111)	Closed
Ω_7	$\lambda_n \geq 0$ for all n: archimedean ceiling $\frac{n}{2}\log n$ dominates the prime oscillation	**terminal residue**: open; $\equiv$ RH (Li)

Every Ω-link Ω_1–Ω_6 is proved above. The terminal residue Ω_7 is the classical Li–Keiper positivity in the sharp uniform-in-n form isolated by the chain. Two proved facts about Ω_7 delimit what a closing argument may look like.

Proposition 9.115 (no positive prime truncation (computational) — the terminal positivity is a global cancellation)**.** *Fix an interior gauge and $N \geq 6$. For the whitened truncated form $m_N(X) := \lambda_{\min}\big(A_N^{-1/2}(A_N - \mathcal{T}_\Lambda^{\leq X})A_N^{-1/2}\big)$, where $\mathcal{T}_\Lambda^{\leq X} = \sum_{n\leq X}\Lambda(n)n^{-1/2}(S_{\log n} + S_{\log n}^*)$, one has (to the stated working precision) $m_N(X) < 0$ for a range of finite X even though the full defect $m_N(\infty) = \delta_N > 0$. Consequently $A_N - \mathcal{P}_\Lambda$ is* not *of the form (positive truncated part) − (small tail): no finite prime cutoff yields a positive operator to perturb from.*

Proof (computational, with the harness). Driver `omega_g1g2_truncation.py` evaluates $m_N(X)$ and the whitened tail norm at $t_0 \approx 297.24$, $y = 1$. Already at $X = 50$ one finds $m_6 = -0.53$, $m_8 = -0.66$, $m_{10} = -0.73$, while $\delta_6 = 8.0\cdot 10^{-5}$, $\delta_8 = 7.7\cdot 10^{-8}$, $\delta_{10} = 1.8\cdot 10^{-11}$; the tail norm exceeds $|m_N(X)|$ at every listed X. Thus the truncated operator is indefinite while the full one is positive: the positivity is restored only by the entire prime tail. $\square$

Remark 9.116 (why this fixes the method, not just a method). Theorem 9.115 says the archimedean part does *not* dominate the prime part termwise or up to a controllable tail; positivity is an exact cancellation across all primes, equivalently the exact evaluation of the explicit formula at the (real) zeros. The same obstruction defeats the two perturbative backups we tested: mollifying the symbol (a deeper gauge) is the $X \to \infty$ end of the same truncation and inherits the indefiniteness at finite scale (driver `omega_g1g2_truncation.py`), and the backward-heat-flow route (driver `omega_g3_backflow.py`, from $t > \Lambda_{\mathrm{dBN}}$ where Polymath15 [84] gives real zeros) propagates the factorially small margin $\delta_N \asymp (\kappa L/N)^{2N}$ against a polynomial Gronwall drift, hence closes only a range $N \le N_*(L)$. Both confirm, from independent directions, that Ω_7 carries the full arithmetic content: a proof must supply the exact archimedean-over-prime domination of the Li coefficients uniformly in n, not a dominated inequality. The chain reduces RH to precisely that classical statement and to nothing softer.

Ω_7 up close: exact structure, the archimedean trend, and the detection mechanism. We record what a direct assault on Ω_7 meets, with exact numbers (drivers `omega7_li_decomp.py`, `omega7_li_inject.py`). Two structural facts organize everything.

Proposition 9.117 (the pairing and the on-line sum-of-squares)**.** *Nontrivial zeros pair as $\rho \leftrightarrow 1-\rho$, and the Li variables are reciprocal: $w_{1-\rho} = 1/w_\rho$, $w_\rho = 1 - 1/\rho$. Moreover*

$$|w_\rho| = \frac{|\rho - 1|}{|\rho|} \begin{cases} = 1, & \operatorname{Re}\rho = \frac{1}{2}, \\ > 1, & \operatorname{Re}\rho < \frac{1}{2}, \\ < 1, & \operatorname{Re}\rho > \frac{1}{2}. \end{cases}$$

Consequently, if RH holds then, pairing $\rho = \frac{1}{2} + i\gamma$ with $\bar\rho = 1 - \rho$ and writing $w_\rho = e^{i\theta_\gamma}$,

$$\lambda_n = \sum_{\gamma > 0} \big(2 - 2\cos n\theta_\gamma\big) = 4 \sum_{\gamma>0} \sin^2 \tfrac{n\theta_\gamma}{2} \;\ge\; 0,$$

a manifest sum of squares. Thus Ω_7 is trivial given *RH; its content is establishing the same nonnegativity from the arithmetic definition of λ_n, in which the zeros do not appear.*

Proof. $w_\rho w_{1-\rho} = \frac{\rho-1}{\rho} \cdot \frac{-\rho}{1-\rho} = 1$; and $|w_\rho| = |\rho-1|/|\rho|$ with $|\rho-1|^2 - |\rho|^2 = 1 - 2\operatorname{Re}\rho$ gives the trichotomy. For $\operatorname{Re}\rho = \frac{1}{2}$, $1 - \rho = \bar\rho$, so the pair contributes $[1 - w^n] + [1 - \overline{w}^{\,n}] = 2 - 2\operatorname{Re} w^n = 2 - 2\cos n\theta_\gamma$. Summing over $\gamma > 0$ and using $\theta_\gamma = O(1/\gamma)$ (whence the terms are $O(n^2/\gamma^2)$ for $\gamma \gg n$) gives an absolutely convergent nonnegative series. $\square$

Remark 9.118 (the exact trend and the measured margin). Splitting $\log\xi = \big[\log s - \frac{s}{2}\log\pi + \log\Gamma(\frac{s}{2})\big] + \log\big((s-1)\zeta(s)\big)$ gives, through $\lambda_n = n\sum_{k=1}^{n}\binom{n-1}{k-1}\sigma_k$ with $\sigma_k = [(s-1)^k]\log\xi(s)$ (closed form: polygamma for the archimedean part, Stieltjes constants γ_j for the ζ part), an *unconditional* exact decomposition $\lambda_n = \lambda_n^{\mathrm{arch}} + \lambda_n^{\mathrm{prime}}$. Computed to $n = 120$: every $\lambda_n > 0$; the archimedean part follows the Voros trend $\lambda_n \sim \frac{n}{2}\log\frac{n}{2\pi}$ (ratio $0.823, 0.845, 0.873$ at $n = 40, 80, 120$, increasing to 1); and the prime oscillation is a small relative fluctuation, $|\lambda_n^{\mathrm{prime}}|/\lambda_n^{\mathrm{arch}}$ falling to 0.002–0.04 for $n \ge 30$ (still 0.083 at $n = 20$). Under RH this smallness is forced by $|w_\rho| = 1$; unconditionally it is exactly what is at stake.

Remark 9.119 (the detection mechanism, exhibited). The obstruction is a single geometric term. A putative off-line zero ρ_0 ($\operatorname{Re}\rho_0 \ne \frac{1}{2}$) has a functional-equation partner with $|w| > 1$, and its quartet contributes $-2\operatorname{Re} w^n + \cdots$ with $|w|^n$ growing geometrically; for large n this overturns the $\frac{n}{2}\log n$ trend and drives $\lambda_n < 0$. Injecting a low off-line quartet at $\rho_0 = 0.9 + 3i$ (with $\max|w| = 1.0434$ over the quartet) into the true λ_n turns it negative first at $n = 97$ ($\lambda_{97}^{\mathrm{inj}} = -5.95$ against the true $\lambda_{97} = 113.4$), while an on-line control $\rho_0 = \frac{1}{2} + 3i$ keeps every $\lambda_n > 0$ (it only adds $4\sin^2$). This is the Li-coordinate form of the terminal forensic: positivity is destroyed precisely, and only, by the geometric growth $|w| > 1$ of an off-line zero. Closing Ω_7 therefore means bounding $|\lambda_n^{\mathrm{prime}}| < \lambda_n^{\mathrm{arch}}$ for all n *unconditionally*, i.e. forbidding $|w_\rho| > 1$ — which is RH itself, now displayed as a single explicit inequality between an archimedean trend of size $\frac{n}{2}\log n$ and an arithmetic oscillation whose only possible unbounded source is an off-line zero.

Which inputs cannot close Ω_7: an insufficiency lemma and the height–range correspondence. We explored bounding $\lambda_n^{\mathrm{prime}}$ against $\lambda_n^{\mathrm{arch}}$ for all n using inputs weaker than RH. The outcome is an *input-class restriction*, not an impossibility: it rules out one family of hypotheses (zero-location statements weaker than RH) and leaves the direct arithmetic attack fully open.

Proposition 9.120 (insufficiency of zero-location inputs)**.** *If ξ has a zero ρ_0 with $\operatorname{Re}\rho_0 \neq \frac{1}{2}$, then $\lambda_n < 0$ for infinitely many n (this is the known converse direction of the Li criterion* [81, 82]*). Consequently no hypothesis* consistent with *the existence of a single off-line zero can imply $\lambda_n \geq 0$ for all n. In particular neither the classical zero-free region $\beta < 1 - c/\log(|\gamma| + 2)$ nor any zero-density estimate $N(\sigma, T) \ll T^{a(1-\sigma)}$ suffices as the sole input for Ω_7: both permit off-line zeros.*

Proof sketch, with attribution. The full statement is Bombieri–Lagarias [82] applied to the zero multiset of ξ; we record the mechanism for a single dominant quartet. By the functional equation and conjugation, ρ_0 belongs to a quartet containing a zero ρ with $\operatorname{Re}\rho < \frac{1}{2}$, for which $r := |w_\rho| = |\rho - 1|/|\rho| > 1$ (Proposition 9.117). Its quartet contributes $-(r^n + r^{-n})\cos(n\varphi) + O(1)$, $\varphi = \arg w_\rho$, of amplitude r^n exceeding, for large n along a positive-density set where $\cos(n\varphi) \geq \frac{1}{2}$ (Weyl equidistribution if $\varphi/2\pi$ is irrational, periodicity if rational), both the trend $\lambda_n^{\mathrm{arch}} \sim \frac{n}{2}\log\frac{n}{2\pi}$ and the contribution of the remaining zeros *provided* ρ realizes the maximal $|w|$ alone. When several off-line zeros share the maximal modulus, cancellations among their phases must be excluded; this is exactly what the Bombieri–Lagarias argument handles in general, and we rely on their theorem rather than the sketch for the full claim. □

Remark 9.121 (scope: what this does and does not say)*.* Proposition 9.120 restricts the *input class*: Ω_7 cannot be closed by citing a zero-location statement weaker than RH, because any such statement tolerates the zero that destroys positivity. It does *not* say that Ω_7 is unprovable, and the slogan "the only sufficient input is RH itself" would be circular: since $\lambda_n \geq 0\ \forall n$ is *equivalent* to RH [81], any proof of Ω_7 tautologically yields RH — that is the goal of the chain, not an obstruction to it. The open route is the one the decomposition itself exhibits: bound $|\lambda_n^{\mathrm{prime}}| < \lambda_n^{\mathrm{arch}}$ directly on the *prime side* (explicit-formula / Weil-positivity inputs, test-function constructions, or the operator-theoretic mechanisms M1–M3 discussed below), none of which passes through a zero-location hypothesis and none of which is excluded here. Ω_7 therefore remains **open**, with this lemma serving only to prune dead-end input families.

Remark 9.122 (height–range correspondence: the achievable unconditional traction)*.* The geometric growth is slow for high zeros, and this is quantitative. A putative off-line zero at height γ and displacement $\delta = \frac{1}{2} - \beta$ has $\log|w_\rho| = \dfrac{\delta}{\beta^2 + \gamma^2}\,(1 + o(1))$, so it can force $\lambda_n < 0$ only once $r^n \gtrsim \lambda_n^{\mathrm{arch}}$, i.e. for

$$n \;\gtrsim\; n_*(\gamma, \delta) \;\approx\; \frac{\beta^2 + \gamma^2}{\delta}\,\log\!\Big(\tfrac{n}{2}\log\tfrac{n}{2\pi}\Big).$$

The measured thresholds obey $n_* \cdot \log|w|$ nearly constant (values $4.85, 4.92, 5.73, 5.98, 7.50, 6.71, 8.62$ across $\gamma = 3-10$, driver `omega7_li_threshold.py`), confirming $n_* \asymp (\beta^2 + \gamma^2)\delta^{-1}\log(\cdots)$. Contrapositively, if RH is verified up to height H — no off-line zeros with $|\gamma| \leq H$ — then the lowest possible off-line zero sits at $\gamma > H$ and cannot overturn the trend until $n \gtrsim H^2\log H$: the single-zero analysis gives $\lambda_n > 0$ unconditionally for $n \lesssim H^2\log H$. With RH verified to $H \approx 3 \times 10^{12}$ this is $\lambda_n > 0$ for n up to about 10^{25}. The step from the single-zero bound to a fully rigorous combined bound over all $\gamma > H$ requires a zero-density input to control the phases of many high zeros; that is the exact, and only, place where unconditional work on

Ω_7 via *zero-location* inputs can still gain ground: such inputs yield a finite range of n, never the all-n statement (Proposition 9.120). The all-n statement must come from the prime side (Remark 9.121), where no such barrier is proved.

The prime side made exact: Ω_7 as an inequality about an explicit prime sum. Remark 9.121 leaves one open front: bound $|\lambda_n^{\mathrm{prime}}|$ against $\lambda_n^{\mathrm{arch}}$ directly on the prime side. The first step is to make $\lambda_n^{\mathrm{prime}}$ an *explicit* prime sum. Both identities below were numerically gated before being written (driver `omega7_prime_kernel.py`).

Proposition 9.123 (Laguerre kernel of the prime side). *For every $n \geq 1$ and $x > 0$,*

$$n \sum_{k=1}^{n} \binom{n-1}{k-1} \frac{(-x)^k}{k!} \;=\; -\,x\, L_{n-1}^{(1)}(x),$$

with $L_{n-1}^{(1)}$ the generalized Laguerre polynomial. Consequently, for $\operatorname{Re} s_0 > 1$, applying the Li binomial transform to the Taylor coefficients $\sigma_k^{\zeta}(s_0) = [(s-s_0)^k] \log \zeta(s)$ gives the absolutely convergent representation

$$T_n(s_0) \;:=\; n \sum_{k=1}^{n} \binom{n-1}{k-1} \sigma_k^{\zeta}(s_0) \;=\; -\sum_{m \geq 2} \frac{\Lambda(m)}{m^{s_0}}\, L_{n-1}^{(1)}(\log m),$$

and the prime part of the Li coefficient is the boundary value

$$\lambda_n^{\mathrm{prime}} \;=\; \lim_{\varepsilon \downarrow 0} \Big[\, n \sum_{k=1}^{n} \binom{n-1}{k-1} \frac{(-1)^{k-1}}{k\, \varepsilon^k} \;-\; \sum_{m \geq 2} \frac{\Lambda(m)}{m^{1+\varepsilon}}\, L_{n-1}^{(1)}(\log m) \Big],$$

the divergences of the pole and prime pieces cancelling in the limit.

Proof. The kernel identity follows from the generating function $\sum_{n\geq 0} L_n^{(1)}(x) z^n = (1-z)^{-2} e^{-xz/(1-z)}$ together with $\sum_n \lambda_n^{\bullet} z^{n-1}$ applied to $m^{-s} = m^{-s_0} e^{-(s-s_0)\log m}$; it was verified to 25 digits at mixed (n, x). The representation for $T_n(s_0)$ is then term-by-term rearrangement of the absolutely convergent double sum ($\operatorname{Re} s_0 > 1$). The boundary statement is the same computation for $\log\big((s-1)\zeta(s)\big) = \log(s-1) + \log\zeta(s)$ at $s_0 = 1+\varepsilon$, whose left side is regular at $\varepsilon = 0$ with Taylor data the Stieltjes constants. Numerical gates: $T_n(s_0)$ agrees with the exact log-ζ derivative route to 10^{-9} at $s_0 = 2$ and 10^{-6} at $s_0 = 1.5$ (prime powers to 2×10^6 plus PNT tail; the residual has exactly the size of the $\psi(y) - y$ fluctuation, i.e. the zero term); the ε-limit converges linearly to the exact Stieltjes values of $\lambda_n^{\mathrm{prime}}$. □

Remark 9.124 (what the representation buys, and what it costs). Ω_7 is now the single explicit inequality

$$\Big|\lim_{\varepsilon \downarrow 0} \big[\mathrm{pole}_n(\varepsilon) - \textstyle\sum_m \Lambda(m) m^{-1-\varepsilon} L_{n-1}^{(1)}(\log m)\big]\Big| \;<\; \lambda_n^{\mathrm{arch}} \sim \frac{n}{2} \log \frac{n}{2\pi} \qquad (\forall n),$$

with two structural assets not available to zero-location inputs: the weights $\Lambda(m) \geq 0$ are nonnegative, and the kernel $L_{n-1}^{(1)}$ is classical (Plancherel–Rotach: oscillatory for $\log m \lesssim 4n$, monotone beyond). The cost is equally visible: the kernel's effective support reaches $m \approx e^{4n}$, so the inequality demands cancellation in $\psi(y) - y$ over exponentially long ranges — the measured cancellation ratios $|\sum|/\sum|\cdot|$ at $s_0 = 1.5$ are 0.28–0.49 for $n \leq 12$ and must, at $\varepsilon = 0$, come from the oscillation of the prime counting error against the Laguerre oscillation. This is the precise prime-side face of Ω_7.

Step 2: pointwise prime-error bounds are insufficient — the $\Theta(n)$ cancellation deficit. With Proposition 9.123 in hand, the natural first attack is partial summation: integrate by parts (exactly, as Stieltjes integrals; the pole term is the average-Λ compensation, $\mathrm{pole}_n(\varepsilon) = \int_0^\infty L_{n-1}^{(1)}(x)e^{-\varepsilon x}dx$, verified to 10^{-20}) to get

$$\lambda_n^{\mathrm{prime}} = -\lim_{\varepsilon\downarrow 0}\int_1^\infty \big(\psi(y)-y\big)\,\frac{d}{dy}\Big[\frac{L_{n-1}^{(1)}(\log y)}{y^{1+\varepsilon}}\Big]\,dy,$$

and then insert a pointwise bound $|\psi(y)-y| \le B(y)$. We measured the resulting ceiling $A_n[B] := \int_1^\infty B(y)\,\big|\frac{d}{dy}[L_{n-1}^{(1)}(\log y)/y]\big|\,dy$ against $\lambda_n^{\mathrm{arch}}$ (driver `omega7_partial_summation.py`; true $|\psi(y)-y|$ from a sieve to 10^7, tails by model):

n	$\lambda_n^{\mathrm{arch}} \sim \frac{n}{2}\log\frac{n}{2\pi}$	A_n (true+RH tail)	A_n (RH model $\frac{\sqrt{y}\log^2 y}{8\pi}$)	A_n (VK shape)
8	0.97	45.3	37.8	2.8×10^{11}
20	11.6	334	311	4.6×10^{23}
35	30.1	1221	1178	1.0×10^{33}

Remark 9.125 (second insufficiency: pointwise bounds on $\psi(y)-y$). The ceiling exceeds the archimedean budget at *every* $n \ge 2$ — even for the RH-optimal pointwise bound $B(y) = \sqrt{y}\,\log^2 y/(8\pi)$. Unconditional shapes are hopeless outright: the Vinogradov–Korobov saving $e^{-c(\log y)^{3/5}}$ cannot suppress the y-size of the trivial bound against the polynomial Laguerre tail, and A_n explodes exponentially. But the RH-model deficit is only *polynomial*: across $2 \le n \le 35$ the measured law is $A_n \approx 0.27\,n^2\log n$ (the ratio $A_n/(n^2\log n)$ is 0.26–0.27 throughout), against a budget $\lambda_n^{\mathrm{arch}} \approx \frac{n}{2}\log n$. So partial summation with absolute values overshoots by a factor $\Theta(n)$, no more: closing Ω_7 requires extracting exactly one factor n of sign-cancellation from the oscillation of $\psi(y)-y$ against the oscillation of the Laguerre kernel (both oscillate in $\log m \lesssim 4n$; their relative phase is where the remaining content of RH sits). Together with Proposition 9.120 this sharpens the target of the prime-side programme: not pointwise magnitude, not zero location — *phase*. The structural asset $\Lambda(m) \ge 0$ and the operator mechanisms M1–M3, which see signs, are the remaining instruments (Step 3 of the attack plan).

Step 3: the $\Lambda \ge 0$ asset in whitened coordinates — factorial per-term loss. The last instrument of the prime-side plan is the nonnegativity of the von Mangoldt weights. At an interior gauge $z_0 = t_0 - iy$ with $y > \frac{1}{2}$ (so $\mathrm{Re}\,s_0 = \frac{1}{2}+y > 1$) the prime jet decomposes absolutely per prime power, $\mathcal{P}_\Lambda = \sum_m P_m$, each P_m carrying the weight $\Lambda(m) \ge 0$ (decomposition cross-checked against the exact ζ-derivative jet: error 3×10^{-4} at $N=4$, 2.5×10^{-2} at $N=8$; $N=12$ is truncation-limited at $X = 2\times 10^6$). Whitening by $S = A_N^{-1/2}$, we measured the per-term ceiling $S_{\mathrm{abs}}(N) = \sum_m \Lambda(m)\,\|SP_m^{(1)}S\|$ (the best any term-by-term absolute argument can achieve using only $\Lambda \ge 0$) against the true one-sided extremes of $S\mathcal{P}_\Lambda S$ (driver `omega7_operator_interference.py`):

t_0	N	$\lambda_{\max}(S\mathcal{P}S)$	$\lambda_{\min}$	S_{abs}	interference I
10	4	0.99999993	-0.097	219	2.2×10^2
10	8	$1-10^{-12}$	-0.369	1.9×10^4	1.9×10^4
30	4	0.9982	-3.34	59.5	17.8
30	8	$1-2\times 10^{-12}$	-7.62	4.5×10^3	5.8×10^2
30	12	1^-	-11.90	$\gtrsim 2.9\times 10^5$	$\gtrsim 2.5\times 10^4$

Remark 9.126 (third insufficiency, and the sharp shape of H8). Three structural facts, each measured above. (i) *The one-sided domination is razor-sharp:* $\lambda_{\max}(S\mathcal{P}_\Lambda S)$ saturates at 1 to twelve digits already at $N=8$ — the H8 inequality holds with margin $\to 0$ exponentially fast in

N, as difficulty-equivalence with RH demands. (ii) *Per-term positivity is factorially insufficient:* the ceiling S_{abs} grows factorially in N (e.g. $219 \to 1.9 \times 10^4$ from $N = 4$ to 8 at $t_0 = 10$), so the $\Lambda(m) \geq 0$ asset used absolutely — any argument that bounds prime contributions one at a time, in any whitened jet coordinates — loses *factorially*, even worse than the $\Theta(n)$ deficit of the Li diagonal (Remark 9.125). The surviving positivity is carried entirely by cross-prime phase interference, the decoherence mechanism of Remark 9.15. (iii) *The reduction is one-sided only:* the bottom of the whitened prime jet grows like $-N$, so no proof route through two-sided norm bounds (contraction arguments, unitary dilations of $\mathcal{T}$) can be correct as stated; the target is exactly $\lambda_{\max} \leq 1$. Taken together with the first two insufficiencies (Proposition 9.120: zero-location inputs; Remark 9.125: pointwise error bounds), the prime-side programme has now excluded every input class that ignores phase: what remains of RH in Ω_7/H8 is precisely a phase-coherence statement about $\{m^{-it_0}\}$ against the Laguerre oscillation, sharp to factor 1, with zero margin. This is the honest terminal shape of the exercise.

9.8. **Step 5 substructure (continued): closure attempts on Ω_7 and a closure specification.** The terminal residue Ω_7 — equivalently $\delta_N \geq 0$ for all N — is the single open mathematical input of the whole route. We record here, for the audit trail, the strongest attacks executed against it, each run to its exact failure point, together with the unconditional byproducts those failures leave behind. Difficulty conservation (H0) is not an editorial convention but modus ponens: any purported proof contains exactly one step that is either RH-strength new mathematics or an error. The productive form of the exercise is therefore to choose the best route by elimination, execute it, and perform the autopsy of the death point — which is itself a theorem.

Route elimination. The proved walls exclude fixed margins (Corollary 9.85), losses $e^{o(N \log N)}$ (the factorial wall), zero-location inputs (Proposition 9.120), pointwise bounds on $\psi(y) - y$ (the $\Theta(n)$ deficit of Remark 9.125), and term-by-term $\Lambda \geq 0$ positivity (the measured factorial loss of Remark 9.126). Three further mechanisms are dismissed quickly. *Krein extension*: short-window positivity (supp $< \log 2$, no primes) is provable, and Krein extends any interval-positive-definite function to a positive-definite function on $\mathbb{R}$; but the route needs *our* kernel to be the positive extension, not the mere existence of one. *Fejér–Riesz*: factoring $\mathcal{A} - \mathcal{T}_\Lambda = B^*B$ requires the boundary symbol to be nonnegative $\iff$ RH (circular, already recorded). *Reflection positivity*: the per-prime symbol $2\,\mathrm{Re}[x/(1-x)]$, $|x| = p^{-1/2}$, changes sign, so positivity lives in cross-prime interference. What remains unexplored is the *gauge structure*: unconditional positivity at large height for each fixed N (band closure) against geometric negativity near an off-line zero ($\neg$RH). Between them lies only a quantifier exchange, whose natural tool is almost-periodicity of the prime side.

The attack: almost-periodic gauge transport.

Lemma 9.127 (gauge transport of the arithmetic form). *Fix $c \in \mathbb{C}^N$ and gauges $z_0 = t_0 - iy$, $z_0'' = t_0'' - iy$ with $N \leq N_*(\min(t_0, t_0''))$. Then*

$$c^* J_N(t_0'')c = c^* J_N(t_0)c + (L'' - L)\, c^* S_N c + \mathcal{E}_{\mathrm{arch}} + \mathcal{E}_{\mathrm{prime}},$$

where $|\mathcal{E}_{\mathrm{arch}}| \leq C(y)\big(\frac{\log t_0}{t_0} + \frac{\log t_0''}{t_0''}\big)c^*(LS_N)c$ [modulo Lemma 9.81], *and*

$$|\mathcal{E}_{\mathrm{prime}}| \leq \Big(\sup_{n \leq X} \big|n^{-it_0''} - n^{-it_0}\big|\Big)\, C_1(N)\, \|c\|^2 + C_2\, X^{-(y-\frac{1}{2})}\mathrm{poly}(N)\, \|c\|^2.$$

Lemma 9.128 (geometric detection)**.** *Under $\neg RH$ with an off-line zero $z_w = \gamma_w - i\delta_w$, at the gauge $t_0 = \gamma_w$ there is a witness c with $c^* J_N c \le -m_N$, $m_N \asymp |b|^{2N}$, $|b| = \frac{y+\delta_w}{y-\delta_w} > 1$ — geometric growth in N (Proposition 9.86, validated to five figures on Davenport–Heilbronn).*

The intended contradiction runs as follows. By Kronecker, the recurrence times of the phases $\{n^{-it}\}_{n\le X}$ to precision ε are syndetic with gap $\ell(X,\varepsilon)$. Transport the negative witness of Lemma 9.128 to a gauge t_0'' with paired phases, away from the quartet: there the zero side gives $c^* J_N(t_0'')c \ge -C$ (the local real zeros give a Gram form $\succeq 0$; the quartet from distance $\Delta \ge \sqrt{N}$ contributes $e^{2Ny\delta_w/\Delta^2} = O(1)$), while Lemma 9.127 gives $c^* J_N(t_0'')c \le -m_N + (L''-L)c^* S_N c + \mathcal{E}$. If the geometric m_N dominates the polynomial errors and the controlled term $(L''-L)c^* S_N c$ (chosen via $t_0'' \in [t_0, 2t_0]$), the two estimates collide and RH follows.

Autopsy: the recurrence–detection imbalance. The attack dies at a quantified point. The N-jets see primes up to $X = e^{cN}$, and the margin requires $\varepsilon \le m_N e^{-CN}$. The Kronecker recurrence gap is

$$\ell(X,\varepsilon) \;\asymp\; \varepsilon^{-\pi(X)} \;=\; \exp\!\big(e^{cN}\cdot C'N\big) \;=\; \mathrm{Tower}(N),$$

while the detection gain $m_N = e^{c\delta_w N}$ is only geometric. The admissible window $[t_0, 2t_0]$ contains a recurrence time only if $t_0 \ge \ell(X,\varepsilon)$, i.e. $\gamma_w \ge \mathrm{Tower}(\log \gamma_w/\delta_w)$, unsatisfiable for every γ_w: full phase matching costs a tower, detection pays geometric, and the two never meet. One need not match all phases, only force a single scalar $F(t) := c^* P_N(t) c$ to return near its value at t_0 — a level-return of an almost-periodic function, cheap (positive density) if the value is not extremal. Whether $F(t_0)$ is extremal is decided by the following rigidity.

Proposition 9.129 (ceiling rigidity)**.** *Fix c and let $F(t) := c^* P_N(t) c$, an almost-periodic–structured function of the gauge with frequency module $\{\log n\}$. Whenever the zeros within the detection window of t are real,*

$$F(t) \;\le\; c^* A_N(t) c \;+\; O(\text{far-pair tails}),$$

because $c^* J_N(t) c = \sum_\gamma |R_c^{(t)}(\gamma)|^2 \ge 0$ is a Gram form. The event $F(t) > c^* A_N(t) c$ — the prime form exceeding the archimedean ceiling — occurs exactly *at gauges detecting an off-line zero. Hence the super-ceiling level set of F coincides with the (off-line) exceptional set itself.*

Thus $F(t_0) = c^* A_N c + m_N$ sits *above* the ceiling: it is not a typical fluctuation of F but the detection event itself, and the returns of the required super-ceiling deviation are, one for one, other off-line zeros. The recurrence argument is exactly equivalent to producing the object whose nonexistence it seeks to prove. This is difficulty conservation with an explicit mechanism.

Theorem 9.130 (W4: recurrence–detection imbalance)**.** *No proof of Ω_7 can proceed by gauge transport of the arithmetic form. Full phase-matching transport requires recurrence times of size $\exp(\pi(e^{cN})\log(1/\varepsilon))$, super-exponentially larger than the geometric detection gain $|b|^{2N}$; and scalar-level transport requires returns of a super-ceiling deviation of $c^* P_N(t) c$, which by Proposition 9.129 occur exactly at off-line zeros. Gauge-recurrence arguments are therefore excluded alongside* W1–W3 *(margins, truncations, zero-location and pointwise inputs). In the classification of §4 this is $I(\mathcal{I}; C)$ with $\mathcal{I}$ = gauge-transport/recurrence arguments and $C = \Omega_7$.*

The classical landing is the most useful by-product: the question "how often can $F(t) = c^* P_N(t) c$ be large?" is, in classical coordinates, the theory of *large values of Dirichlet polynomials* (Montgomery, Huxley, and the current front Guth–Maynard 2024). Large-value estimates bound the *measure* of the bad-gauge set; by this route the apparatus re-derives zero-density–type theorems, not RH — exactly what difficulty conservation predicts, and now with the precise classical problem attached to each closing move (scalar transport $\Longleftarrow$ optimal large values $\Longleftarrow$ a density hypothesis $\Longleftarrow \cdots \Longleftarrow$ RH). This yields a concrete achievable target, sharper than the deterministic $\log\log$ range of Corollary 9.79:

$$\delta_N(t_0) \geq 0 \text{ for all } t_0 \in [T, 2T] \text{ except measure } O(T^{1-c}), \quad \text{uniformly for } N \leq (\log T)^{\theta},$$

provable with existing large-value technology and offered here as the most immediately achievable incremental result of this line of attack.

Round two: transplanting the finite-field proof. RH for curves over finite fields is proved (Weil) through positivity of the Hodge index on $C \times C$ — the Castelnuovo inequality — and Weil positivity for ζ is its numerical shadow. Executing the dictionary in the present coordinates, every piece transfers except one, and it is always the same piece.

Function field (proved)	Present framework
Correspondence algebra, $\mathrm{Fr}_n \circ \mathrm{Fr}_m = \mathrm{Fr}_{nm}$	shift semigroup $S_{\log n} S_{\log m} = S_{\log nm}$ (exists)
Lefschetz fixed-point formula	explicit formula (Theorem 9.8) (exists)
Trivial classes H^0, H^2 (fibre/plane)	the two pole atoms at $\pm i/2$ (Lemma 9.72) (exists)
Projecting off the trivial plane	Schur shorting of the pole vector (exists)
Intersection form on H^1	the form J_N
Ample class θ	the archimedean form $\mathcal{A}$
Riemann–Roch with effectivity (Castelnuovo engine)	$\varnothing$ — does not exist

Definition 9.131 (closure specification)**.** A *positivity-bearing realization* is a graded object $H = H^0 \oplus H^1 \oplus H^2$ with a pairing $\langle \cdot, \cdot \rangle$ and an algebra action of the shift semigroup, such that:

(X1) the semigroup $\{S_{\log n}\}$ acts as correspondences with $S_{\log n} \circ S_{\log m} = S_{\log nm}$;
(X2) the pairing of correspondences is computed by the Weil explicit formula (Theorem 9.8), with $\dim H^0 = \dim H^2 = 1$ realized by the pole atoms at $\pm i/2$;
(X3) (effectivity/Riemann–Roch) every class $\alpha \perp H^0 \oplus H^2$ with $\langle \alpha, \alpha \rangle > 0$ admits an effective representative, and effective classes pair nonnegatively against the archimedean ample class $\theta = \mathcal{A}$;
(X4) the finite jets $J_N(z_0)$ are compressions of the H^1-pairing.

Theorem 9.132 (conditional closure)**.** *If a realization satisfying* (X1)–(X4) *exists, then* $\delta_N(z_0) \geq 0$ *for every* N*, hence* Ω_7*, hence* RH.

Proof (Castelnuovo transplant). Suppose $c^* J_N c < 0$ for some N, c. By (X4) this lifts to a class $\alpha \in H^1$ with $\langle \alpha, \alpha \rangle < 0$ after shorting the trivial plane (X2); the Hodge-index argument of Weil, powered by (X3) exactly as effectivity powers the Castelnuovo–Severi inequality on $C \times C$, forces $\langle \alpha, \alpha \rangle \geq -\langle \alpha, \theta \rangle^2 / \langle \theta, \theta \rangle$ with the trivial-plane contribution already removed, yielding $\langle \alpha, \alpha \rangle \geq 0$; contradiction. ☐

The consistency tests pass non-trivially: Davenport–Heilbronn fails *exactly* at (X1) (no Euler product, no correspondence semigroup), so the falsifier discriminates at the correct axiom rather than by accident; the sine model satisfies (X1)–(X2) trivially with H^1 of controlled rank (empty cascade); and any (X3) candidate is falsifiable in hours against the measured Cholesky flow and factorial defect law. Two cross-coordinate attempts confirm the same gap. The *Nyman–Beurling* pincer (RH $\iff d_N \to 0$, a twin arithmetic Gram of Vasyunin sums) collapses because every inter-frame inequality factors through the common spectral measure, i.e. through the zeros; it does, however, supply the Möbius mollifier as an explicit falsification test (a certificate must reproduce Burnol's rate $d_N^2 \sim C/\log N$). The *multiplicative-independence/large-values* attempt reproduces the achievable measure-theoretic target above, not RH.

Creating (X3): the collapse obstruction and the θ-square.

Lemma 9.133 (the collapse obstruction)**.** *In commutative rings, $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} = \mathbb{Z}$, hence $\operatorname{Spec}\mathbb{Z} \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec}\mathbb{Z} = \operatorname{Spec}\mathbb{Z}$: the diagonal is an isomorphism and the square degenerates to the curve itself. Any realization of* (X3) *must therefore first change the base category (a base "below" $\mathbb{Z}$). The gap is not theorem-shaped but* definition-shaped*: before proving Riemann–Roch on the square, the square must exist.*

Weil had the surface $C \times_{\mathbb{F}_q} C$ for free; here it must be manufactured. Three known substrates de-collapse the product, and the falsification battery discriminates among them.

Substrate	X1	X2	X3 (effectivity)	Test vs. our data
Connes–Consani (scaling site, char. 1)	$\surd$	$\surd$ (trace formula)	partial: tropical RR exists, but idempotent semirings have no subtraction $\Rightarrow$ no *signature* (Hodge index needs an abelian category)	restricted-support positivity = our finite-N regime
Borger (λ-rings / Witt: the square as $W(\mathbb{Z})$ with two Frobenius families)	$\surd\surd$ (the ψ_p commute: our shift semigroup)	unproved	no finite cohomology theory	right structure, no engine
Deninger (foliated dynamical system)	$\surd$ (periodic orbits of length $\log p$ = our lags)	conjectural	needs leafwise Hodge theory	his flow = our gauge translation; ceiling rigidity (W4) = his recurrence structure

The construction starts from the one thing that already exists exactly in dimension one: arithmetic Riemann–Roch by theta invariants (van der Geer–Schoof, Bost). For a metrized line bundle $\bar{L}$ over $\overline{\operatorname{Spec}\mathbb{Z}}$,

$$h^0_\theta(\bar{L}) := \log \sum_{x \in L} e^{-\pi \|x\|^2}, \qquad h^0_\theta(\bar{L}) - h^0_\theta(\bar{\omega} - \bar{L}) = \deg \bar{L} \quad \text{(exact)},$$

where Serre duality is literally Poisson summation and the effectivity engine is Minkowski (deg large $\Rightarrow$ short vector $\Rightarrow$ effective section). In dimension one the full Weil dictionary is thus already proved, with Poisson as duality and geometry of numbers as effectivity; (X3) is its lift to the square. The divisors of the square have a natural candidate here: correspondences are the Dirichlet polynomials (finite shift combinations), the ample class is $\theta = \mathcal{A}$, and the (X2) pairing is Poisson along the diagonal. Three definitions of "effective" were tested. *(i) The cone of real-point evaluation atoms* is circular (it presupposes real zeros). *(ii) The measured Cholesky factor $B_N(t_0)$* exists $\iff$ positivity (presupposes the conclusion), though it records the shape an effective section would take. *(iii) The prime-shift cone $E := \operatorname{cone}\{\Lambda(n)n^{-1/2}S_{\log n}\} \cup$*

(positive archimedean cone) survives the circularity tests: it is defined from $\Lambda(n) \geq 0$ alone, with no zeros, and discriminates Davenport–Heilbronn at the correct axiom. With this cone, (X3) becomes a single statement of geometry-of-numbers type.

Conjecture 9.134 (M2: Minkowski for correspondences)**.** *Let D be a correspondence class orthogonal to the trivial plane $H^0 \oplus H^2$ (the pole atoms at $\pm i/2$) with $\langle D, D\rangle_W > 0$ under the Weil pairing. Then the class of D, modulo the trivial plane, contains an effective representative in the shift cone E; and every element of E pairs nonnegatively against the ample class $\theta = \mathcal{A}$.*

Theorem 9.135 (conditional closure, sharpened)**.** (X1)*,* (X2)*,* (X4) (all existing) *together with* M2 *imply* (X3)*, hence $\delta_N \geq 0$ for all N, hence Ω_7, hence* RH *— by the Castelnuovo transplant of Theorem 9.132.*

Problem 9.136 (M2$_N$: finite falsifiable shadow). In the N-jet compression at gauge z_0: does every $c \perp$ (pole plane) with $c^* J_N c > 0$ admit a decomposition $c = e_+ - e_-$ with $e_\pm$ in the compressed shift cone E_N and $\langle e_+, e_+\rangle \geq \langle c, c\rangle - \varepsilon_N$? This is a conic-feasibility problem (SDP), decidable at $N \leq 12$ in hours. Protocol: (i) verify the ampleness pairings $\langle \Lambda(n) n^{-1/2} S_{\log n}\, v,\ \mathcal{A}\, v\rangle \geq 0$ numerically on the Laguerre frame; (ii) run feasibility on ζ across gauges; (iii) confirm structural failure on Davenport–Heilbronn at the cone-existence step; (iv) check that the extremal e_+ reproduces the measured Cholesky flow.

A candidate construction of (X3): thermal Riemann–Roch (Gibbs effectivity). We record one concrete proposal for the missing effectivity engine (X3). Everything in this subsubsection has *conjectural* status; its value is that it is precise enough to be attacked, and it comes with the map of where to attack it.

The idea. Every attempt to build the square $\operatorname{Spec}\mathbb{Z} \times_{\mathbb{F}_1} \operatorname{Spec}\mathbb{Z}$ as a *geometric* object dies in the collapse $\mathbb{Z} \otimes \mathbb{Z} = \mathbb{Z}$ (Lemma 9.133). The proposal here is that the square is not a space but a *thermodynamic system*: its points are KMS states, its Serre duality is temperature inversion $\beta \mapsto 1/\beta$, its trivial plane $H^0 \oplus H^2$ is a phase transition, and — the core — effectivity is not a count of sections but *positivity of free energy.* The role played by Minkowski in dimension one (and by Riemann–Roch in Weil's proof) is taken by the unconditional Gibbs variational principle; the positivity engine ceases to be spectral (where every argument is circular) and becomes entropic (where positivity is structural — a variance).

The objects. (D1) The Frobenius semiring $\{[n]\}_{n \geq 1}$, $[m][n] = [mn]$, is the shift semigroup; its analytic carrier already exists as the Bost–Connes system, whose KMS$_\beta$ states, twisted by the archimedean weight κ (below), are the proposed family ω_β. Bost–Connes has a phase transition at $\beta = 1$ with partition function $\zeta(\beta)$.

Definition 9.137 (thermal sections and effectivity)**.** For a finitely supported real class $D = \sum_n c_n [n]$, define the free energy

$$F_\beta(D) := \log\, \omega_\beta\big(e^D\big)\,, \qquad e^D := \sum_{k \geq 0} \frac{D^k}{k!} \ \ \text{(well-defined by multiplicativity of } [n]\text{)}.$$

D is *effective* iff $F_\beta(D) \geq 0$ for all β in the self-dual window. The effective cone is multiplicatively closed by construction (Hölder on states).

(D3) Riemann–Roch as a theta flip: the duality $\beta \mapsto 1/\beta$ is implemented not as a symmetry of Bost–Connes (it is not one) but by the adelic Fourier transform inside the GNS space — Tate's local functional equations as local RR, assembled into

$$F_\beta(D) - F_{1/\beta}(\kappa - D) = \deg D, \qquad \kappa = \mathcal{A} \text{ (the ample archimedean class)},$$

with deg read off the residue of the Bost–Connes transition.

Conjecture 9.138 (Gibbs realization of the Weil pairing)**.** *There is a normalization $c(\beta) > 0$ and a heat-regularization $D \mapsto \widehat{D}$ such that, for every real class D orthogonal to the trivial plane,*

$$\langle D, D\rangle_W \;=\; -\lim_{\beta\to 1^+} c(\beta)\,\mathrm{Var}_{\omega_\beta}\big(\widehat{D}\big) \;\le\; 0.$$

That is: on primitive classes, the negative of the Weil pairing is a variance *of the twisted KMS state. Combined with the two-dimensional positive trivial plane, this is the Hodge index statement, and* RH *follows by the Castelnuovo transplant (Theorem 9.132).*

The structural non-circularity, which distinguishes this from the buried mechanisms, is that a variance is ≥ 0 by Cauchy–Schwarz, not by spectral information, and ω_β is defined from purely multiplicative and thermal data (the Bost–Connes algebra plus the KMS condition) with no zeros in any input: it passes W2 by design.

Three structural coincidences suggesting novelty. (i) *Temperature dictionary.* Bost–Connes KMS weights scale as $n^{-\beta}$, our effective weights as $n^{-1/2-y}$, giving $\beta = \frac{1}{2} + y$; the boundary limit $y \to \frac{1}{2}^+$ where all the programme's difficulty lives is *exactly* the Bost–Connes phase transition $\beta \to 1^+$, and the pole of ζ as a divergent partition function coincides with the pole pair $\pm i/2 = H^0 \oplus H^2$ appearing at symmetry breaking. (ii) *Native multiplicativity.* The measured $\sim 1/\delta_N$ cancellation in the *linear* cone (sums do not buy effectivity) is repaired by $F_\beta = \log \omega(e^D)$, which is log-convex and multiplicatively closed — the closure the data demanded. (iii) *Torus lift.* The characters of $\mathbb{Q}^*_{>0}$ form the infinite torus $\mathbb{T}^\infty$ (one angle per prime), and the explicit-formula line is the Kronecker flow inside it; the conjecture is equivalent to the statement that the Weil distribution is the restriction to the Kronecker line of a positive kernel on $\mathbb{T}^\infty$. Since the vanishing of Λ at mixed integers $p^a q^b$ forces the mixed cumulants to vanish, the kernel is quasi-product (a Gaussian field over independent coordinates), and the whole content compresses into a single lemma: the archimedean term (the only non-product coupling) lifts to the torus diagonal dominating the per-prime negatives.

The load map. (L1, pencil and paper, start here) Compute the cumulants of the regularized $[p]$ in ω_β from the known Bost–Connes KMS formulae and compare with the explicit-formula coefficients; kill-test: if the candidate's mixed cumulants do not vanish at non-prime-powers, the state family is wrong and is discarded in an afternoon. (L2, the core, where circularity would hide) The twisted state $\omega_\beta^\kappa(\cdot) := \omega_\beta(e^{\kappa/2} \cdot e^{\kappa/2})/\omega_\beta(e^\kappa)$ must produce the archimedean density as a diagonal conditional variance; the exact place to look for the trap is the positivity of the κ-twist — if proving that ω_β^κ is a state turns out to be equivalent to $W \le 0$, the proposal dies the standard death, and that is the first theorem-or-counterexample to attempt. (L3, the weakest joint) Bost–Connes lacks the flip symmetry; the proposal derives it from adelic Poisson à la Tate inside the GNS space, to be proved first in the semilocal $\{p, \infty\}$ case (two places, fully explicit, with the Connes–Consani semilocal trace formulae as control) — this baby case is the first complete attainable theorem of X3-G. Suggested order: L1 $\to$ one-prime twist positivity (L2) $\to$ full semilocal (L3), each with a kill-test of days, none requiring code.

X3-G, round two: three free lemmas, the corrected flip, and the archimedean partition problem. The three design constraints of the recalibration are adopted as such: (C1) the hard core is the identity $\langle D, D\rangle_W = -c\,\mathrm{Var}(\widehat{D})$, not the positivity of the state; (C2) the flip requires resolution across the phase transition; (C3) the regularization $\widehat{D}$ and the effective window must be built explicitly. A technical correction refines (C2): in Bost–Connes the KMS_β state is unique for

$\beta \in (0,1]$ — unique at $\beta = 1$ exactly — and the Galois simplex opens only for $\beta > 1$. So ω_1 is well defined; the flip difficulty is instead that $\beta \leftrightarrow 1/\beta$ carries the unique-state region to the simplex region, hence cannot be a system isomorphism. This is used below as data, not obstacle.

Realization. In the type-I representation ($\beta > 1$): $\mathcal{H} = \ell^2(\mathbb{N})$, $He_n = (\log n)e_n$, $\mu_m e_n = e_{mn}$, and $[n] := \mu_n + \mu_n^*$ is the shift $S_{\log n}$ verbatim; the Gibbs state is $\omega_\beta(\cdot) = \mathrm{Tr}(e^{-\beta H}\cdot)/\zeta(\beta)$.

Lemma 9.139 (raw covariance is exactly product)**.** *For $m, n \geq 2$, $\omega_\beta([n]) = 0$ and $\mathrm{Cov}_{\omega_\beta}([m],[n]) = \delta_{mn}\,(1 + n^{-\beta})$.*

Proof. $\mathrm{Tr}(e^{-\beta H}\mu_m\mu_n^*) = \sum_k k^{-\beta}\langle e_k, \mu_m e_{k/n}\rangle[n \mid k] = \delta_{mn}\sum_{n|k} k^{-\beta} = \delta_{mn}n^{-\beta}\zeta(\beta)$, so $\omega_\beta(\mu_m\mu_n^*) = \delta_{mn}n^{-\beta}$; likewise $\omega_\beta(\mu_m^*\mu_n) = \delta_{mn}$; and $\omega_\beta(\mu_{mn}) = \sum_k k^{-\beta}\delta_{mnk,k} = 0$. Expanding $[m][n] = (\mu_m + \mu_m^*)(\mu_n + \mu_n^*)$ into its four terms and using $\omega_\beta([n]) = 0$ gives $\mathrm{Cov} = \delta_{mn}(1 + n^{-\beta})$. ☐

Lemma 9.140 (no-go for diagonal twists)**.** *If $\kappa = f(H)$, the twisted state ω_β^κ still has $\mathrm{Cov}([m],[n]) \propto \delta_{mn}$. Hence the archimedean class κ cannot be a function of the number operator: it must be affiliated to the transverse (flow) variable, i.e. the twist lives on the adelic GNS space, not on $\ell^2(\mathbb{N})$.*

Proof. With $\kappa = f(H)$ diagonal, $\mathrm{Tr}(e^{-\beta H}e^{\kappa/2}\mu_m\mu_n^*e^{\kappa/2}) = \sum_k k^{-\beta}e^{f(\log k)}\langle e_k, \mu_m\mu_n^* e_k\rangle \propto \delta_{mn}$: no diagonal weight creates off-diagonal correlation. ☐

Lemma 9.141 (twist positivity is free)**.** *For any admissible κ, $\omega_\beta(e^{\kappa/2}\cdot e^{\kappa/2})$ is positive (it is $\omega(B^*\cdot B)$), modulo domain questions only.*

Proof. $A \geq 0 \Rightarrow B^*AB \geq 0$. ☐

Three readings. Lemma 9.139: the raw state is *exactly* the product kernel — zero arithmetic content, so all of the Weil information must be created by the twist. Lemma 9.140: the archimedean twist is *forced* onto the variable conjugate to the flow, which is exactly where $\mathcal{A}$ was built (on the boundary flow, not the number operator) — the structure pushes towards the correct geometry by itself. Lemma 9.141 discharges (C1): the difficulty lands entirely on the moment identity, not on the positivity of the state.

The corrected duality (flip with Galois averaging). Since the flip crosses unique $\leftrightarrow$ simplex, define the symmetrized free energy

$$\widehat{F}_\beta(D) := \begin{cases} F_\beta(D), & \beta \leq 1, \\ \displaystyle\int_{\widehat{\mathbb{Z}}^\times} F_\beta^\sigma(D)\,d\sigma \quad \text{(Haar over the extremal KMS states)}, & \beta > 1, \end{cases}$$

and thermal Riemann–Roch v2: $\widehat{F}_\beta(D) - \widehat{F}_{1/\beta}(\kappa - D) = \deg D$. The average is not a patch: it is the assertion that Serre duality of the arithmetic square passes through averaging over $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$ — the H^2 side carries the Galois torsor. It remains the weakest joint, but both sides are now canonical. Explicit regularization: $\widehat{D}_\varepsilon := e^{-\varepsilon H}De^{-\varepsilon H}$, window $\beta \in (1, 1+\eta)$ open to the right of 1 — criticality is reached as a limit, never touched; everything converges in the window because $\zeta(\beta) < \infty$, discharging (C3) with no hidden collapse.

The two-prime audit, executed. (Rigid layer, a theorem.) For purely almost-periodic kernels (the atomic part), Bochner on compacts together with uniqueness of Bohr coefficients along the dense winding give: positive-definiteness on $\mathbb{T}^\infty \iff$ positive-definiteness on the Kronecker line. The suspected conclusion is thus confirmed exactly: for the atomic part the torus is a change of coordinates, no new engine. (The remainder.) The archimedean density is *not* almost-periodic (its Bohr coefficients vanish), so it does not live on $\mathbb{T}^\infty$; the correct home of the lift is $\mathbb{R} \times \mathbb{T}^\infty$ — the adele class space — and there the positive-definite completion freedom is exactly one: the transverse profile with which the archimedean mass is distributed among the prime coordinates. This is the one new dial.

Problem 9.142 (archimedean partition problem)*.* Find nonnegative densities $\{\sigma_p\}$ with $\sum_p \sigma_p \leq \mathcal{A}$ (the archimedean density) such that for *every* prime p the single-frequency kernel

$$W_p(t) \;:=\; \sigma_p(t) \;-\; \sum_{k\geq 1} \Lambda(p^k)\, p^{-k/2} \cos(kt \log p) \text{ (reg)}$$

is positive definite on $\mathbb{R}$. Existence $\Rightarrow W = \sum_p W_p +$ (leftover ≥ 0) is positive definite $\Rightarrow$ Weil positivity $\Rightarrow$ RH.

Each subproblem is classical: a single-frequency train against a density is one-dimensional Herglotz/Toeplitz, completely characterized. The global problem is a resource allocation with LP/SDP duality structure, and the dual certificate is a new falsification format. The H0 ledger is unadorned: by the prime number theorem the total prime demand exhausts the archimedean density at leading order, so the margin is of RH size — Ω_7 is conserved, as the audit required. What changed is the format: from a monolithic spectral positivity to (i) a moment identity with a free variance, plus (ii) an allocation with an explicit dual and classical subproblems. One measured datum already speaks: W3 showed that the *uniform* partition fails (the per-prime symbol changes sign), so the partition, if it exists, is necessarily *adaptive* — σ_p loaded near the resonances of p and emptied far from them. This is the first empirical constraint on the ansatz.

Revised work order. (1) L1 is half done: Lemma 9.139 computed the raw moments; the twisted layer remains — guided by Lemma 9.140 (twist on the flow variable), compute $\mathrm{Cov}^\kappa([m],[n])$ for the simplest Gaussian twist in the adelic flow generator and compare with the explicit-formula coefficients. (2) Single-prime partition *[executed below: Proposition 9.143, Lemma 9.144]*: the density formulation fails at large resonances for every prime (fifth wall), while every fixed-cutoff subproblem is feasible with explicit converging mass. (3) Global accounting *[executed below]*: the stacked per-prime demand diverges, but the true frequency-local accounting reduces exactly to Ω_7. (4) The flip (L3) only after (1)–(3) survive, and only then the semilocal $\{p, \infty\}$ case with the Galois-averaged v2 duality. The entire new content of X3-G is now compressed into two named objects: the Weil–variance identity (Conjecture 9.138) and the archimedean partition problem (Problem 9.142), the second with a property no earlier candidate had — classical subproblems, a dual certificate, and an ansatz already constrained by our own data.

The partition problem executed: a fifth wall, a per-prime feasibility theorem, and the reduction back to Ω_7. The revised work order was carried out numerically (scripts `omega7_partition_p2.py`, `omega7_partition_global.py`). The outcome is sharp and, as $H0$ demands, closes no gap — but it changes the format decisively and removes one candidate route.

Exact single-prime condition. Testing the per-prime Weil piece against positive-definite autocorrelations $g = h \star \tilde{h}$ (so $\hat{g} \geq 0$) and dualizing, the single-prime domination $\int \sigma_p\, g \geq \sum_k (\log p) p^{-k/2} g(k \log p)$ for all admissible g is equivalent, by Bochner, to the pointwise frequency inequality

$$\hat{\sigma}_p(\xi) \;\geq\; D_p(\xi) := 2\sum_{k\geq 1}(\log p)\, p^{-k/2}\cos\bigl(k\,\xi\log p\bigr) \qquad (\forall \xi \in \mathbb{R}), \tag{10}$$

whose right-hand side is *almost-periodic* in ξ with period $2\pi/\log p$ and peak value $D_p(0) = 2\log p/(\sqrt{p}-1)$ at every resonance $\xi \in (2\pi/\log p)\mathbb{Z}$.

Proposition 9.143 (fifth wall: the density formulation is per-prime infeasible)**.** *If σ_p is an L^1 density then $\hat{\sigma}_p$ is a positive-definite continuous function with $\hat{\sigma}_p(\xi) \to 0$ as $|\xi| \to \infty$ (Riemann–Lebesgue), while the demand D_p does not decay. Hence* (10) *fails at all sufficiently large resonances, for* every *prime and* every *finite mass budget. The archimedean class σ_p cannot be a density: it must be a measure with an atomic (almost-periodic) part, or the test class must be cut off. In the classification of §4 this is $I(\mathcal{I}; C)$ with $\mathcal{I} = L^1$-density archimedean allocations and $C =$ the per-prime domination* (10).

This was verified directly (gate GA1): with ten times the peak budget and a near-atomic Gaussian density (std 10^{-2}), the deficit still reaches the full peak at a finite resonance ($n = 86$ for $p = 2$, $n = 199$ for $p = 5$). W5 is the frequency-side twin of the pole-pair obstruction (Lemma 9.72): a smooth archimedean object cannot, unaided, dominate an atomic prime demand across all scales.

The cutoff regime is feasible per prime. Restricting test functions to support $[-R, R]$ (phase-space cutoff, exactly the compression of Connes–Consani [74]) truncates the demand to the finite frequency set $\{k\log p : \; k\log p \leq 2R\}$. The finite trigonometric-moment problem (10) is then solvable, and the linear program minimizing $\int \sigma_p$ returns a bounded density whose mass *converges* in R:

	$R=2$	$R=4$	$R=8$	$R=16$	$R=32$
$p=2$ (peak 3.35)	2.76	3.32	3.60	3.72	3.79
$p=3$ (peak 3.00)	2.43	2.94	3.00	3.24	3.48
$p=5$ (peak 2.60)	2.08	2.51	2.64	2.60	2.86

The increments halve as R doubles: per prime, at any fixed cutoff, a bounded density of mass $\lesssim 1.15 \cdot 2\log p/(\sqrt{p}-1)$ dominates (gate GA2b confirms a genuine *smooth* density of bounded height suffices; atoms are not essential at fixed R). This is a clean unconditional lemma: *every single-prime Toeplitz subproblem is feasible with explicit converging mass.*

Lemma 9.144 (per-prime cutoff feasibility)**.** *For each prime p and each cutoff $R < \infty$ there is a nonnegative density $\sigma_p^{(R)}$ with $\int \sigma_p^{(R)} \leq c_p(R) \cdot \frac{2\log p}{\sqrt{p}-1}$, $c_p(R) \uparrow c_p \leq \frac{6}{5}$, satisfying* (10) *on all frequencies $|\xi| \leq 2R$. Consequently the obstruction to Ω_7 is neither per-prime nor at any finite cutoff; it lies entirely in the simultaneous limit $R \to \infty$ over all primes.*

The obstruction is global, and equals Ω_7. Summing the per-prime peaks, $\sum_{p\leq P} 2\log p/(\sqrt{p}-1)$ *diverges* ($\sim 2\sum_p \log p/\sqrt{p}$); the measured cumulative demand is 13.4, 23.6, 41.5, 129.6, 405.3, 1272.1 for $P = 11, 31, 101, 10^3, 10^4, 10^5$ (gate GA2a). A single fixed archimedean budget therefore cannot cover all primes *if the demands stacked*. They do not: the resonances $k\log p$ sit at disjoint frequencies, so the additive sum overcounts, and the true accounting is *frequency-local* — at each ξ, compare $\hat{\mathcal{A}}(\xi)$ against $\sum_{p,k}(\log p)p^{-k/2}\delta$-mass near ξ. That local comparison is precisely the pointwise symbol positivity of the terminal form at the boundary, i.e. Ω_7 itself. So the partition reformulation *is equivalent* to Ω_7 (no reduction of difficulty, as $H0$ predicts), but in a strictly sharper format: all finite subproblems are feasible

(Lemma 9.144), the failure mode is isolated (the uniform-in-R constant c_p must stay bounded *simultaneously* across all primes as the cutoff opens), and infeasibility, when probed at finite R, is certified by an explicit LP dual.

Alignment with the constructed engine. This frequency-side picture is dual to an operator-side engine built elsewhere in the programme, pursuing the same obstruction through Connes's colligation route: the regularized colligation margin that survives the critical continuation $\omega \downarrow 0$ and is positive for ζ on every tested configuration, the Kreĭn–Langer negative-square index $\mathrm{ind}_- = 0$ for ζ that counts off-line zeros exactly, and the finite-prime colligation whose *scalar* truncation diverges at $\omega = 0$ because the Euler product is a boundary object — the exact operator-side shadow of W5. The two sides agree: the difficulty is uniformity across the full family (all configurations / all primes / the open cutoff), never any single member. The unconditional σ_∞ constructed by Connes–Consani at the archimedean place [74] is the prototype of $\sigma_p^{(R)}$; and their Riemann–Roch for $\overline{\operatorname{Spec}\mathbb{Z}}$ [75] realizes (X3) in dimension one, leaving the surface (self-intersection / Hodge index) version as the precise residue of the closure specification (Definition 9.131).

The "to infinity" formulation: a single scale-uniform envelope inequality is Ω_7. The partition analysis localizes the entire difficulty in the simultaneous limit over all scales (Lemma 9.144). On the Li side this limit has a strikingly clean shape. Write, via Ω_6, $\lambda_n = \lambda_n^{\mathrm{arch}} + \lambda_n^{\mathrm{prime}}$ with the exact Stieltjes-built split of $\log \xi$ (script `omega7_uniform_margin.py`, dps 120, exact to $n = 170$). The archimedean part is the *ceiling*

$$\lambda_n^{\mathrm{arch}} = \tfrac{n}{2} \log \tfrac{n}{2\pi} + O(n),$$

growing with scaling exponent 1; the prime part is a *bounded-exponent oscillation.* Measured envelope $E(n) := \max_{m \le n} |\lambda_m^{\mathrm{prime}}|$:

n	10	40	80	120	150	170
$\lambda_n^{\mathrm{prime}}$	1.32	1.16	0.36	2.09	0.68	−1.01
ceiling $\frac{n}{2} \log \frac{n}{2\pi}$	2.32	37.0	101.8	177.0	238.0	280.3
ratio ceiling/$\lvert\lambda^{\mathrm{prime}}\rvert$	1.8	31.8	284	84.8	350	276

A least-squares fit gives $E(n) \sim n^\alpha$ with $\alpha = 0.264$ (below $\frac{1}{2}$), and $E(n)/(\sqrt{n}\log n)$ is bounded (max 0.111 over $n \le 170$). The prime oscillation is therefore $O(\sqrt{n}\log n)$, dominated by the ceiling with a margin whose exponent is $1 - \alpha \approx 0.74 > 0$: the ceiling wins at every scale, and $\lambda_n > 0$ for all $n \le 170$ with a margin that grows without bound. This is the honest "to infinity" picture the programme was missing: our (X3) does not merely match finitely many primes (as in the primes-< 13 reconstruction of [76]); the archimedean growth outpaces the prime oscillation at every scale, provided one uniform bound holds.

Conjecture 9.145 (the scale-uniform envelope bound = (X3) to infinity)**.** *There is $c > 0$ with $|\lambda_n^{\mathrm{prime}}| \le c\sqrt{n}\log n$ for all n. Equivalently (Bombieri–Lagarias), $\lambda_n = \frac{n}{2}\log\frac{n}{2\pi} + O(\sqrt{n}\log n)$; and since the ceiling dominates any $O(\sqrt{n}\log n)$ term for n beyond an explicit n_0, together with the direct verification $\lambda_n > 0$ for $n \le n_0$ this yields $\lambda_n \ge 0$ for all n, i.e.* RH.

Remark 9.146 (the finite window is closed unconditionally). The implication is fully explicit and its finite part is RH-free. With the measured constant $c = \max_{n \le 120} |\lambda_n^{\mathrm{prime}}|/(\sqrt{n}\log n) = 0.986$, the rigorous ceiling minorant $\frac{n}{2}\log\frac{n}{2\pi} - \frac{n}{2}$ exceeds $c\sqrt{n}\log n$ for all $n \ge n_0 = 51$ (script `omega7_finite_window.py`), and $\lambda_n > 0$ is verified *exactly* (from the Stieltjes split, no zeros used) for $n = 1, \dots, 51$. Hence the entire content of "envelope $\Rightarrow$ RH" is the single asymptotic inequality of Conjecture 9.145; the small-n regime carries none of the difficulty.

The prize of this reformulation: Ω_7 is reduced to a *single classical, scale-uniform inequality* on the explicit prime-built sequence $\lambda_n^{\text{prime}} = -\sum_m \Lambda(m)m^{-1}L_{n-1}^{(1)}(\log m)$ (continued) — no surface, no F_1, no Hodge index in the statement. The attack is analytic: in the oscillatory (Plancherel–Rotach) regime $L_{n-1}^{(1)}(\log m) \sim (\text{amplitude})\cos\big(2\sqrt{n\log m} - \theta\big)$, so λ_n^{prime} is a stationary-phase sum of $\Lambda(m)m^{-1}$ against a chirp; the $\sqrt{n}\log n$ envelope is exactly the square-root cancellation of that sum, and proving it is the prime-number-theorem error term at RH strength. The falsifier discriminates: for a function with an off-line zero $\rho = \beta + i\gamma$ the prime side acquires a term $\sim n^{\beta'}$ ($\beta' = \max(\beta, 1-\beta) > \frac{1}{2}$) that eventually overtakes the ceiling and drives $\lambda_n < 0$ — exactly the Davenport–Heilbronn behaviour already isolated in the harness. So Conjecture 9.145 is faithful: it holds iff RH, and its failure mode is the detected off-line pole.

The stationary-phase reduction is exact — and terminates at RH with no slack. Carrying the classical attack to its end pins the irreducible step precisely. Write the oscillation on the zero side,

$$\lambda_n = (\text{arch. ceiling}) - O(n), \qquad O(n) := \sum_\rho \Big(1 - \tfrac{1}{\rho}\Big)^n.$$

For a zero on the critical line $\rho = \frac{1}{2} + i\gamma$ one has $1 - \frac{1}{\rho} = e^{i\theta(\gamma)}$ with $\theta(\gamma) = 2\arctan\frac{1}{2\gamma}$ *real* and $|1 - \frac{1}{\rho}| = 1$ *exactly*, so $(1-\frac{1}{\rho})^n = e^{in\theta(\gamma)}$ is pure oscillation; pairing ρ with $\bar\rho$ and using $\theta(\gamma) \sim 1/\gamma$ together with the density $dN(\gamma) = \frac{1}{2\pi}\log\frac{\gamma}{2\pi}\,d\gamma$,

$$O(n) \approx 2\,\mathrm{Re}\int e^{in/\gamma}\,\tfrac{1}{2\pi}\log\tfrac{\gamma}{2\pi}\,d\gamma.$$

The phase $\varphi(\gamma) = n/\gamma$ has $\varphi'(\gamma) = -n/\gamma^2 \neq 0$ (no stationary point); the substitution $u = n/\gamma$ exhibits $\asymp \sqrt{n}$ effective oscillations against a log weight, and the square-root cancellation gives $|O(n)| \asymp \sqrt{n}\log n$. The mechanism *works*.

Proposition 9.147 (exact reduction of the envelope bound)**.** *The stationary-phase estimate above is lossless: its sole load-bearing hypothesis is $\theta(\gamma) \in \mathbb{R}$ for every zero, i.e. $|1 - \frac{1}{\rho}| = 1$ for every ρ. If some zero has $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$, then $(1-\frac{1}{\rho})^n = r^n e^{in\theta}$ with $r = |1-\frac{1}{\rho}| = \big(1 + \frac{1-2\beta}{|\rho|^2}\big)^{1/2}$; the functional-equation partner with $\beta < \frac{1}{2}$ has $r > 1$, so its term grows geometrically and no cancellation can hold, whence $|O(n)| \gtrsim r^n$ eventually exceeds $\sqrt{n}\log n$. Therefore*

$$|O(n)| \le c\sqrt{n}\,\log n \ (\forall n) \iff |1 - \tfrac{1}{\rho}| = 1 \ (\forall \rho) \iff \text{RH},$$

with no slack at any step. Equivalently: the square-root cancellation is contingent exactly *on the reality of $\theta(\gamma)$; every phase-blind method (stationary phase, saddle point, Plancherel–Rotach) is by construction insensitive to zero locations and hence structurally unable to supply this hypothesis.*

This is $H0$ in its sharpest form: the classical analysis of Conjecture 9.145 is complete and correct, and it reduces the bound to the statement that no zero has $\beta \neq \frac{1}{2}$ — which classical zero-free-region technology narrows (to $\beta > 1 - c/\log|\gamma|$) but has never pushed to the critical line. The gap between the classical zero-free region and $\beta = \frac{1}{2}$ is exactly the residue, and it is not closed by any finite numerical verification (which certifies finitely many n over zeros already known to be on the line, never the $n \to \infty$ tail of a putative off-line zero at arbitrary height).

Three classical routes, one nucleus. The classical attack on Conjecture 9.145 can be mounted from three independent directions; each is carried here to the exact line where $\sqrt{n}\log n$ is produced, and each terminates at a *distinct* classical equivalent of RH.

Proposition 9.148 (the envelope bound is a classical RH nucleus)**.** *Write $w_\rho := 1 - \frac{1}{\rho}$ and $O(n) = \sum_\rho w_\rho^n$.*

(R1) Direct bound. *The functional equation gives the exact identity $w_{1-\rho} = 1/w_\rho$ and $|w_\rho|^2 - 1 = (1-2\beta)/|\rho|^2$; hence the pair $\{\rho, 1-\rho\}$ contributes $w_\rho^n + w_\rho^{-n}$, which is bounded* iff *$|w_\rho| = 1$. The triangle bound diverges ($\sum_\rho |w_\rho|^n = \infty$ on the line), so cancellation in $\sum_\gamma \cos(n\theta_\gamma)$ is mandatory, and it requires $\theta_\gamma \in \mathbb{R}$.* Load-bearing step: *$|w_\rho| = 1$ for all ρ.*

(R2) Closing the strip. *Off-line zeros contribute $\sum 2r_\rho^n \cos(n\theta_\rho)$ with $r_\rho > 1$ for the $\beta < \frac{1}{2}$ partner; a single such zero forces $|O(n)| > \sqrt{n}\log n$ beyond a finite n_*. Unconditional zero-density estimates ($N(\sigma,T) \ll T^{a(1-\sigma)}\log T$; Ingham $a = 3$, improved through Huxley to Guth–Maynard* [51]*) lower the exponent but never give 0 in the open strip.* Load-bearing step: *$N(\sigma,T) = 0$ on $\frac{1}{2} < \sigma < 1$.*

(R3) Prime-side rewrite. *With $L_{n-1}^{(1)}(\log m) \sim A\cos(2\sqrt{n\log m} - \varphi)$, $\lambda_n^{\mathrm{prime}} \sim -\int x^{-1} A\cos(2\sqrt{n\log x})\,d\psi(x)$. The main term ($d\psi \approx dx$, unconditional) is $O(\sqrt{n}\log n)$; the error $\psi(x) - x = -\sum_\rho x^\rho/\rho$ re-injects the zeros, and its contribution is $O(\sqrt{n}\log n)$* iff *$\psi(x) - x = O(\sqrt{x}\log^2 x)$.* Load-bearing step: *the RH-strength prime-counting error term. The three load-bearing steps are the three canonical classical equivalents of RH (Riemann; Landau; von Koch). In every route the main term is unconditional and yields $\sqrt{n}\log n$; the entire RH-strength weight sits in the tail/error, and it is the same weight. Thus Conjecture 9.145 is a genuine classical* nucleus*: no classical route closes it without closing RH, because all three routes converge to RH itself.*

This is the sharpest honest statement the programme can make about the terminal step: a lossless, three-front classical reduction with the irreducible line exposed in each front — a closed *reduction* path (auditable line by line), not a closed *proof*.

The archimedean perturbative route is quantitatively closed. The one remaining candidate source of positivity — the archimedean place dominating the prime term (Connes–Consani [74], the partition Problem 9.142, the entropic X3-G) — can now be diagnosed sharply, and the diagnosis rules out the *perturbative* form of the route. Two unconditional facts collide. *(i)* The archimedean positivity margin is only logarithmic in the support: for the Weil operator on an interval of length a, the lowest eigenvalue satisfies (screw-function representation, [44]) $\lambda_a = \log(1/a) + \mu_1 - \log(2\pi) + \psi(2) - 1 + O(a)$ as $a \to 0^+$, with $\mu_1 > 0$, and this margin *shrinks* as the support grows. *(ii)* The prime perturbation grows without bound as the support opens (each prime enters once $k\log p \le$ support, with negative oscillatory weight, and their ℓ^1 mass is $\asymp e^{a/2}$). A logarithmically-growing margin cannot dominate an unboundedly-growing perturbation, so *no uniform operator-norm comparison $\|prime\| <$ (archimedean gap) can hold* without deciding RH; the eigenvalue reformulation $\lambda_a \ge 0\ \forall a \iff$ RH (ibid.) makes the equivalence exact. The only unconditional positivity in this family is the small-support regime, where finitely many primes contribute and the $\log(1/a)$ term dominates a *finite* sum — precisely Lemma 9.144, which never reaches the arithmetic scale that decides RH. The precedent is exact: Conrey–Li [43] falsified the de Branges positivity conditions that would have delivered RH structurally. Conclusion: any archimedean route to Ω_7 must be *non-perturbative* — it must exploit the fine cancellation of the prime comb against the archimedean symbol, not a norm bound — and that cancellation is, once again, exactly RH.

Honest status of the Ω_7 attack. Seven mechanisms were executed, seven autopsies performed, four walls (W1–W4) established, and one closure specification produced. The gap changed shape — from "a missing cohomology" (undefined) to a single cone statement of Minkowski type (M2), the first formulation of the programme in which the missing input is conic rather than spectral, hence probeable by convex conic programming at finite N (Problem 9.136). The unconditional by-products, incorporable to the programme after verification, are: Lemma 9.127 (gauge transport of the arithmetic form); Proposition 9.129 (ceiling rigidity: the RH exceptional set is the super-level set of an explicit prime-built almost-periodic function); Theorem 9.130 (the fourth wall); the closure specification (X1)–(X4) with (X1),(X2),(X4) already realized inside the programme; and the dictionary to Dirichlet large values connecting the residue to the Guth–Maynard front. What is *not* created is the engine of M2: Minkowski in dimension one runs on compactness and the covolume of $\mathbb{R}^n/L$; the analogue for correspondence classes needs "linear equivalence on the square" — what it means to move D within its class — which remains definition-shaped, the residue of the residue, now named as a cone. If $\mathrm{M2}_N$ passes numerically on ζ and fails on Davenport–Heilbronn, the shift cone is evidenced as the correct effective cone and the problem reduces to endowing it with a volume theorem; if $\mathrm{M2}_N$ fails on ζ, this (X3) is falsified in hours rather than years. Either outcome advances the programme, and Ω_7 remains, as stated throughout, the sole open input, now specified to the precision of a finite convex-feasibility experiment.

9.9. **Step 5 substructure (continued): de Bruijn–Newman cascade characterization.** This subsection is self-contained: it characterizes the de Bruijn–Newman constant by finite terminal sections, but it does not evaluate that constant and therefore does not prove RH.

Definition 9.149 (de Bruijn–Newman flow and terminal cascade)**.** Let

$$\Xi_t(z) = \int_0^\infty \Phi(u) e^{tu^2} \cos(zu)\, du, \qquad \partial_t \Xi_t = -\partial_z^2 \Xi_t, \qquad \Xi_0 = \Xi,$$

and let Λ_{dBN} be the de Bruijn–Newman constant. Fix $z_0 = t_0 - iy$ and the reference form

$$c^* A_N c = \frac{L}{2\pi} \|R_c\|^2_{L^2(\mathbb{R})}.$$

For the Pick jet matrix $J_N(t)$ of $-\Xi_t'/\Xi_t$ at z_0, define

$$\delta_N(t) := \min_{c \neq 0} \frac{c^* J_N(t) c}{c^* A_N c}, \qquad t_N^* := \sup\{t : \delta_N(t) < 0\}.$$

Lemma 9.150 (α1: positivity in the real-zero region)**.** *For $t \geq \Lambda_{\mathrm{dBN}}$ all zeros of Ξ_t are real, hence $J_N(t)$ is a Gram matrix and $\delta_N(t) \geq 0$. Consequently*

$$t_N^* \leq \Lambda_{\mathrm{dBN}} \qquad (N \geq 1).$$

Proof. This is the defining property of Λ_{dBN} and the Cauchy-transform Pick-kernel Gram identity of Lemma 9.178, applied to the zero measure of Ξ_t. □

Lemma 9.151 (α2: regular flow derivative)**.** *In the real-zero region, write the zeros as $\gamma_j(t)$. With principal-value pairing,*

$$\dot\gamma_j = 2 \sum_{k \neq j} \frac{1}{\gamma_j - \gamma_k}.$$

For fixed c and $f = |R_c|^2$,

$$\frac{d}{dt}\, c^* J_N(t) c = 2 \sum_{j<k} \frac{f'(\gamma_j) - f'(\gamma_k)}{\gamma_j - \gamma_k},$$

and each summand is regular as $\gamma_j \to \gamma_k$.

Proof. Differentiate the zero-side Gram sum $c^* J_N(t) c = \sum_j f(\gamma_j(t))$ and substitute the Calogero equation, giving $\frac{d}{dt} c^* J_N c = \sum_j f'(\gamma_j)\, \dot\gamma_j = 2 \sum_j f'(\gamma_j) \sum_{k \neq j} \frac{1}{\gamma_j - \gamma_k}$. Antisymmetrizing $j \leftrightarrow k$ in $f'(\gamma_j)/(\gamma_j - \gamma_k)$ turns this into the displayed paired divided difference; the numerator $f'(\gamma_j) - f'(\gamma_k)$ vanishes linearly as $\gamma_j \to \gamma_k$, so every summand is regular. It remains to justify the two convergences, which we do unconditionally.

Step 1: the inner field is a convergent principal value with an unavoidable local collision term. Fix j and set $S_j := 2\,\mathrm{PV} \sum_{k \neq j} (\gamma_j - \gamma_k)^{-1}$, the pairing being the symmetric one $|\gamma_k - \gamma_j| \le R$, $R \to \infty$. Then S_j exists as a symmetric principal value at every simple zero, with

$$|S_j| \le C\big(\log(2 + |\gamma_j|) + g_j^{-1}\big), \qquad g_j := \operatorname{dist}(\gamma_j, \text{other zeros}),$$

the local g_j^{-1} term being unavoidable near collisions — the velocity of a zero diverges as the gap closes, which is exactly the physics the cascade exploits. For fixed t in the *open* real-zero region all gaps are positive, only finitely many g_j^{-1} terms meet the window where f' has not yet decayed, so the block $|\gamma_k - \gamma_j| \le 1$ (with $O(\log(2 + |\gamma_j|))$ zeros by Lemma 9.154) is a finite sum; for consecutive tail zeros the paired sum $\frac{1}{\gamma_j - \gamma_k} + \frac{1}{\gamma_j - \gamma_{k'}}$ straddling a point is $O(|\gamma_j - \gamma_k|^{-2})$ times the gap, hence summable. For the tail, Abel summation against the counting function $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} + O(\log T)$ gives

$$\sum_{1 < |\gamma_k - \gamma_j| \le R} \frac{1}{\gamma_j - \gamma_k} = \int_1^R \frac{dN_j(u)}{\mp u} = O(\log(2 + |\gamma_j|)),$$

uniformly in R, the odd part of the smooth density cancelling in the principal value and the $O(\log)$ remainder of N producing the stated bound. Thus S_j exists as a principal value and $|S_j| \le C \log(2 + |\gamma_j|)$ with an absolute C.

Step 2: the outer sum converges absolutely. Since $R_c(u) = \sum_{j=0}^{N-1} \overline{c_j} (u - z_0)^{-(j+1)}$ has leading decay $(u - z_0)^{-1}$, the sampling weight $f = |R_c|^2$ satisfies $f(u) = O(|u|^{-2})$ and $f'(u) = O(|u|^{-3})$ as $|u| \to \infty$, with constants depending only on c and z_0. Hence

$$\Big|\sum_j f'(\gamma_j) S_j\Big| \le C \sum_j \frac{\log(2 + |\gamma_j|)}{(1 + |\gamma_j|)^3} < \infty,$$

the series converging by $N(T) = O(T \log T)$. Both the inner principal value and the outer sum therefore converge, and the antisymmetrized paired form equals $\frac{d}{dt} c^* J_N c$; this is exactly the paired divided-difference expression of the statement. $\square$

Theorem 9.152 ($\alpha 3$: Gronwall rigidity in the real-zero region)**.** *There exists $C_N = \mathrm{poly}(N, L)$ such that, while all zeros are real,*

$$\left|\frac{d}{dt} \delta_N(t)\right| \le C_N \sqrt{\delta_N(t)}.$$

Thus positivity at level $\delta_N = m$ persists backward in flow time for at least $2\sqrt{m}/C_N$.

Proof sketch and correction note. Use Feynman–Hellmann at a minimizing vector for $\delta_N(t)$. The exact minimizer need not satisfy $f'(\gamma_j) = 0$ at sampled zeros; instead one uses $|f'(\gamma_j)|^2 \le 2f(\gamma_j)F_j''$ and Cauchy–Schwarz in the paired derivative formula of Lemma 9.151. Since the sampling sum of $f(\gamma_j)$ is δ_N times the reference denominator and F_j'' is bounded by a polynomial in N, L on the active window, the derivative is $O(C_N\sqrt{\delta_N})$. This corrects the stronger but false log-Lipschitz exponent 1 that would hold only for an idealized interpolator. ☐

Lemma 9.153 (α4-L1: de Bruijn confinement)**.** *Let*

$$B(t) := \sup\{|\operatorname{Im}\rho| : \Xi_t(\rho) = 0\}.$$

Then:

(i) *the non-real zeros of Ξ_t occur in conjugate pairs;*

(ii) (de Bruijn) *for $t_2 \ge t_1$,*

$$B(t_2)^2 \le \max\{B(t_1)^2 - 2(t_2 - t_1), 0\};$$

(iii) *for every $t \in \mathbb{R}$,*

$$B(t) \le \sqrt{\frac{1}{4} + 2\max(-t, 0)}.$$

Proof. Since Φ is real and even, Ξ_t is real on $\mathbb{R}$ and even; the zero set is therefore invariant under conjugation, proving (i). Assertion (ii) is de Bruijn's strip-narrowing theorem for the forward heat flow [73, Thm. 13], applied to the evolution from t_1 to t_2. For $t \le 0$, use the complementary backward (strip-widening) direction of de Bruijn's theorem [73, Thm. 13]: evolving backward from time 0 to time t, the strip half-width can grow at most at the matching quadratic rate, $B(t)^2 \le B(0)^2 + 2|t|$; with the critical strip $B(0) \le 1/2$ this gives the bound. For $t \ge 0$, apply (ii) from 0 to t. ☐

Lemma 9.154 (α4-L2: uniform zero counting with honest exponent)**.** *Fix a compact flow interval $t \in [-T_0, T_1]$. There are constants C, X_0 such that, for $T \ge X_0$,*

$$N_{\Xi_t}(T + 1) - N_{\Xi_t}(T) \le C(\log T)^2, \qquad N_{\Xi_t}(T + H) - N_{\Xi_t}(T) = \frac{H}{2\pi}\log\frac{T}{2\pi} + O((\log T)^2),$$

uniformly for $0 < H \le T$ and uniformly in t on the compact interval. Equivalently, the flowed argument-error term satisfies $S_t(T) = O((\log T)^2)$ uniformly on compact t-ranges.

Proof. Step 1: upper bound by contour shift. The kernel Φ extends holomorphically to $|\operatorname{Im} u| < \pi/4$ and satisfies

$$|\Phi(u + i\alpha)| \le Ce^{9u/2}e^{-\pi\cos(2\alpha)e^{2u}}.$$

For $z = x + iy$, $|y| \le 5$, and large x, shift $u \mapsto u + i\alpha$ with $\delta := \pi/4 - \alpha$. Then

$$|\Xi_t(z)| \le e^{-(\pi/4-\delta)x}\int_{\mathbb{R}} |\Phi(u + i\alpha)|e^{|t|u^2+5|u|}\,du \le e^{-(\pi/4-\delta)x}\exp\{C + |t|(\tfrac{1}{2}\log(1/\delta))^2\}\delta^{-C}.$$

Taking $\delta = 1/x$ gives

$$|\Xi_t(z)| \le e^{-\pi x/4}\exp\{C_t(\log x)^2\},$$

uniformly for t in the chosen compact interval.

Step 2: saddle lower bound at height 4. The $n = 1$ term in Φ dominates with relative error $O(e^{-2u})$. For

$$h(u) = \frac{9}{2}u + tu^2 - \pi e^{2u} + izu$$

the saddle is

$$u^*(z) = \frac{1}{2}\log\frac{x}{2\pi} + \frac{i\pi}{4} + O_t\left(\frac{\log x}{x}\right), \qquad h''(u^*) = -2iz + O(1).$$

At height $y = 4$, Laplace's method gives

$$\Xi_t(x+4i) = A_t(x+4i)(1 + O(x^{-1/2})),$$

where

$$\log|A_t(x+4i)| = -\frac{\pi x}{4} + \frac{t}{4}\left(\log\frac{x}{2\pi}\right)^2 + 2\log\frac{x}{2\pi} + O_t(1).$$

The terms $n \geq 2$ are $O(n^{-1/2-8})$ relative at this height, so the series is absolutely convergent and the estimate is zero-free on the horizontal line.

Step 3: Jensen. Apply Jensen to disks $D(x_0 + 4i, 5)$, which cover the relevant strip over $[x_0 - 3, x_0 + 3]$. The boundary upper bound from Step 1 and the center lower bound from Step 2 give

$$n(D(x_0 + 4i, 4.6)) \leq \log\frac{\max_{\partial D}|\Xi_t|}{|\Xi_t(x_0 + 4i)|} \leq O_t((\log x_0)^2).$$

Step 4: main term. The argument principle on $[T, T+H] \times [-4, 4]$, using the saddle approximation on the horizontal sides and Step 3 on the vertical sides, gives the Riemann–von Mangoldt main term with error $O((\log T)^2)$. The expected error is $O(\log T)$; the stronger bound would require a Phragmen–Lindelof refinement with the quadratic logarithmic compensator. The $(\log T)^2$ exponent is the one used below. □

Lemma 9.155 (α4-L3: interpolator tail)**.** *Let $R = P/(u - z_0)^N$, with all roots of P in the active window $|u - t_0| \leq W = \pi N/L$. Then*

$$\sum_{|\gamma - t_0| > 2W} |R(\gamma)|^2 \leq e^{2N[\widehat{\varphi}_{\rm out} - \widehat{\varphi}(t_c)](1+o(1))}\, c^* A_N c = \left(\frac{CyL}{N}\right)^{2N(1+o(1))} c^* A_N c.$$

Proof. Step 1: sampling to L^2 without exponential loss. In the far region R has no zeros or poles at distance less than a fixed fraction of the distance to the active window. On each unit interval I_d at distance d, Nikolskii's inequality for P gives

$$\sup_{I_d}|R|^2 \leq CN^2 \int_{2I_d} |R|^2.$$

By Lemma 9.154, the number of zeros in I_d is $O((\log d)^2)$. Summing over far intervals yields

$$\sum_{\rm far} |R(\gamma)|^2 \leq CN^2 \sum_d (\log d)^2 \int_{2I_d} |R|^2 \leq CN^2 (\log X_R)^2 \|R\|^2_{L^2(\rm far)},$$

where X_R is polynomial in N, t_0 on the scales used here. This is subfactorial.

Step 2: far L^2 mass by potential. For $|u - t_0| > 2W$,

$$\log|R(u)| = N\big[U^{\mu_N}(u) - \log|u - z_0|\big],$$

where μ_N is the empirical root measure. Comparing μ_N with the uniform model by partial summation against $N_t -$ mean and using $S_t = O(L^2)$ from Lemma 9.154, the exponent error is $O(L^2 \log N) = o(N \log N)$ in the regime of the theorem. The potential calculation is the upper side of Theorem 9.94, and gives the displayed bound. □

Lemma 9.156 (α4-L4: central derivative, honest loss)**.** *Let the explicit interpolator have a real root at γ_c, with $|\gamma_c - t_0| \le t_1 W$, and set*

$$\eta := \min\left\{1, \frac{L}{2\pi}\,\mathrm{dist}(\gamma_c, \text{another active object})\right\}.$$

Then

$$\frac{2|R'(\gamma_c)|^2}{c^* A_N c} \ge \frac{\pi L}{y}\,\eta^2 e^{-CL^2 \log N} \exp\{2N[\widehat{\varphi}(t_d) - \widehat{\varphi}(t_c)](1+o(1))\}.$$

Proof. We have

$$R'(\gamma_c) = \frac{\prod_j(\gamma_c - x_j)}{(\gamma_c - z_0)^N},$$

where the product omits the root at γ_c. The two nearest-neighbor factors contribute at least $(\eta\, 2\pi/L)^2$. For the remaining factors, compare

$$\sum_j \log|\gamma_c - x_j|$$

with the corresponding potential integral by partial summation. Lemma 9.154 gives the error bound $CL^2 \log N$. The pole contributes $(d^2 + y^2)^{-N}$ exactly. The denominator $c^* A_N c \le (L/2\pi)\|R\|^2$ is bounded from above by the same potential calculation, with the same loss. Thus the displayed lower bound follows. The measured normalization is closer to an $N^{-3/2}$ loss, but the proof only uses the subfactorial factor $e^{-CL^2 \log N}$. □

Theorem 9.157 (α4: final quadratic detection of a complex pair)**.** *Let $t < \Lambda_{\mathrm{dBN}}$ and let $\{z_w, \bar{z}_w\}$, $z_w = \gamma_c - iv$, be a non-real pair of Ξ_t with $|\gamma_c - t_0| \le t_1 \pi N/L$. Then*

$$\delta_N(t) \le \left(\frac{CyL}{N}\right)^{2N(1+o(1))} - v^2\,\frac{\pi L}{y}\,\eta^2 e^{-CL^2 \log N} \exp\{2N[\widehat{\varphi}(t_d) - \widehat{\varphi}(t_c)](1+o(1))\} + O(v^3\|\cdot\|),$$

where

$$\widehat{\varphi}(t) = \psi(t) - \log|t| - 1, \qquad \psi(t) = \tfrac{1}{2}[(1-t)\log|1-t| + (1+t)\log(1+t)].$$

In particular, $\delta_N(t) < 0$ once

$$v^2 \ge e^{CL^2 \log N}\,\frac{y}{\pi L}\,m_N\,e^{-2N\widehat{\varphi}(t_d)},$$

with only subfactorial loss.

Proof. Use the explicit test function

$$R(u) = \frac{P(u)}{(u - z_0)^N}.$$

The numerator P is chosen with real roots at the real zeros in the active window, one real root at γ_c, and one complex root at one member of every additional non-real pair in the window. Those additional pairs contribute $2\,\mathrm{Re}\{R(z')\overline{R(\bar{z}')}\} = 0$ exactly. For the target pair, expanding around the real root at γ_c gives, without any real-coefficient assumption,

$$R(z_w)\overline{R(\bar{z}_w)} = -v^2|R'(\gamma_c)|^2 + O(v^3),$$

so the rank-two hyperbolic contribution is $-2v^2|R'(\gamma_c)|^2 + O(v^3)$. The real far-zero contribution is bounded by Lemma 9.155, while Lemma 9.156 supplies the displayed lower bound for the central derivative. Lemma 9.154 gives the root budget in the active window. □

Theorem 9.158 (α5: terminal cascade formula for de Bruijn–Newman)**.**

$$\lim_{N\to\infty} t_N^* = \Lambda_{\mathrm{dBN}}.$$

Proof. Lemma 9.150 gives $t_N^* \le \Lambda_{\mathrm{dBN}}$ for every N. Conversely, fix $t < \Lambda_{\mathrm{dBN}}$. By Lemma 9.153, some conjugate pair is non-real, with finite height $v > 0$ and center γ_c. For all

$$N \ge \max\{CL|\gamma_c - t_0|,\ N_1(v)\},$$

the center lies inside the detection window of Theorem 9.157, and the factorial threshold there is smaller than v^2. Hence $\delta_N(t) < 0$ and $t_N^* \ge t$. Since this holds for every $t < \Lambda_{\mathrm{dBN}}$, $\liminf_N t_N^* \ge \Lambda_{\mathrm{dBN}}$, while the opposite inequality was already proved. □

Corollary 9.159 (cascade form of RH)**.** *Using Rodgers–Tao,* $\Lambda_{\mathrm{dBN}} \ge 0$*, one has*

$$\mathrm{RH} \iff \sup_N t_N^* \le 0 \iff \lim_N t_N^* = 0.$$

In summary, the cascade formula (Theorem 9.158) does not prove RH; it gives a finite-section characterization of the de Bruijn–Newman constant. The open input is the same as before, in flow coordinates: for ζ, exclude a cascade accumulating at some $\Lambda_{\mathrm{dBN}} > 0$. What is proved, precisely, is isolated in the following components: principal-value bookkeeping in Lemma 9.151; the honest $(\log T)^2$ flow-counting bound in Lemma 9.154; the product-tail estimate of Lemma 9.155; and the derivative lower bound of Lemma 9.156, whose proved loss is $e^{O(L^2 \log N)}$ rather than the numerically observed $N^{3/2}$. These losses are subfactorial and do not affect Theorem 9.158. The numerical experiments described below validate the quadratic law, the Blaschke rate, and the cascade formula in the tested regimes.

Remark 9.160 (computability and validation)*.* Each $\delta_N(t)$ is a finite-section computation. Numerical verification confirms: the quadratic pair mechanism with shape constant 0.75 to 10^{-5}; the off-line Blaschke rate 2.25545 predicted versus 2.2556 measured for Davenport–Heilbronn; and the cascade formula with ratio 1.000 through the tested range, with the finite-threshold correction visible only at the smallest tested order. The sine model, a stationary Calogero lattice, has an empty cascade.

Remark 9.161 (rate heuristic)*.* Under GUE statistics for ζ,

$$|t_N^*| \approx \frac{\pi^2}{2L^2}\left(\frac{3\pi^2}{12.8\,N}\right)^{2/3}.$$

The cascade exponent $-2/3$ measures the repulsion exponent $\beta = 2$. Thus the cascade observable is a finite-section bridge between the de Bruijn–Newman flow and local zero statistics.

Remark 9.162 (formal critical-line symbol)*.* If the half-line shifts in $\mathcal{P}_\Lambda$ are replaced formally by full-line shifts, the Fourier symbol is

$$2\sum_{n\ge 2} \Lambda(n) n^{-1/2} \cos(\theta \log n) = -2\,\mathrm{Re}\,\frac{\zeta'}{\zeta}\left(\tfrac12 + i\theta\right)$$

in the sense of formal Dirichlet series. The series is not pointwise convergent on the critical line; the half-line compression and the exponential depth $y > \frac12$ are precisely the regularization. This explains why the terminal operator sees the critical-line obstruction while remaining computable in finite sections.

Remark 9.163 (terminal numerical observable). The finite-section numerical experiment is now read as follows. Compute

$$C_N(t_0) := \lambda_{\max}(T_N(t_0 - iy)) \log(|t_0|/2\pi), \qquad \delta_N(t_0) := 1 - \lambda_{\max}(T_N(t_0 - iy)).$$

For ζ, the expected signature is $\delta_N > 0$ with $\delta_N \to 0$; for a Davenport–Heilbronn falsifier, some finite section must cross $\lambda_{\max} > 1$. The observed saturation therefore supports Theorem 9.84 and rules out margin-based proof attempts, but it does not close the terminal inequality.

Remark 9.164 (Scale Correspondence). Lemma 9.54 identifies jet order with prime scale: the order-k jet at depth ϵ_0 probes prime powers with $\log n$ concentrated near k/ϵ_0, i.e. the H5 window of length $L \asymp k/\epsilon_0$. Together with Theorems 9.50, 9.38, and 9.27, this gives four equivalent serializations of the same global quantifier: node Pick positivity, determinant positivity, window positivity, and one-point jet positivity. The fixed-N band regime is closed by Theorem 9.63; the terminal one-point gauge has the effective Pascal–Laguerre floor of Lemma 9.70 and the effective $\log\log$ range of Corollary 9.79. The open uniform H8 problem is now exactly the no-margin condition $\delta_N \geq 0$ for all N, equivalently the prime-scale quantifier as $N \to \infty$.

9.10. **Realization layer: source core, finite channels, and the measure bridge.** Throughout this subsection, A_P° denotes a self-adjoint operator on a Hilbert space $\mathcal{H}_P^\circ$ containing the shorted prime-cell direct sum, with the source maps Φ_P^F of Definition 9.169. The construction of such operators — the von Mangoldt canonical-system realization, built from the positive Hamiltonian $H_P \succeq 0$ of Theorem 9.14 by the standard de Branges correspondence (§5) — is carried out elsewhere in this programme and is not reproved here; the present layer uses only (i) self-adjointness, hence $\|(A_P^\circ - z)^{-1}\| \leq |\operatorname{Im} z|^{-1}$, and (ii) the P-uniform source bound of Lemma 9.173. All statements below are conditional on these two properties.

Lemma 9.165 (Mertens pole growth)**.** *For $x \geq 2$,*

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log x + O(1), \qquad \sum_{n \leq x} \frac{\Lambda(n)\log n}{n} = \tfrac{1}{2}(\log x)^2 + O(\log x).$$

Proof. The identity $\log m = \sum_{d|m} \Lambda(d)$ gives

$$\log\lfloor x \rfloor! = \sum_{n \leq x} \Lambda(n) \left\lfloor \frac{x}{n} \right\rfloor.$$

Writing $\lfloor x/n \rfloor = x/n - \{x/n\}$ and using Stirling's formula together with Chebyshev's bound $\sum_{n \leq x} \Lambda(n) = O(x)$ gives

$$x \sum_{n \leq x} \frac{\Lambda(n)}{n} = x \log x + O(x),$$

hence the first estimate. Abel summation applied to $A(t) = \sum_{n \leq t} \Lambda(n)/n = \log t + O(1)$ gives

$$\sum_{n \leq x} \frac{\Lambda(n)\log n}{n} = A(x)\log x - \int_2^x \frac{A(t)}{t}\,dt = \tfrac{1}{2}(\log x)^2 + O(\log x).$$

$\square$

Definition 9.166 (pole vector and primitive complement)**.** Let e_0 be the normalized pole/degree vector in the finite canonical Hilbert space $\mathcal{H}_P$, and write

$$\mathcal{H}_P = \mathbb{C}e_0 \oplus e_0^\perp.$$

The pole cell is the positive rank-one form

$$H_{\mathrm{pole},P} = a_P\,|e_0\rangle\langle e_0|, \qquad a_P = \sum_{n\le P^2} \frac{\Lambda(n)\log n}{n} + O(\log P).$$

The primitive complement is $\mathcal{H}_P^\circ = e_0^\perp$ equipped with the Schur-shorted form.

Lemma 9.167 (rank-one pole isolation)**.** *The only divergent direction in the finite von Mangoldt Gram form is the pole vector e_0. Its self-pairing satisfies*

$$[e_0, e_0]_P = a_P = \tfrac{1}{2}(\log P^2)^2 + O(\log P),$$

and every vector $\phi \in \mathcal{S}_a^\circ$ has uniformly bounded unshorted primitive mass away from the e_0 direction.

Proof. By Definition 9.166, the pole contribution is a positive rank-one form supported on e_0. Lemma 9.165 gives

$$a_P = \sum_{n\le P^2} \frac{\Lambda(n)\log n}{n} + O(\log P) = \tfrac{1}{2}(\log P^2)^2 + O(\log P).$$

All non-pole prime cells are evaluated on the pole-shorted source profile. For $\phi \in \mathcal{S}_a^\circ$,

$$\sum_{p,k} (\log p) p^{-k/2} |\widehat{\phi}(k\log p)|^2 \le C \sum_{p,k} (\log p) p^{-k(1/2+2a_\phi)} < \infty$$

because $a_\phi > \frac{1}{2} > \frac{1}{4}$. Thus no source direction in the primitive core carries the Mertens divergence. $\square$

Lemma 9.168 (Schur shorting closes A3)**.** *The divergent pole contribution is a positive rank-one block, and Schur shorting it preserves the negative-square index. More precisely, if*

$$K_P = \begin{pmatrix} \alpha_P & \beta_P^* \\ \beta_P & C_P \end{pmatrix}, \qquad \alpha_P > 0,$$

then

$$K_P^\circ = C_P - \beta_P \alpha_P^{-1} \beta_P^*$$

satisfies

$$\mathrm{sq}_-(K_P) = \mathrm{sq}_-(\alpha_P) + \mathrm{sq}_-(K_P^\circ) = \mathrm{sq}_-(K_P^\circ).$$

Proof. Lemma 9.167 identifies the pole block as the positive scalar $\alpha_P = [e_0, e_0]_P > 0$ on the rank-one space $\mathbb{C}e_0$. Haynsworth inertia additivity follows from the congruence

$$\begin{pmatrix} I & 0 \\ -\beta_P\alpha_P^{-1} & I \end{pmatrix}^* K_P \begin{pmatrix} I & 0 \\ -\beta_P\alpha_P^{-1} & I \end{pmatrix} = \begin{pmatrix} \alpha_P & 0 \\ 0 & K_P^\circ \end{pmatrix}.$$

Since $\alpha_P > 0$, it contributes no negative square. $\square$

Definition 9.169 (finite source embedding)**.** For a finite cutoff P and $\phi \in \mathcal{S}_a^\circ$, define the unshorted primitive source vector $\widetilde{\iota}_P\phi$ in the finite prime-cell Hilbert space by its cell coordinates

$$(\widetilde{\iota}_P\phi)_{p,k} := \big((\log p)p^{-k/2}\big)^{1/2} \widehat{\phi}(k\log p), \qquad p^k \le P.$$

Equivalently, if the primitive cell carriers $u_{p,k}$ are normalized, then

$$\widetilde{\iota}_P\phi = \sum_{p^k\le P} \big((\log p)p^{-k/2}\big)^{1/2} \widehat{\phi}(k\log p)\, u_{p,k}.$$

The primitive source vector used in the resolvent compression is the Schur-shorted projection

$$\iota_P\phi := \Pi_P^\circ \widetilde{\iota}_P\phi \in \mathcal{H}_P^\circ.$$

For $F = \operatorname{span}\{\phi_1, \dots, \phi_m\}$, the source map is

$$\Phi_P^F(c_1, \dots, c_m) = \sum_{\alpha=1}^m c_\alpha \, \iota_P\phi_\alpha.$$

Lemma 9.170 (source embedding norm identity)**.** *Before Schur shorting,*

$$\|\widetilde{\iota}_P\phi\|^2 = \sum_{p^k \le P} (\log p) p^{-k/2} |\widehat{\phi}(k\log p)|^2.$$

After shorting,

$$\|\iota_P\phi\|^2_{\mathcal{H}_P^\circ} \le \|\widetilde{\iota}_P\phi\|^2.$$

Proof. The first identity is the norm in the orthogonal finite prime-cell direct sum of Definition 9.169. The second inequality is the monotonicity of Schur shorting, Lemma 9.172, applied to the positive pole block. □

Lemma 9.171 (compact logarithmic tests belong to the source core)**.** *Every $\phi \in C_c^\infty(0,\infty)$ belongs to $\mathcal{S}_a^\circ$.*

Proof. Let $\widehat{\phi}$ be the logarithmic Mellin profile and assume $\operatorname{supp}\widehat{\phi} \subset [\alpha, \beta] \subset (0,\infty)$. Fix any $a_\phi > \frac{1}{2}$. For each N,

$$e^{a_\phi t}(1+t)^N |\widehat{\phi}(t)|$$

is continuous on the compact interval $[\alpha, \beta]$ and is zero outside it, hence has a finite supremum C_N. Therefore

$$|\widehat{\phi}(t)| \le C_N e^{-a_\phi t}(1+t)^{-N}.$$

The pole-shorted condition is obtained by projecting away the single pole vector of Definition 9.166; this changes only a finite-dimensional component and leaves the compact logarithmic test in the primitive source space.

For the strip estimate, write $\operatorname{supp}\phi \subset [0,T]$ and fix $\varepsilon_\phi \in (0, a_\phi - \frac{1}{2})$. If $|\operatorname{Im} w| \le \frac{1}{2} + \varepsilon_\phi/2$, integration by parts gives, for every N,

$$|\ell_w(\phi)| = \left| \int_0^T \phi(t) e^{-iwt}\, dt \right| \le C_{N,\phi}(1 + |\operatorname{Re} w|)^{-N},$$

because all boundary terms vanish for a compactly supported smooth test and $d^N(e^{-iwt})/dt^N = (-iw)^N e^{-iwt}$, while $|e^{-iwt}| \le e^{(\frac{1}{2}+\varepsilon_\phi/2)T}$ on the enlarged closed substrip. Thus $\phi \in \mathcal{S}_a^\circ$. □

Lemma 9.172 (shorting does not increase primitive source mass)**.** *Let $K_P \ge 0$ be written in the pole/primitive splitting as in Lemma 9.168. For every primitive vector $x \in e_0^\perp$,*

$$\langle K_P^\circ x, x\rangle = \langle C_P x, x\rangle - \langle \alpha_P^{-1}\beta_P^* x, \beta_P^* x\rangle \le \langle C_P x, x\rangle.$$

Consequently passing from the unshorted primitive block to the Schur-shorted primitive form cannot increase the norm of a source vector.

Proof. Since $\alpha_P > 0$, the second term $\langle \alpha_P^{-1}\beta_P^* x, \beta_P^* x\rangle$ is nonnegative. The displayed inequality is immediate. This is the order-theoretic form of Schur shorting: $0 \le K_P^\circ \le C_P$ as quadratic forms on the primitive complement. □

Lemma 9.173 (uniform finite-channel source bound)**.** *For every $\phi \in \mathcal{S}_a^\circ$,*

$$\sup_P \|\iota_P \phi\|_{\mathcal{H}_P^\circ} < \infty.$$

Proof. By Lemma 9.170, before shorting the primitive diagonal mass is

$$\sum_{p,k} (\log p) p^{-k/2} |\widehat{\phi}(k \log p)|^2 \le C \sum_{p,k} (\log p) p^{-k(1/2+2a_\phi)}.$$

The series converges because $1/2 + 2a_\phi > 1$. The bound is independent of P. By Lemma 9.170, Schur shorting the pole mode can only decrease this primitive quadratic form. Therefore the same P-independent bound holds in $\mathcal{H}_P^\circ$. □

Corollary 9.174 (finite-channel normal family)**.** *For every finite source plane $F \subset \mathcal{S}_a^\circ$, the functions*

$$M_P^F(z) = (\Phi_P^F)^* (A_P^\circ - z)^{-1} \Phi_P^F$$

satisfy the uniform bound required in Theorem 9.186.

Proof. Combine self-adjointness of A_P° with Lemma 9.173:

$$\|M_P^F(z)\| \le \|\Phi_P^F\|^2 |\operatorname{Im} z|^{-1} \le C_F |\operatorname{Im} z|^{-1}.$$

□

Lemma 9.175 (subsequential spectral-measure compactness)**.** *Fix a finite source plane $F \subset \mathcal{S}_a^\circ$. For every sequence of cutoffs P_j there is a subsequence P_{j_ν} and a positive matrix measure Σ^F on $\mathbb{R}$, with finite total mass, such that*

$$M_{P_{j_\nu}}^F(z) \longrightarrow \int_{\mathbb{R}} \frac{d\Sigma^F(t)}{t - z}$$

locally uniformly on $\mathbb{C} \setminus \mathbb{R}$.

Proof. Let E_P be the spectral measure of the self-adjoint operator A_P°. Define the positive matrix-valued measure

$$\Sigma_P^F(B) := (\Phi_P^F)^* E_P(B) \Phi_P^F, \qquad B \subset \mathbb{R} \text{ Borel}.$$

The spectral theorem gives

$$M_P^F(z) = \int_{\mathbb{R}} \frac{d\Sigma_P^F(t)}{t - z}.$$

Corollary 9.174 and Lemma 9.173 give a P-independent bound on the total mass:

$$\Sigma_P^F(\mathbb{R}) = (\Phi_P^F)^* \Phi_P^F \le C_F I_F.$$

For each pair of basis vectors e_α, e_β in F, the scalar measures $\langle \Sigma_P^F e_\beta, e_\alpha \rangle$ therefore have uniformly bounded total variation. By Banach–Alaoglu, equivalently the Helly selection theorem for finite measures on the locally compact space $\mathbb{R}$, a diagonal subsequence converges weak-* against $C_0(\mathbb{R})$ for all finitely many matrix entries. The limit is a positive matrix measure Σ^F because positivity is preserved under weak-* limits: for every vector c, the scalar measures $\langle \Sigma_{P_{j_\nu}}^F c, c \rangle$ are positive and converge to $\langle \Sigma^F c, c \rangle$.

For each $z \in \mathbb{C} \setminus \mathbb{R}$, the function $t \mapsto (t - z)^{-1}$ belongs to $C_0(\mathbb{R})$, hence the displayed Cauchy transforms converge pointwise. On a compact $K \subset \mathbb{C} \setminus \mathbb{R}$, the kernels $(t - z)^{-1}$ are uniformly bounded and uniformly Lipschitz in z by constants depending only on $\operatorname{dist}(K, \mathbb{R})$. A finite-net argument upgrades pointwise convergence to uniform convergence on K. Thus convergence is locally uniform on $\mathbb{C} \setminus \mathbb{R}$. □

Theorem 9.176 (moment criterion for operator-tower convergence). *Fix a point $z_\bullet \in U_-$. For a finite source plane $F \subset \mathcal{S}_a^\circ$, the standing tower convergence*

$$M_P^F(z) \longrightarrow G_\Xi^F(z)$$

locally uniformly on U_- is equivalent to the convergence of all resolvent moments at $z_\bullet$:

$$(\Phi_P^F)^*(A_P^\circ - z_\bullet)^{-k-1}\Phi_P^F \longrightarrow \frac{1}{k!}\,\partial_z^k G_\Xi^F(z_\bullet), \qquad k = 0, 1, 2, \dots.$$

Proof. If $M_P^F \to G_\Xi^F$ locally uniformly on U_-, Cauchy's integral formula gives convergence of all derivatives at $z_\bullet$. Since

$$\partial_z^k (A_P^\circ - z)^{-1} = k!(A_P^\circ - z)^{-k-1},$$

the displayed moment convergence follows.

Conversely, assume all displayed moments converge. Corollary 9.174 makes $\{M_P^F\}_P$ a normal family on Ω_-. Let H be any subsequential locally uniform limit on Ω_-. Cauchy's formula again gives

$$\frac{1}{k!}\,\partial_z^k H(z_\bullet) = \lim_P (\Phi_P^F)^*(A_P^\circ - z_\bullet)^{-k-1}\Phi_P^F = \frac{1}{k!}\,\partial_z^k G_\Xi^F(z_\bullet).$$

The point $z_\bullet$ lies in U_-, where Ξ has no zeros because $\operatorname{Re} s > 1$ and the Euler product for ζ is nonzero there; hence G_Ξ^F is holomorphic near $z_\bullet$. The Taylor series of H and G_Ξ^F therefore agree in a disk around $z_\bullet$. By the identity theorem they agree on the connected component of U_-, and then on Ω_- after removing any apparent poles of the meromorphic target, because H is holomorphic on all of Ω_-. Every subsequential limit is the same; normality then implies the full sequence converges locally uniformly to G_Ξ^F on U_-. □

Theorem 9.177 (measure-form tower bridge). *Assume the standing operator-theoretic tower realizes tower-ARP, i.e. $M_P^F \to G_\Xi^F$ on U_- for every finite source plane F. Then every finite-channel subsequential spectral limit supplied by Lemma 9.175 has Cauchy transform equal to G_Ξ^F on U_- and holomorphically removes all possible poles of G_Ξ^F in Ω_-. Consequently RH follows.*

Proof. Fix F and a cutoff sequence. Lemma 9.175 gives a subsequence whose transforms converge locally uniformly on $\mathbb{C} \setminus \mathbb{R}$ to a Cauchy transform

$$H^F(z) = \int_{\mathbb{R}} \frac{d\Sigma^F(t)}{t - z}.$$

The function H^F is holomorphic on Ω_-. The tower convergence identifies the same subsequential limit with G_Ξ^F on U_-. Hence $H^F = G_\Xi^F$ on U_- and, by the identity theorem, on all of Ω_- after removing the poles of the meromorphic target. Lemma 9.187 detects every non-real zero by some finite channel, so no such pole can exist in Ω_-. The evenness and real symmetry of Ξ reflect the conclusion to the upper half-plane by Lemma 9.1(ii)–(iii): $\Xi(-z) = \Xi(z)$ gives $M_\Xi(-z) = -M_\Xi(z)$ and $\Xi(\bar z) = \overline{\Xi(z)}$ gives $M_\Xi(\bar z) = \overline{M_\Xi(z)}$ away from poles. Therefore all zeros of Ξ lie on the real z-axis, which is RH. □

9.11. **Step 6: positive Cauchy transforms have positive Pick kernels.** Steps 6–15 are carried out formally in Lemma 9.178, Definition 9.179, Theorem 9.181, Lemma 9.182, Theorem 9.185, Lemma 9.187, Theorem 9.189, and Corollary 9.190; the surrounding discussion explains their content but the formal statements are authoritative.

Let Σ be a positive finite matrix measure on $\mathbb{R}$ and

$$g(z) = \int_{\mathbb{R}} \frac{d\Sigma(t)}{t-z}.$$

Then, for real t,

$$\frac{1}{t-z} - \frac{1}{t-\bar{w}} = \frac{z-\bar{w}}{(t-z)(t-\bar{w})}.$$

Hence

$$\frac{g(z)-g(w)^*}{z-\bar{w}} = \int_{\mathbb{R}} \frac{d\Sigma(t)}{(t-z)(t-\bar{w})}.$$

For finite nodes z_i and vectors c_i,

$$\sum_{i,j} c_i^* \left[\int_{\mathbb{R}} \frac{d\Sigma(t)}{(t-z_i)(t-\bar{z}_j)}\right] c_j = \int_{\mathbb{R}} \left(\sum_i \frac{c_i}{t-\bar{z}_i}\right)^* d\Sigma(t) \left(\sum_j \frac{c_j}{t-\bar{z}_j}\right) \geq 0.$$

This proves the Pick positivity of all positive spectral Cauchy transforms.

Lemma 9.178 (Cauchy transforms have positive Pick kernels)**.** *Let Σ be a positive finite matrix measure on $\mathbb{R}$ and*

$$g(z) = \int_{\mathbb{R}} \frac{d\Sigma(t)}{t-z}.$$

Then, for all $z, w \in \mathbb{C} \setminus \mathbb{R}$ with $z \neq \bar{w}$ (in particular for all z, w in the same half-plane)*, the integral defining g converges absolutely since $|t-z| \geq |\operatorname{Im} z|$ and $\Sigma(\mathbb{R}) < \infty$, and*

$$\frac{g(z)-g(w)^*}{z-\bar{w}} = \int_{\mathbb{R}} \frac{d\Sigma(t)}{(t-z)(t-\bar{w})}$$

as a positive matrix kernel.

Proof. Since t is real,

$$\frac{1}{t-z} - \frac{1}{t-\bar{w}} = \frac{z-\bar{w}}{(t-z)(t-\bar{w})}.$$

Dividing by $z-\bar{w}$ gives the displayed identity. If $z_1, \ldots, z_N$ are nodes and $c_1, \ldots, c_N$ are vectors in the channel space, then

$$\sum_{i,j} c_i^* \left[\int_{\mathbb{R}} \frac{d\Sigma(t)}{(t-z_i)(t-\bar{z}_j)}\right] c_j = \int_{\mathbb{R}} \left(\sum_i \frac{c_i}{t-\bar{z}_i}\right)^* d\Sigma(t) \left(\sum_j \frac{c_j}{t-\bar{z}_j}\right) \geq 0.$$

Thus the block Pick matrices are positive semidefinite. ☐

9.12. **Step 7: tower-ARP implies ARP-P.** If a positive tower satisfies tower-ARP, then each finite cutoff produces a positive spectral Cauchy transform M_P^F. By Step 6 every finite Pick matrix of M_P^F is positive semidefinite. Tower-ARP gives pointwise convergence

$$M_P^F(z_i) \longrightarrow g^F(z_i)$$

for each finite node set. The cone of positive semidefinite matrices is closed, so

$$\mathsf{P}[g^F; z_1, \ldots, z_N] \succeq 0.$$

Thus tower-ARP is a stronger possible certificate for ARP-P, but it is not required by the minimal route.

Definition 9.179 (positive tower; tower-ARP)**.** A *positive tower* for the finite channel F is a sequence $\{\Sigma_P^F\}_P$ of positive finite matrix measures on $\mathbb{R}$ with $\sup_P \Sigma_P^F(\mathbb{R}) < \infty$, whose Cauchy transforms

$$M_P^F(z) = \int_{\mathbb{R}} \frac{d\Sigma_P^F(t)}{t-z}$$

converge pointwise on U_- to $g^F(z)$. The channel data satisfy *tower-ARP* if every finite source plane $F \subset \mathcal{S}_a^\circ$ admits a positive tower. Tower-ARP is a convergence property, not a per-cutoff identity.

Remark 9.180 (operator-theoretic towers). Any family of self-adjoint operators A_P on Hilbert spaces $\mathcal{H}_P$ with P-uniformly bounded source maps $\Phi_P^F : F \to \mathcal{H}_P$ produces the measures of Definition 9.179 through the spectral compressions $\Sigma_P^F(B) = (\Phi_P^F)^* E_P(B) \Phi_P^F$, for which $M_P^F(z) = (\Phi_P^F)^*(A_P - z)^{-1}\Phi_P^F$. The von Mangoldt canonical-system realization described above is one candidate family; the realization layer of Section 9.10 records its source bounds. The chain uses nothing beyond Definition 9.179.

Theorem 9.181 (tower-ARP implies ARP-P)**.** *If the channel data satisfy tower-ARP, Definition 9.179, then ARP-P holds.*

Proof. Each M_P^F is, by definition, the Cauchy transform of a positive finite matrix measure. Hence Lemma 9.178 gives

$$\mathsf{P}[M_P^F; z_1, \dots, z_N] \succeq 0$$

for every finite node set in U_-. Tower-ARP gives $M_P^F(z_i) \to g^F(z_i)$ at each node. Since the cone of positive semidefinite matrices is closed, the limit matrix $\mathsf{P}[g^F; z_1, \dots, z_N]$ is positive semidefinite. □

9.13. **Step 8: ARP-P gives Nevanlinna–Pick interpolants.** Fix a finite channel F and a finite node set $z_1, \dots, z_N \subset U_-$. ARP-P says exactly that the matrix Pick condition for the lower half-plane is satisfied. The classical matrix-valued Nevanlinna–Pick theorem therefore supplies a holomorphic matrix function F_N on Ω_- such that

$$F_N(z_j) = g^F(z_j), \qquad j = 1, \dots, N,$$

and

$$\frac{F_N(z) - F_N(w)^*}{z - \bar{w}} \succeq 0.$$

The hypotheses of the classical theorem are precisely the finite node condition, the target matrices $g^F(z_j)$, and the positive semidefiniteness of the displayed block Pick matrix; these are exactly provided by Step 5.

The classical theorem is used in its degenerate form: positive *semi*definiteness of the block Pick matrix, singular cases included, is necessary and sufficient for solvability in the matrix Pick class. The exact statement used, in the half-plane normalization of this paper and with the Cayley congruence written out, is displayed as Proposition 9.183 below; its disk core is the commutant-lifting Schur–Pick theorem [78]. ARP-P supplies exactly this semidefiniteness, so no strict positivity is assumed anywhere in the bridge.

Lemma 9.182 (pinned normality of the matrix Pick class)**.** *Let $\{F_n\}$ be holomorphic $M_m(\mathbb{C})$-valued functions on Ω_- with $\operatorname{Im} F_n \preceq 0$, and suppose $F_n(z_1) = W_1$ for all n, at some fixed $z_1 \in \Omega_-$. Then a subsequence converges locally uniformly on $\Omega_- \setminus Z$, for a discrete set Z, to a function which extends holomorphically to all of Ω_- as a matrix Pick-class function F; moreover $F(z) = \lim_k F_{n_k}(z)$ whenever the right side eventually stabilizes or converges.*

Proof. Cayley step. For W with $\operatorname{Im} W \preceq 0$ one computes, for every vector x,

$$\|(W-iI)x\|^2 = \|Wx\|^2 + \|x\|^2 - 2\operatorname{Im}\langle Wx, x\rangle \geq \|x\|^2, \qquad \|(W+iI)x\|^2 \leq \|(W-iI)x\|^2.$$

Thus $W-iI$ is invertible with $\|(W-iI)^{-1}\| \leq 1$ and $S := (W+iI)(W-iI)^{-1}$ is a contraction, with the exact identities

$$S - I = 2i(W-iI)^{-1}, \qquad W = i(S+I)(S-I)^{-1}.$$

Thus $S_n(z) := (F_n(z)+iI)(F_n(z)-iI)^{-1}$ is a holomorphic contraction-valued family.

Montel and pinning. The entries of S_n are bounded by 1, so a diagonal subsequence converges locally uniformly to a holomorphic contraction-valued S_∞. At the pinned node, $S_\infty(z_1) - I = 2i(W_1 - iI)^{-1}$ is invertible. Hence $\varphi := \det(S_\infty - I)$ is holomorphic and not identically zero, and $Z := \{\varphi = 0\}$ is discrete in Ω_-.

Back-transform off Z. On compacts of $\Omega_- \setminus Z$, $|\varphi|$ is bounded below, so $(S_{n_k} - I)^{-1} \to (S_\infty - I)^{-1}$ uniformly and $F_{n_k} \to F := i(S_\infty + I)(S_\infty - I)^{-1}$ locally uniformly; F is holomorphic with $\operatorname{Im} F \preceq 0$ on $\Omega_- \setminus Z$ as a limit of such.

Removability across Z. Fix $\zeta_0 \in Z$ and a unit vector c, and set $f := c^* F c$ on a punctured disk around ζ_0 inside $\Omega_- \setminus Z$. Since $\operatorname{Im} f \leq 0$, $|f - i| \geq 1$, so $h := (f-i)^{-1}$ is holomorphic and bounded by 1, hence extends across ζ_0 by Riemann's theorem. If $h(\zeta_0) = 0$, then $f = i + 1/h$ would have a pole at ζ_0; near a pole the values of f cover all sufficiently large moduli in every direction of the plane, in particular values with positive imaginary part, contradicting $\operatorname{Im} f \leq 0$. Hence $h(\zeta_0) \neq 0$ and f extends holomorphically. Applying this to $c \in \{e_j,\, e_k,\, e_j + e_k,\, e_j + ie_k\}$ and polarizing recovers every entry of F, which therefore extends holomorphically across Z; the extension is Pick-class by continuity.

Node values. If $F_{n_k}(w) = W$ for all large k at some $w \in \Omega_-$, then when $w \notin Z$ this passes to the limit directly. When $w \in Z$, choose a circle around w inside $\Omega_- \setminus Z$ and use the Cauchy integral:

$$F(w) = \frac{1}{2\pi i}\oint \frac{F(\zeta)}{\zeta - w}\, d\zeta = \lim_k \frac{1}{2\pi i}\oint \frac{F_{n_k}(\zeta)}{\zeta - w}\, d\zeta = W,$$

since each F_{n_k} is holomorphic at w. ☐

Proposition 9.183 (degenerate matrix Nevanlinna–Pick theorem, half-plane form)**.** *Let $z_1, \ldots, z_N \in \Omega_-$ be pairwise distinct and let $W_1, \ldots, W_N \in M_m(\mathbb{C})$. There exists a holomorphic $F : \Omega_- \to M_m(\mathbb{C})$ with $\operatorname{Im} F \preceq 0$ and $F(z_j) = W_j$ for all j if and only if the block Pick matrix*

$$\left[\frac{W_i - W_j^*}{z_i - \bar{z}_j}\right]_{i,j=1}^N$$

is positive semi*definite; singular (degenerate) Pick matrices are included.*

Proof. Reduction to the disk. The Möbius map $\zeta(z) = (z+i)/(z-i)$ takes Ω_- biholomorphically onto the unit disk $\mathbb{D}$, and the node images $\zeta_j := \zeta(z_j)$ are pairwise distinct. If the Pick matrix is positive semidefinite, its diagonal blocks give $(W_j - W_j^*)/(2i \operatorname{Im} z_j) \succeq 0$ with $\operatorname{Im} z_j < 0$, hence $\operatorname{Im} W_j \preceq 0$; by the Cayley computation in the proof of Lemma 9.182, $W_j - iI$ is then invertible and $S_j := (W_j + iI)(W_j - iI)^{-1}$ is a contraction, with $S_j - I = 2i(W_j - iI)^{-1}$ invertible. The algebraic identities

$$I - S_i S_j^* = 2i\,(W_i - iI)^{-1}(W_i - W_j^*)\,(W_j^* + iI)^{-1}, \qquad 1 - \zeta_i \bar{\zeta}_j = \frac{2i\,(z_i - \bar{z}_j)}{(z_i - i)(\bar{z}_j + i)},$$

verified by direct expansion (the factors $(W \pm iI)$ and their inverses commute), give the exact block congruence

$$\left[\frac{I - S_i S_j^*}{1 - \zeta_i \bar{\zeta}_j}\right] = D \left[\frac{W_i - W_j^*}{z_i - \bar{z}_j}\right] D^*, \qquad D = \operatorname{diag}\big((z_i - i)(W_i - iI)^{-1}\big)_{i=1}^N,$$

with D invertible. Hence the half-plane Pick matrix is positive semidefinite exactly when the disk Schur–Pick matrix is.

Sufficiency. Assume the Pick matrix is positive semidefinite. By the classical matrix Schur–Pick theorem via commutant lifting [78] — valid verbatim for positive *semi*definite, possibly singular, Pick matrices — there is a holomorphic contraction-valued $S : \mathbb{D} \to M_m(\mathbb{C})$ with $S(\zeta_j) = S_j$. Since $S(\zeta_1) - I = S_1 - I$ is invertible, $\varphi := \det(S \circ \zeta - I) \not\equiv 0$ and its zero set $Z \subset \Omega_-$ is discrete and avoids every node. On $\Omega_- \setminus Z$ define $F := i(S \circ \zeta + I)(S \circ \zeta - I)^{-1}$. For a contraction S_0 with $S_0 - I$ invertible and $u := (S_0 - I)^{-1}x$,

$$\operatorname{Im}\langle i(S_0 + I)(S_0 - I)^{-1}x,\, x\rangle = \|S_0 u\|^2 - \|u\|^2 \le 0,$$

so $\operatorname{Im} F \preceq 0$ off Z. The removability step in the proof of Lemma 9.182 (scalar compressions c^*Fc, boundedness of $(c^*Fc - i)^{-1}$, and polarization) extends F holomorphically across Z with $\operatorname{Im} F \preceq 0$ on all of Ω_-. At the nodes, the exact inverse-Cayley identity $W = i(S + I)(S - I)^{-1}$ of Lemma 9.182 gives $F(z_j) = W_j$.

Necessity. If such an F exists, then $S := (F + iI)(F - iI)^{-1} \circ \zeta^{-1}$ is a holomorphic contraction on $\mathbb{D}$ by the same Cayley computation, the disk Schur–Pick matrix at $(\zeta_j, S(\zeta_j))$ is positive semidefinite by the classical theorem [78], and the displayed congruence transfers the positivity back to the half-plane Pick matrix. □

Lemma 9.184 (prefix serialization: single-channel Pick positivity removes all poles)**.** *Let G be an $M_m(\mathbb{C})$-valued meromorphic function on Ω_-. Let $\{z_j\}_{j\ge1} \subset \Omega_-$ be pairwise distinct with an accumulation point $z_* \in \Omega_-$, and suppose G is holomorphic at every z_j and in a neighborhood of z_*. If*

$$\mathsf{P}[G; z_1, \dots, z_N] = \left[\frac{G(z_i) - G(z_j)^*}{z_i - \bar{z}_j}\right]_{i,j=1}^N \succeq 0 \qquad \textit{for every } N \ge 1,$$

then every pole of G in Ω_- is removable, and the resulting holomorphic extension is a matrix Pick-class function ($\operatorname{Im} G \preceq 0$) on Ω_-.

Proof. For each N, Proposition 9.183 supplies a holomorphic F_N on Ω_- with $\operatorname{Im} F_N \preceq 0$ and $F_N(z_j) = G(z_j)$ for $j \le N$. Every F_N interpolates the pinned node z_1 with the common value $G(z_1)$, so Lemma 9.182 produces a subsequence converging to a holomorphic matrix Pick-class function H on Ω_- with $H(z_j) = G(z_j)$ for every j, each node being eventually interpolated. Both H and G are holomorphic near z_*, and $\{z_j\}$ accumulates at z_*; the identity theorem gives $H = G$ on a neighborhood of z_*. Since Ω_- is connected and G is meromorphic there, $G = H$ on Ω_- minus the polar set of G; hence every pole of G in Ω_- is removable and the extension is H. □

Theorem 9.185 (ARP-P implies RH)**.** *Assume ARP-P. Then every nontrivial zero of ζ has real part $\frac{1}{2}$.*

Proof. Fix a finite channel F and choose a countable, pairwise distinct node set $\{z_j\}_{j\ge1} \subset U_-$ with an accumulation point $z_* \in U_-$. ARP-P gives $\mathsf{P}[g^F; z_1, \dots, z_N] \succeq 0$ for every N. By Lemma 9.6, G_Ξ^F is meromorphic on $\mathbb{C}$ and equals g^F on U_-; its poles lie in the zero strip $|\operatorname{Im} z| < \frac{1}{2}$ (Lemma 9.1(i)), so it is holomorphic at every node and near z_*. Lemma 9.184

therefore removes every pole of G^F_Ξ in Ω_-. Lemma 9.187 detects every non-real zero in some source channel, so no zero of Ξ can correspond to a point in Ω_-. The symmetries $M_\Xi(-z) = -M_\Xi(z)$ and $M_\Xi(\bar z) = \overline{M_\Xi(z)}$ reflect the conclusion to the upper half-plane. Therefore all zeros of Ξ lie on the real z-axis, which is RH. □

9.14. **Step 9: NP normality and Montel extension.** Choose a countable node set $\{z_j\}_{j\ge1} \subset U_-$ with an accumulation point in U_-. For each N, Step 8 gives a Pick-class interpolant F_N through the first N nodes. By Lemma 9.182, applied with the pinned node z_1 interpolated by every F_N, a subsequence converges to a holomorphic matrix Pick-class function H on Ω_-, and $H(z_j) = g^F(z_j)$ for every j, since each node is eventually interpolated. Since the node set accumulates in U_- and both H and g^F are holomorphic there, the identity theorem gives

$$H = g^F \qquad \text{on } U_-.$$

Thus g^F has a holomorphic extension to all of Ω_-.

Theorem 9.186 (finite-channel Vitali bridge)**.** *Let $M^F_P : \Omega_- \to M_m(\mathbb{C})$ be holomorphic functions satisfying*

$$\|M^F_P(z)\| \le \frac{C_F}{|\operatorname{Im} z|}$$

locally uniformly in P. If $M^F_P(z) \to G^F_\Xi(z)$ pointwise on the nonempty open set U_-, where G^F_Ξ is meromorphic on Ω_-, then G^F_Ξ has no pole in Ω_-.

Proof. The estimate makes $\{M^F_P\}$ a normal family on Ω_- by Montel. Let H be any subsequential locally uniform limit. Then H is holomorphic on Ω_-. On U_- the full sequence converges to G^F_Ξ, hence $H = G^F_\Xi$ on U_-. The identity theorem forces every subsequential limit to coincide on all of Ω_-. The full sequence therefore converges locally uniformly to a holomorphic extension of $G^F_\Xi|_{U_-}$, and all poles of G^F_Ξ in Ω_- are removable. □

9.15. **Step 10: residue detection by channels.** Let z_ρ be a non-real zero of Ξ. The principal part of G^F_Ξ in a one-dimensional channel $F = \operatorname{span}\{\phi\}$ is proportional to

$$\ell_{z_\rho}(\phi)\overline{\ell_{\bar z_\rho}(\phi)}.$$

Because $C_c^\infty(0,\infty) \subset \mathcal{S}_a^\circ$, neither functional ℓ_{z_ρ} nor $\ell_{\bar z_\rho}$ can vanish on the whole source core: if ℓ_a vanished on all compactly supported smooth tests, then the distribution e^{-iat} would vanish on $(0,\infty)$, impossible. The kernels of two nonzero linear functionals cannot cover the whole vector space. Therefore one can choose $\phi \in C_c^\infty(0,\infty)$ with

$$\ell_{z_\rho}(\phi)\overline{\ell_{\bar z_\rho}(\phi)} \ne 0.$$

That channel detects the pole at z_ρ.

Lemma 9.187 (residue detection by the source core)**.** *For the source core $\mathcal{S}_a^\circ$ of Definition 9.3, and for every non-real pole z_ρ of M_Ξ, there is a one-dimensional source plane $F = \operatorname{span}\{\phi\} \subset \mathcal{S}_a^\circ$ such that G^F_Ξ has nonzero residue at z_ρ.*

Proof. At a zero z_ρ of multiplicity m_ρ, M_Ξ has principal part $-m_\rho/(z - z_\rho)$. The residue in a source channel is a scalar multiple of $\ell_{z_\rho}(\phi)\overline{\ell_{\bar z_\rho}(\psi)}$, where in the logarithmic model

$$\ell_a(\phi) = \int_0^\infty \phi(t)e^{-iat}\,dt.$$

By Definition 9.3, $C_c^\infty(0,\infty) \subset \mathcal{S}_a^\circ$. If ℓ_{z_ρ} vanished on all $C_c^\infty(0,\infty)$, the locally integrable distribution $e^{-iz_\rho t}$ would be zero on $(0,\infty)$. This is impossible: on any compact interval it is a continuous function, and a continuous function whose distribution is zero is identically zero, whereas $e^{-iz_\rho t} \neq 0$ for every real t. The same argument applies to $\ell_{\bar{z}_\rho}$. Hence the kernels of ℓ_{z_ρ} and $\ell_{\bar{z}_\rho}$ are two proper linear subspaces of $C_c^\infty(0,\infty)$. Their union cannot be the whole vector space: if u is outside the first kernel and v is outside the second, then either u is outside both kernels or v is outside both kernels or $u+v$ is outside both. Choose $\phi \in C_c^\infty(0,\infty) \subset \mathcal{S}_a^\circ$ outside both kernels. Then $\ell_{z_\rho}(\phi)\overline{\ell_{\bar{z}_\rho}(\phi)} \neq 0$, and for $F = \operatorname{span}\{\phi\}$ the residue of G_Ξ^F is therefore nonzero. ☐

9.16. **Step 11: ARP-P removes off-line poles in Ω_-.** Assume ARP-P. Step 9 gives a holomorphic extension of each $G_\Xi^F|_{U_-}$ to Ω_-. But Step 3 says G_Ξ^F is meromorphic with explicitly known principal parts. Hence every pole of G_Ξ^F in Ω_- is removable. Step 10 says every non-real zero of Ξ lying in Ω_- is visible as a non-removable pole in some finite channel. Therefore no such zero exists.

9.17. **Step 12: symmetries cover all of $\mathbb{C} \setminus \mathbb{R}$.** By Lemma 9.1(ii)–(iii), the completed function satisfies $\Xi(-z) = \Xi(z)$ and is real on the real axis. Consequently

$$M_\Xi(-z) = -M_\Xi(z), \qquad M_\Xi(\bar{z}) = \overline{M_\Xi(z)}$$

away from poles. A non-real zero in the upper half-plane reflects by these symmetries to a non-real zero in Ω_-. Step 11 rules out the latter under ARP-P. Hence ARP-P rules out all non-real zeros of Ξ.

9.18. **Step 13: real zeros of Ξ are RH.** By Step 1, a zero $\rho = \beta + i\gamma$ corresponds to $z_\rho = \gamma - i(\beta - \frac{1}{2})$. This number is real if and only if $\beta = \frac{1}{2}$. Thus all zeros of Ξ being real is exactly the Riemann Hypothesis.

Theorem 9.188 (analytic bridge implies RH)**.** *Assume that for every finite $F \subset \mathcal{S}_a^\circ$ there are functions M_P^F satisfying the hypotheses of Theorem 9.186 and converging to G_Ξ^F on U_-. Then RH holds.*

Proof. If M_Ξ had a pole in Ω_-, Lemma 9.187 would choose a finite channel where that pole is visible, contradicting Theorem 9.186. By Lemma 9.1(ii)–(iii), poles in $\mathbb{C}^+$ are reflected from poles in Ω_-. Explicitly, $\Xi(-z) = \Xi(z)$ gives

$$M_\Xi(-z) = -M_\Xi(z),$$

and the reality of Ξ on $\mathbb{R}$ gives $M_\Xi(\bar{z}) = \overline{M_\Xi(z)}$ away from poles. Hence a non-real pole in any quadrant reflects to a pole in Ω_-. Therefore every pole of $M_\Xi = -\Xi'/\Xi$ is real, so every zero of Ξ is real. By Lemma 9.2, this is RH. ☐

9.19. **Step 14: RH implies ARP-P.** Assume RH. Then every z_ρ is real, the target becomes the Cauchy transform of a positive finite matrix measure supported on the real zero coordinates, and Step 6 gives the Pick positivity of every finite node matrix. The formal statement and proof follow.

Theorem 9.189 (RH implies ARP-P)**.** *Assume RH. Then ARP-P holds.*

Proof. Under RH every zero coordinate z_ρ is real. For each finite source plane F, Lemma 9.6 and the rapid decay of the source functionals give a positive finite matrix measure

$$\Sigma_\Xi^F = \sum_\rho m_\rho \left[\ell_{z_\rho}(\phi_\beta)\overline{\ell_{z_\rho}(\phi_\alpha)}\right]_{\alpha,\beta} \delta_{z_\rho}.$$

The total mass is finite because the entries decay faster than any power along the real zero set and $N(T) = O(T \log T)$. The target is its Cauchy transform: under RH $\bar{z}_\rho = z_\rho$, so the general channel weight $\ell_{z_\rho}(\phi_\beta)\overline{\ell_{\bar{z}_\rho}(\phi_\alpha)}$ in (9) is exactly the displayed rank-one positive atom. Thus

$$g^F(z) = \int_{\mathbb{R}} \frac{d\Sigma_\Xi^F(t)}{t - z}.$$

Lemma 9.178 gives the positive semidefiniteness of every finite Pick matrix. Hence ARP-P holds. □

9.20. **Step 15: ARP-P $\iff$ RH.** Steps 8–13 prove

$$\text{ARP-P} \Longrightarrow \text{RH}.$$

Step 14 proves

$$\text{RH} \Longrightarrow \text{ARP-P}.$$

Therefore the route has the exact structural status

$$\boxed{\text{ARP-P} \iff \text{RH}}.$$

This is not a claim that RH is proved; it is a proof that the remaining mathematical work has been localized to the arithmetic Pick positivity of Step 5.

Corollary 9.190 (the ARP-P equivalence)**.**

$$\text{ARP-P} \iff \text{RH}.$$

Proof. The forward implication is Theorem 9.185. The reverse implication is Theorem 9.189. □

Corollary 9.191 (exact status of the remaining gap)**.** *ARP-P is equivalent to RH and is implied by tower-ARP:*

$$\text{tower-ARP} \Longrightarrow \text{ARP-P} \iff \text{RH}.$$

Thus the remaining essential input of the route is not FCC, determinant normality, or a per-cutoff trace-log identity. It is the arithmetic Pick positivity of Definition 9.10.

10. Epilogue

It is worth saying plainly what made this hard, and where, honestly, it still is hard.

The programme's real product is not a finished proof but a *map*: a precise account of where the difficulty of the Riemann Hypothesis lives. Part I is the negative geography — the no-gos and the seven structural walls, each carrying its explicit logical classification (equivalence, framework obstruction, or insufficiency theorem; §4), all unified by the single master quantifier (the passage from average, almost-all, or interior to specific, uniform, or boundary). Part II is the proof path those walls shaped: the ARP-P chain, which does not try to cross any wall but instead *reduces* the Hypothesis, step by step, to one inequality — the terminal-defect positivity Ω_7, $\delta_N \geq 0$ for all N, equivalently $\lambda_n \geq 0$ for all n. Fourteen of the fifteen Steps of the implication chain are closed in the precise sense recorded in the status map (§7.1); the sole unresolved input occurs within Step 5. The internal reduction of Step 5 terminates at Ω_7, which is equivalent to Li–Keiper positivity and hence to RH; we do not close it, and we say so.

The value of the reduction is that it names the residue exactly. When the roads explored in this programme repeatedly ended at the same wall, the task was not to find another road but to understand the wall so precisely that one can state it as a single inequality, and then to attack *that*. We attacked it, from four independent directions — an archimedean partition, a stationary-phase saddle, a positive de Branges kernel, and a perturbative archimedean domination — and each returned, provably, to the walls of Part I: the partition to the uniform-norm wall, the perturbative bound to the wrong-sign wall (a logarithmic margin against an unbounded perturbation), the positive kernel to the Weil-positivity wall (the de Branges positivity conditions being falsified by Conrey–Li), and the saddle to the same magnitude-blind cancellation the programme met at the start. That Ω_7 resists all four in the *same* way is not a failure of effort; it is the sharpest statement the programme can make about why the last step is the whole problem.

The road to this map was long and, for most of its length, negative. The author recalls printing a particular computation three times because each printing revealed a sign error introduced in correcting the previous one; recalls confirming, once again, that positivity is orthogonal to magnitude, and later that it is a *sign* property no size functional can see; recalls a great many letters and emails, some of them stern, every one of them useful, in which an objection one had hoped was pedantic turned out to be fatal and, in being fatal, pointed at the next door. The chain in Part II is built almost entirely out of the wreckage of those objections.

A last word, in the same register of honesty in which the programme was conducted. We are aware of how many careful arguments for this problem have failed at a step their authors believed secure, and we do not exempt ourselves — two of our own results are recorded above as withdrawn. We have written every closed step as a proved statement and named the classical facts it rests on; we have isolated the one place where all the weight is carried, Ω_7, and asked, explicitly, that it be attacked. What we offer is a complete *reduction* and an honest open problem — not a settled theorem. A result of this kind is not finished when its authors are persuaded; it is finished when the community is. The standard we ask the reader to hold this paper to is the one recorded in §3.9: read each claim at the exact strength of its classification, and where a claim fails that reading, treat it as we treated our own withdrawn results. We submit the programme and the path in that spirit.

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